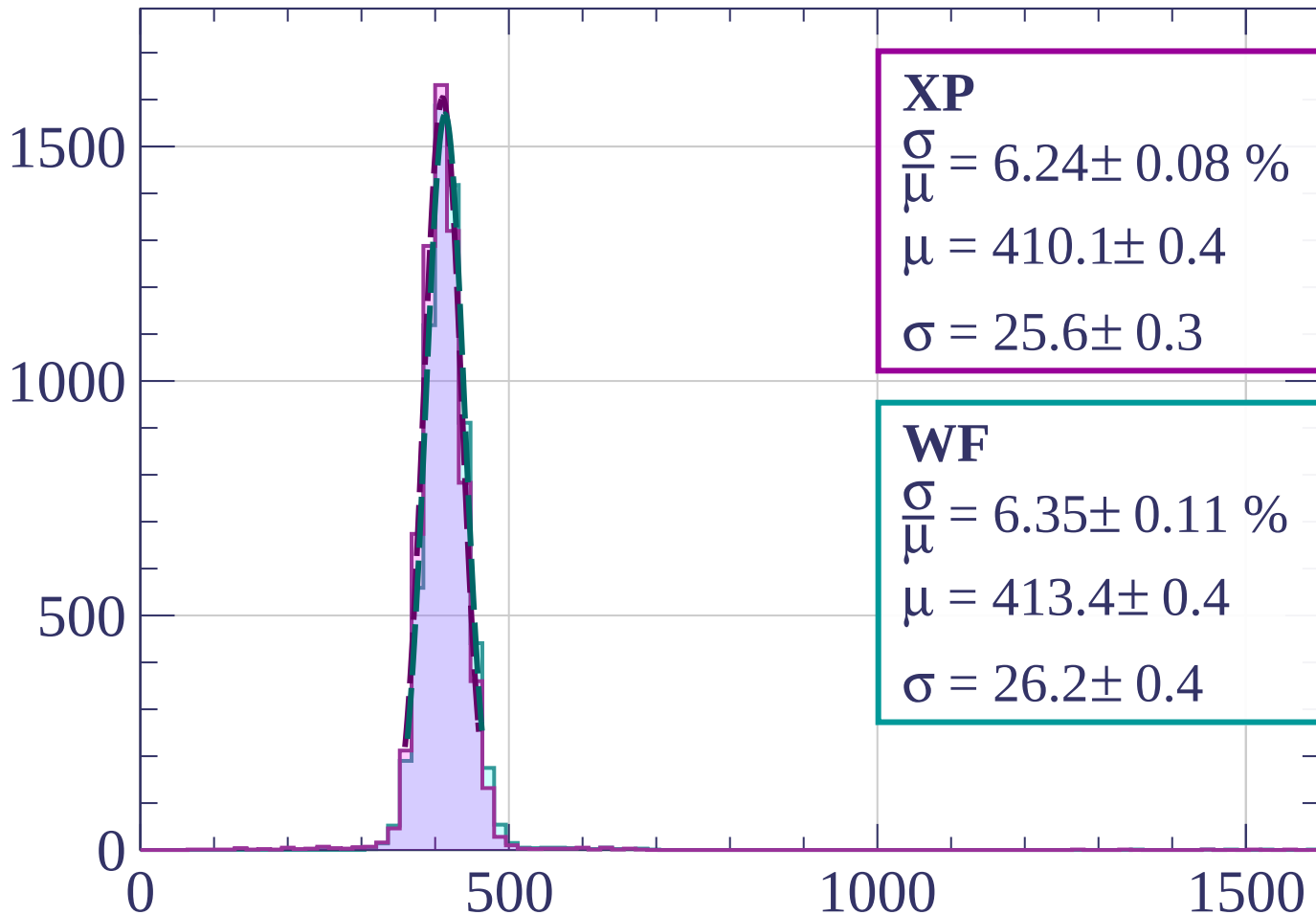


Count



XP

$$\frac{\sigma}{\mu} = 6.24 \pm 0.08 \%$$

$$\mu = 410.1 \pm 0.4$$

$$\sigma = 25.6 \pm 0.3$$

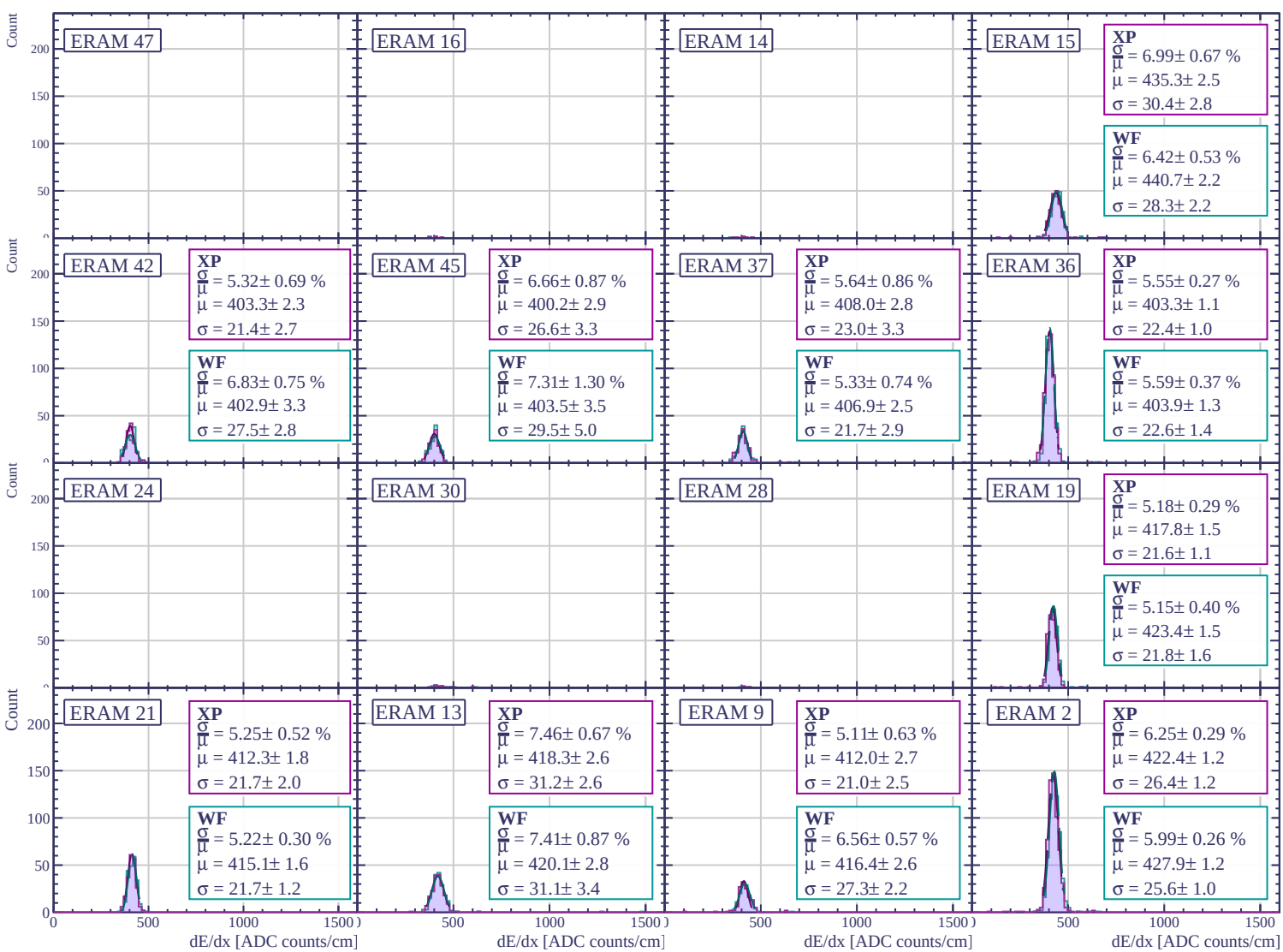
WF

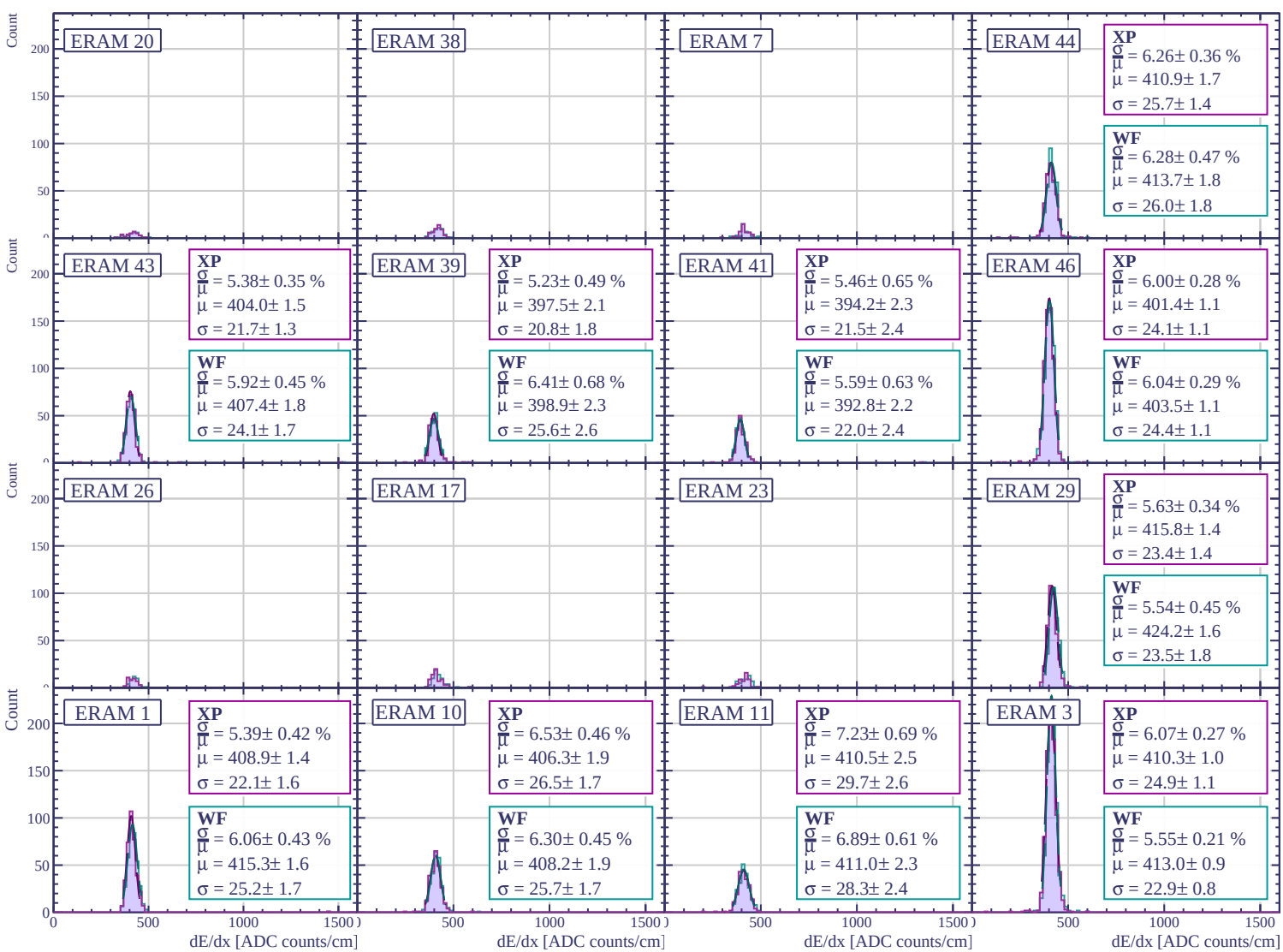
$$\frac{\sigma}{\mu} = 6.35 \pm 0.11 \%$$

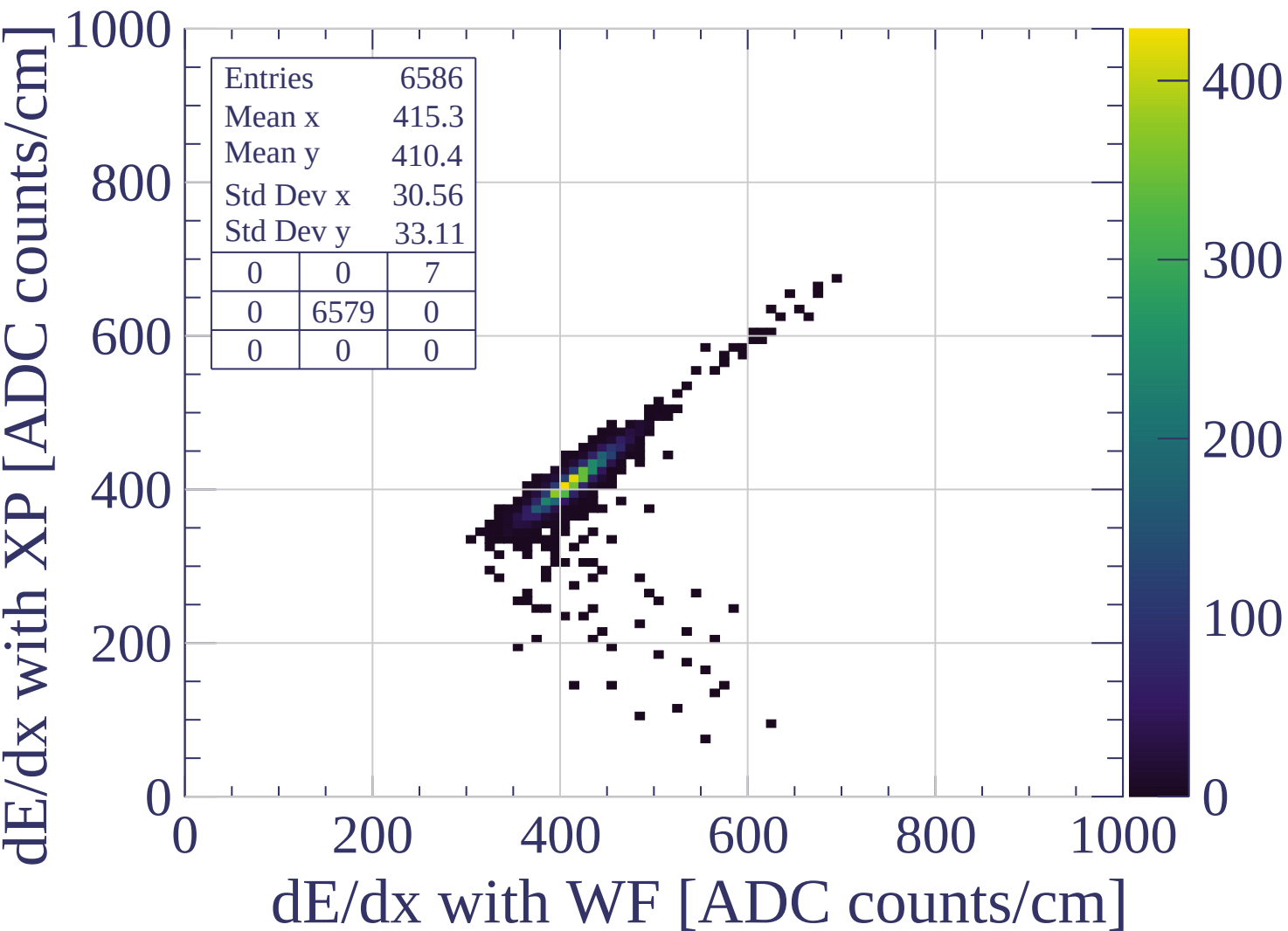
$$\mu = 413.4 \pm 0.4$$

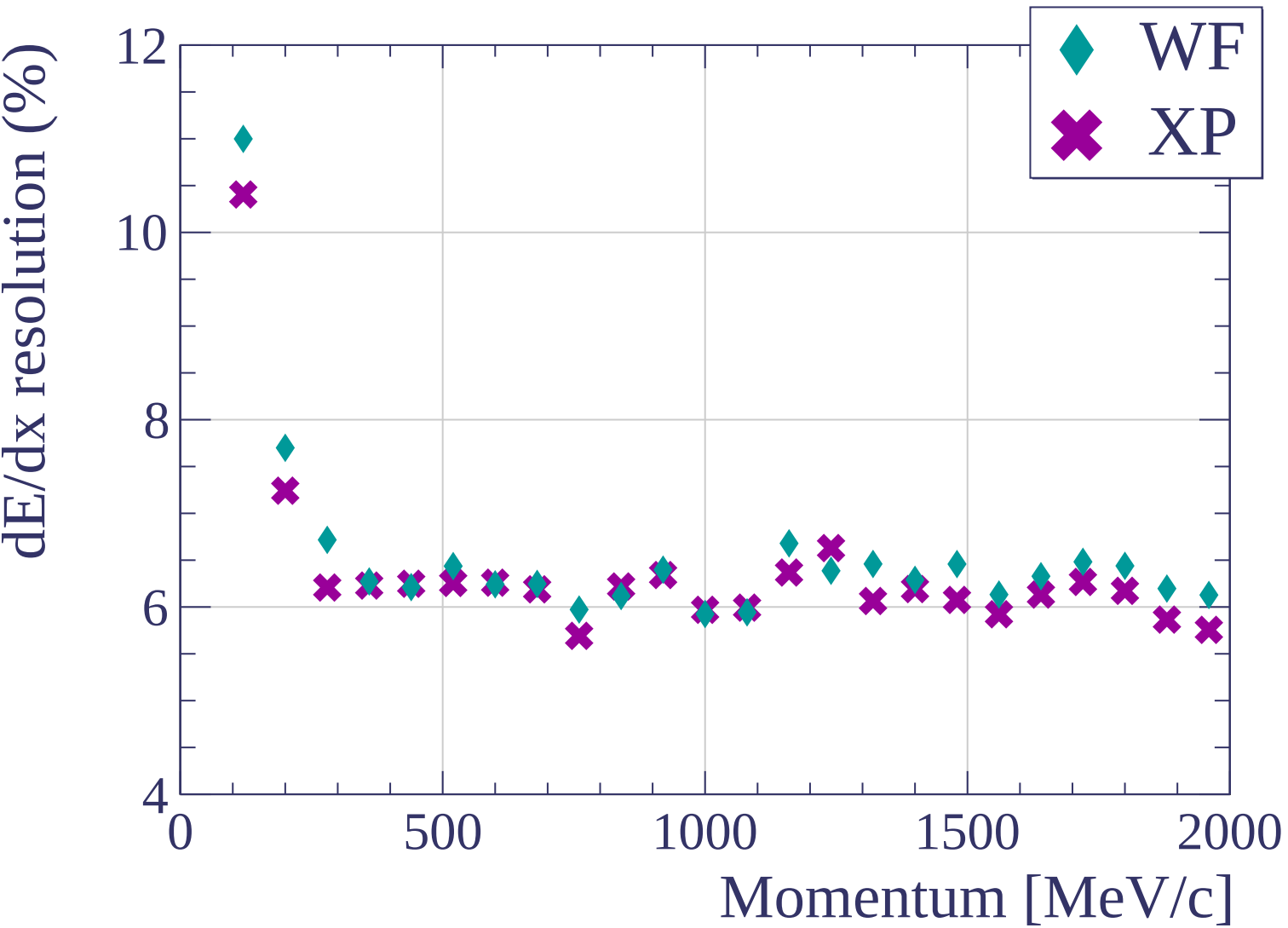
$$\sigma = 26.2 \pm 0.4$$

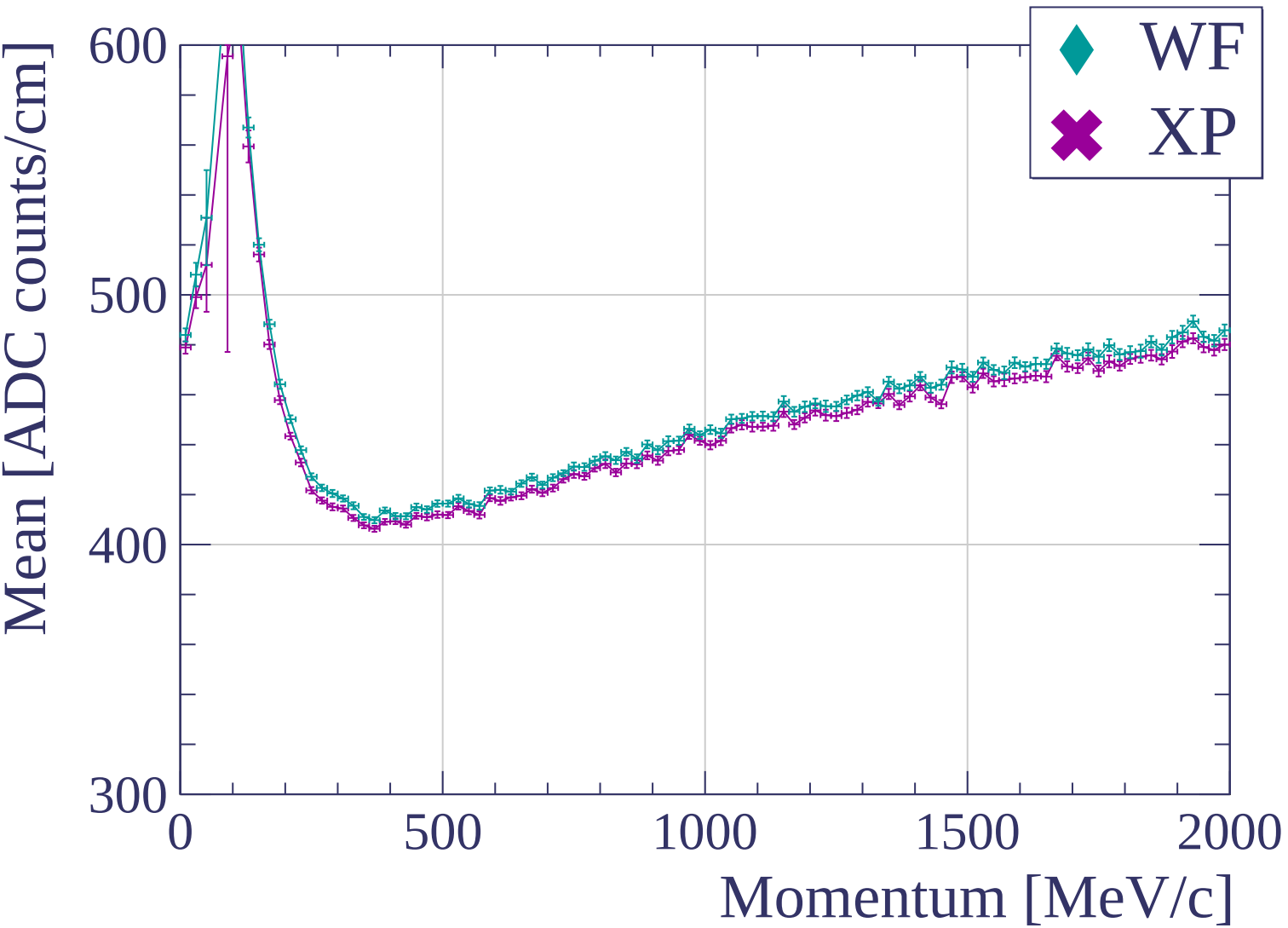
dE/dx [ADC counts/cm]

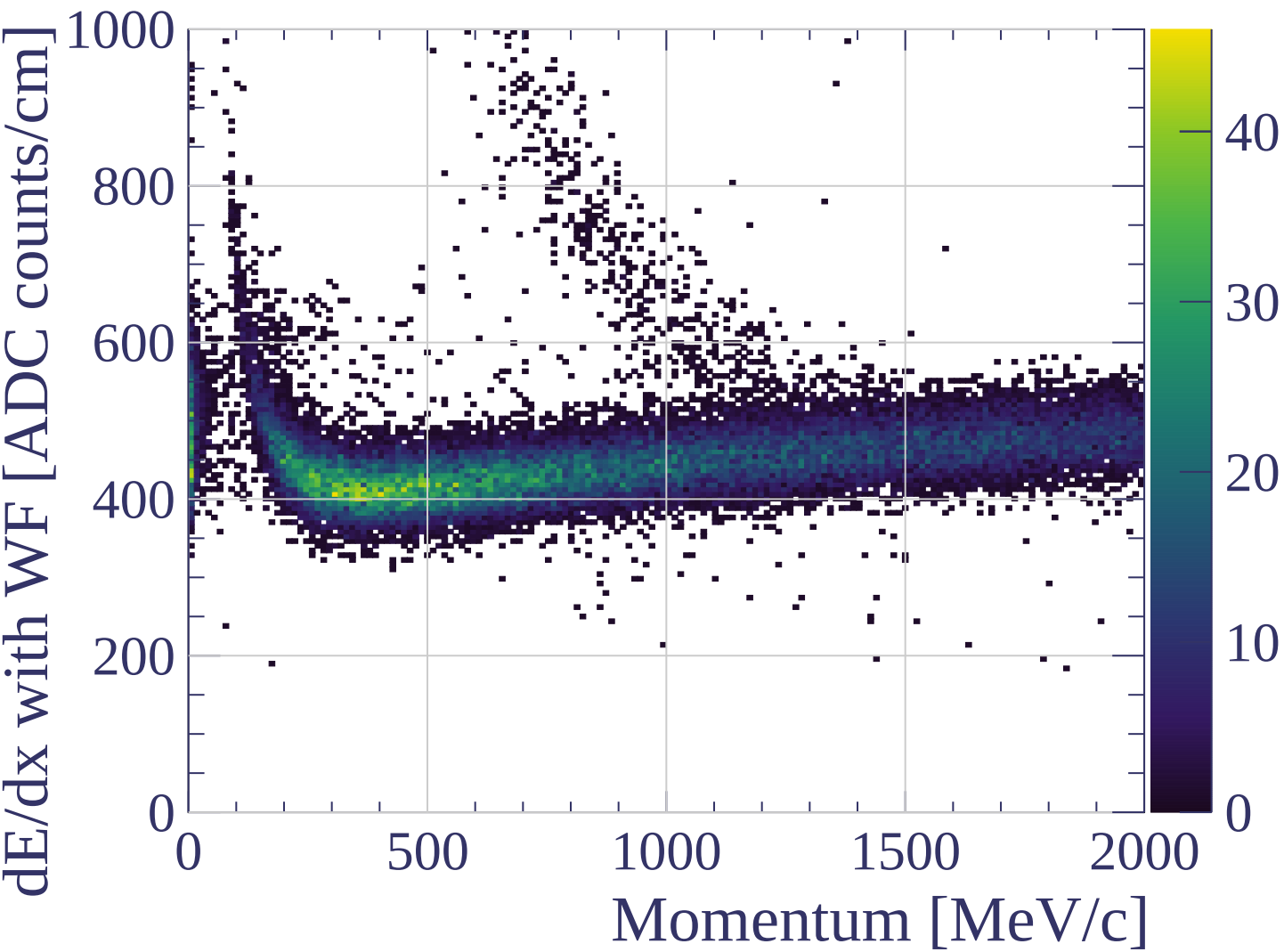


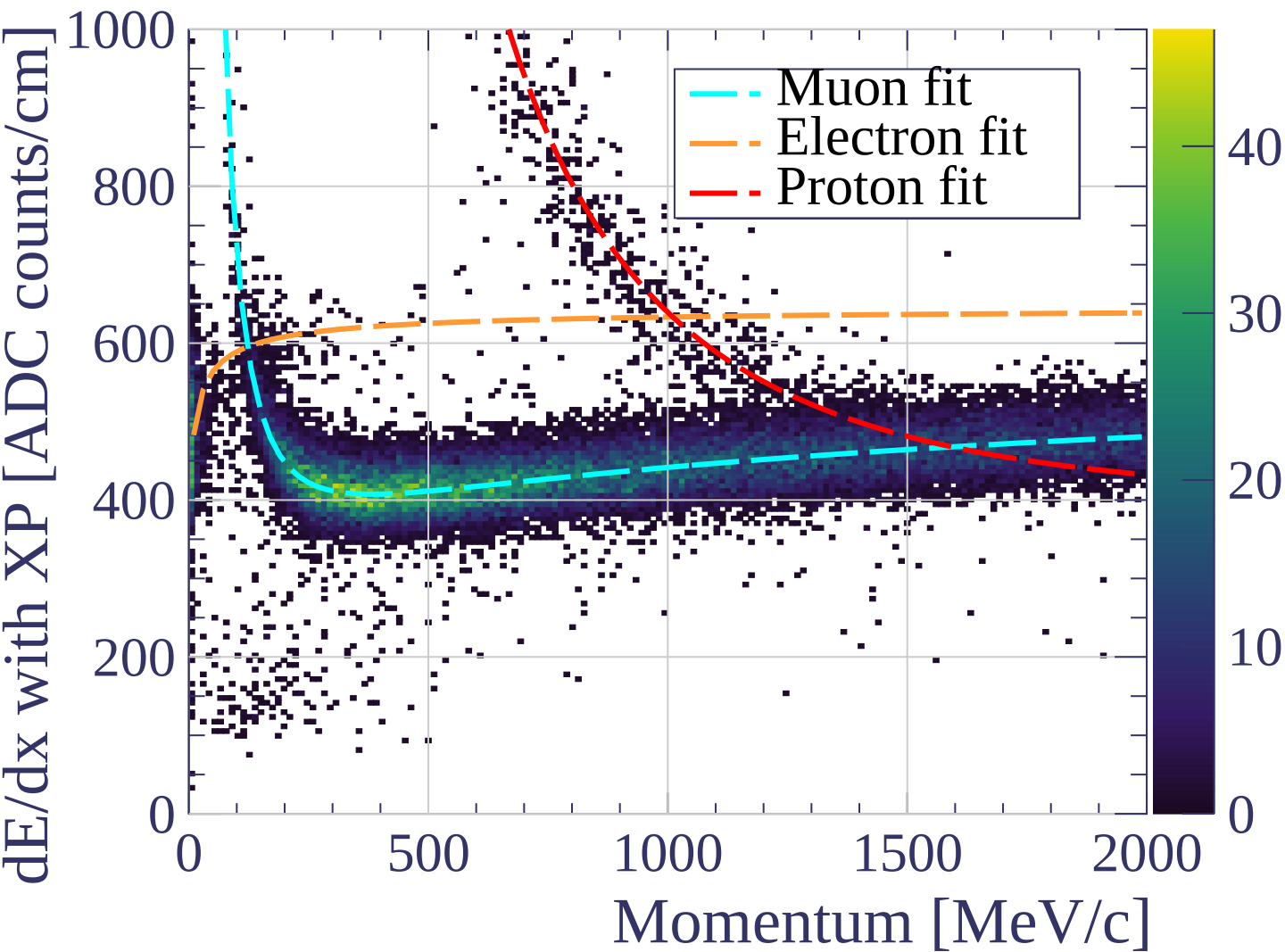


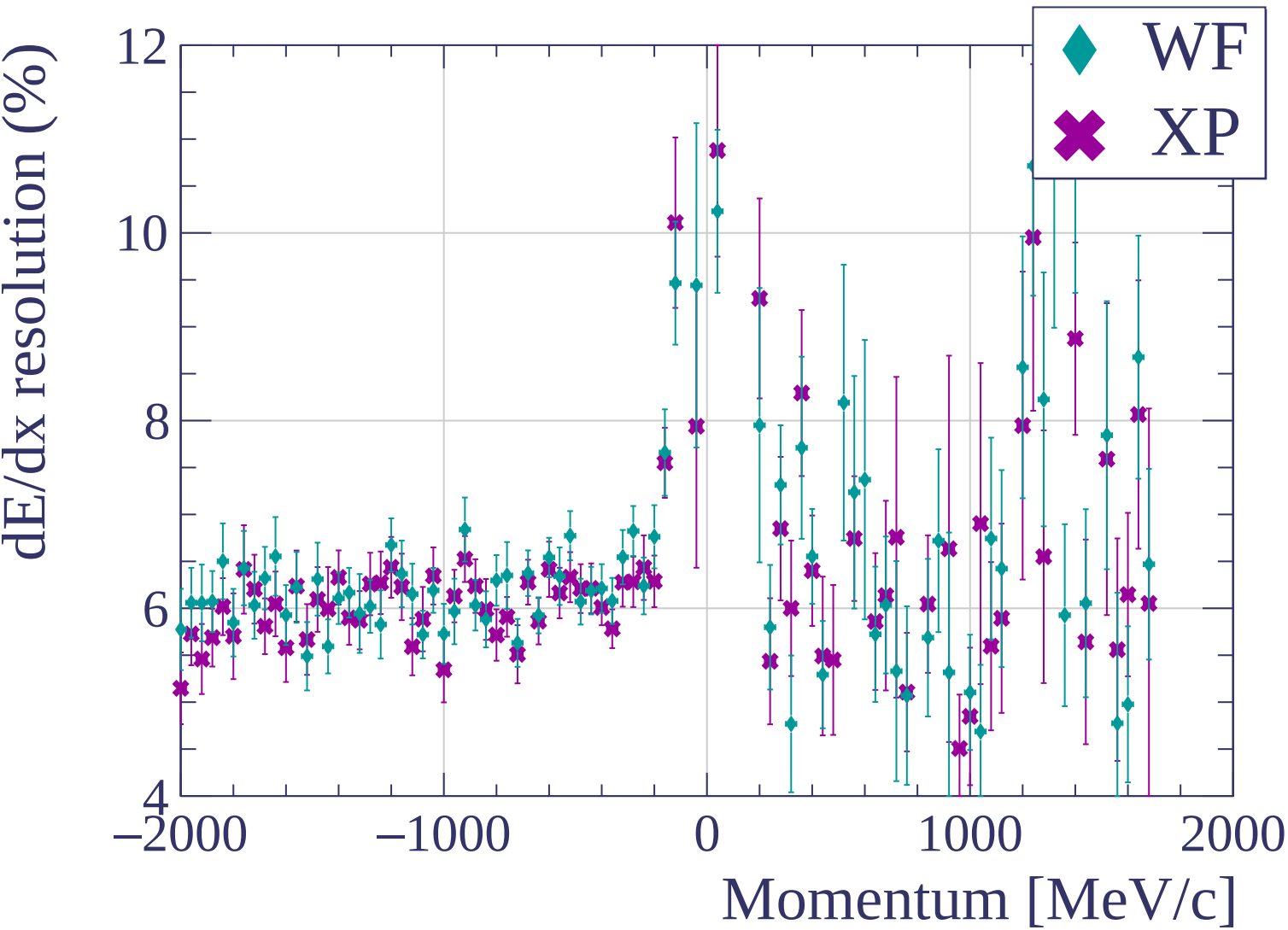


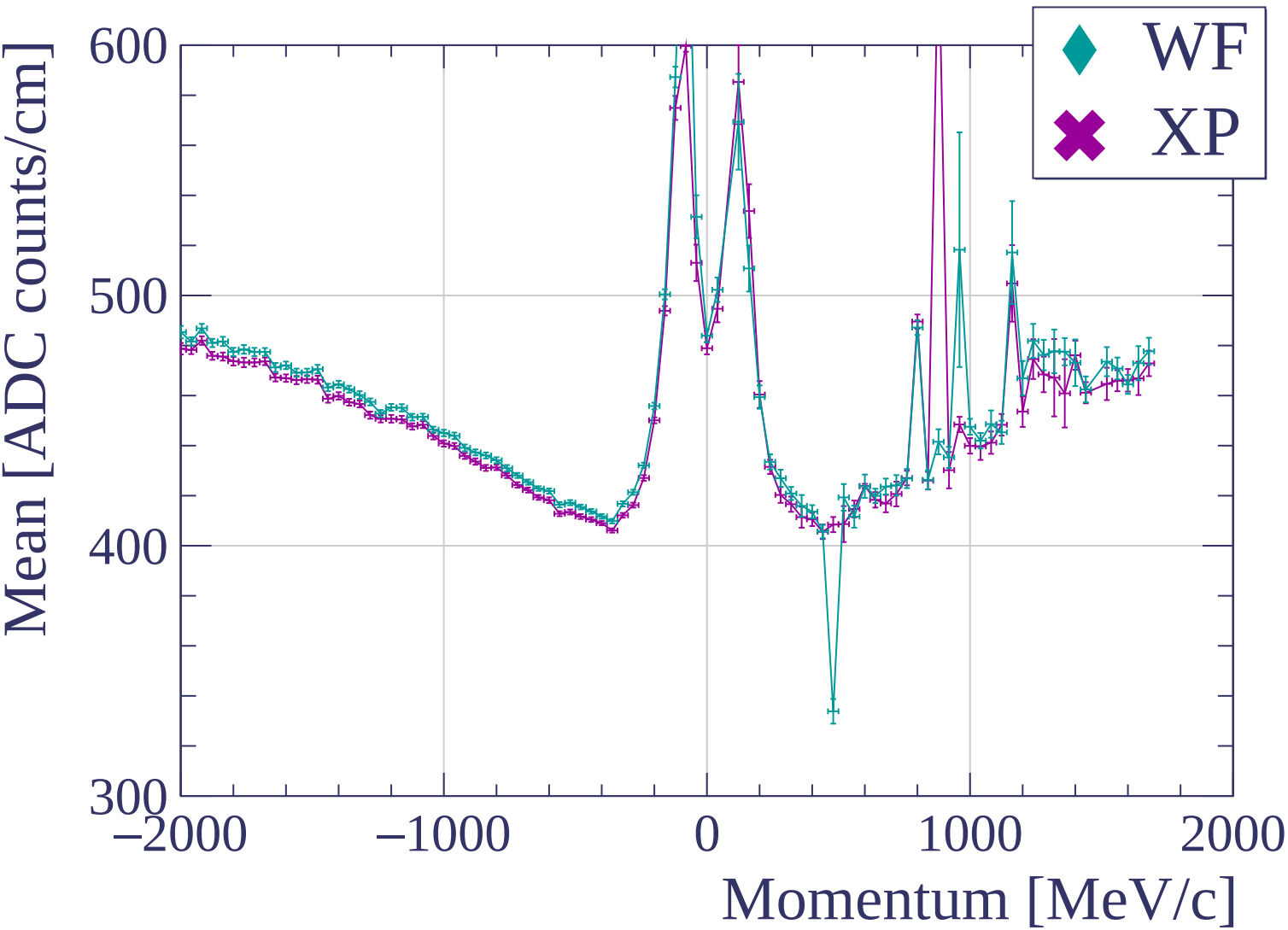


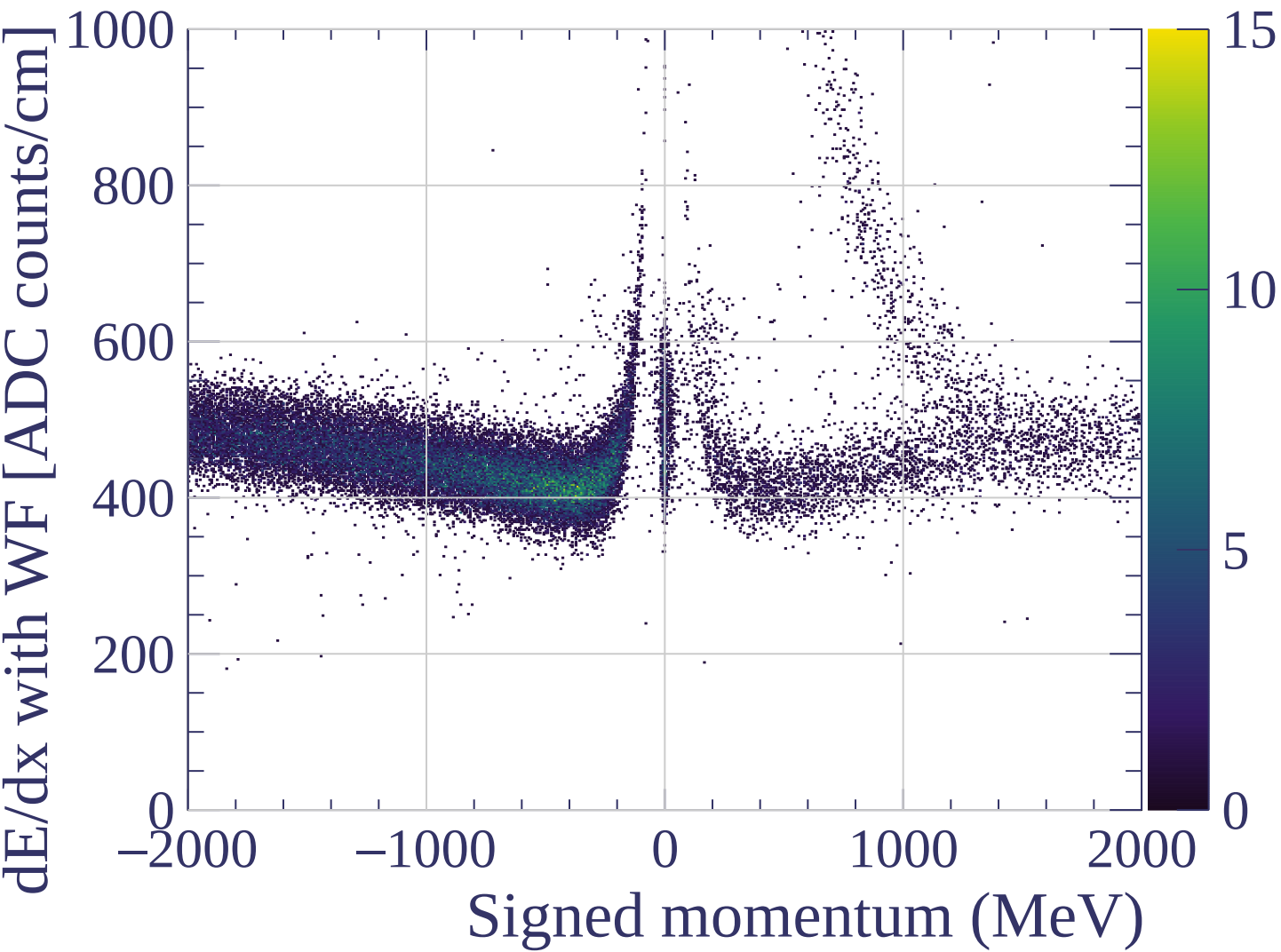


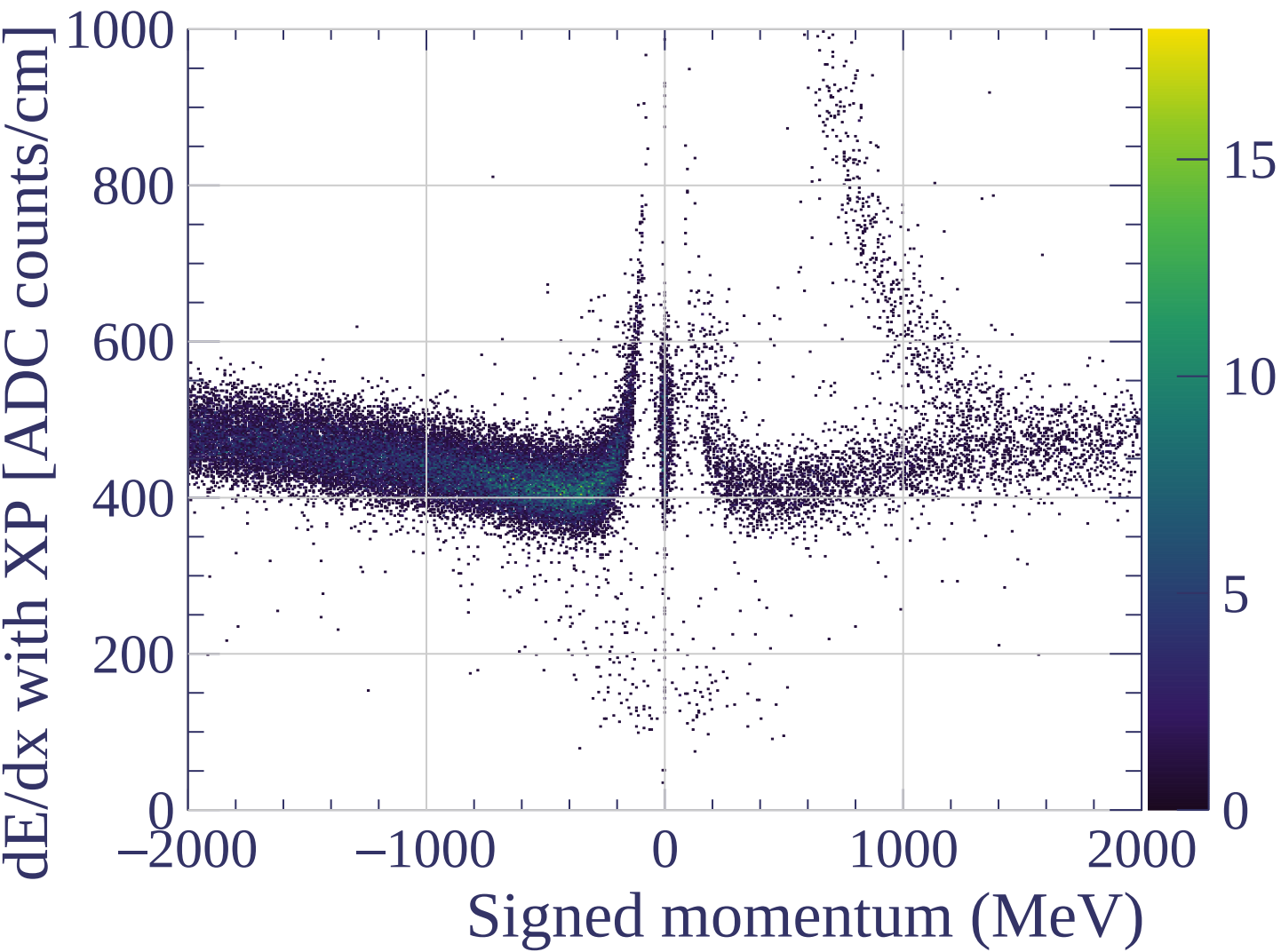




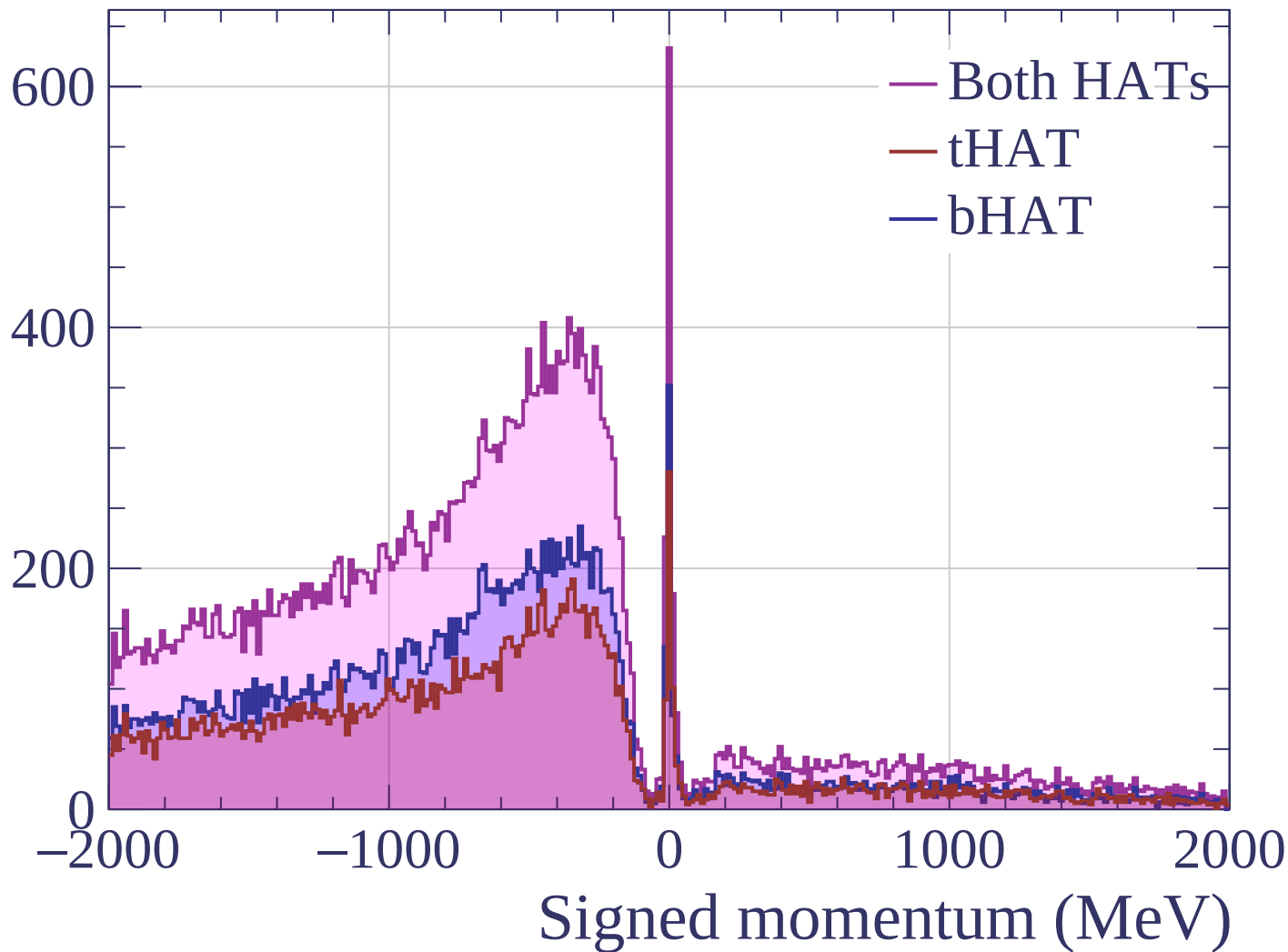


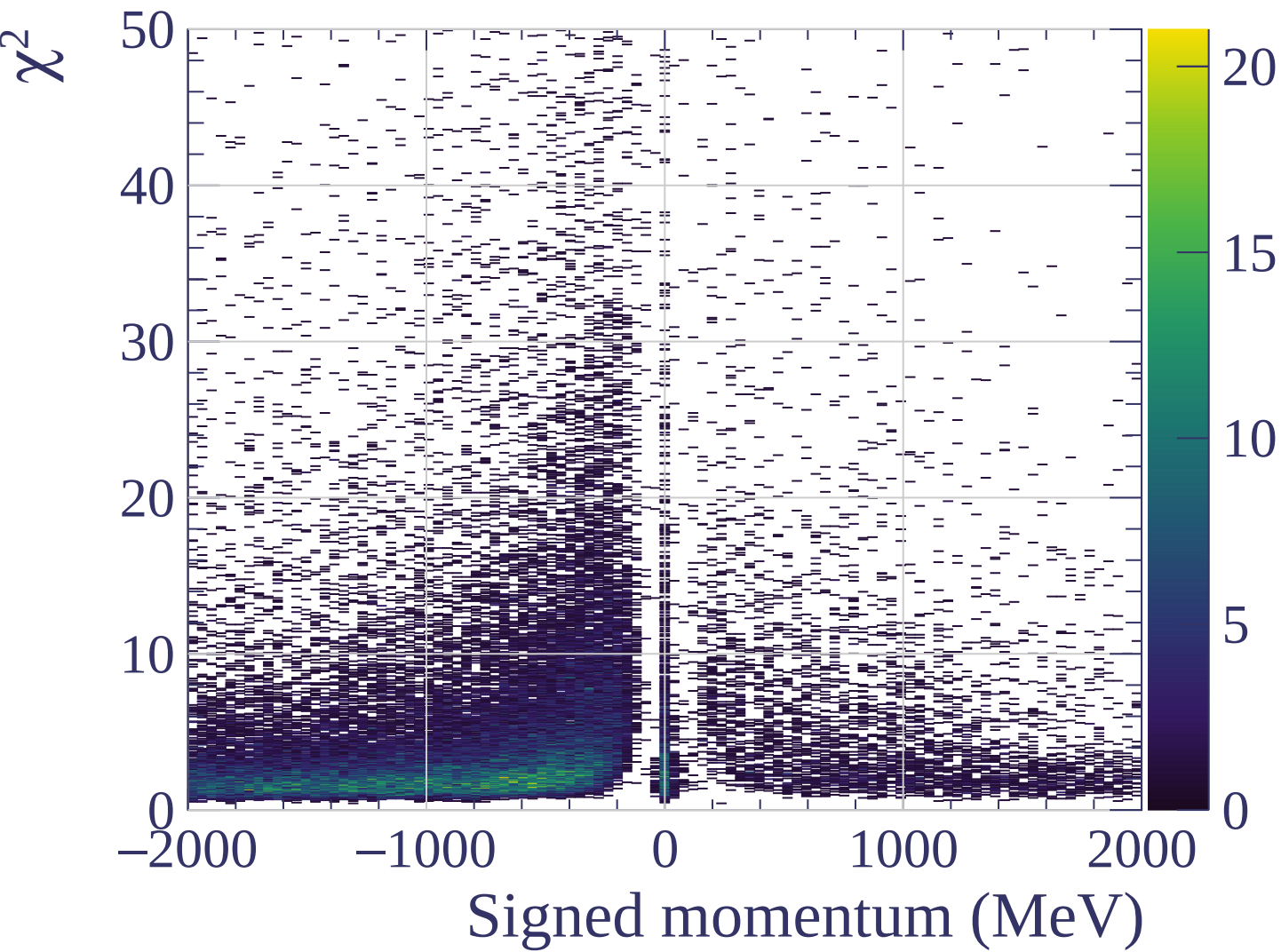


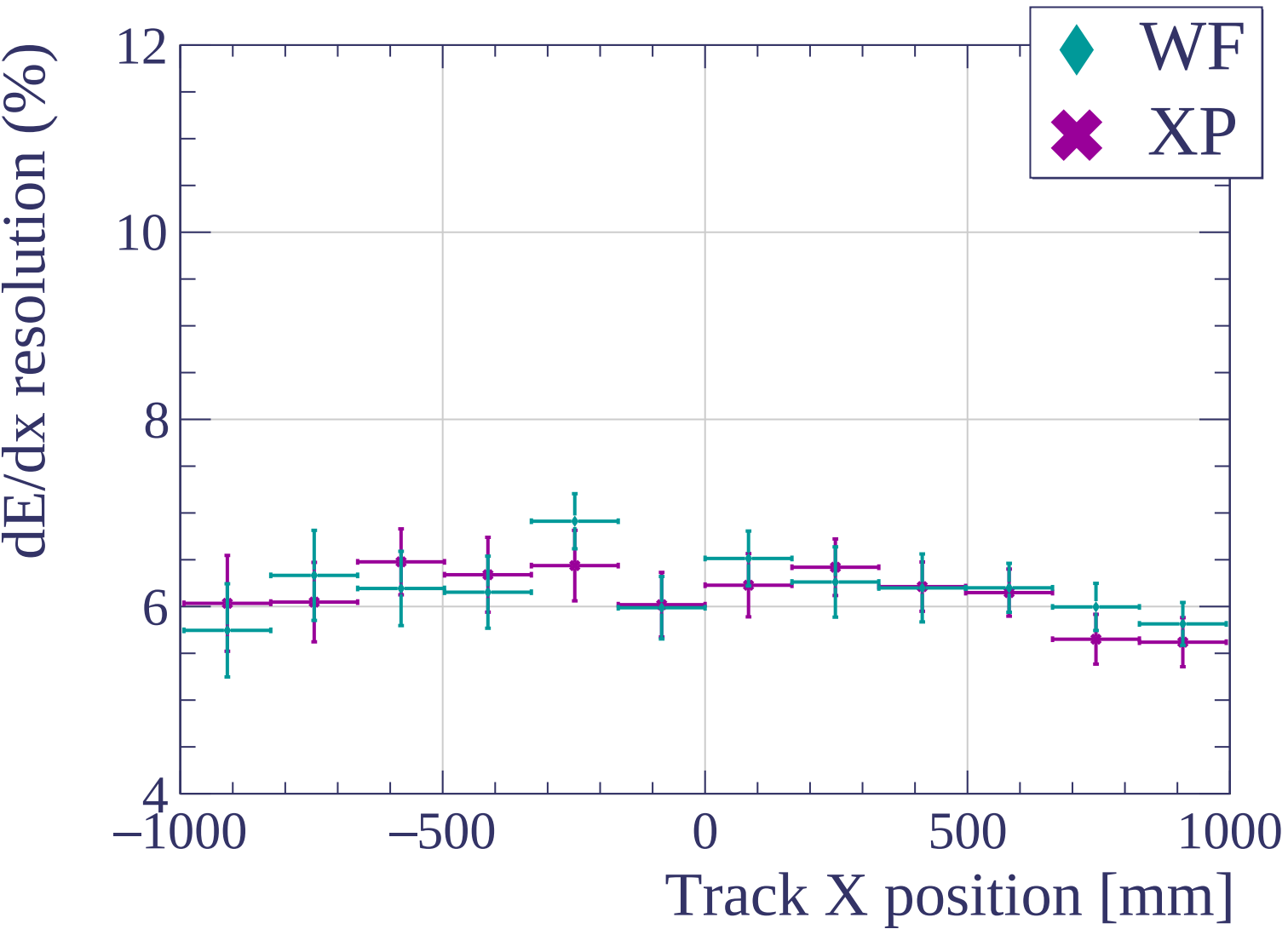


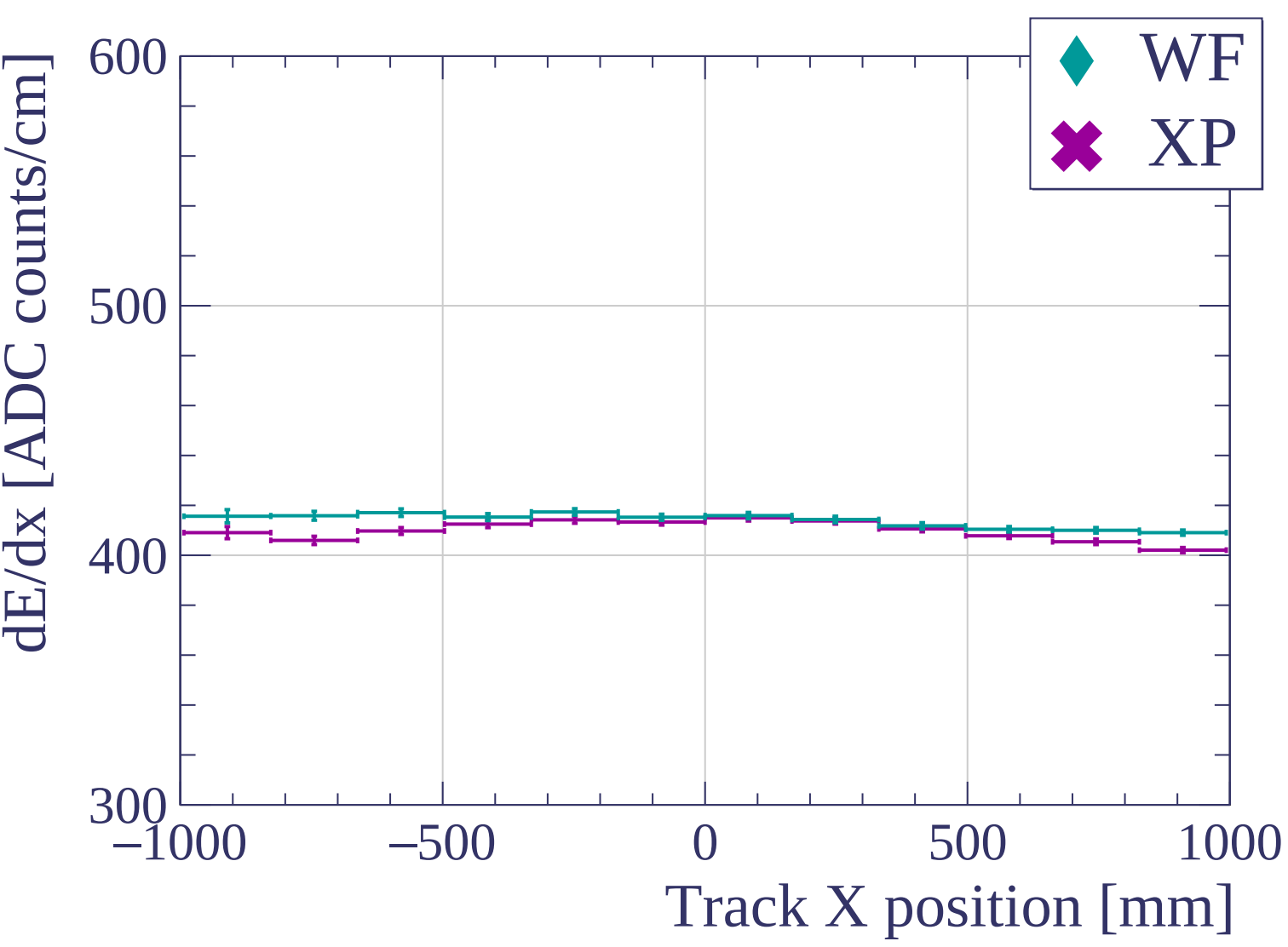


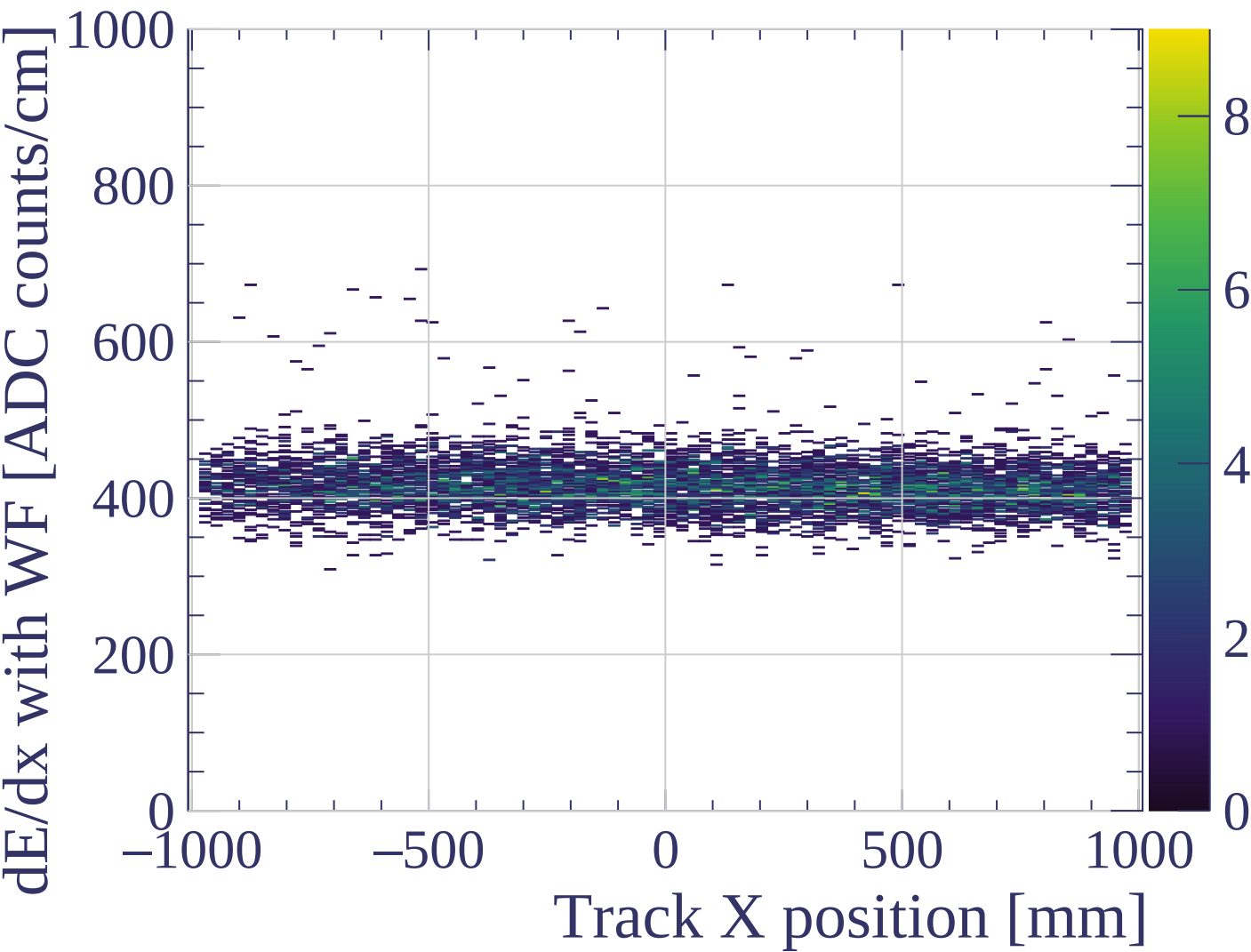
Count

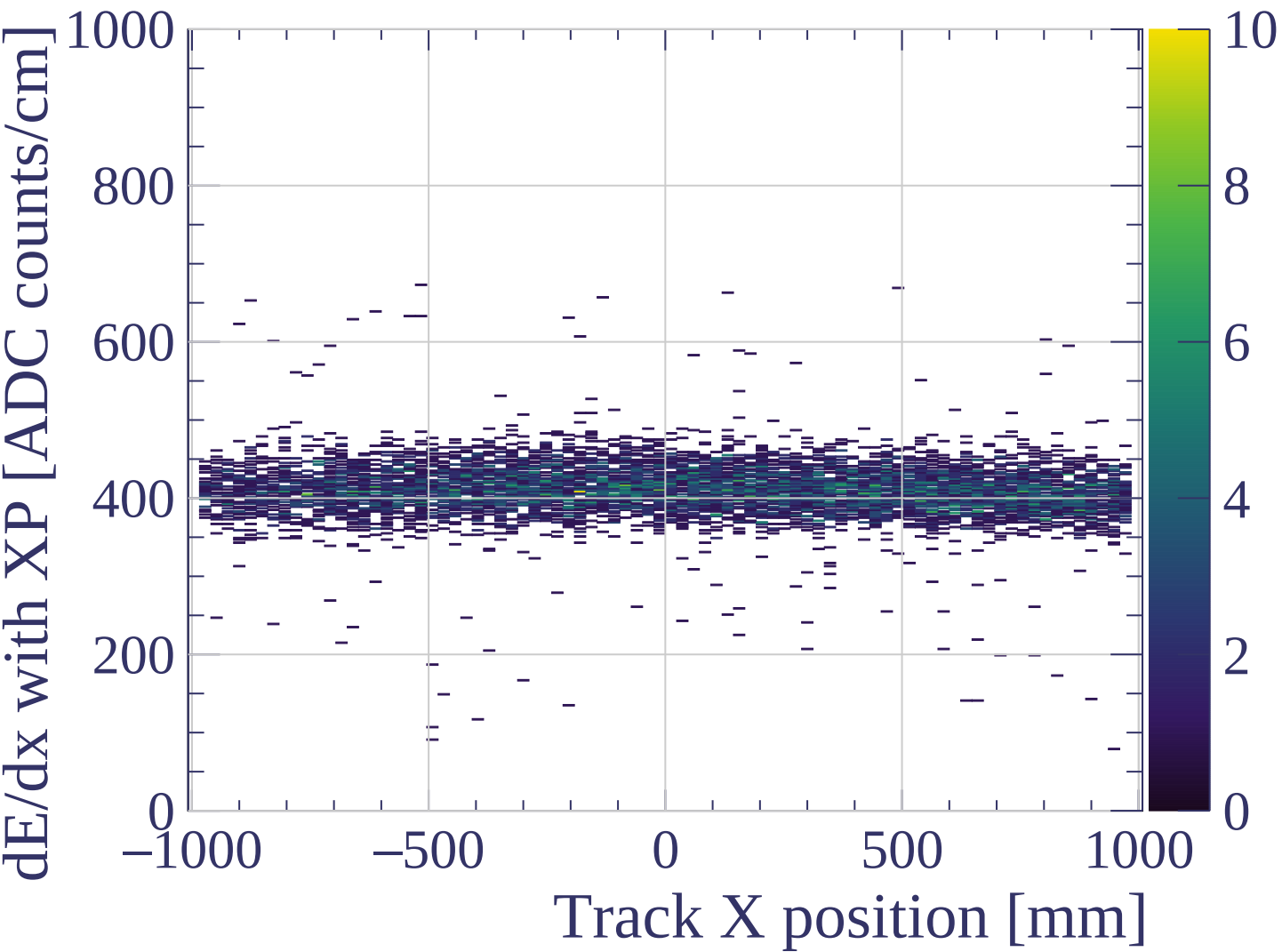


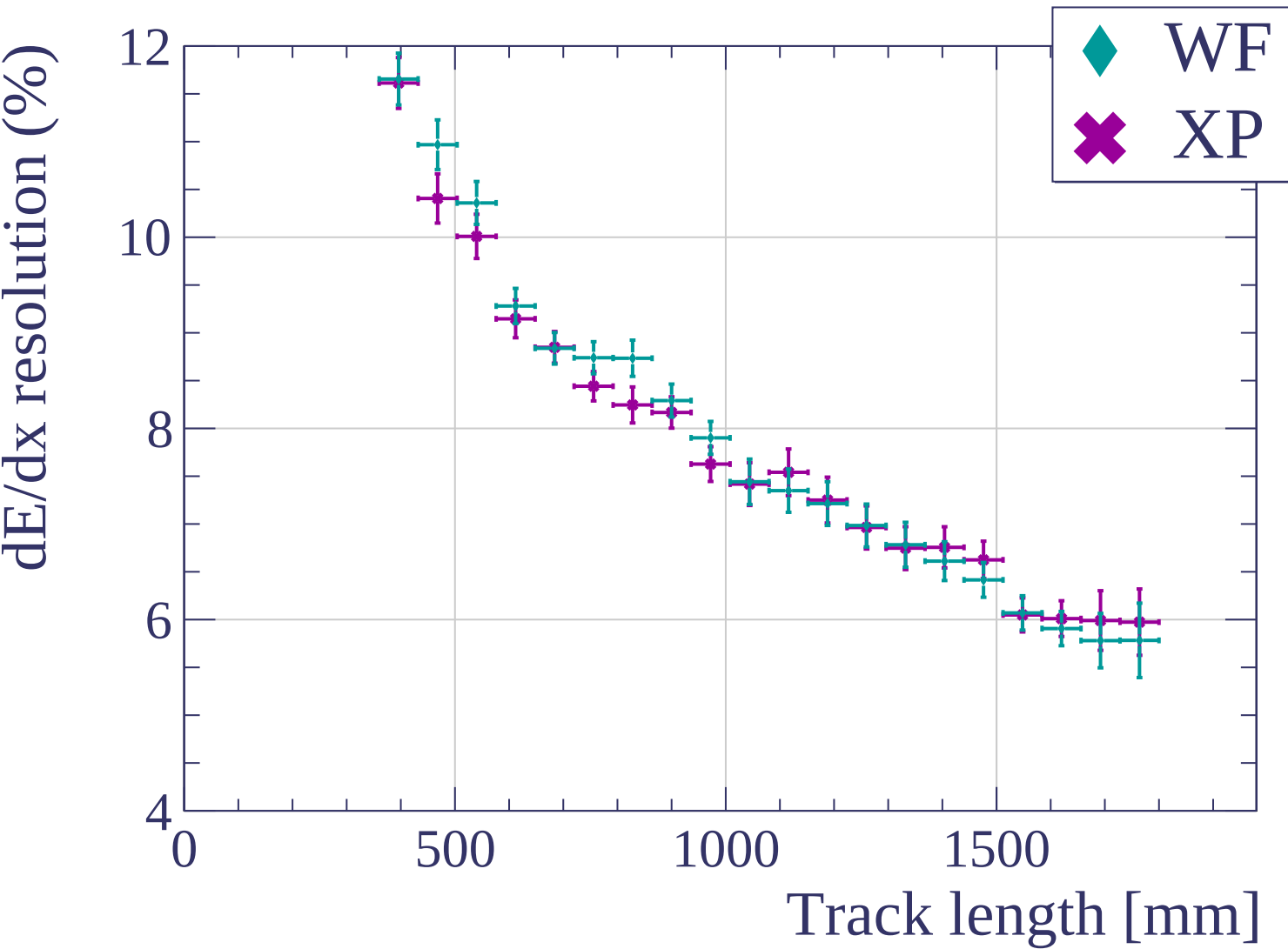


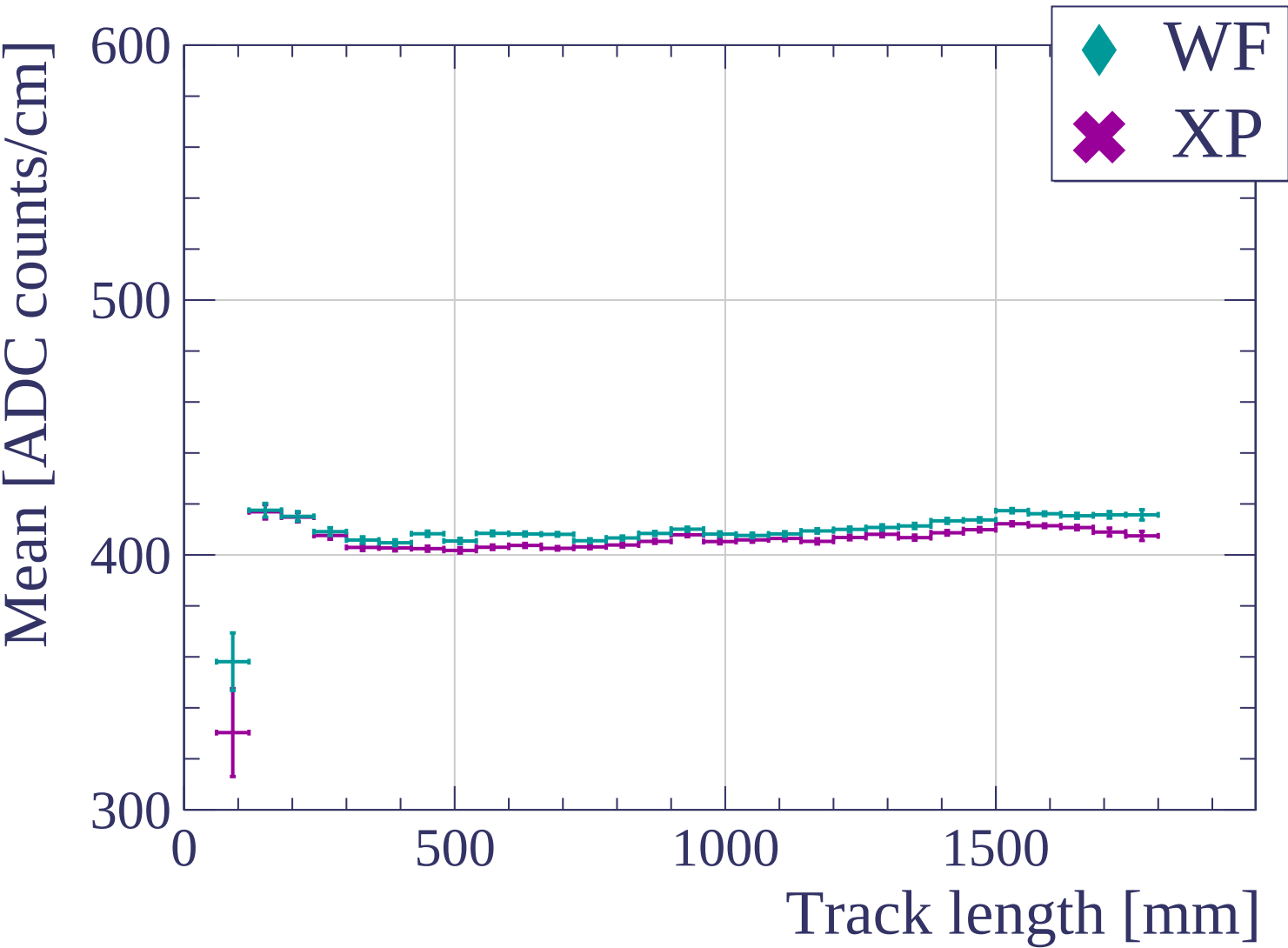


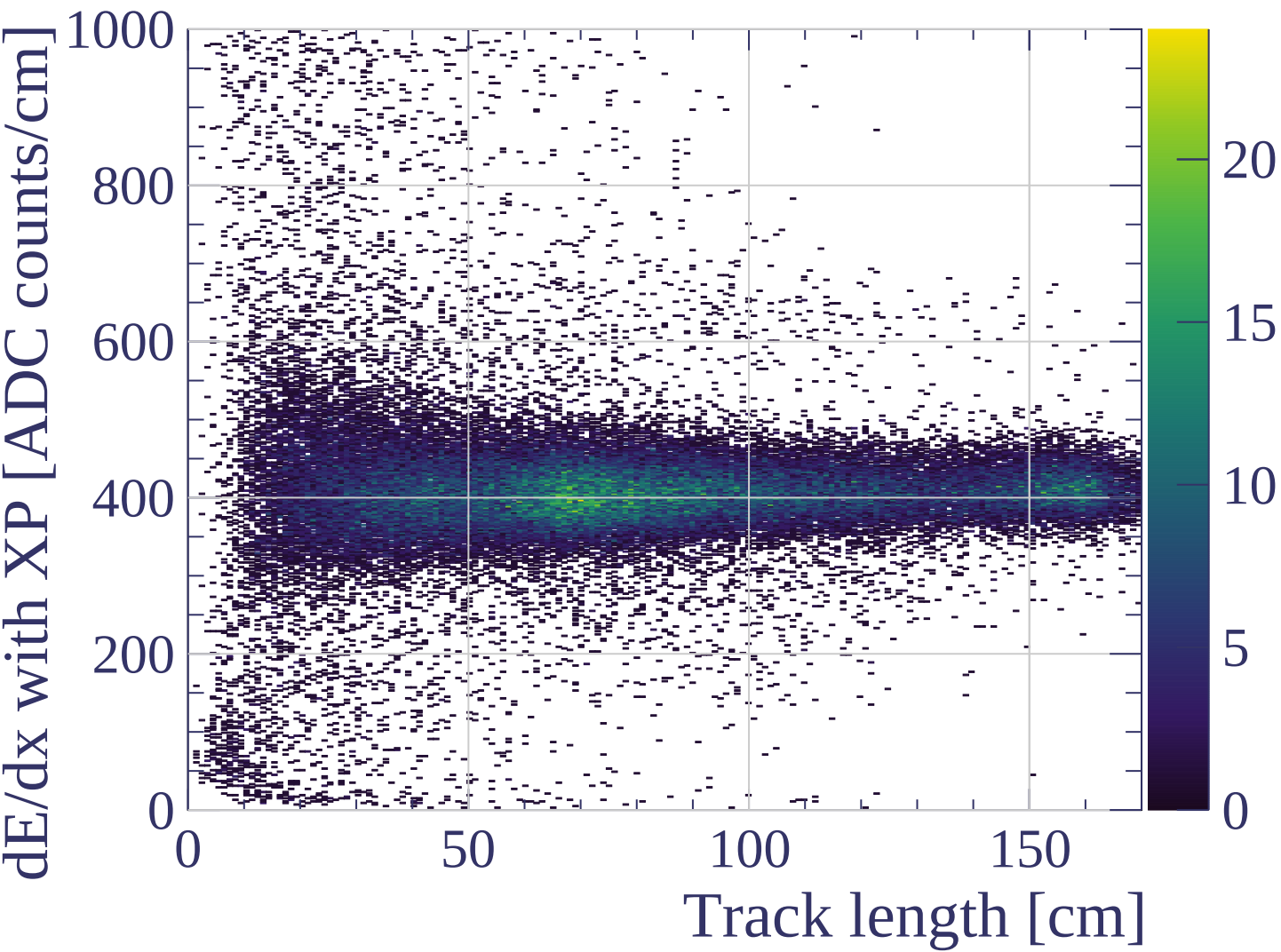


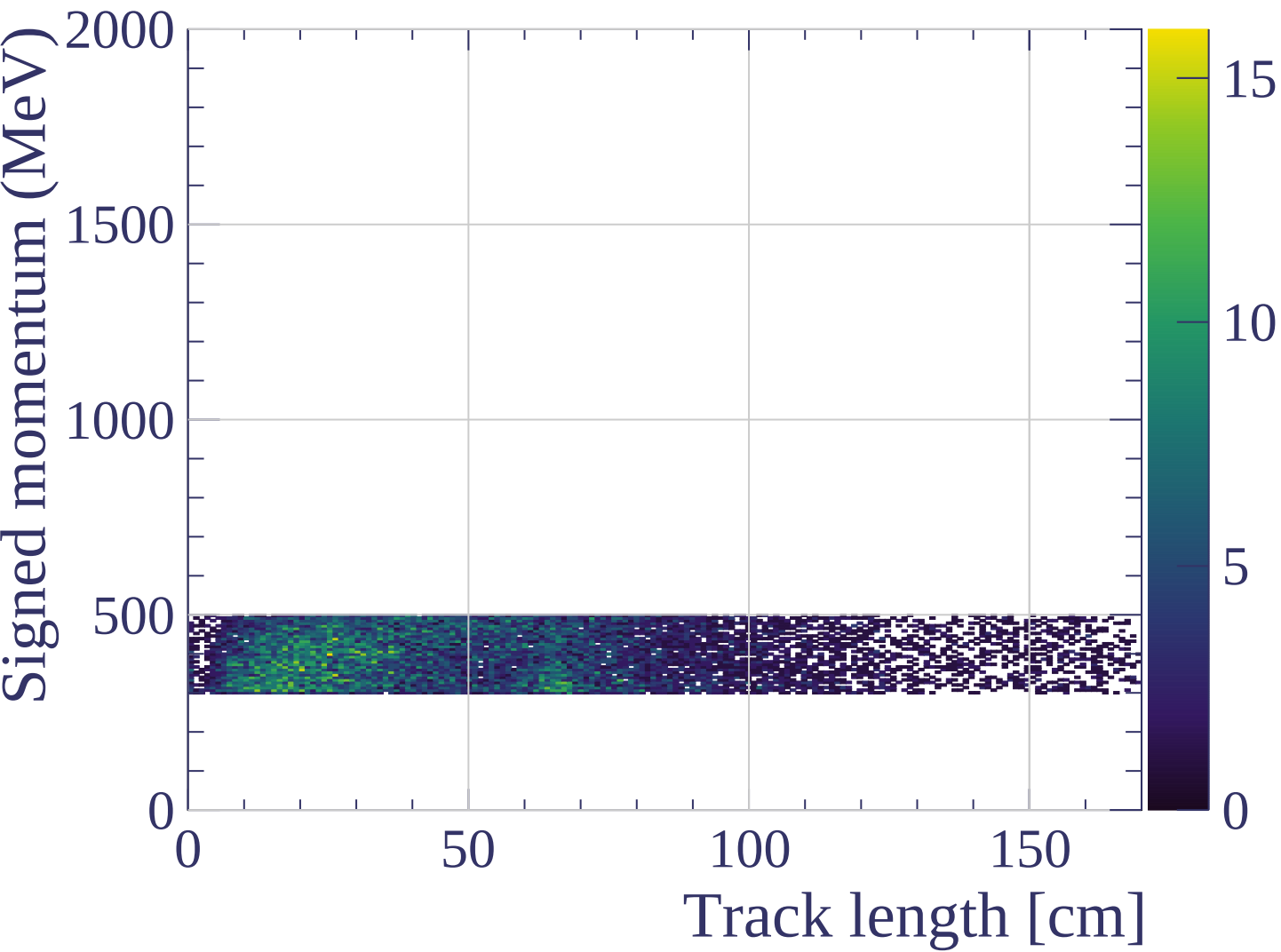


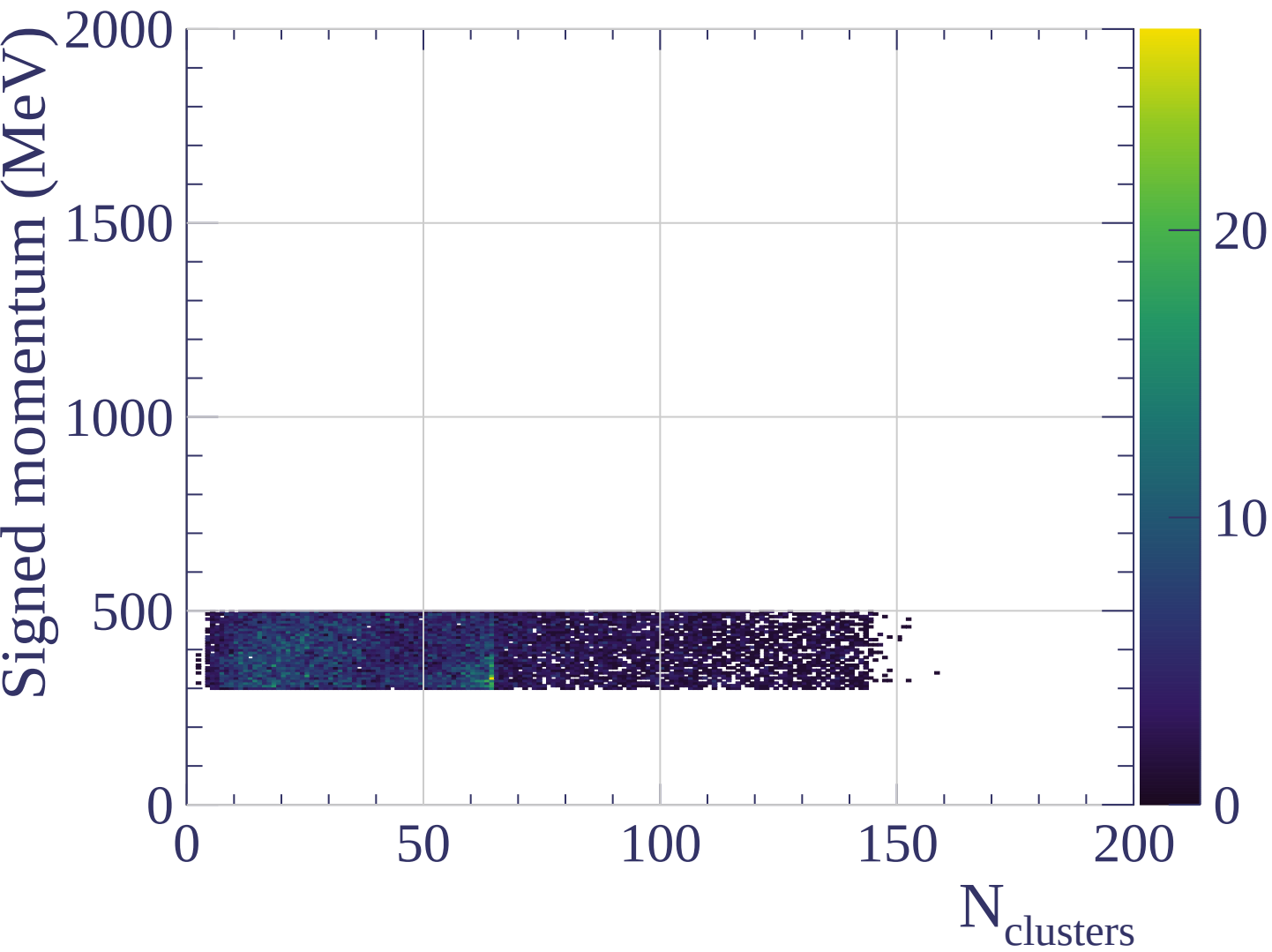


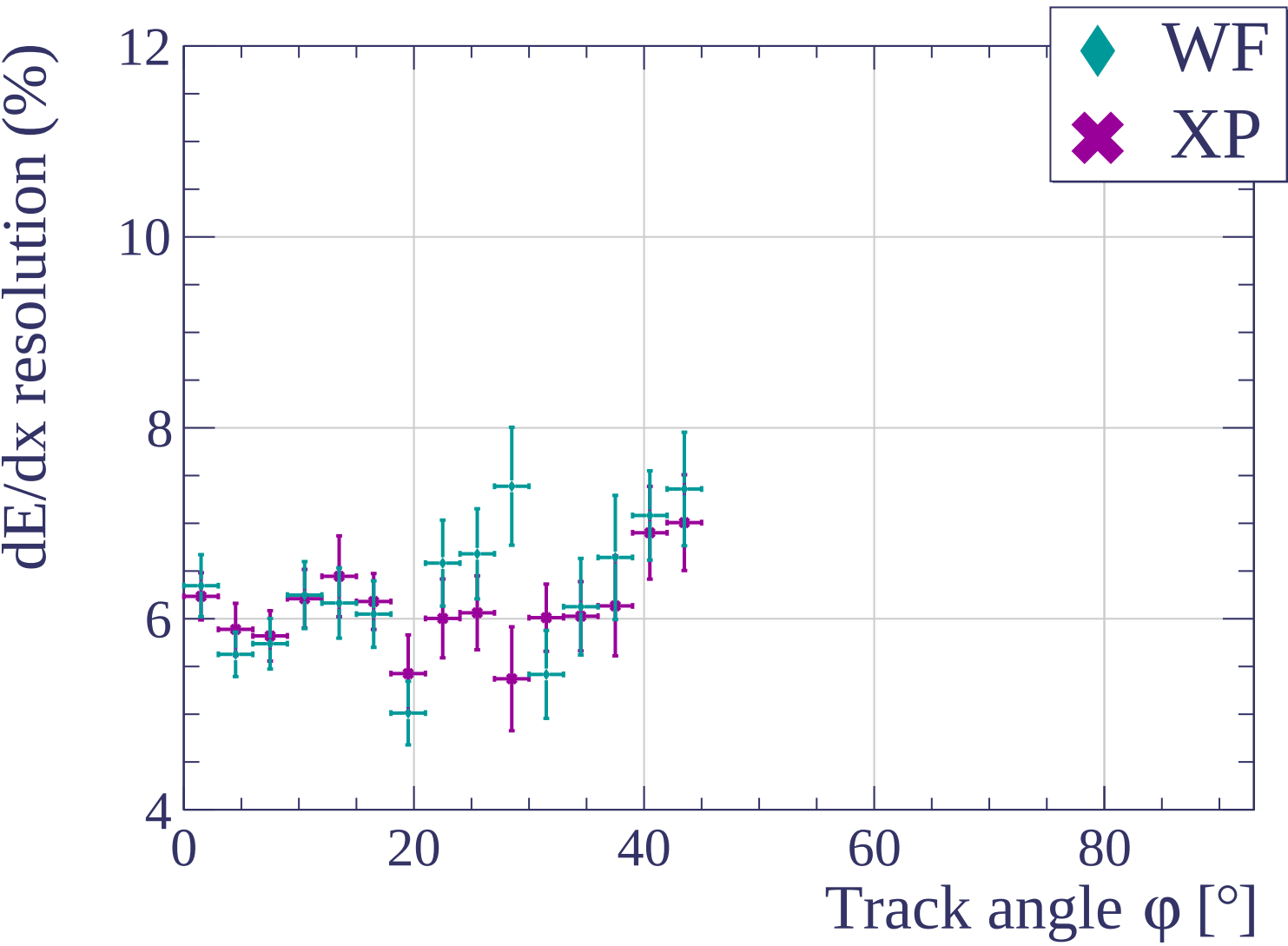


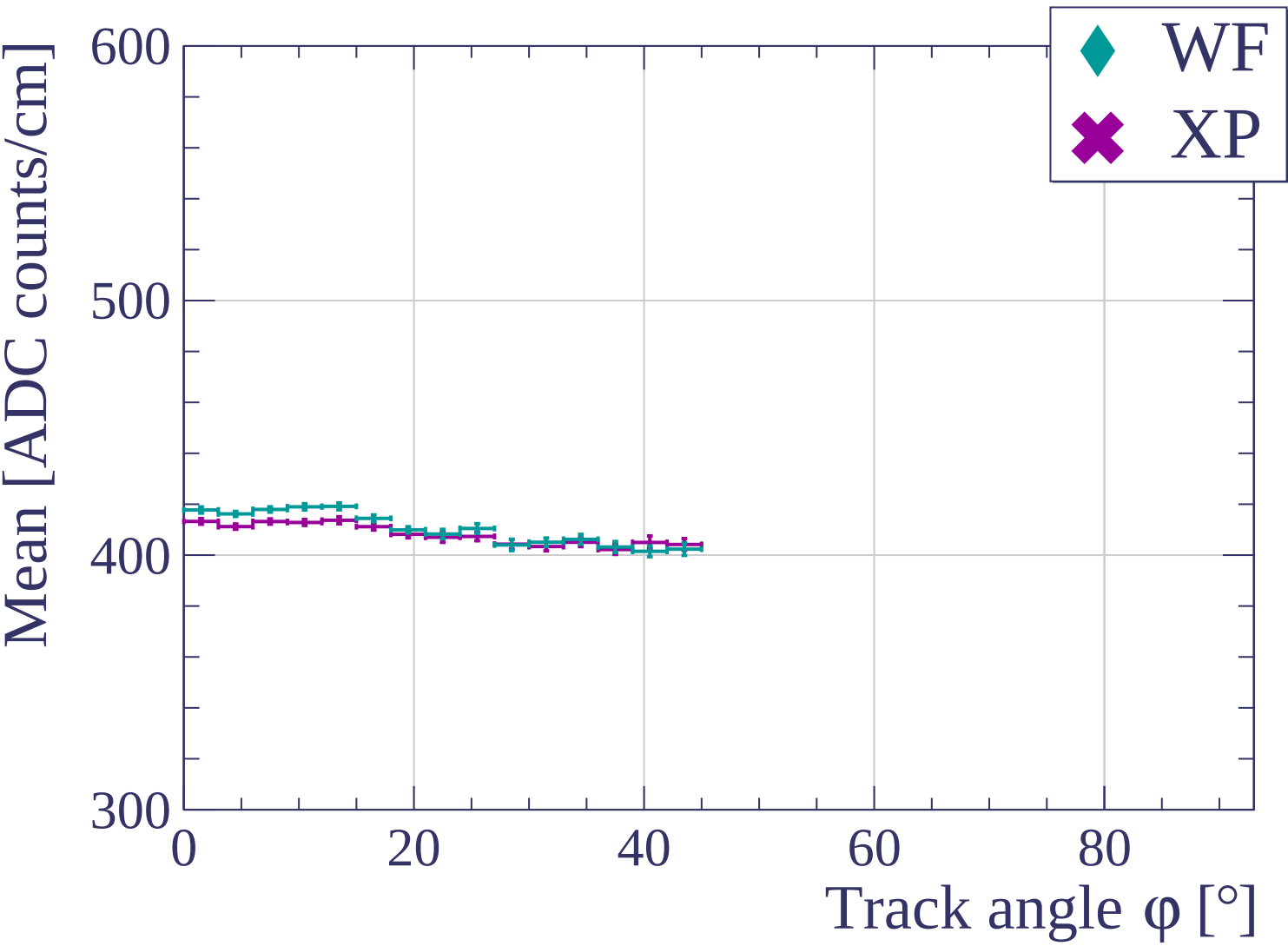


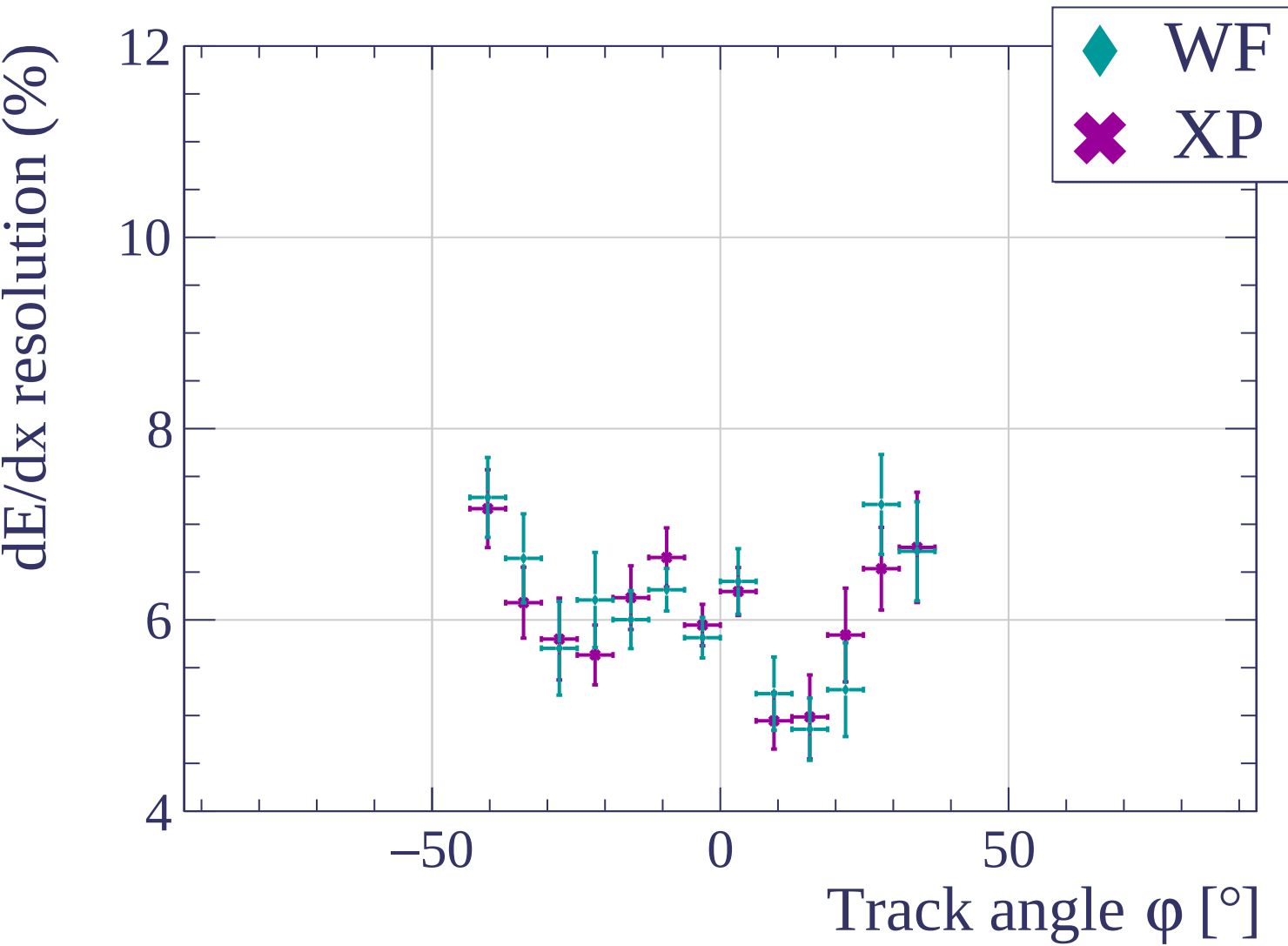


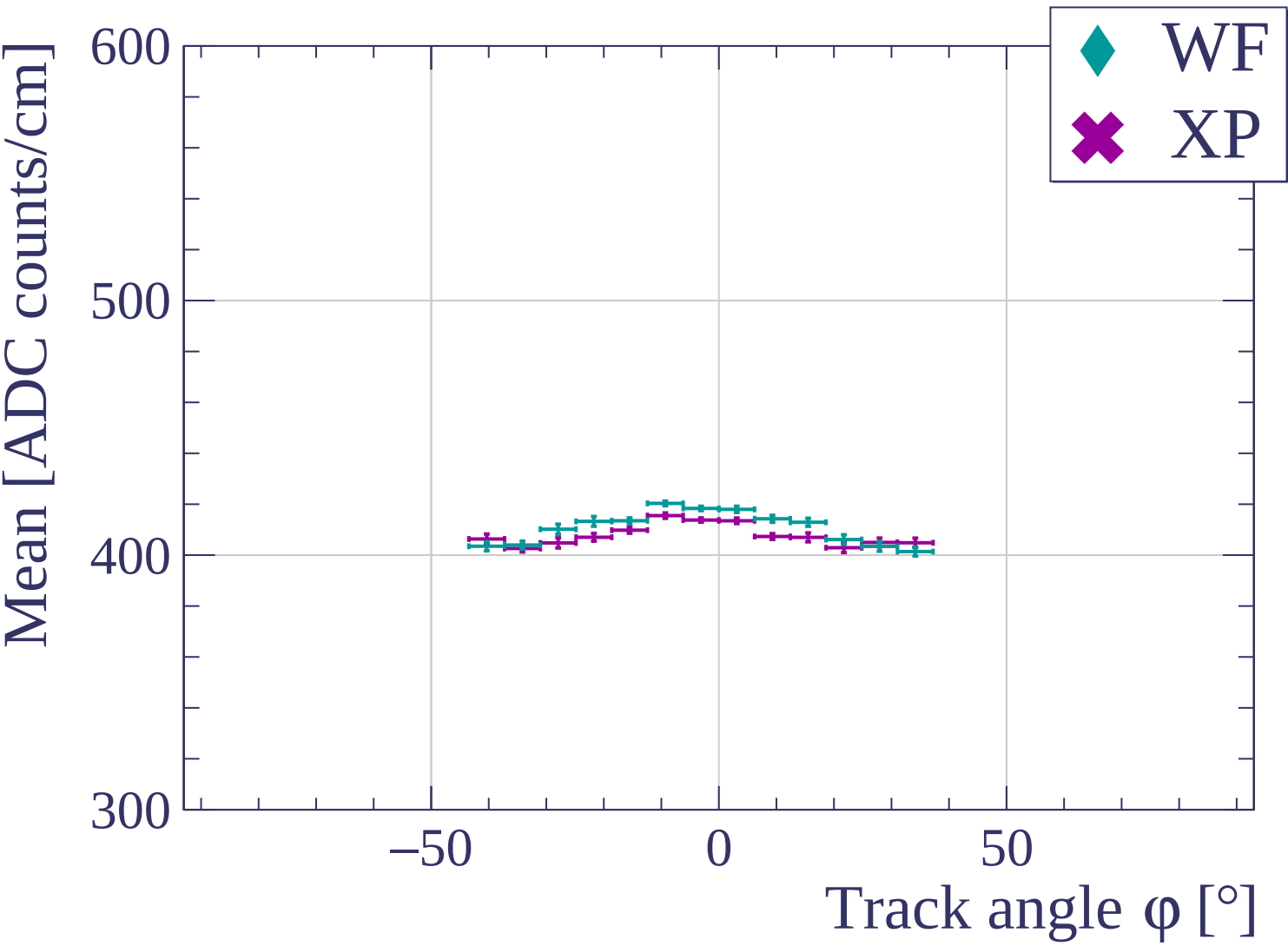


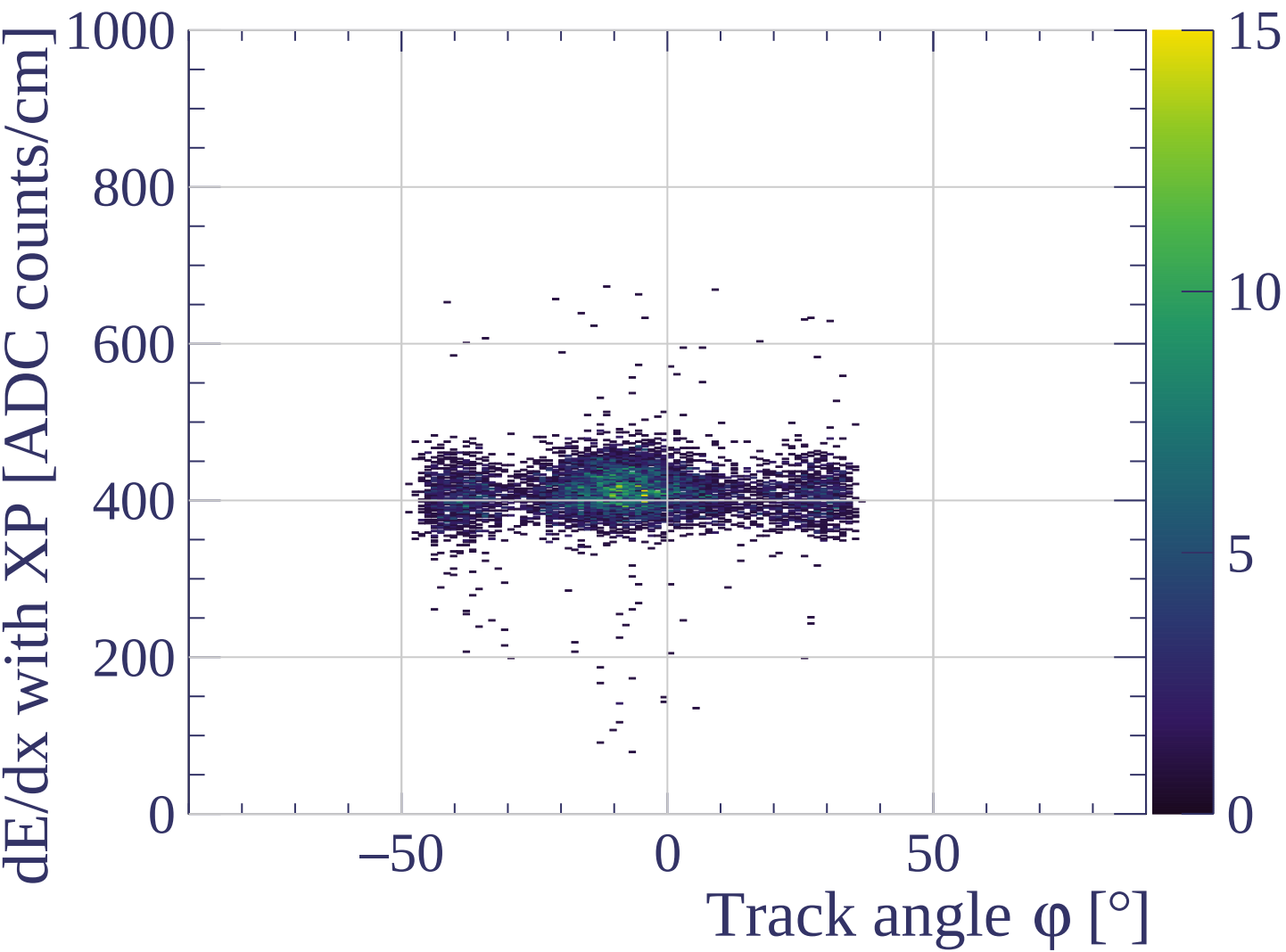




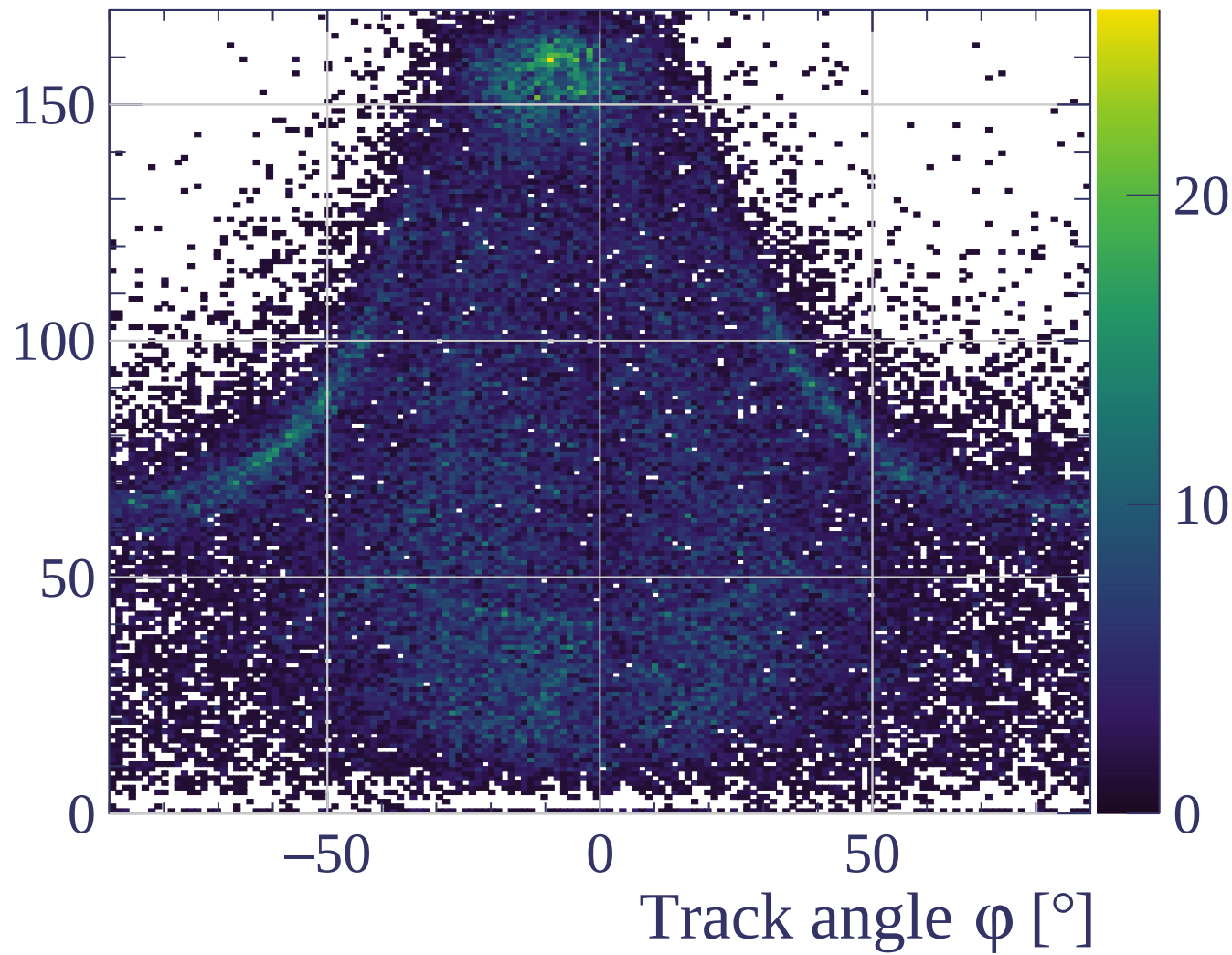


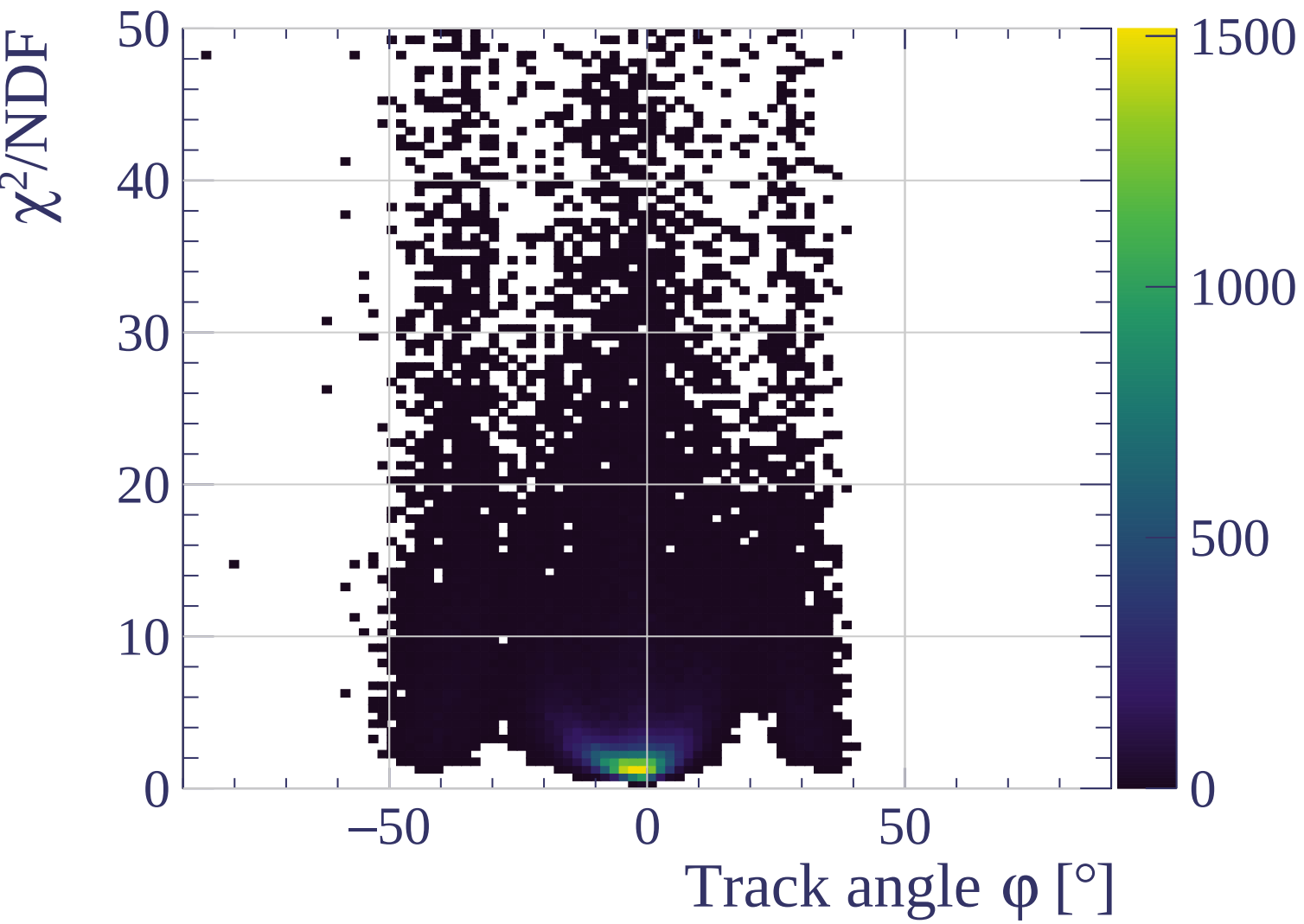


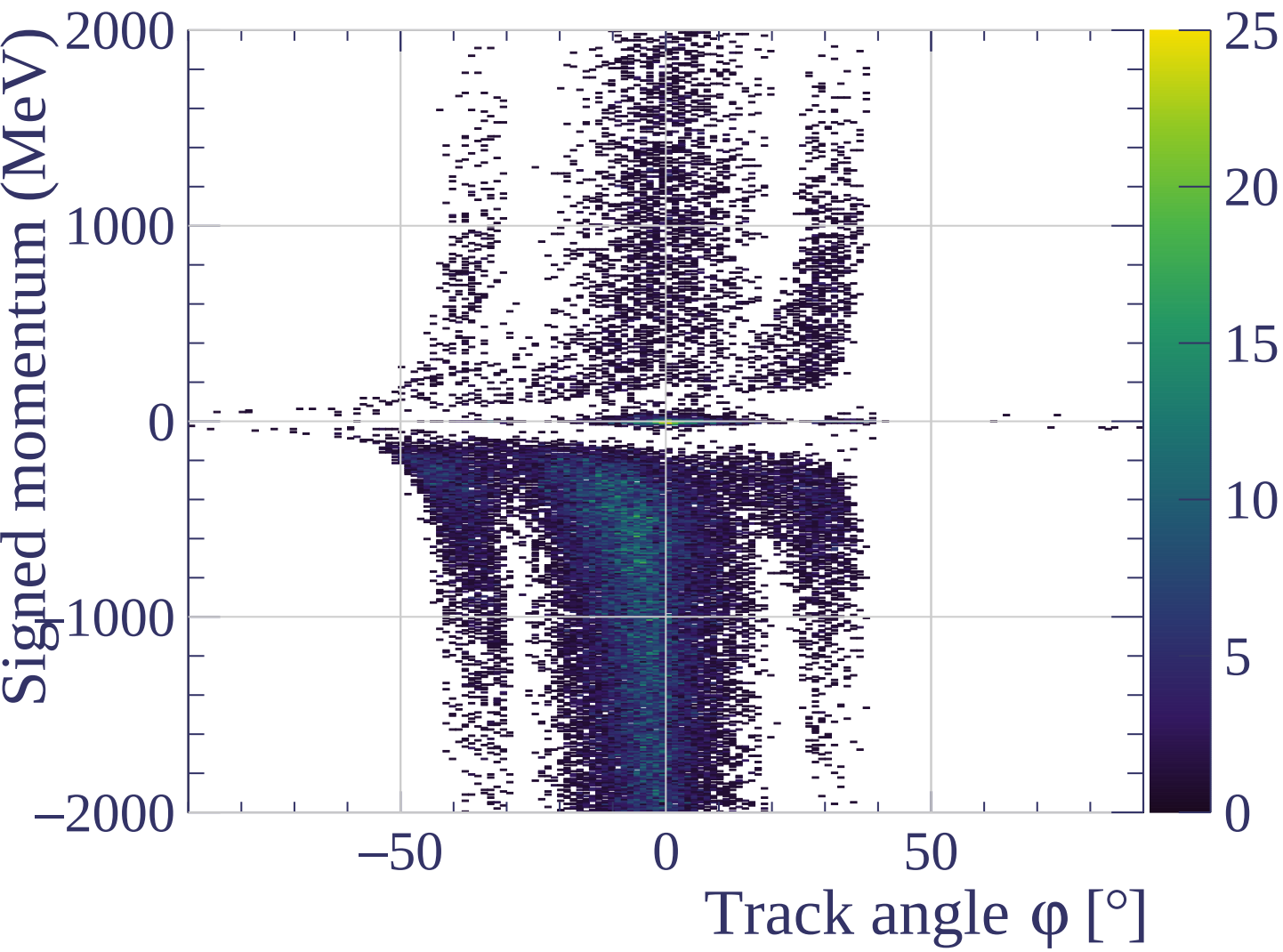




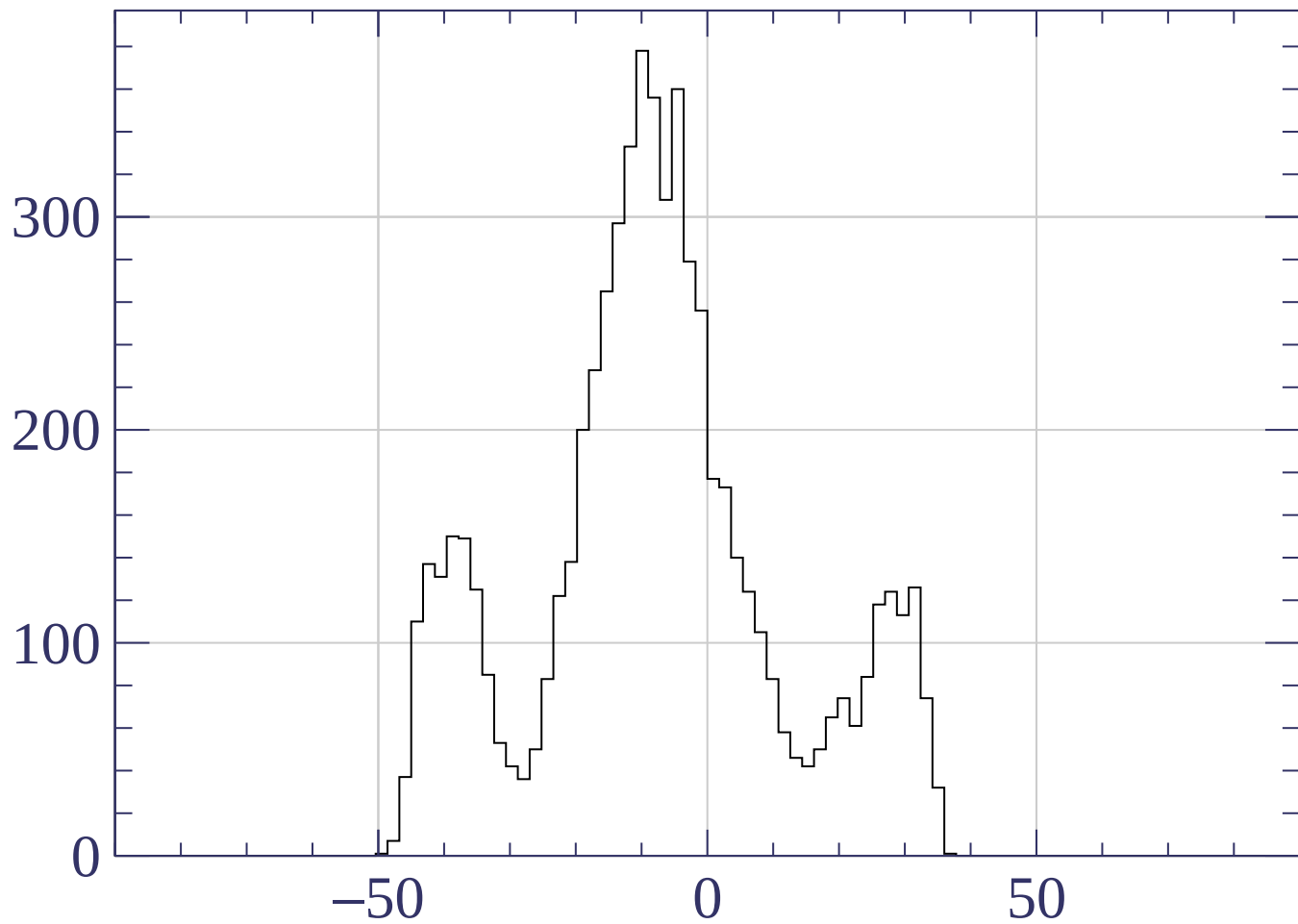
Track length [cm]



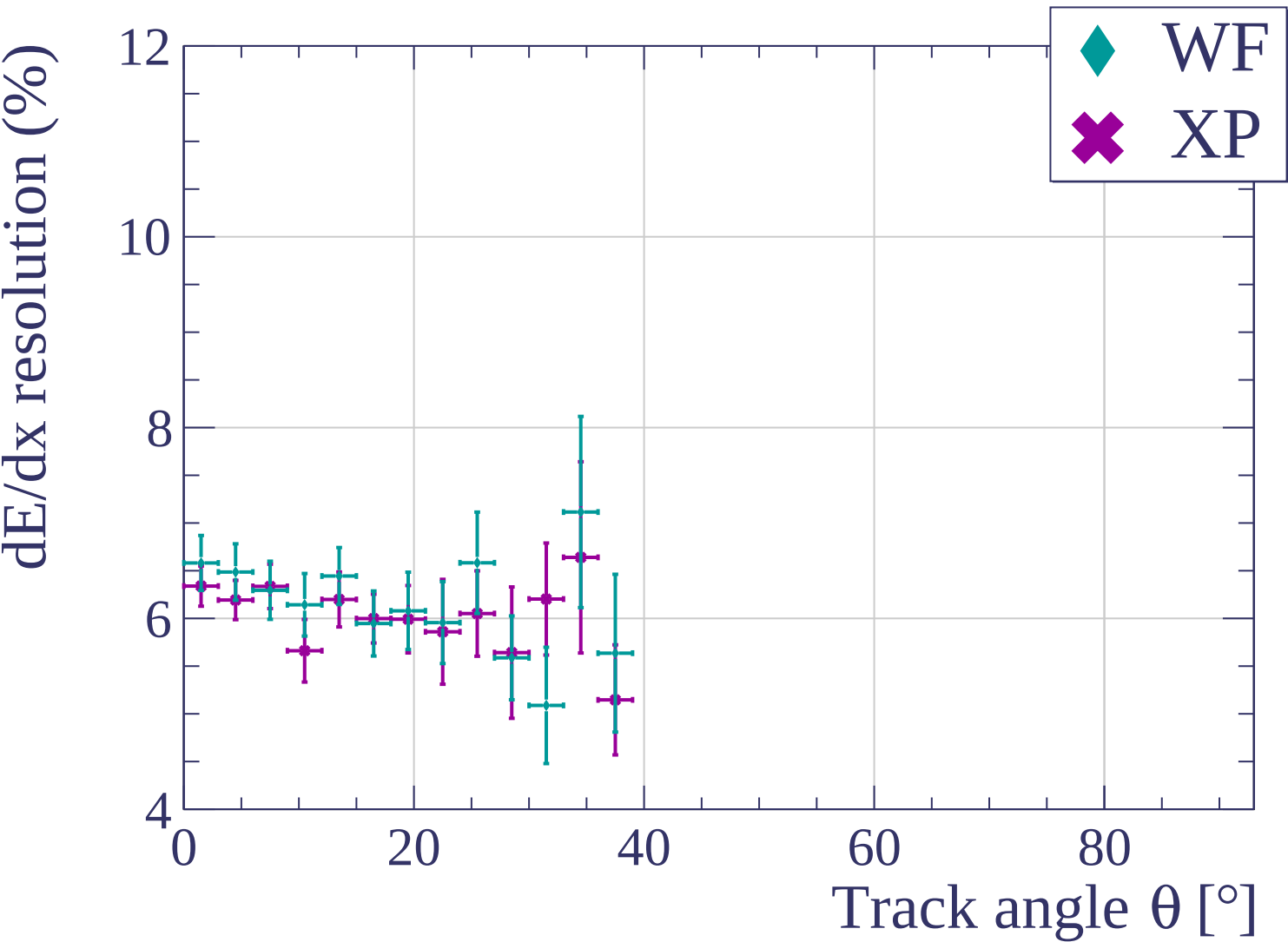


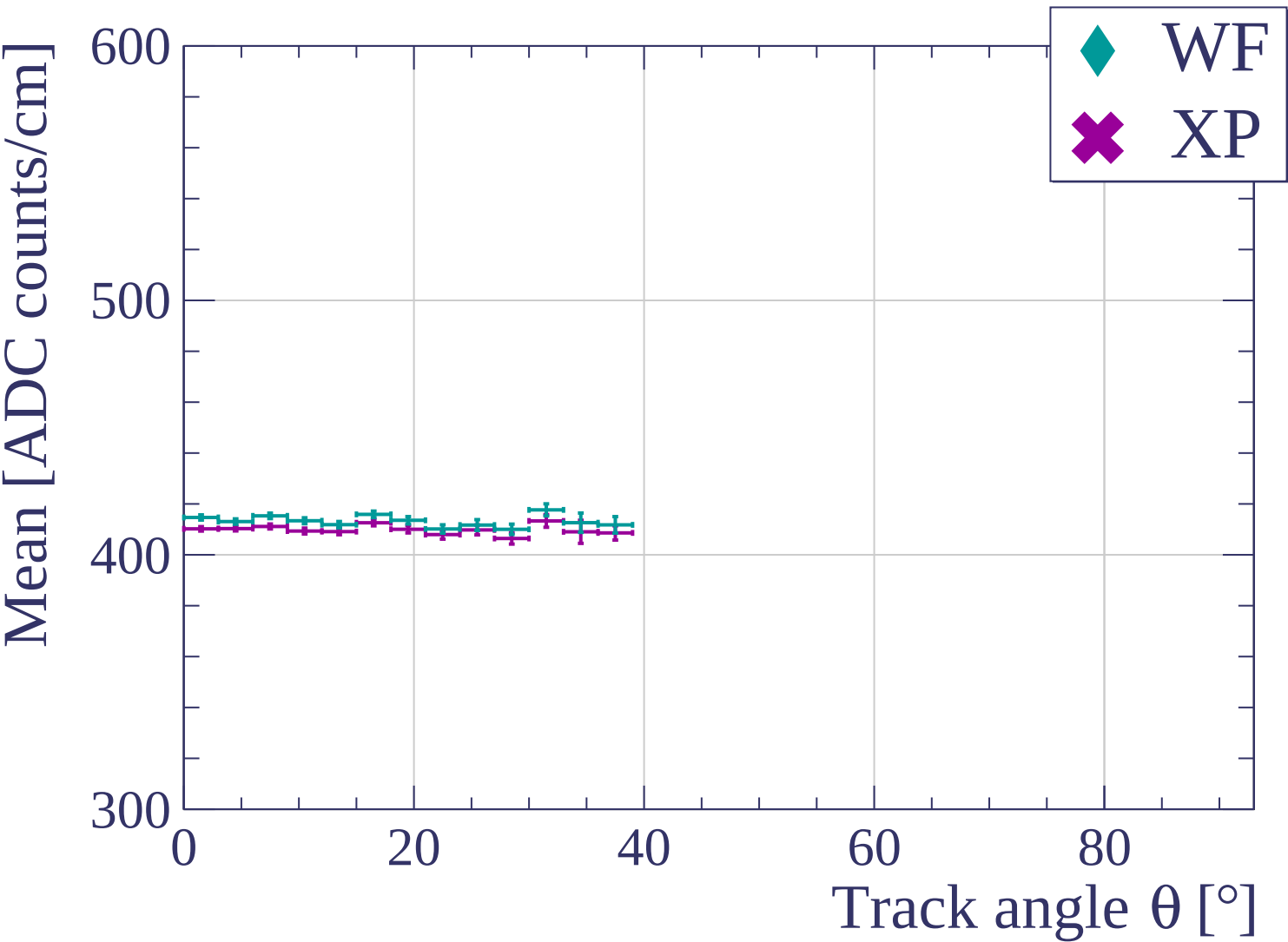


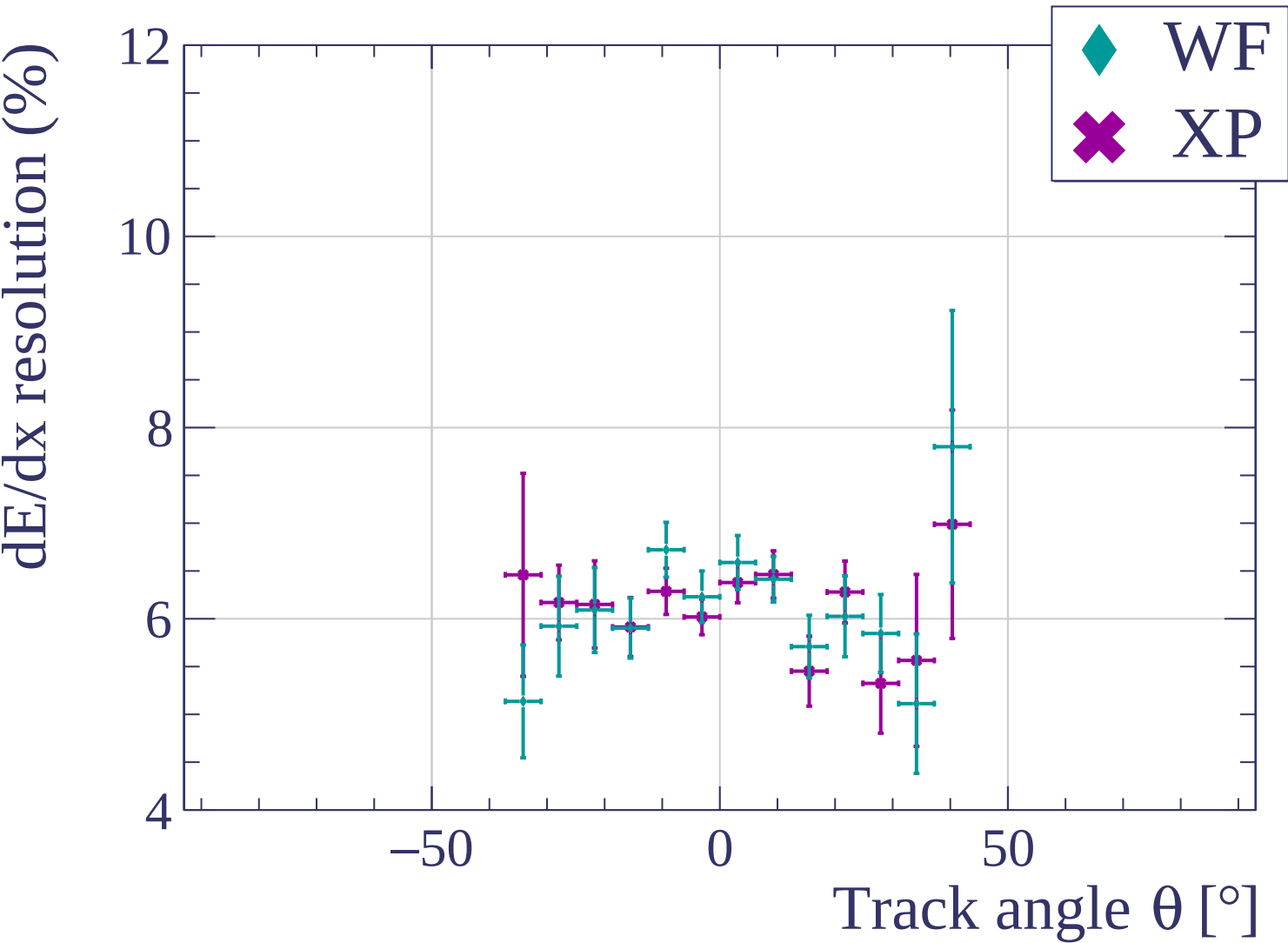
Count

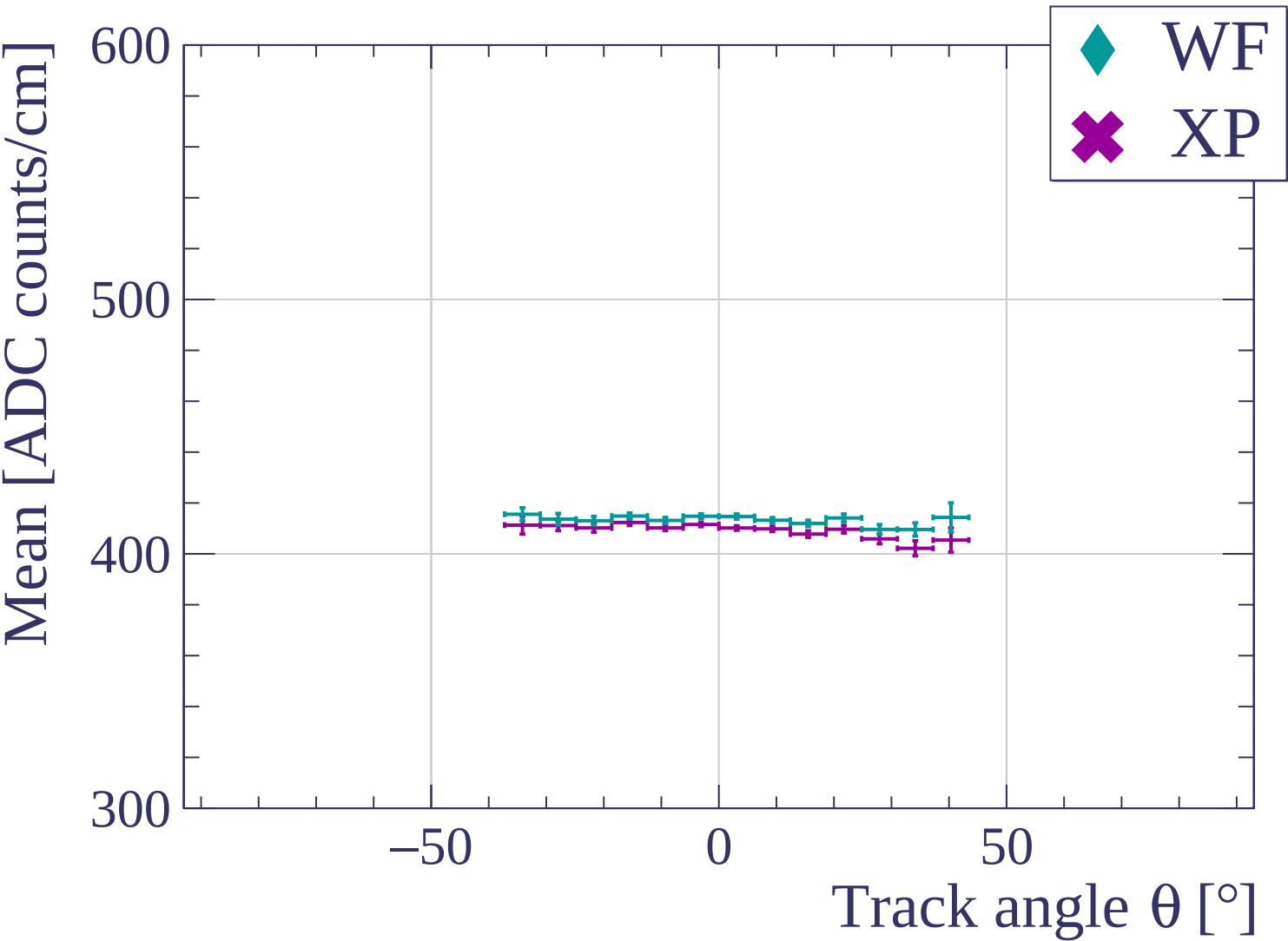


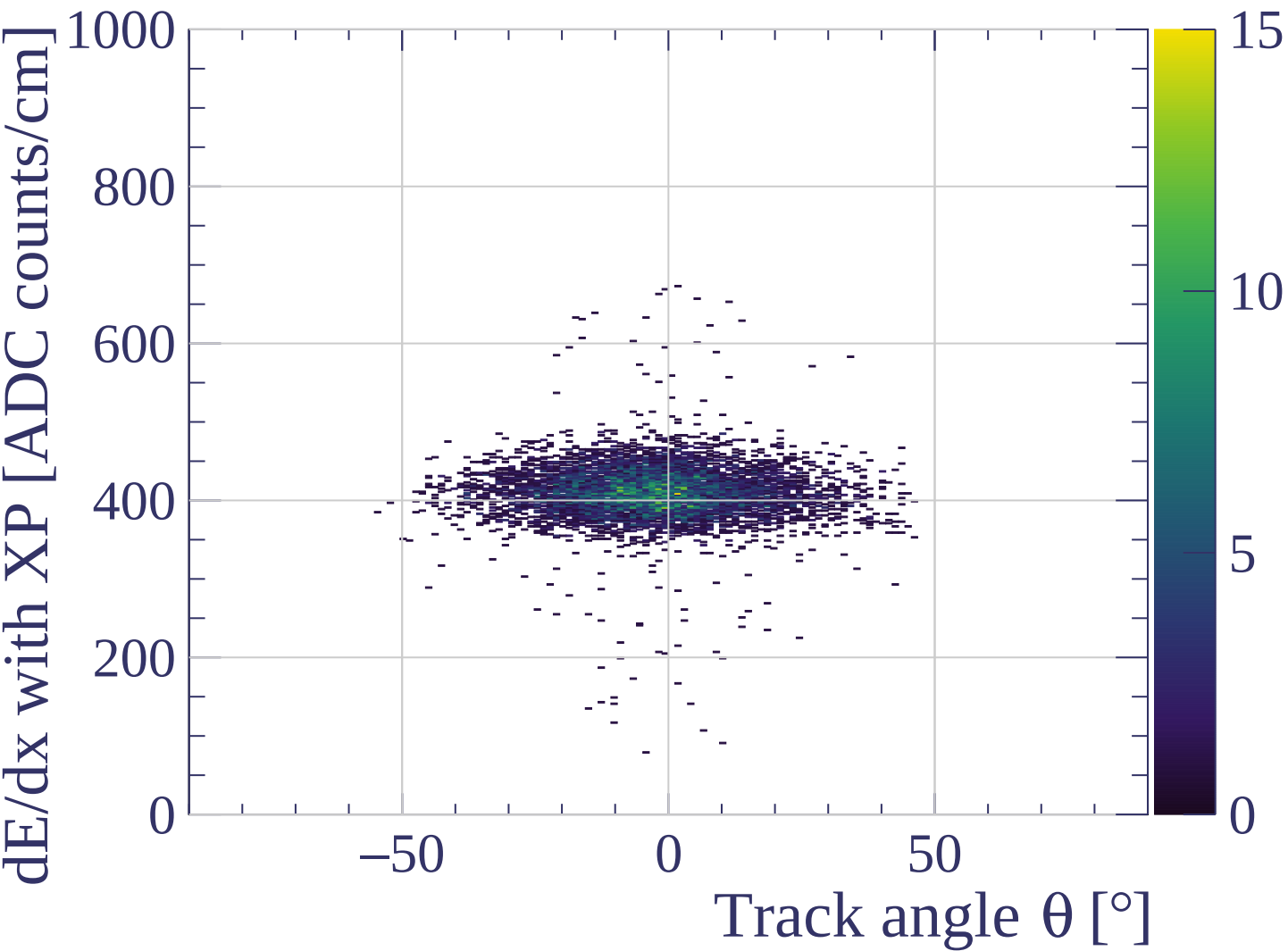
Track angle ϕ [°] angle

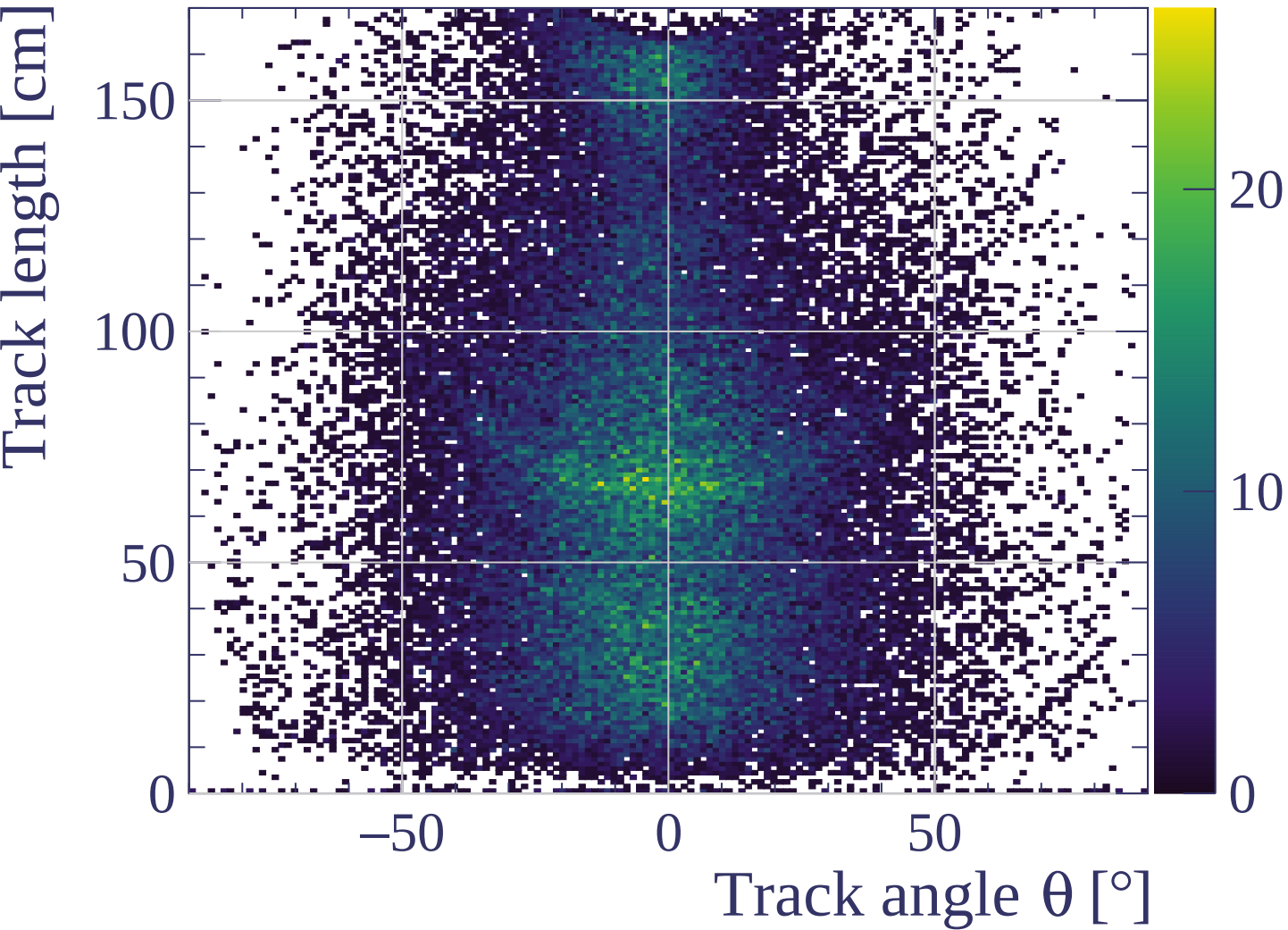


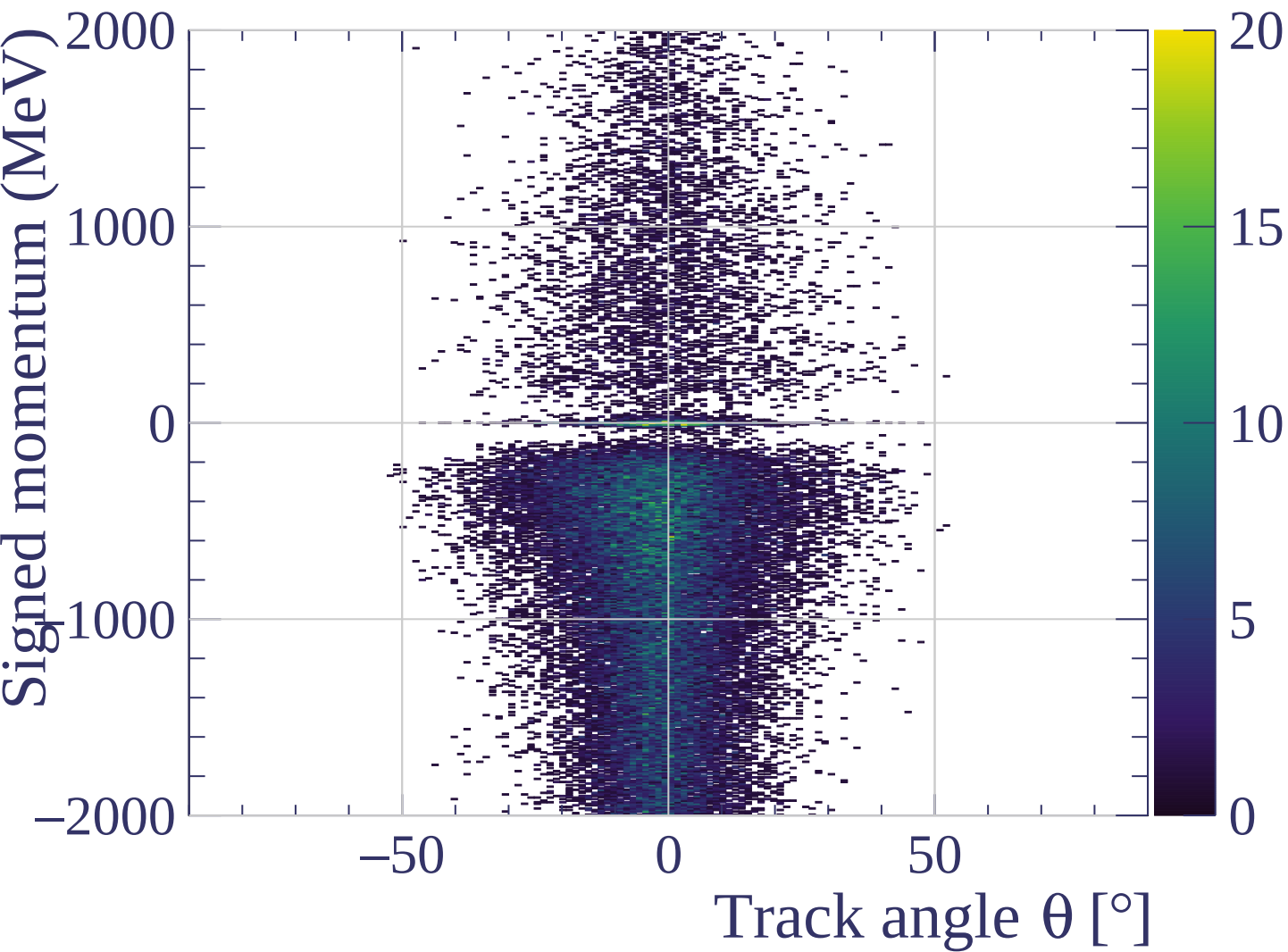




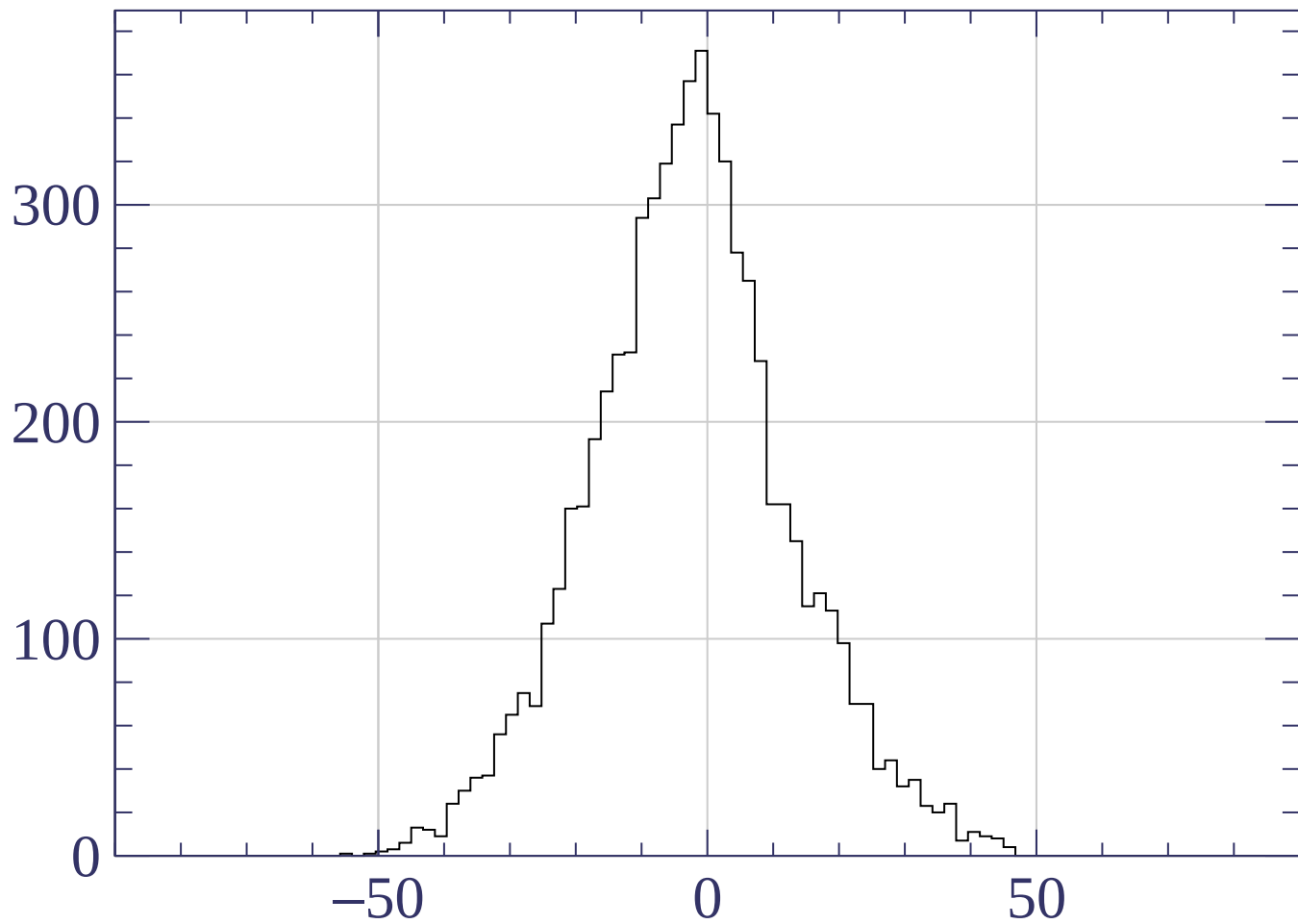






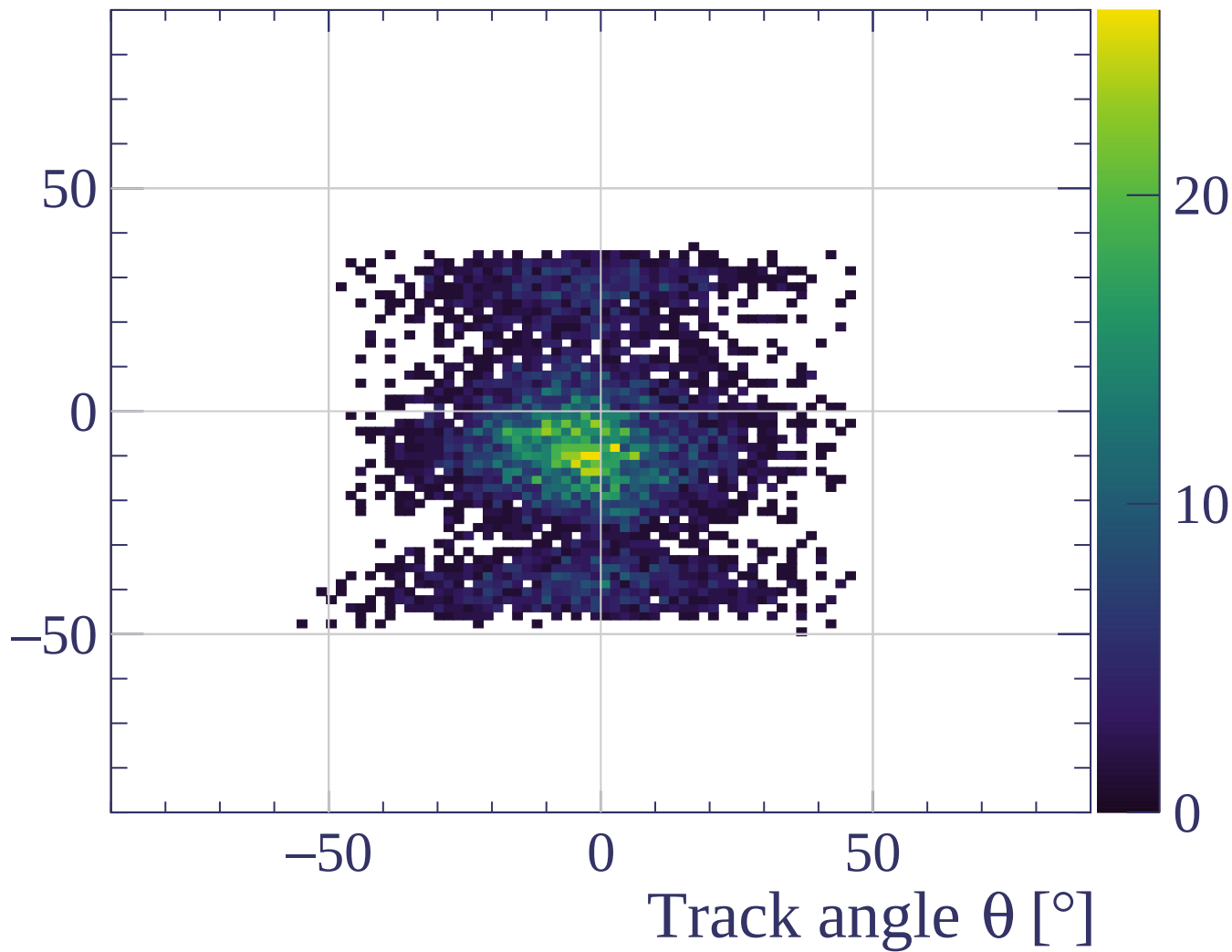


Count

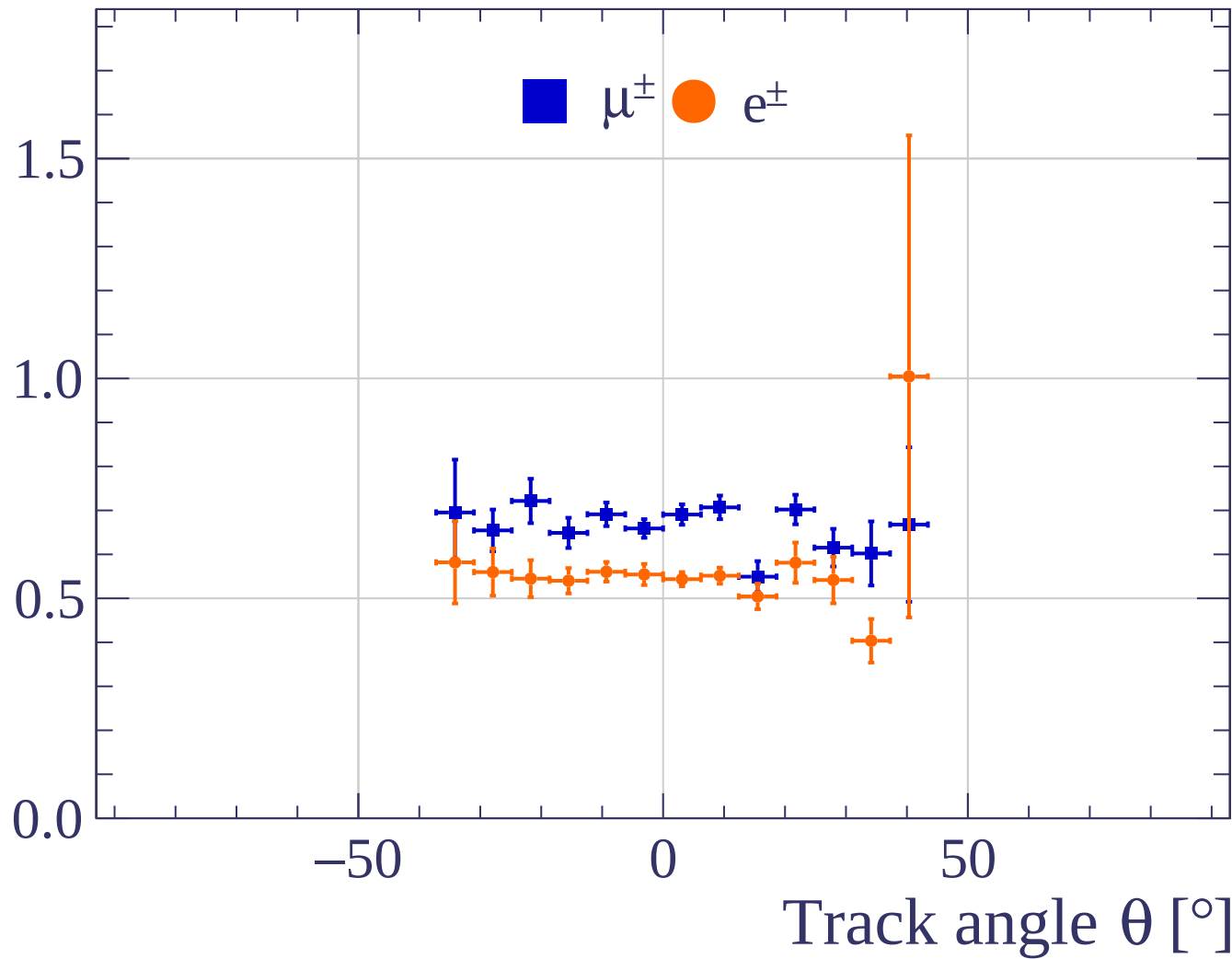


Track angle θ [°]

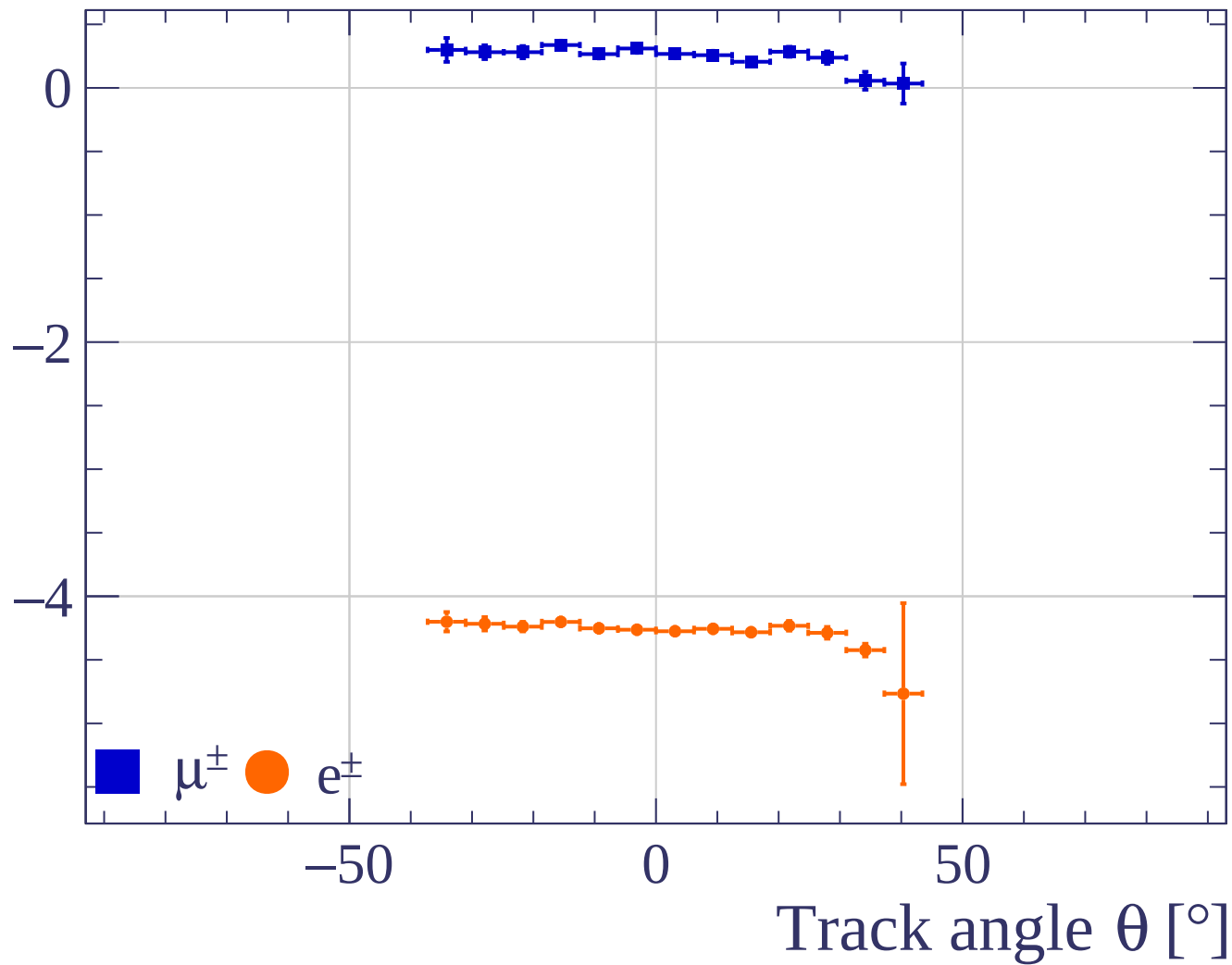
Track angle ϕ [°]



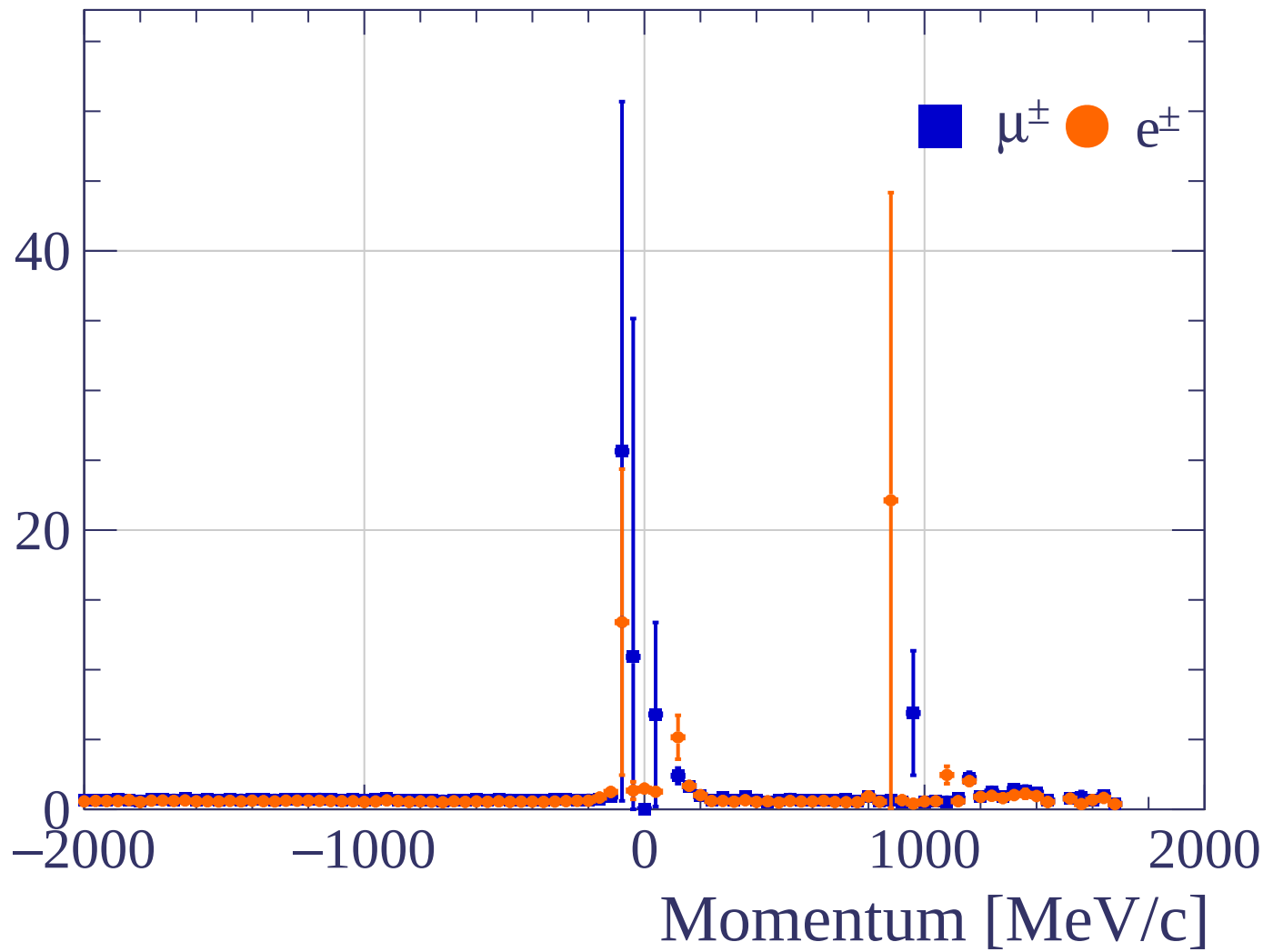
Pulls standard deviation



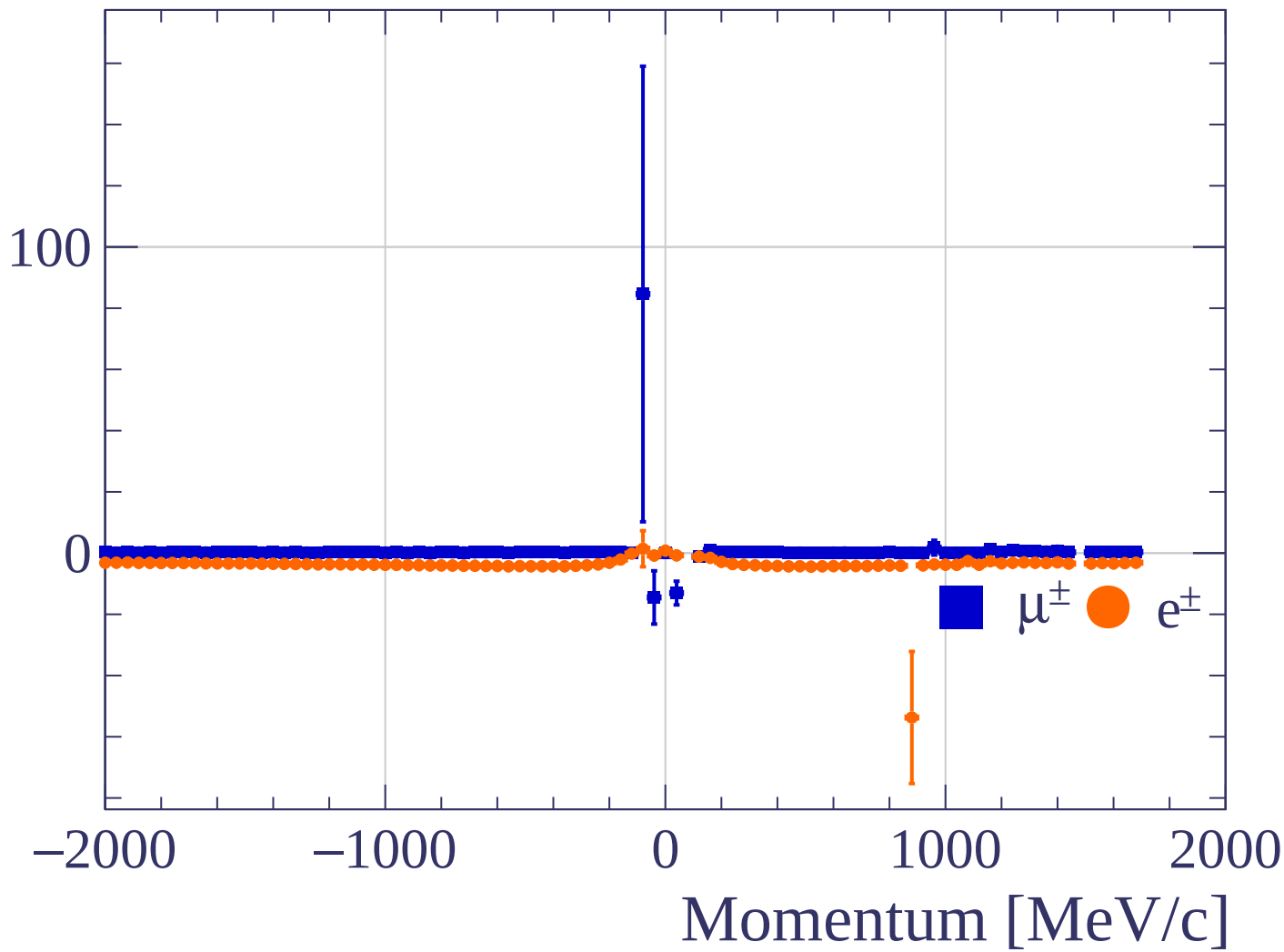
Pulls mean



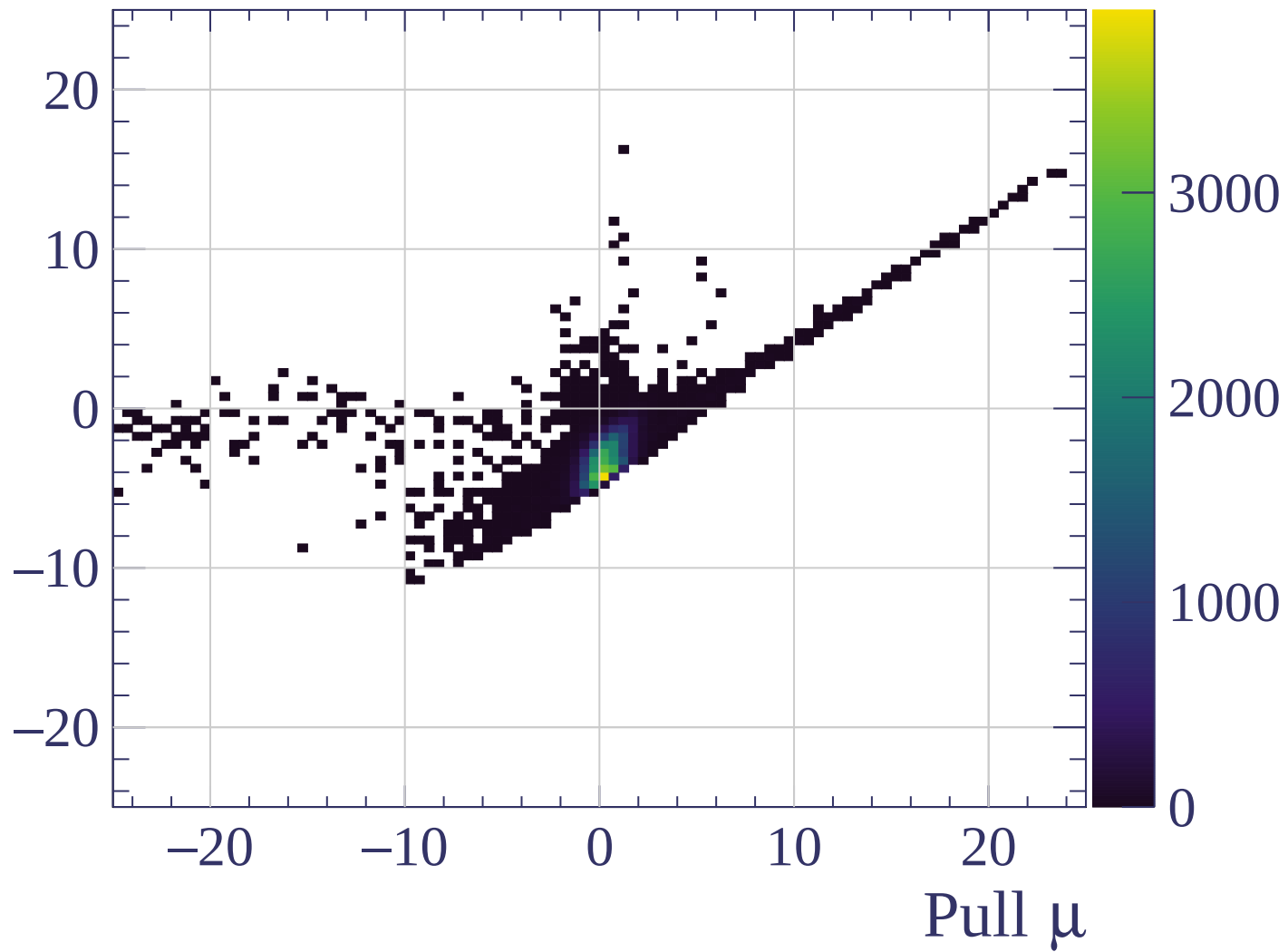
Pulls standard deviation

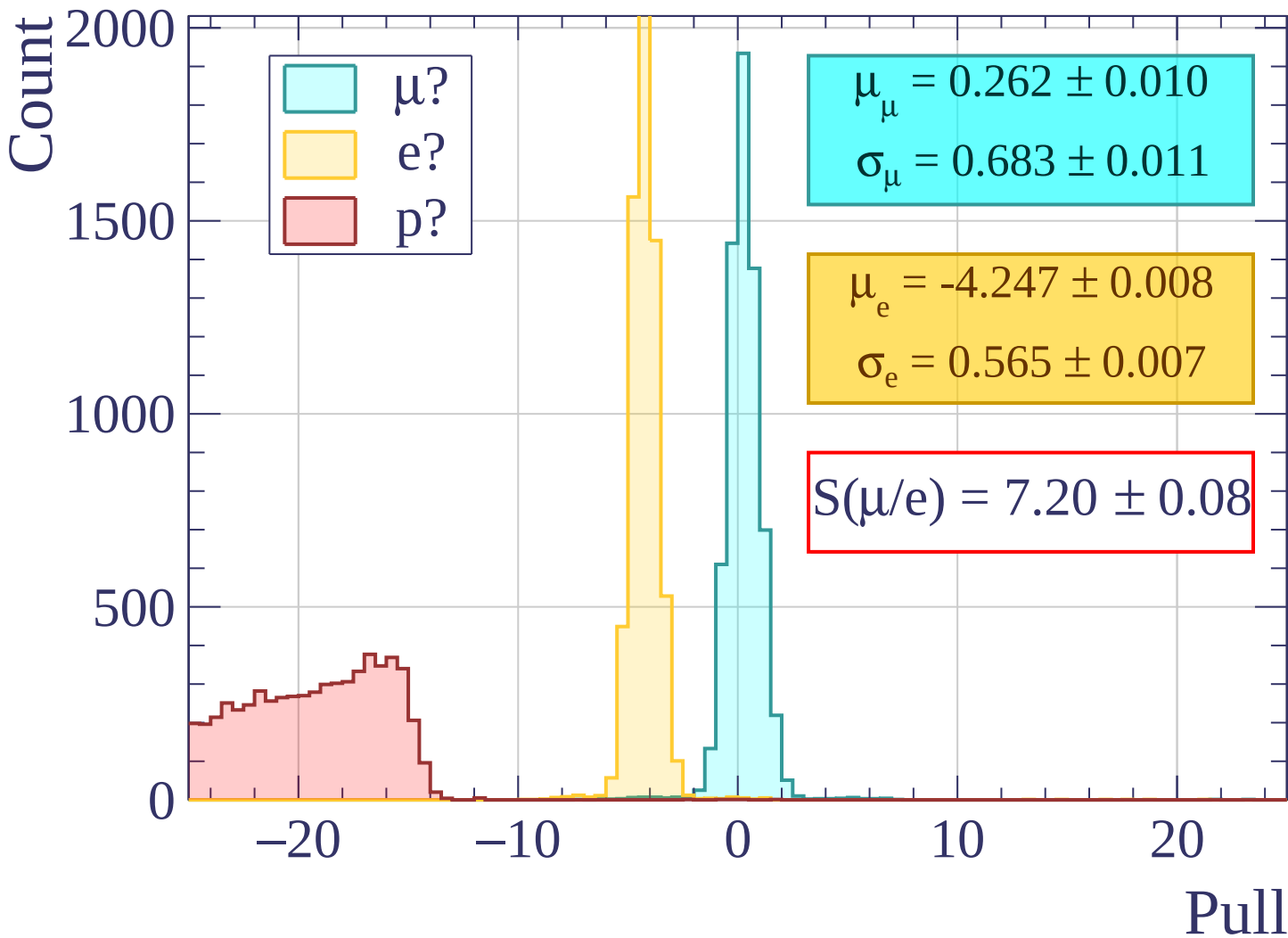


Pulls mean

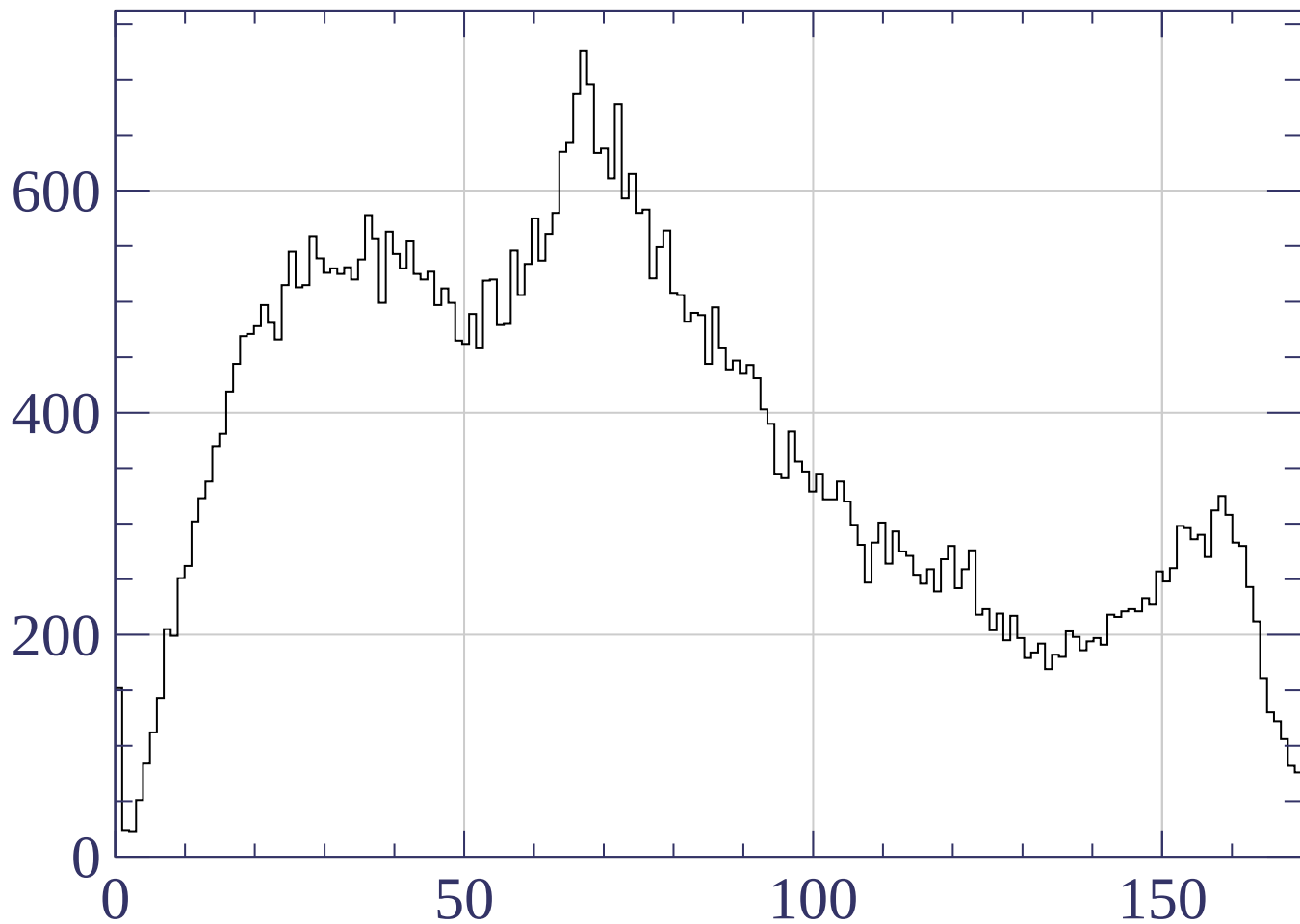


Pull e



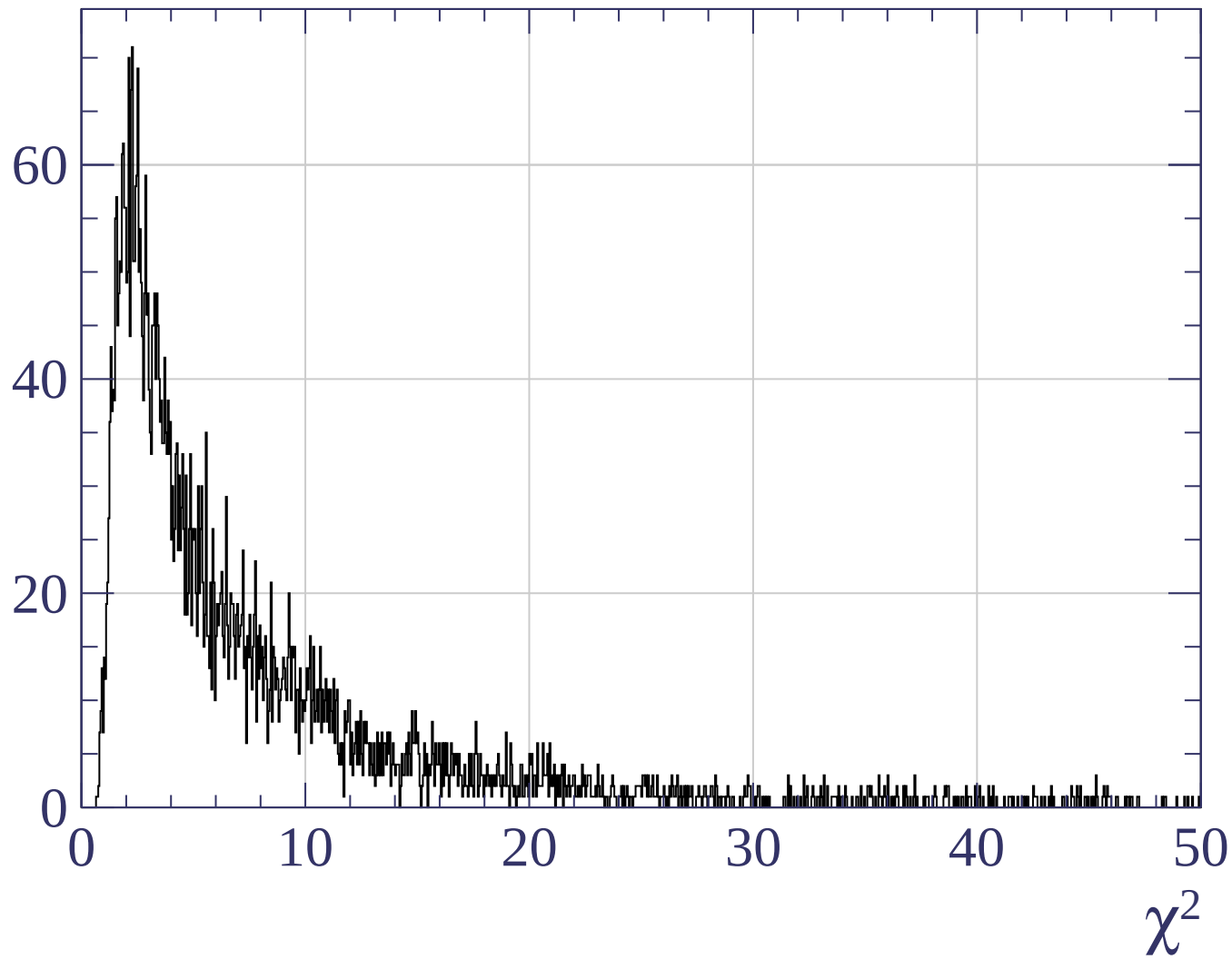


Count

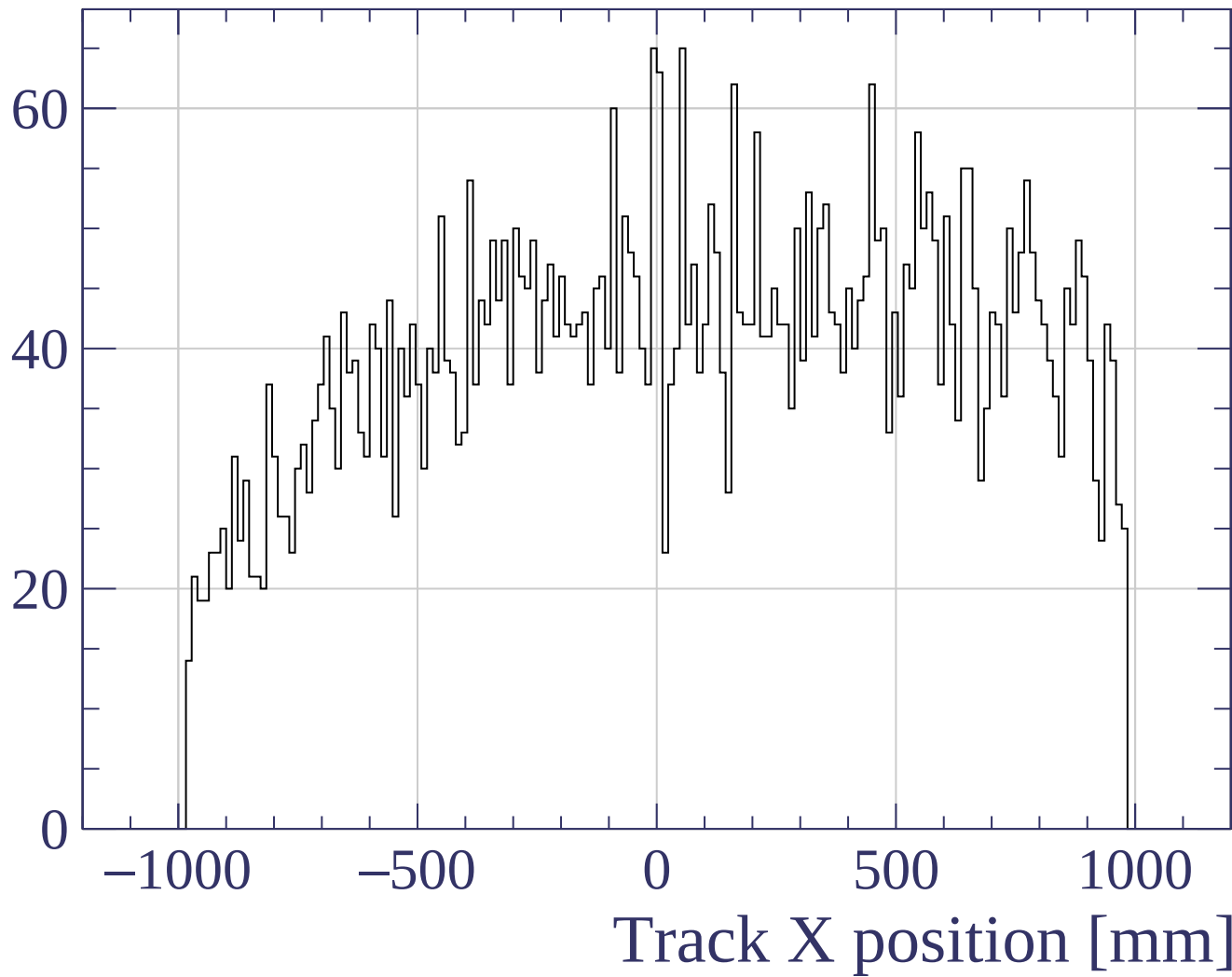


Track length [cm]

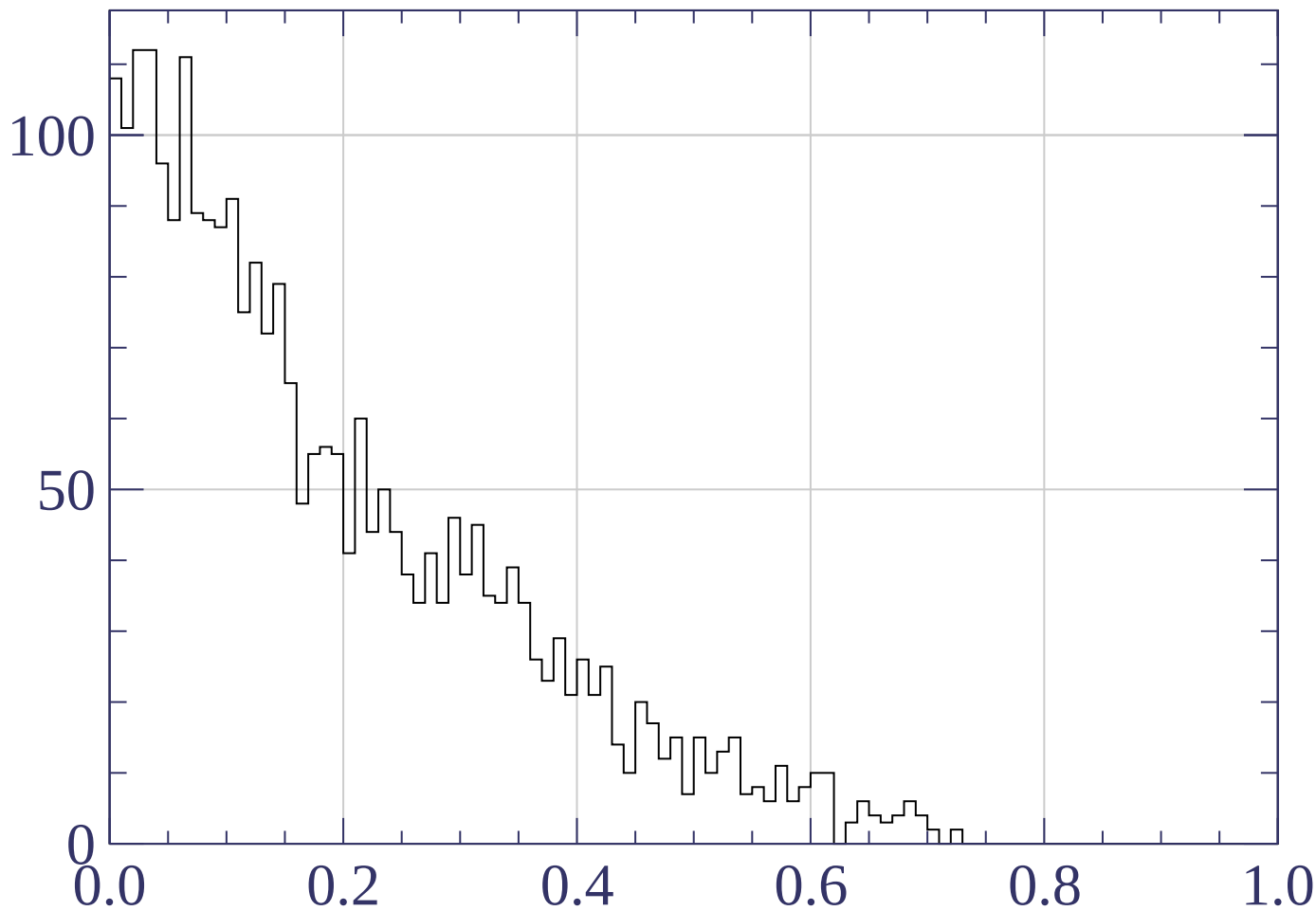
Count



Count

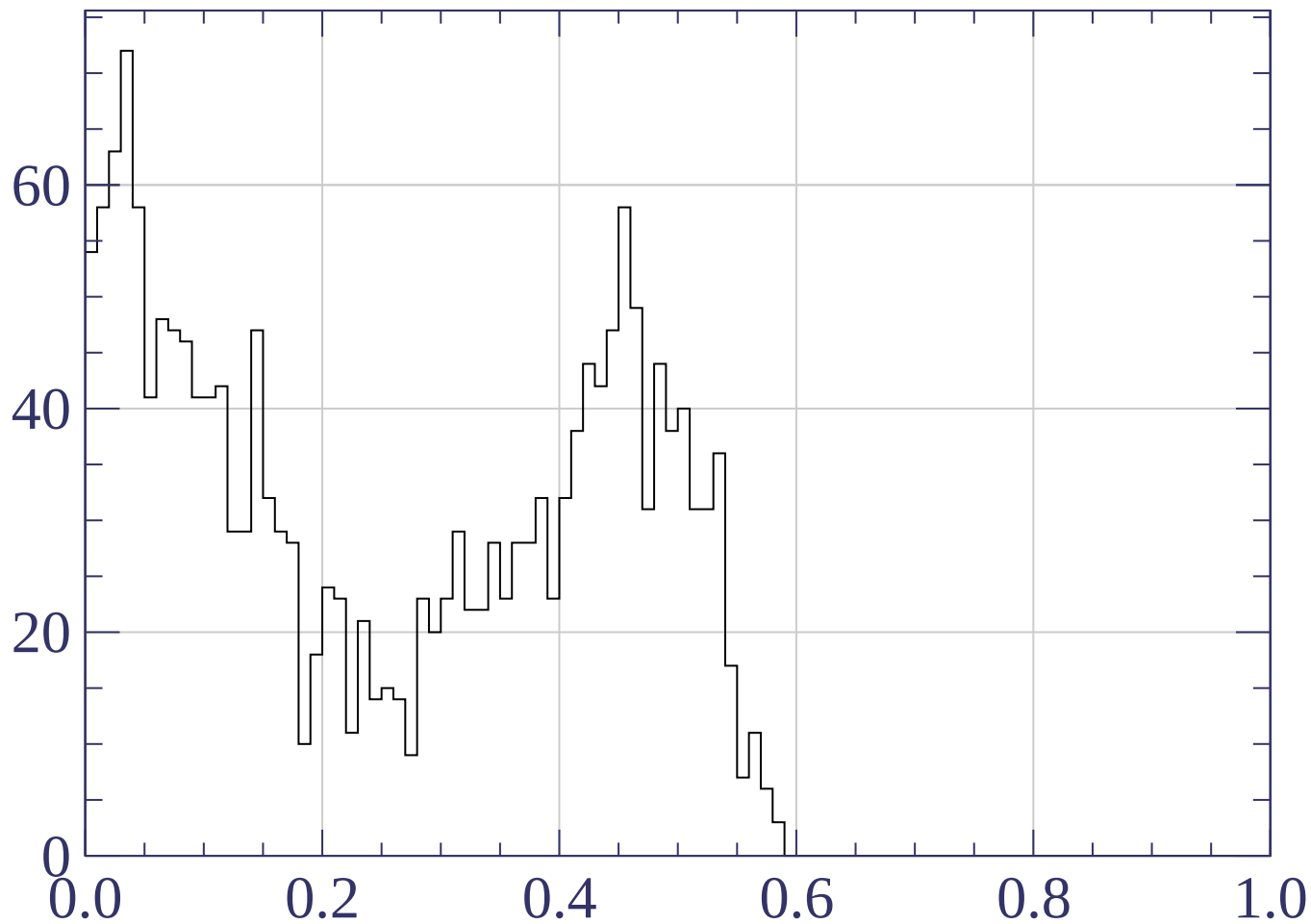


Counts



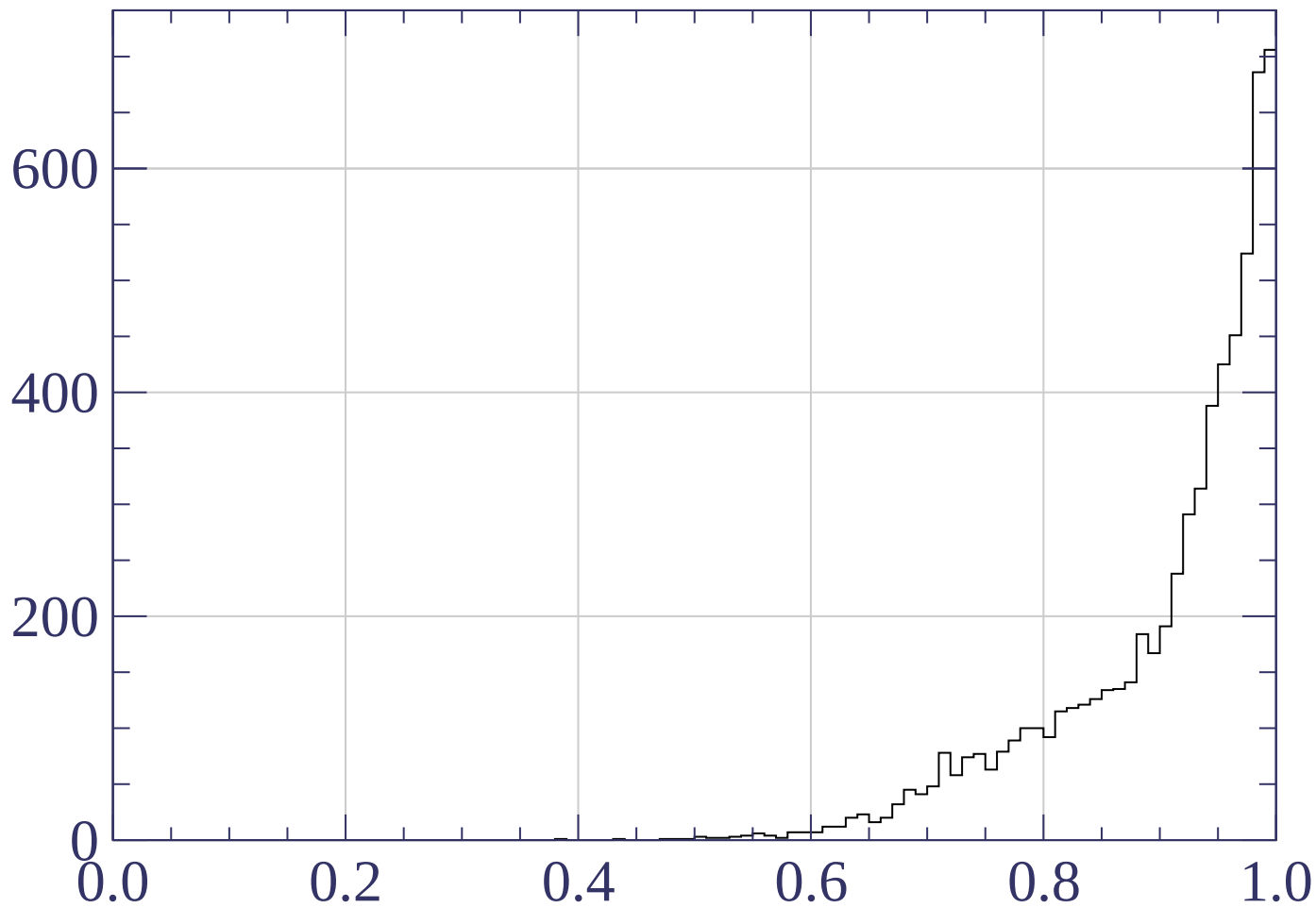
X direction

Counts

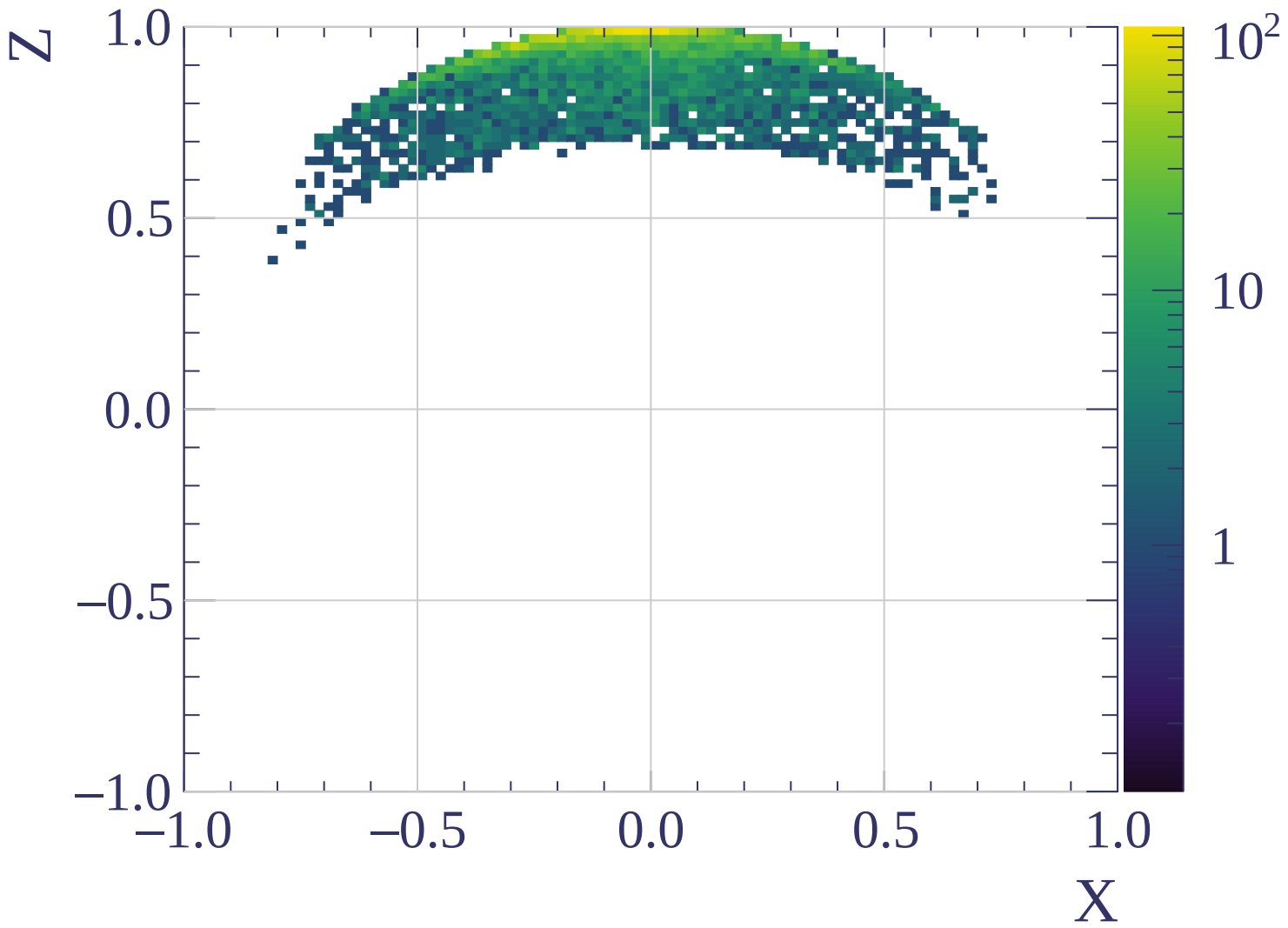


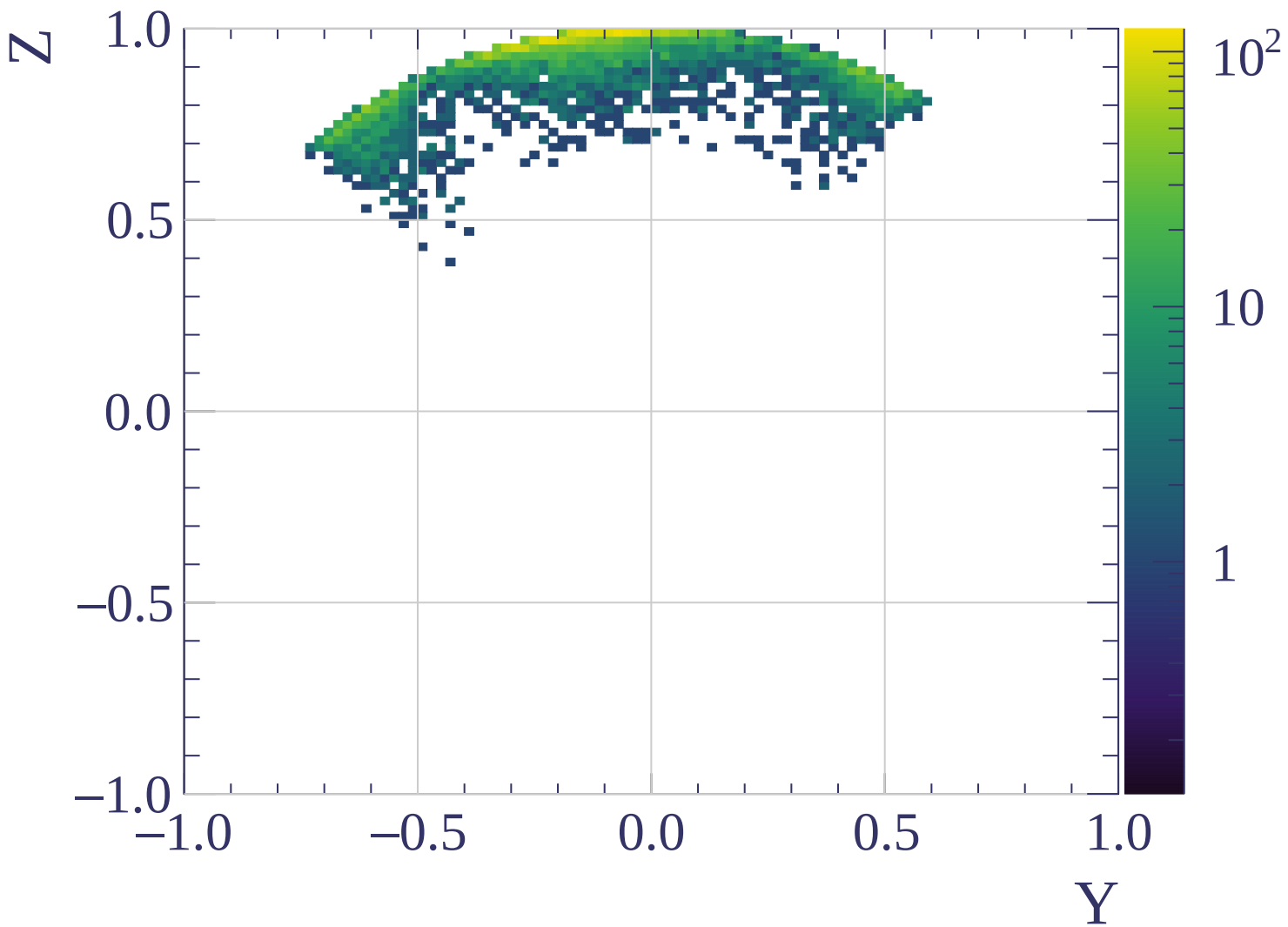
Y direction

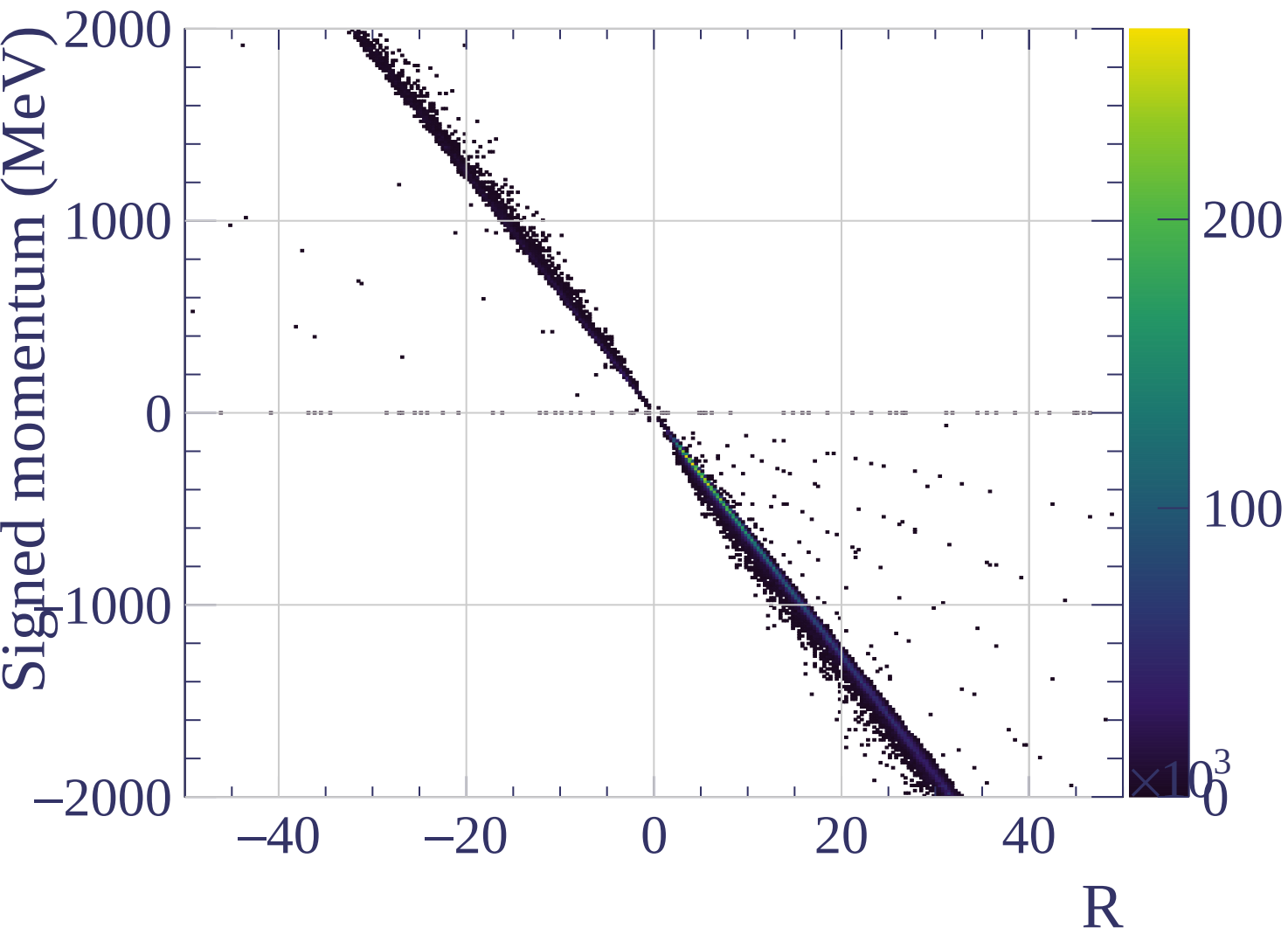
Counts

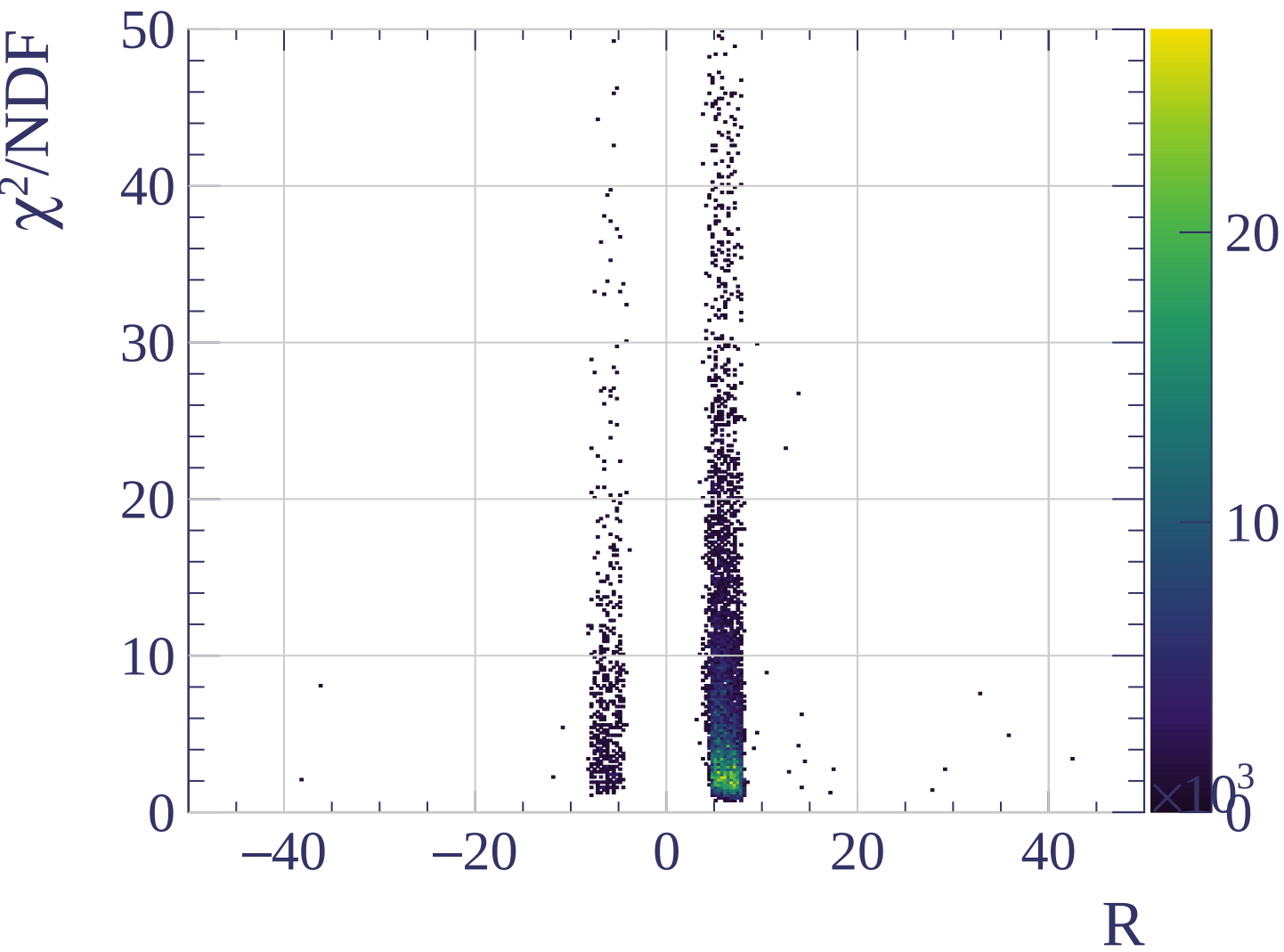


Z direction



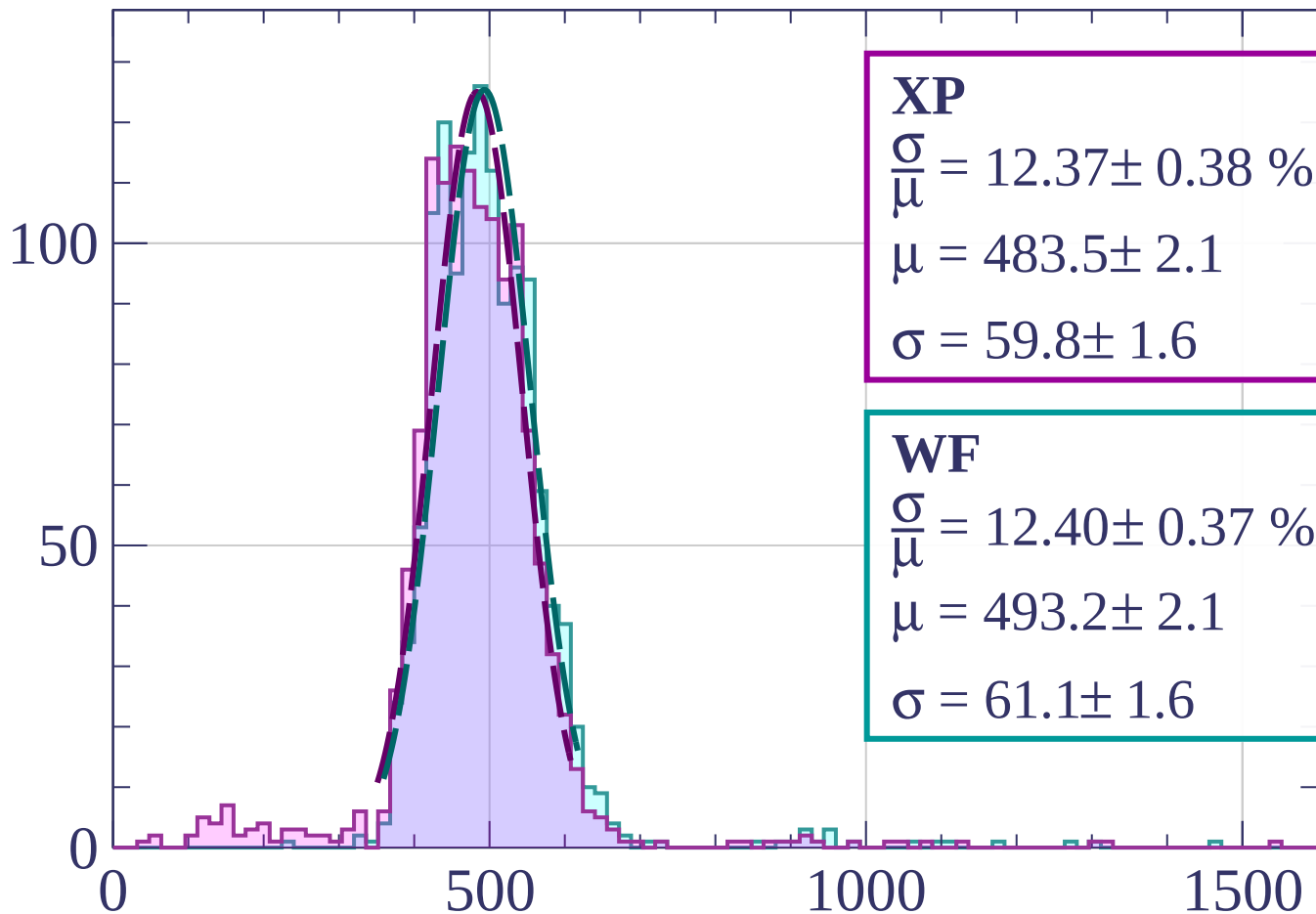






Energy loss | $0 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 12.37 \pm 0.38 \%$$

$$\mu = 483.5 \pm 2.1$$

$$\sigma = 59.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 12.40 \pm 0.37 \%$$

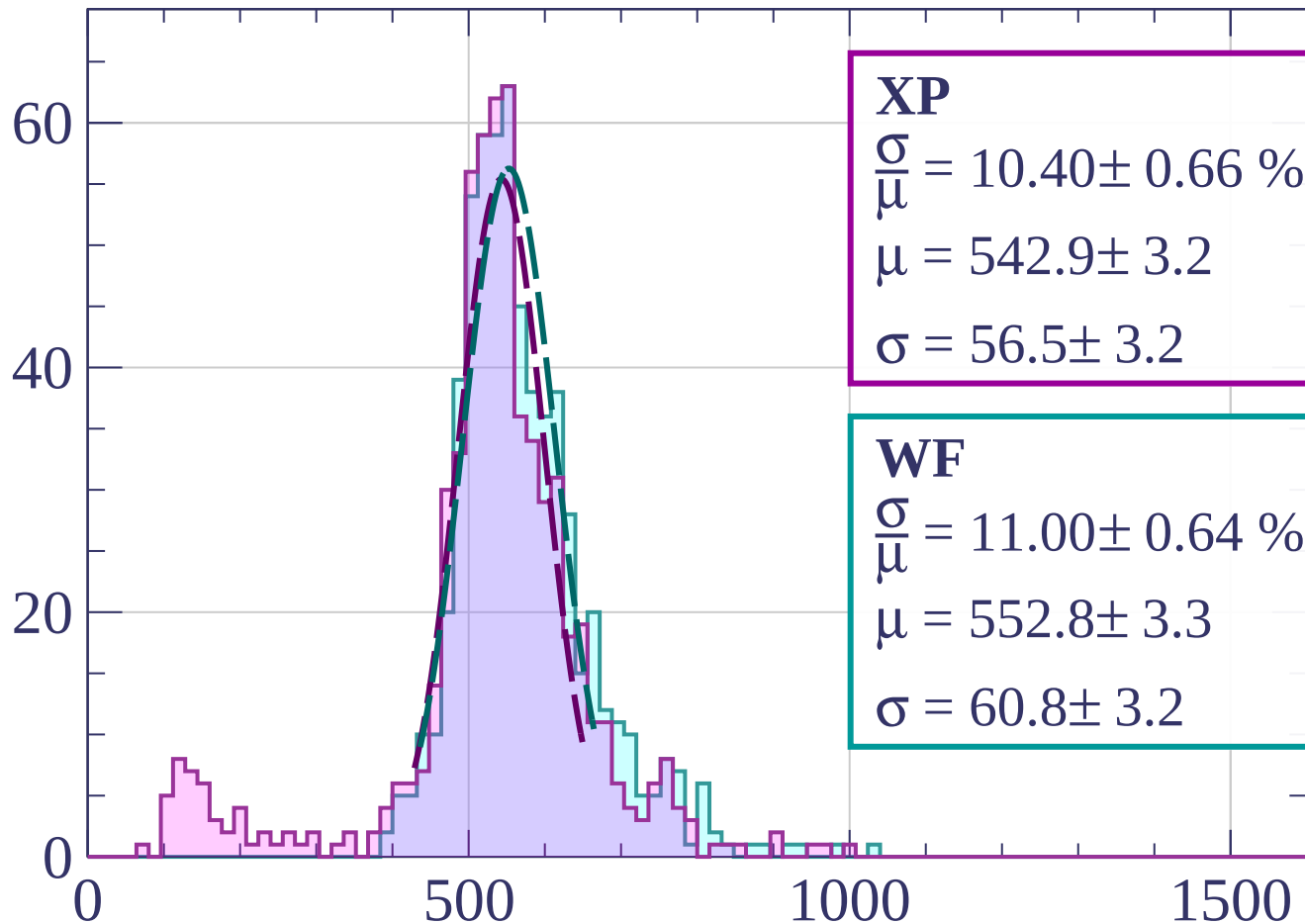
$$\mu = 493.2 \pm 2.1$$

$$\sigma = 61.1 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.66 \%$$

$$\mu = 542.9 \pm 3.2$$

$$\sigma = 56.5 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 11.00 \pm 0.64 \%$$

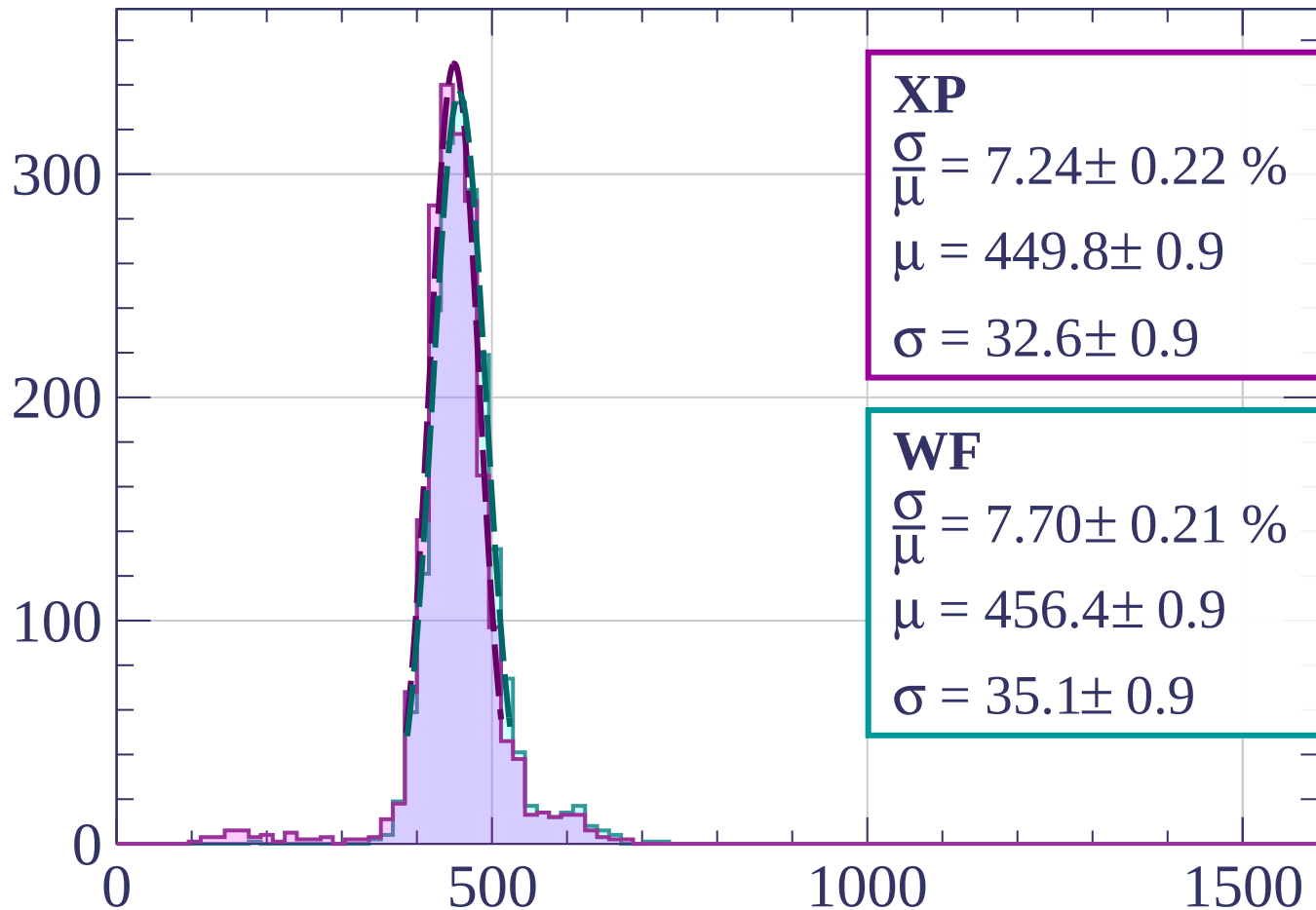
$$\mu = 552.8 \pm 3.3$$

$$\sigma = 60.8 \pm 3.2$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 7.24 \pm 0.22 \%$$

$$\mu = 449.8 \pm 0.9$$

$$\sigma = 32.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.21 \%$$

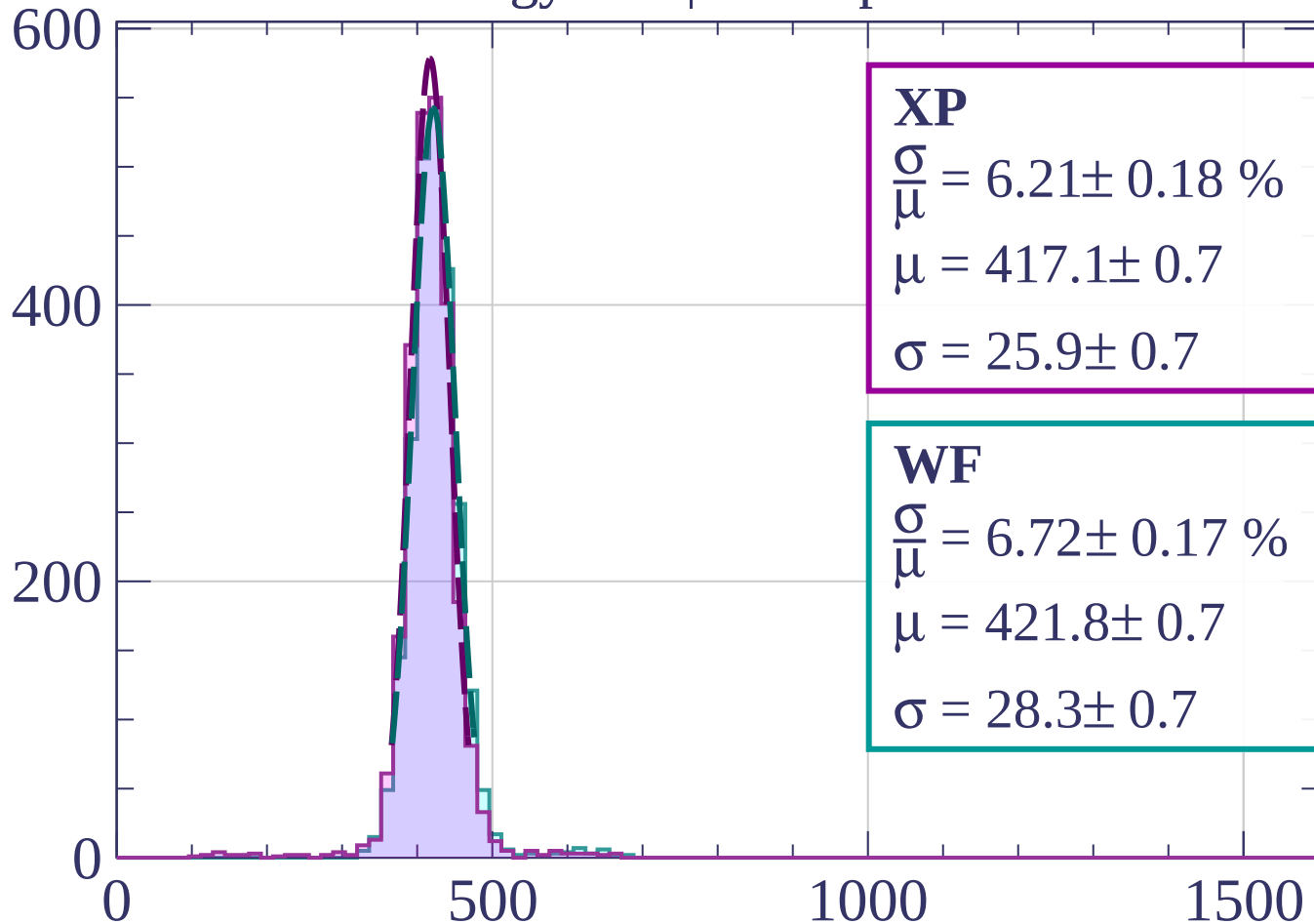
$$\mu = 456.4 \pm 0.9$$

$$\sigma = 35.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 6.21 \pm 0.18 \%$$

$$\mu = 417.1 \pm 0.7$$

$$\sigma = 25.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.72 \pm 0.17 \%$$

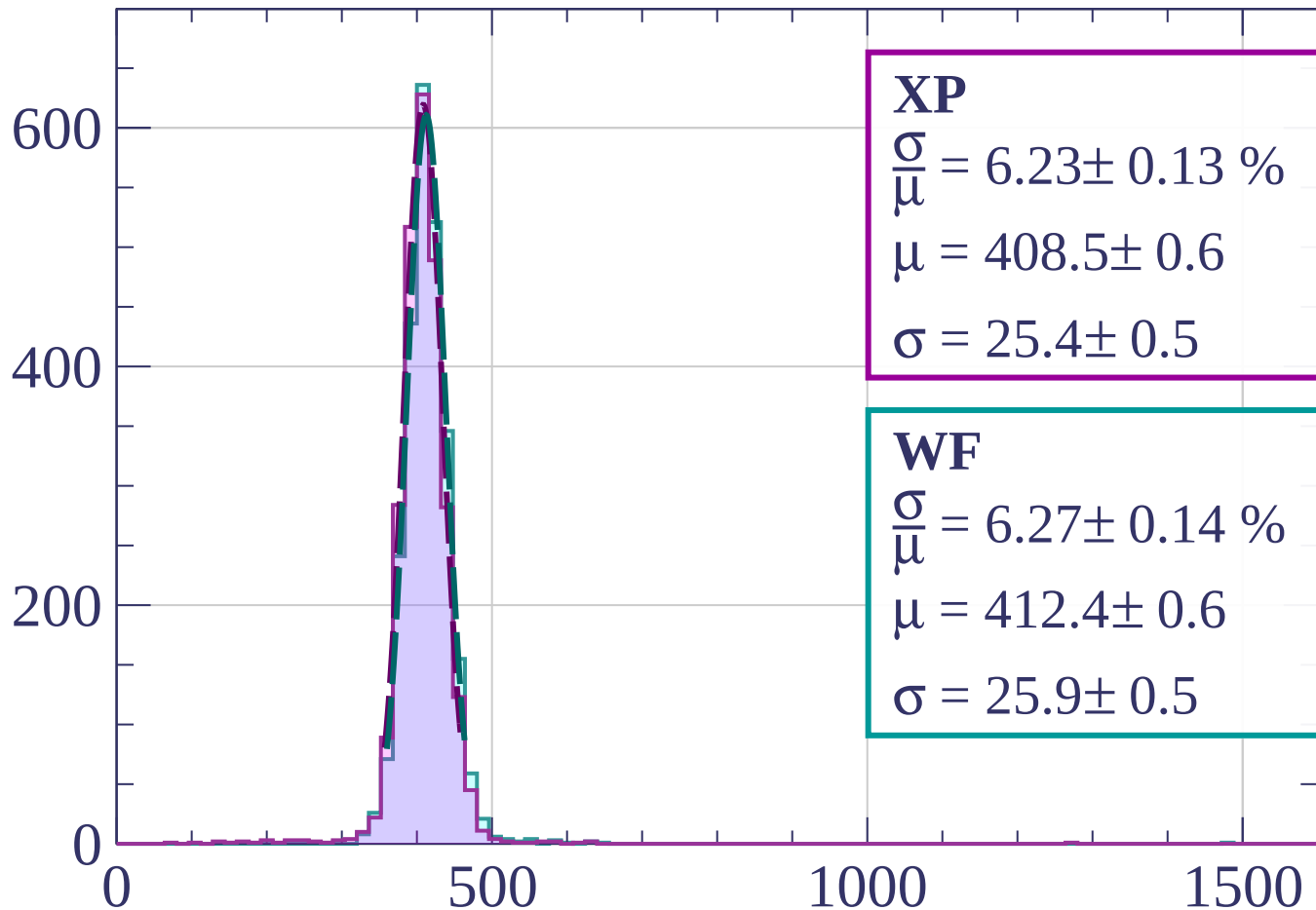
$$\mu = 421.8 \pm 0.7$$

$$\sigma = 28.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

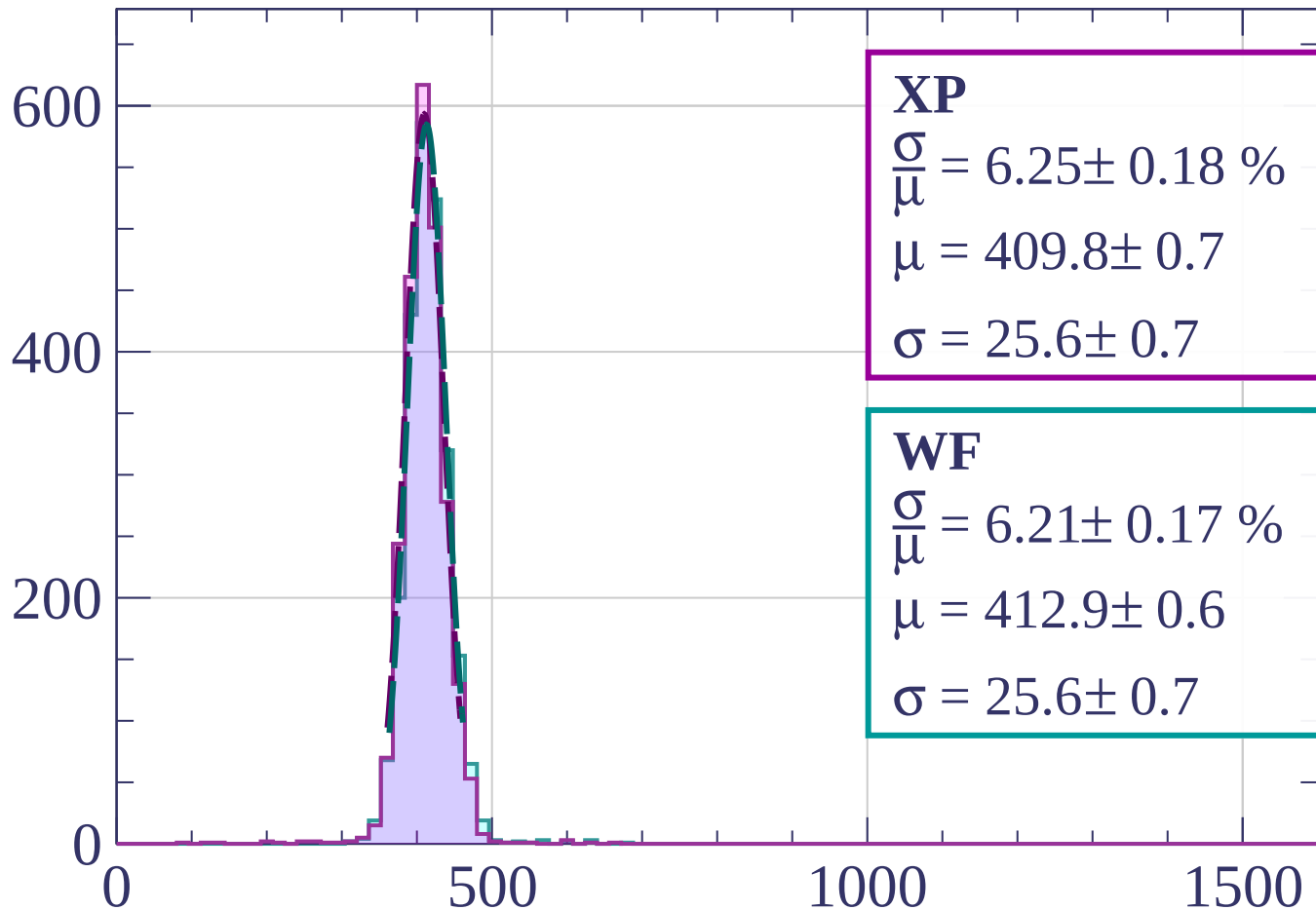
Count



dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 6.25 \pm 0.18 \%$$

$$\mu = 409.8 \pm 0.7$$

$$\sigma = 25.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.21 \pm 0.17 \%$$

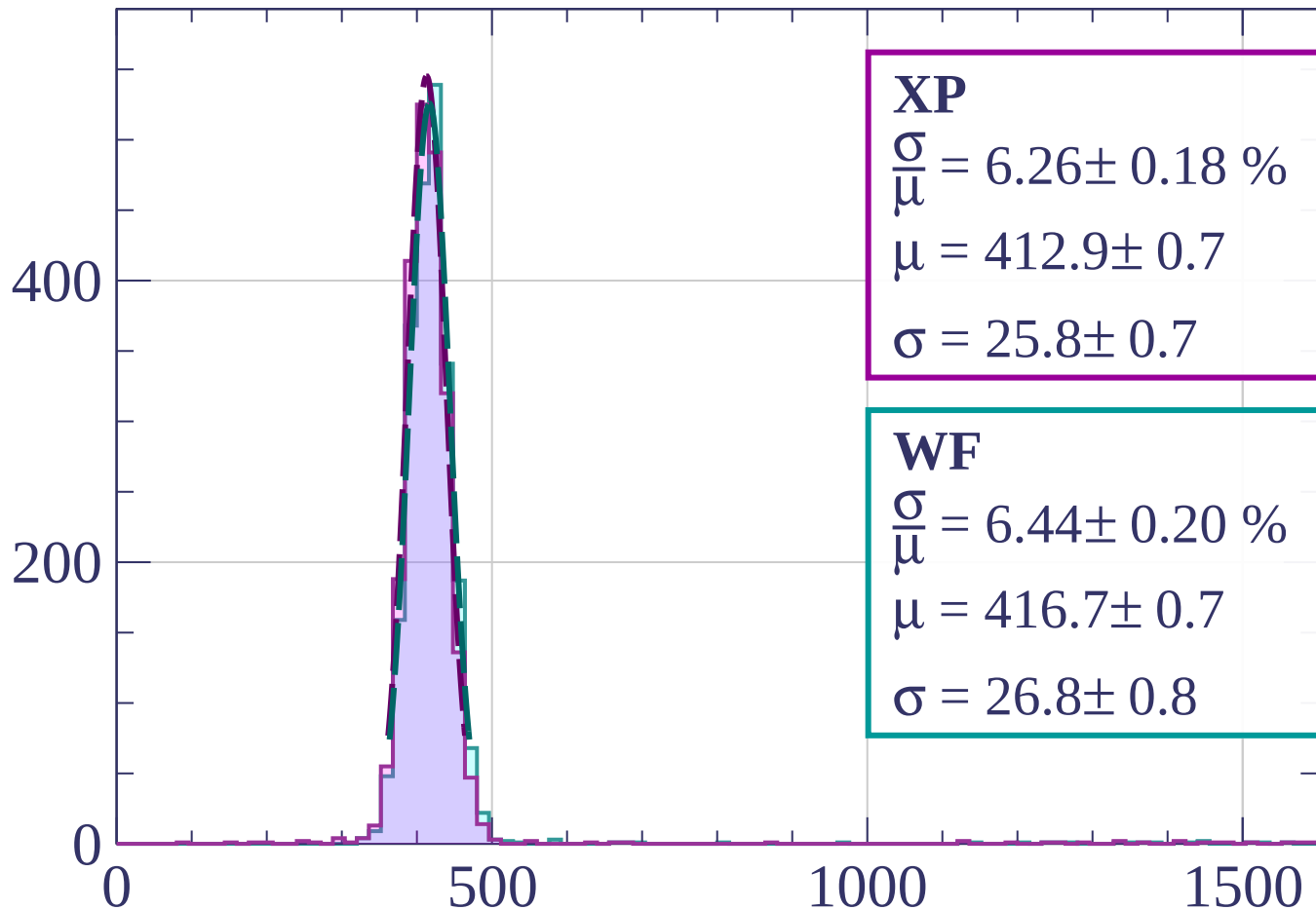
$$\mu = 412.9 \pm 0.6$$

$$\sigma = 25.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.18 \%$$

$$\mu = 412.9 \pm 0.7$$

$$\sigma = 25.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.20 \%$$

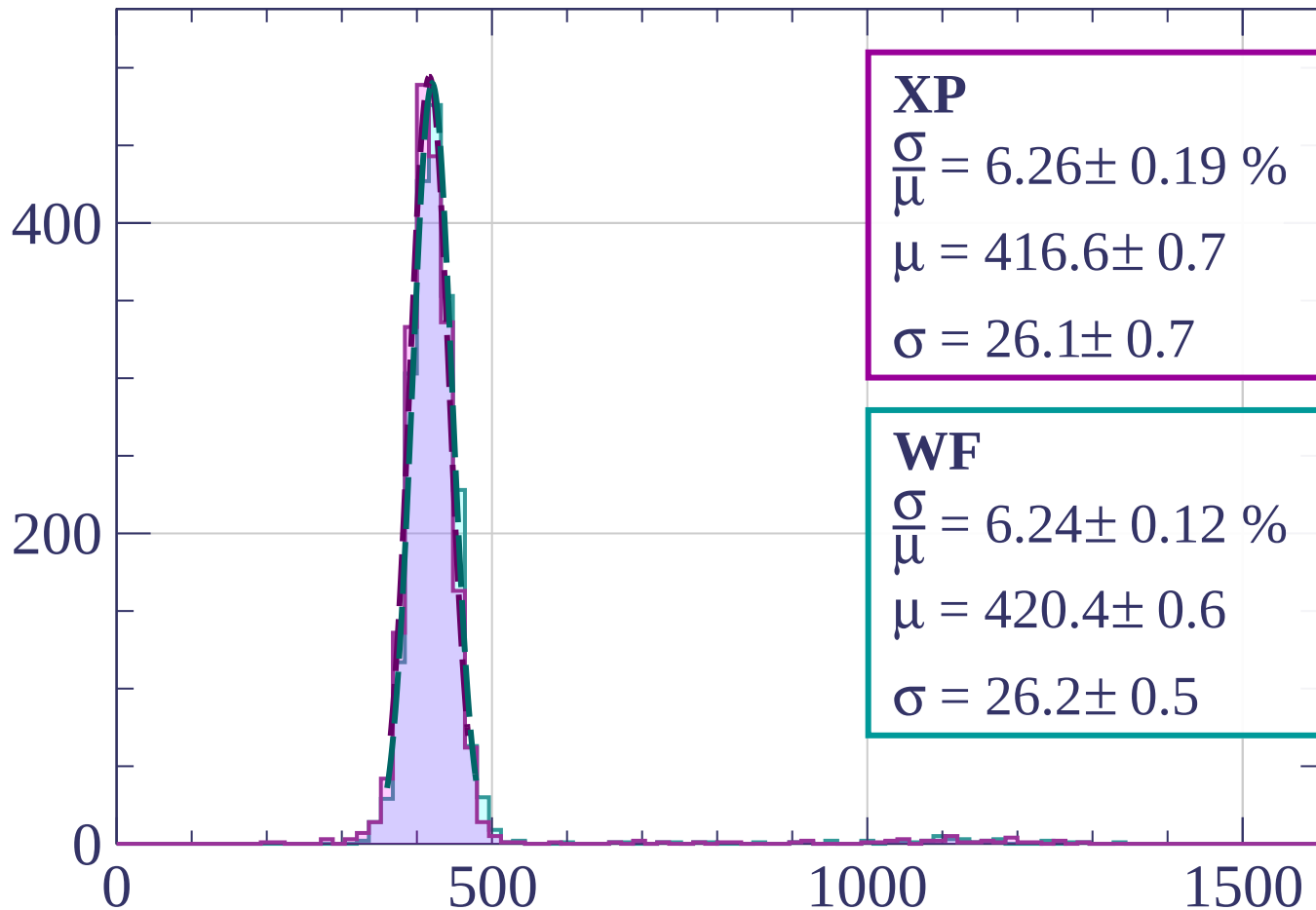
$$\mu = 416.7 \pm 0.7$$

$$\sigma = 26.8 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.19 \%$$

$$\mu = 416.6 \pm 0.7$$

$$\sigma = 26.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.24 \pm 0.12 \%$$

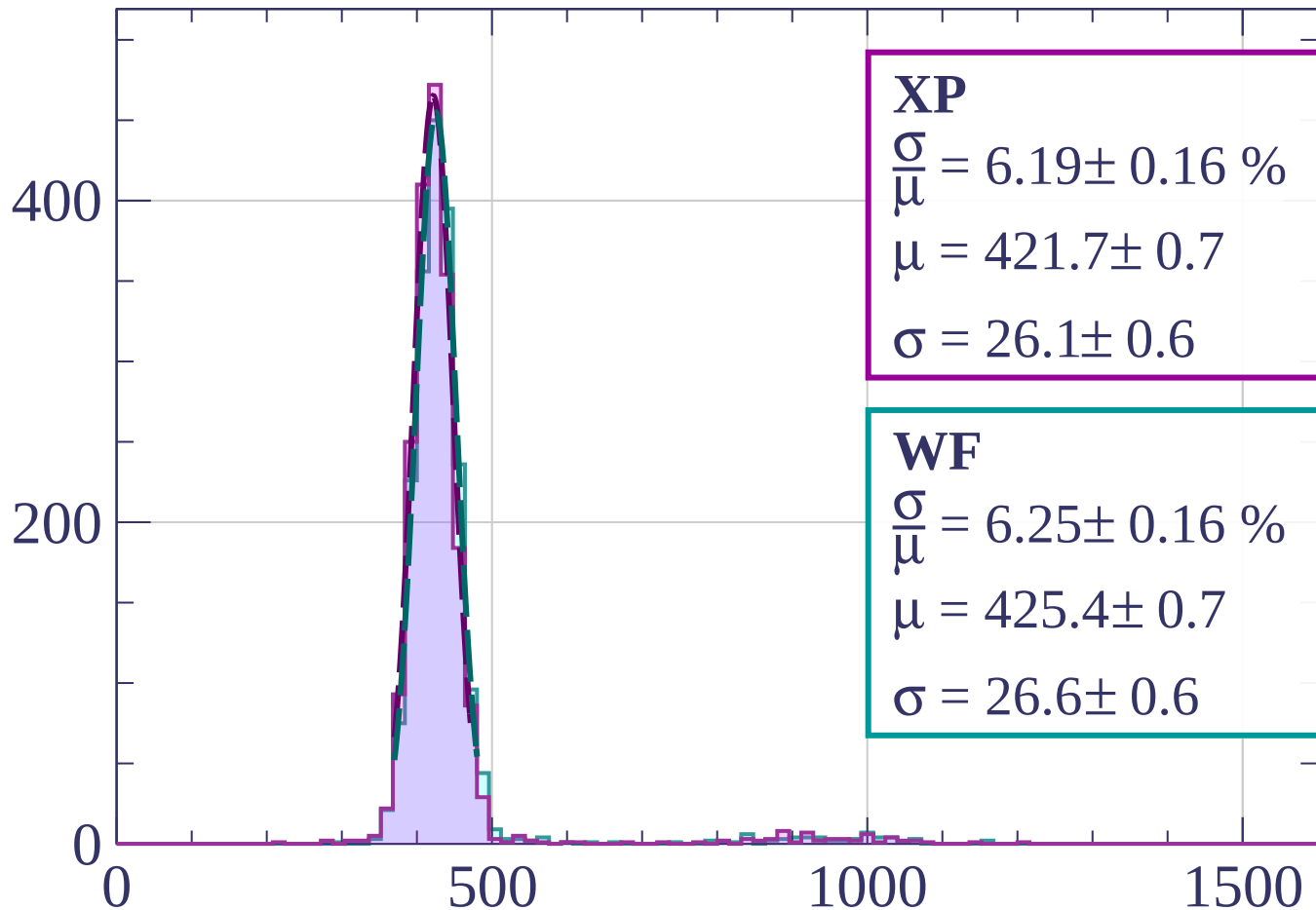
$$\mu = 420.4 \pm 0.6$$

$$\sigma = 26.2 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.16 \%$$

$$\mu = 421.7 \pm 0.7$$

$$\sigma = 26.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.25 \pm 0.16 \%$$

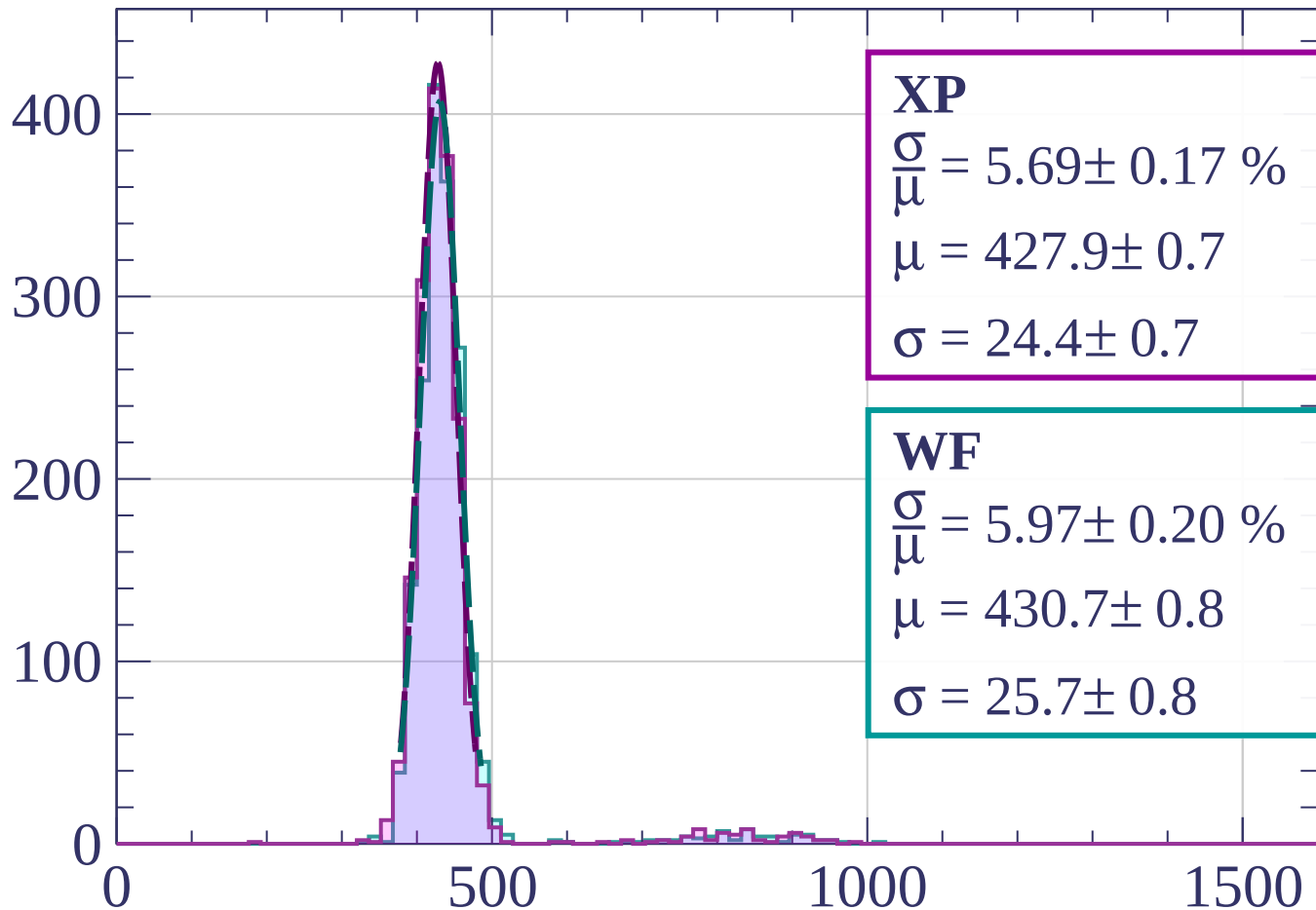
$$\mu = 425.4 \pm 0.7$$

$$\sigma = 26.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 5.69 \pm 0.17 \%$$

$$\mu = 427.9 \pm 0.7$$

$$\sigma = 24.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.97 \pm 0.20 \%$$

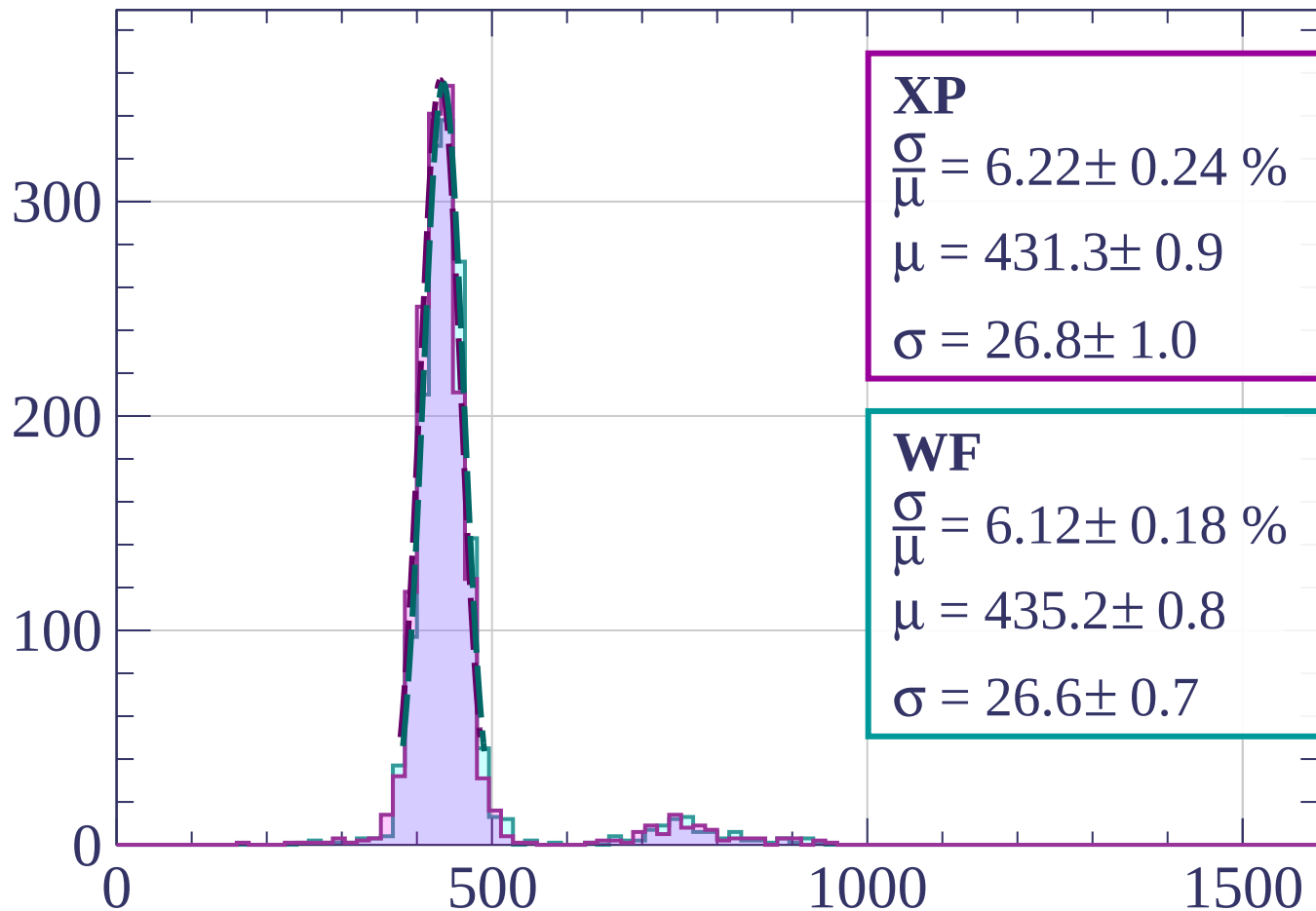
$$\mu = 430.7 \pm 0.8$$

$$\sigma = 25.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.24 \%$$

$$\mu = 431.3 \pm 0.9$$

$$\sigma = 26.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.12 \pm 0.18 \%$$

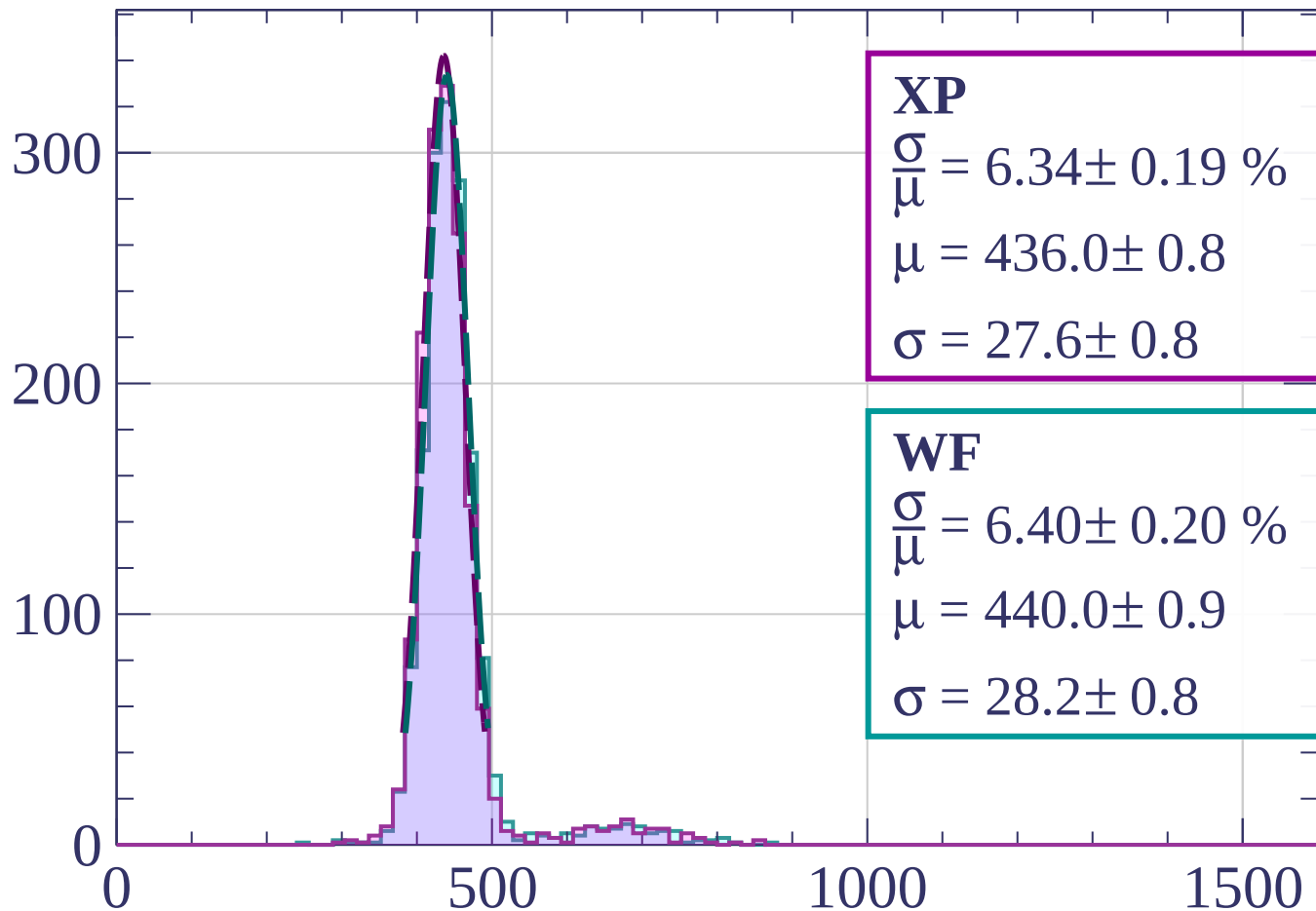
$$\mu = 435.2 \pm 0.8$$

$$\sigma = 26.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 6.34 \pm 0.19 \%$$

$$\mu = 436.0 \pm 0.8$$

$$\sigma = 27.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.40 \pm 0.20 \%$$

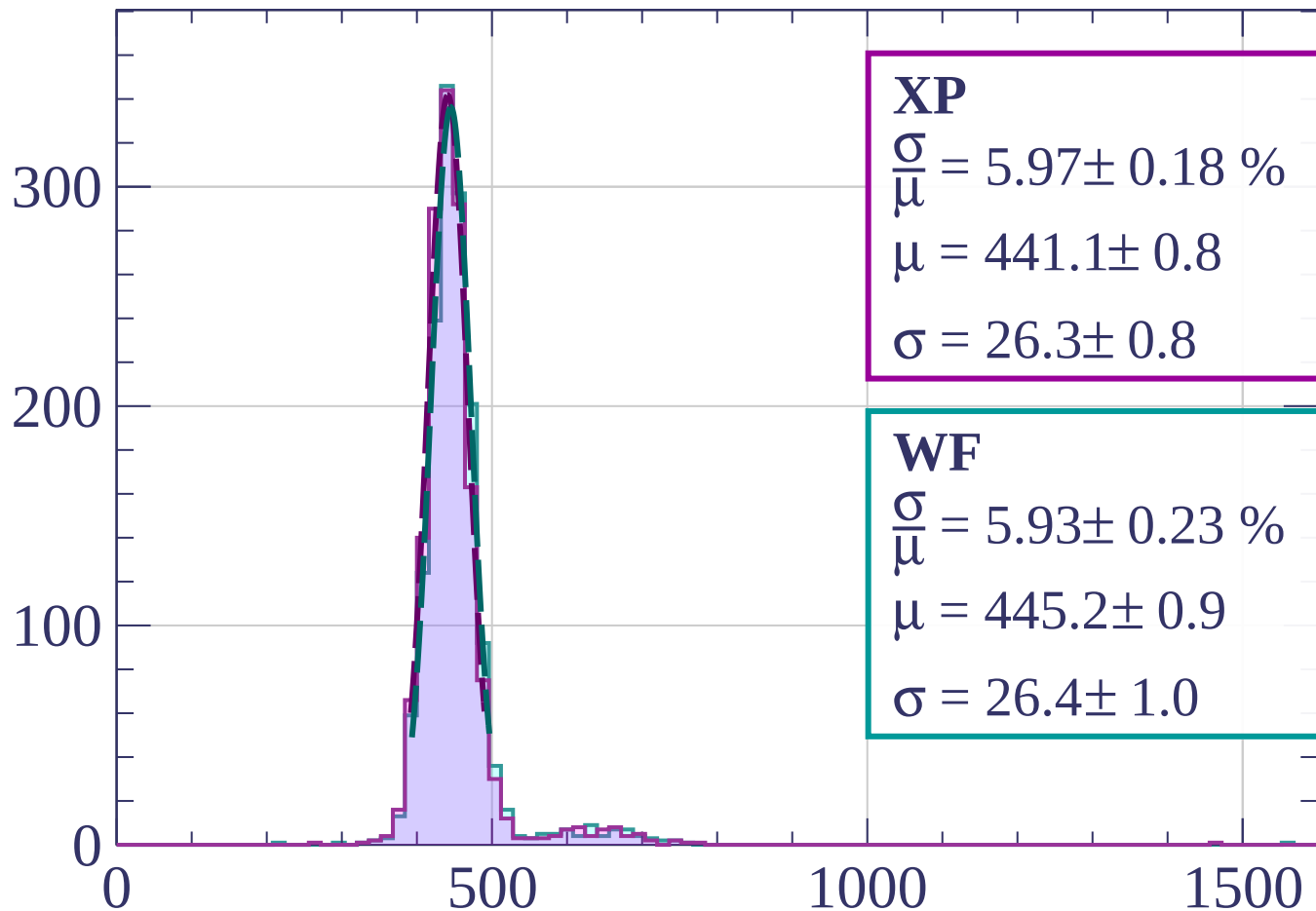
$$\mu = 440.0 \pm 0.9$$

$$\sigma = 28.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 5.97 \pm 0.18 \%$$

$$\mu = 441.1 \pm 0.8$$

$$\sigma = 26.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.93 \pm 0.23 \%$$

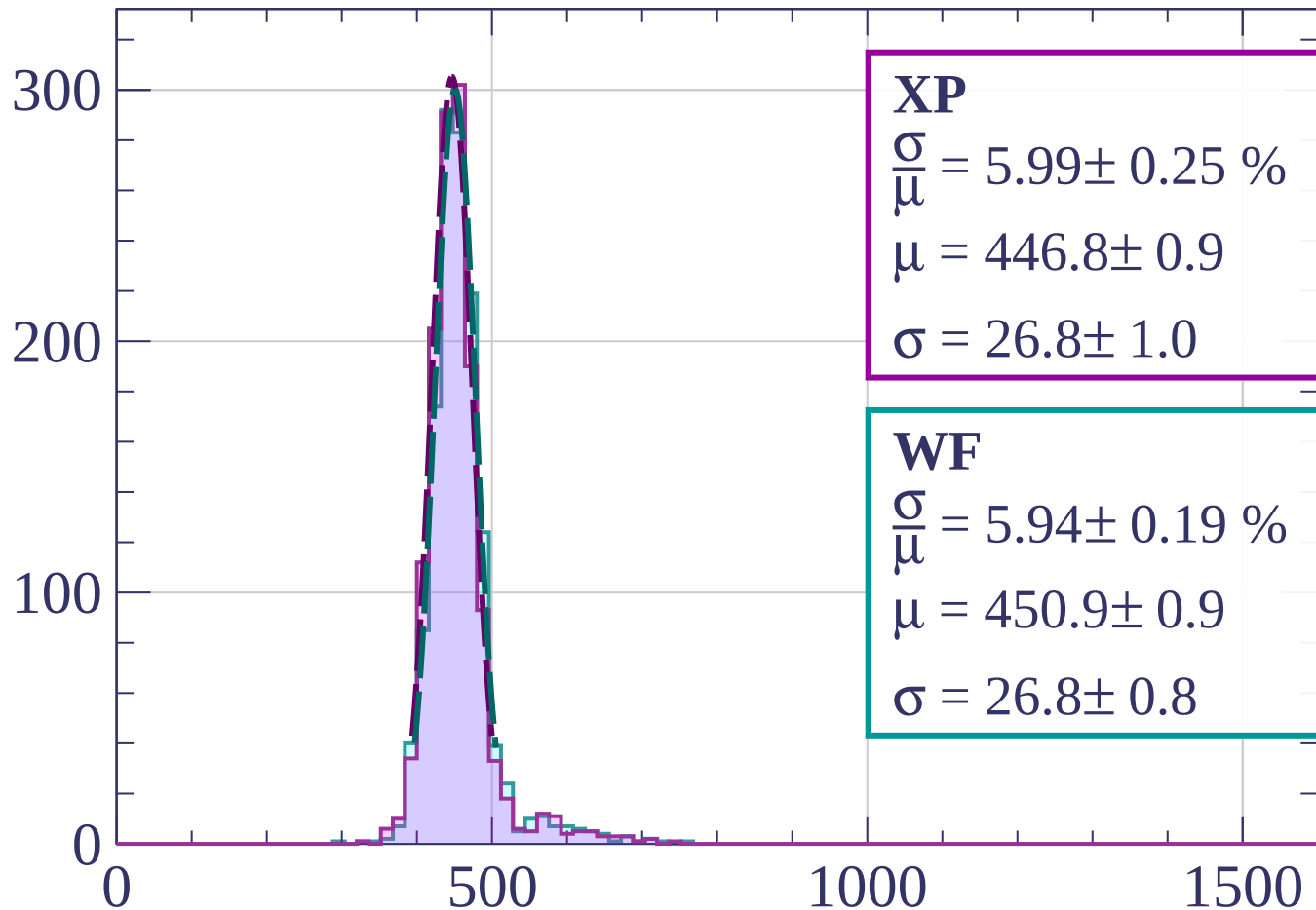
$$\mu = 445.2 \pm 0.9$$

$$\sigma = 26.4 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 5.99 \pm 0.25 \%$$

$$\mu = 446.8 \pm 0.9$$

$$\sigma = 26.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.94 \pm 0.19 \%$$

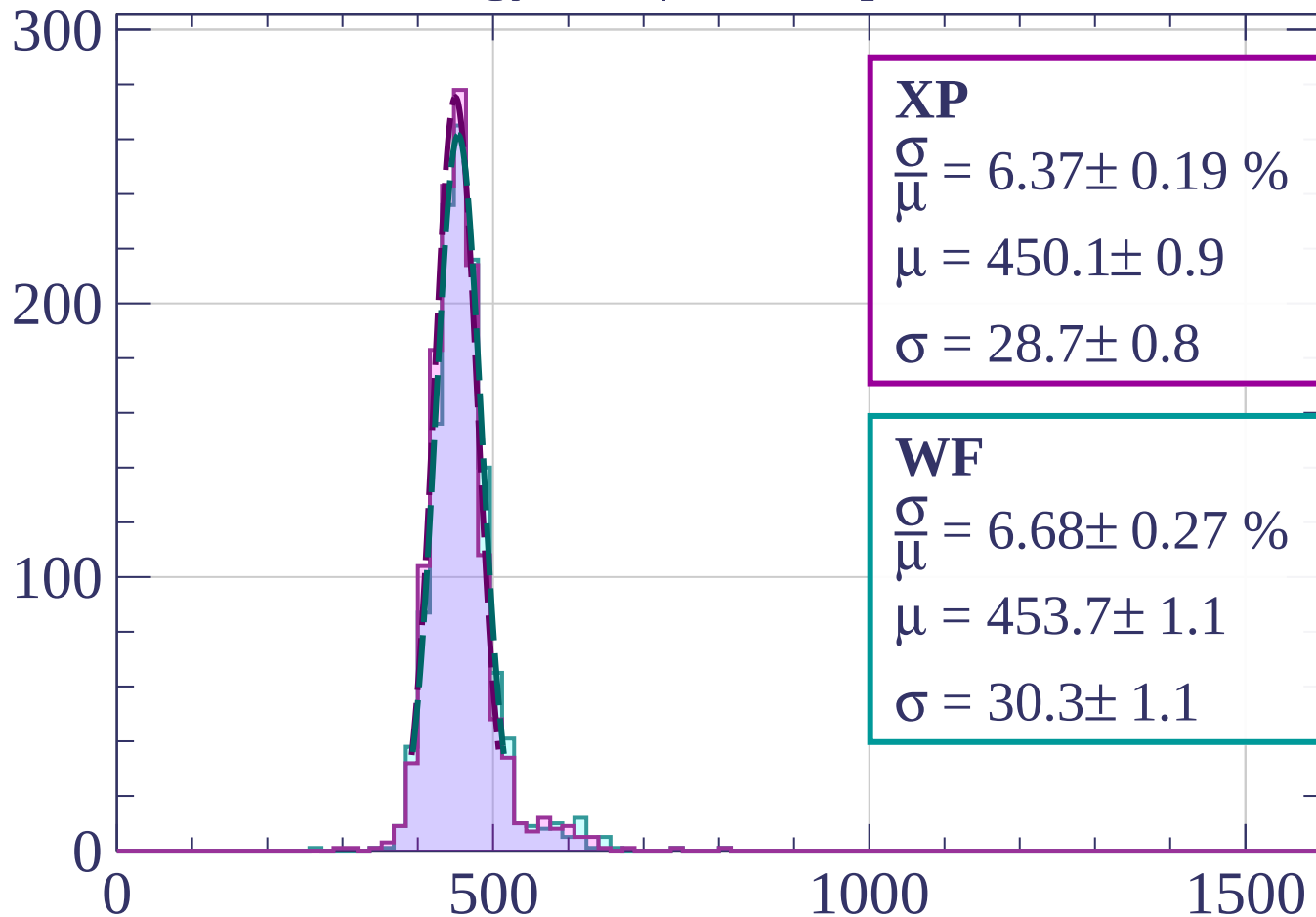
$$\mu = 450.9 \pm 0.9$$

$$\sigma = 26.8 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 6.37 \pm 0.19 \%$$

$$\mu = 450.1 \pm 0.9$$

$$\sigma = 28.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.27 \%$$

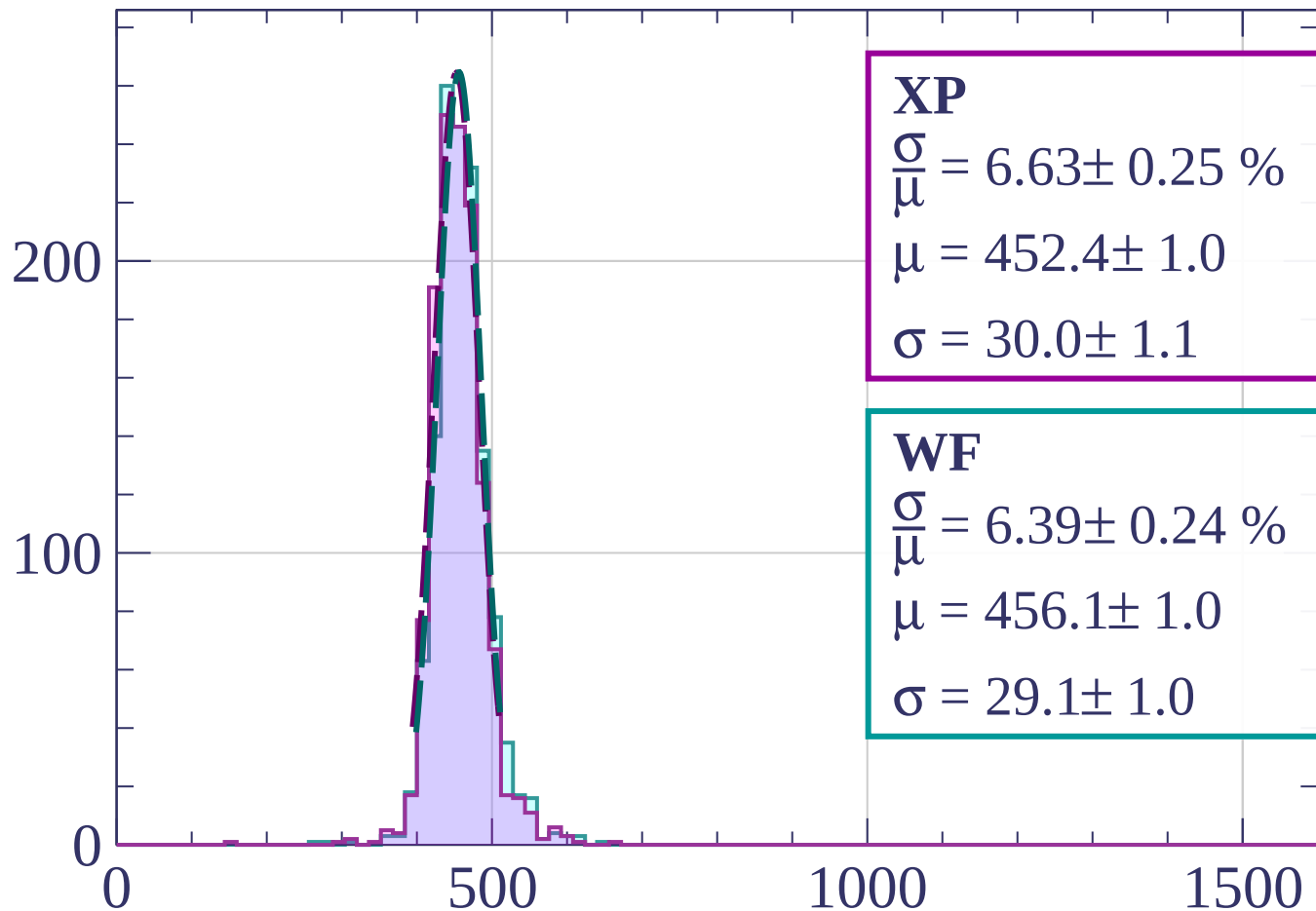
$$\mu = 453.7 \pm 1.1$$

$$\sigma = 30.3 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 6.63 \pm 0.25 \%$$

$$\mu = 452.4 \pm 1.0$$

$$\sigma = 30.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.39 \pm 0.24 \%$$

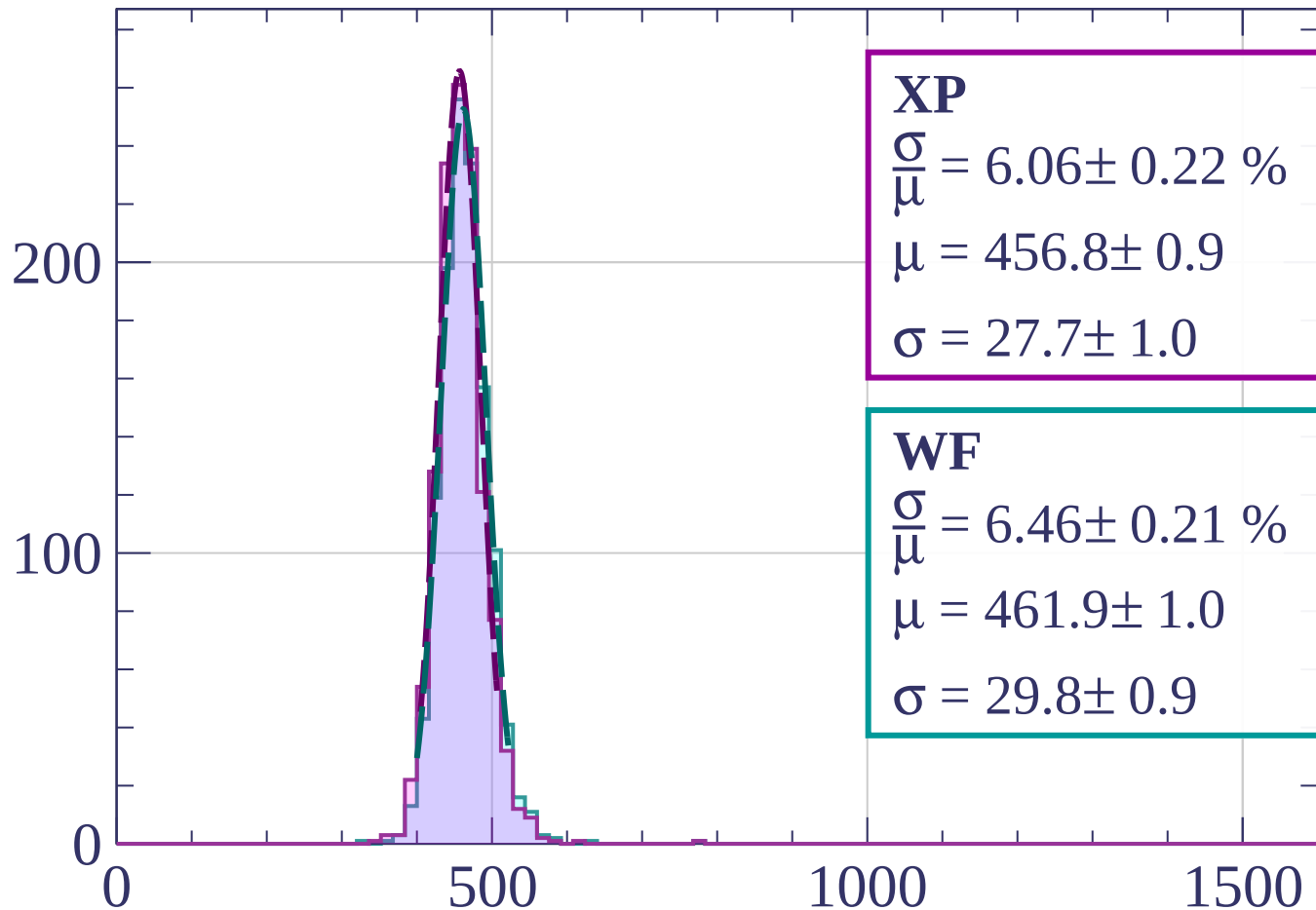
$$\mu = 456.1 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.22 \%$$

$$\mu = 456.8 \pm 0.9$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.21 \%$$

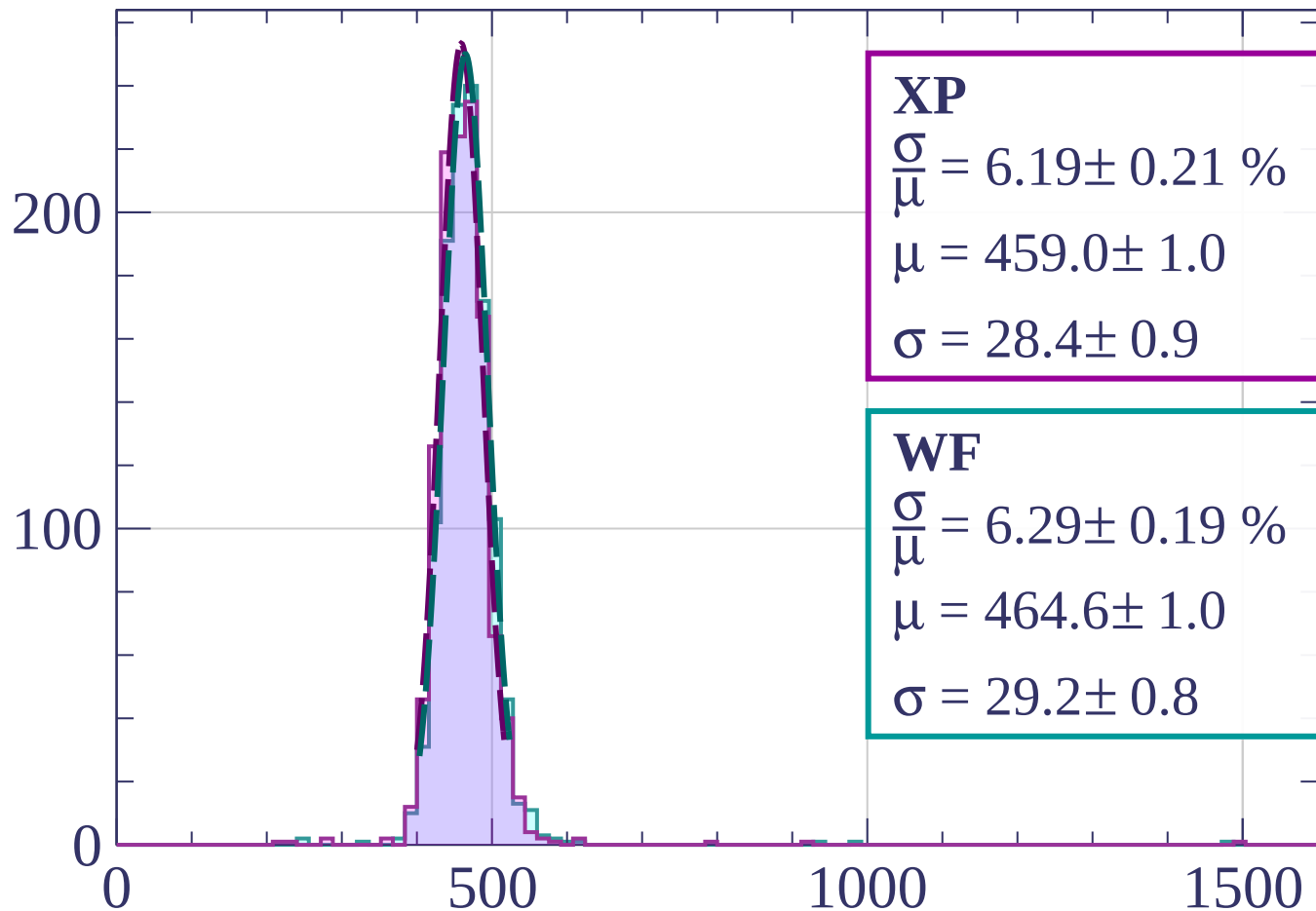
$$\mu = 461.9 \pm 1.0$$

$$\sigma = 29.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.21 \%$$

$$\mu = 459.0 \pm 1.0$$

$$\sigma = 28.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.29 \pm 0.19 \%$$

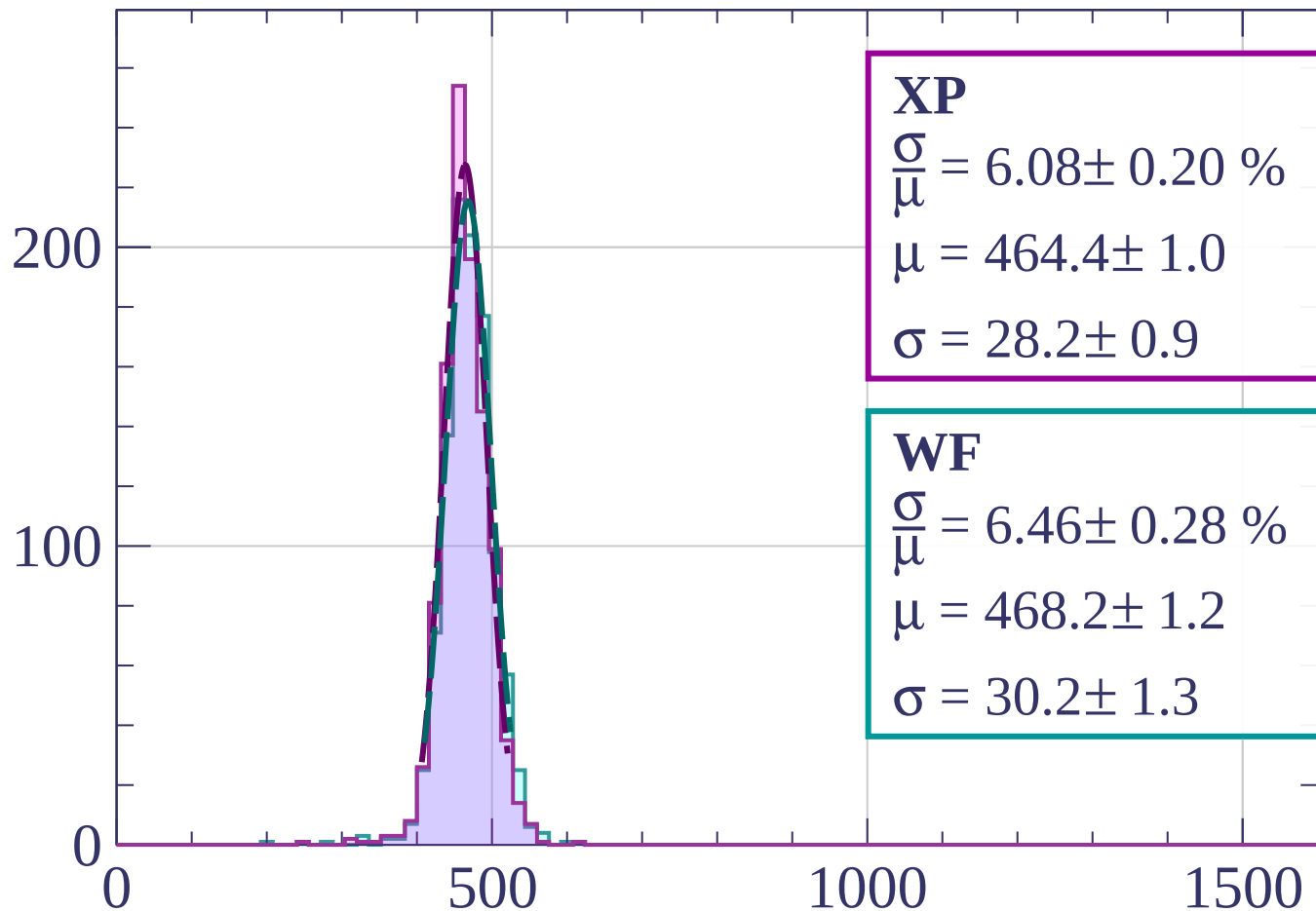
$$\mu = 464.6 \pm 1.0$$

$$\sigma = 29.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 6.08 \pm 0.20 \%$$

$$\mu = 464.4 \pm 1.0$$

$$\sigma = 28.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.28 \%$$

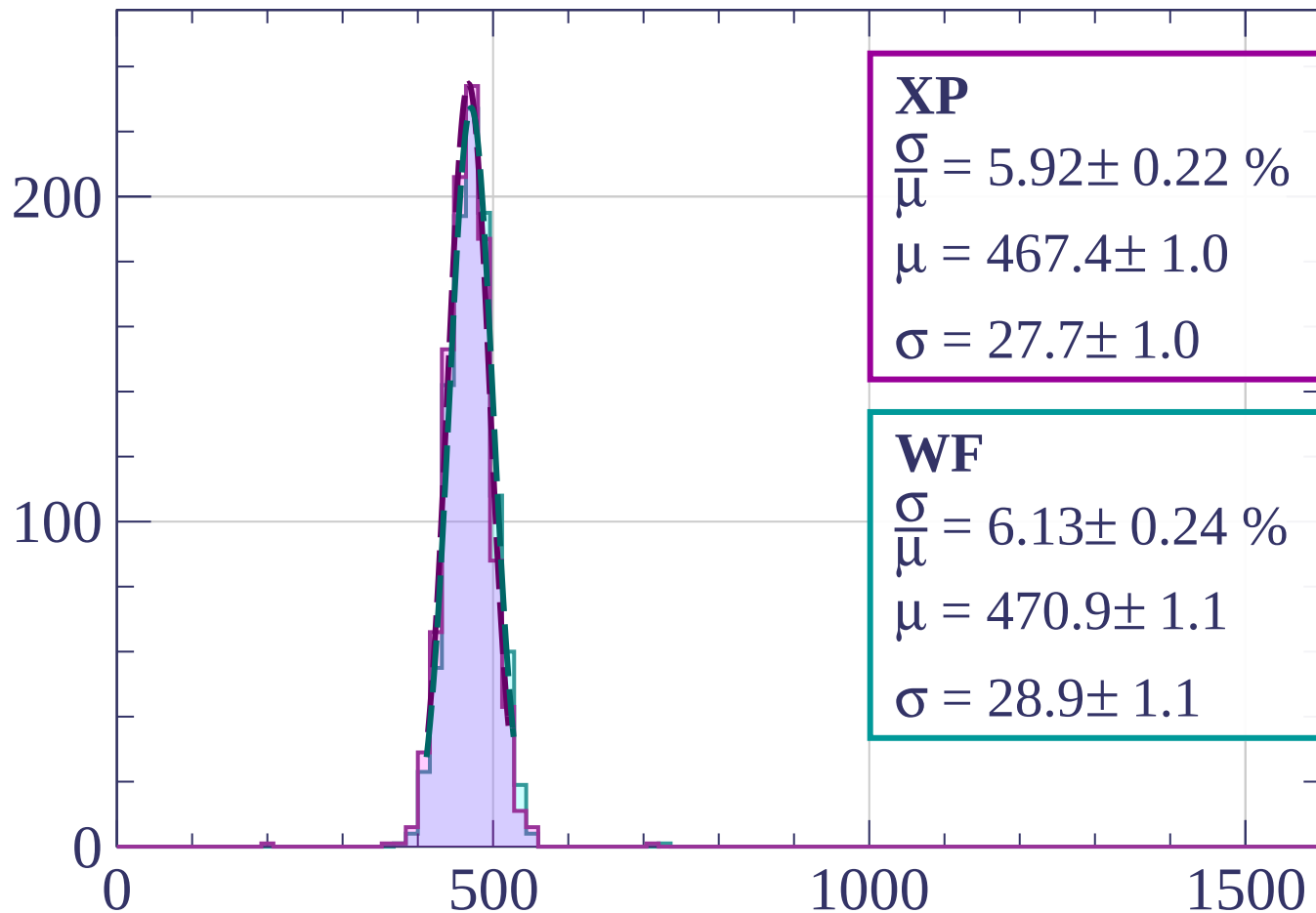
$$\mu = 468.2 \pm 1.2$$

$$\sigma = 30.2 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 5.92 \pm 0.22 \%$$

$$\mu = 467.4 \pm 1.0$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.13 \pm 0.24 \%$$

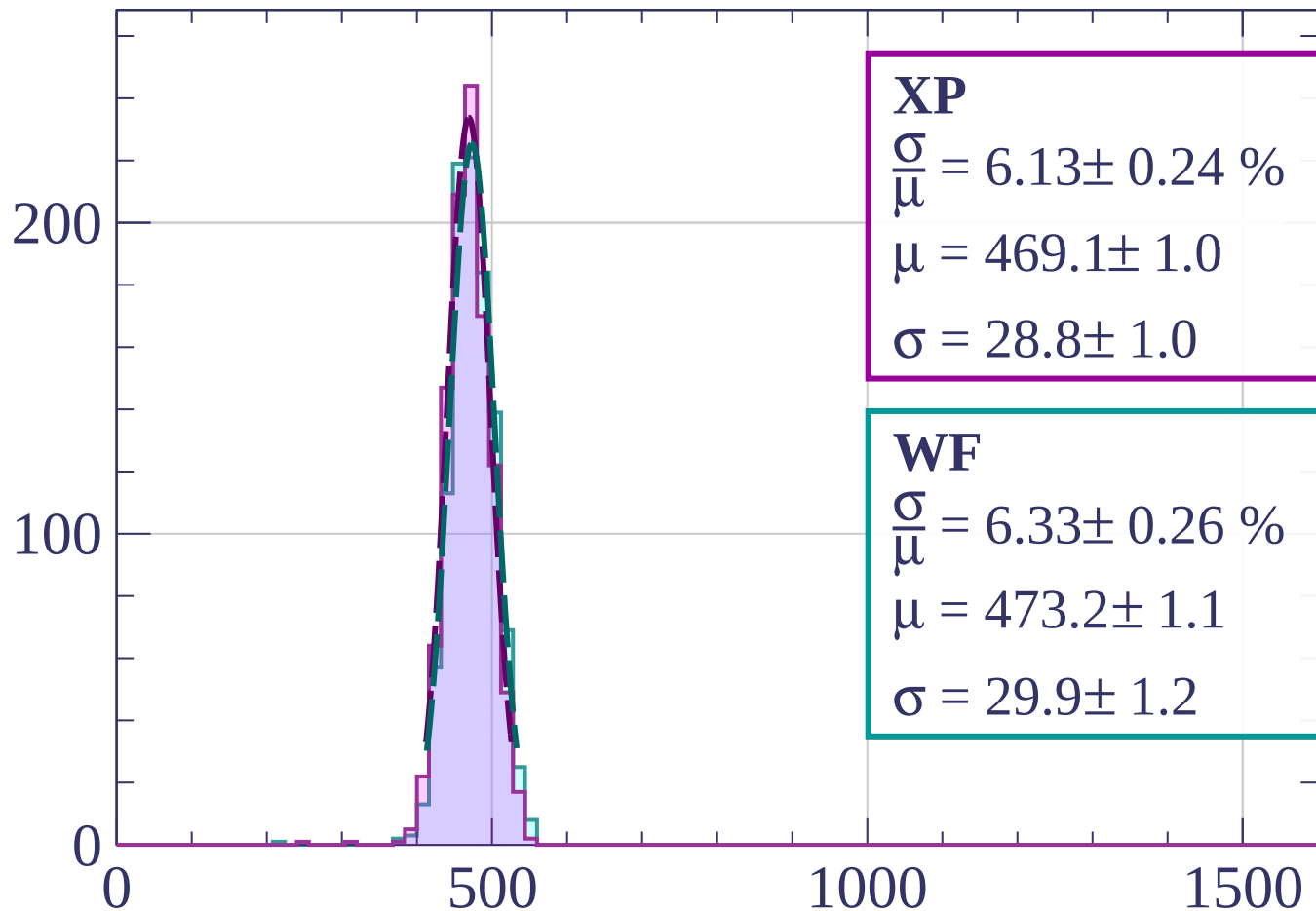
$$\mu = 470.9 \pm 1.1$$

$$\sigma = 28.9 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.24 \%$$

$$\mu = 469.1 \pm 1.0$$

$$\sigma = 28.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.33 \pm 0.26 \%$$

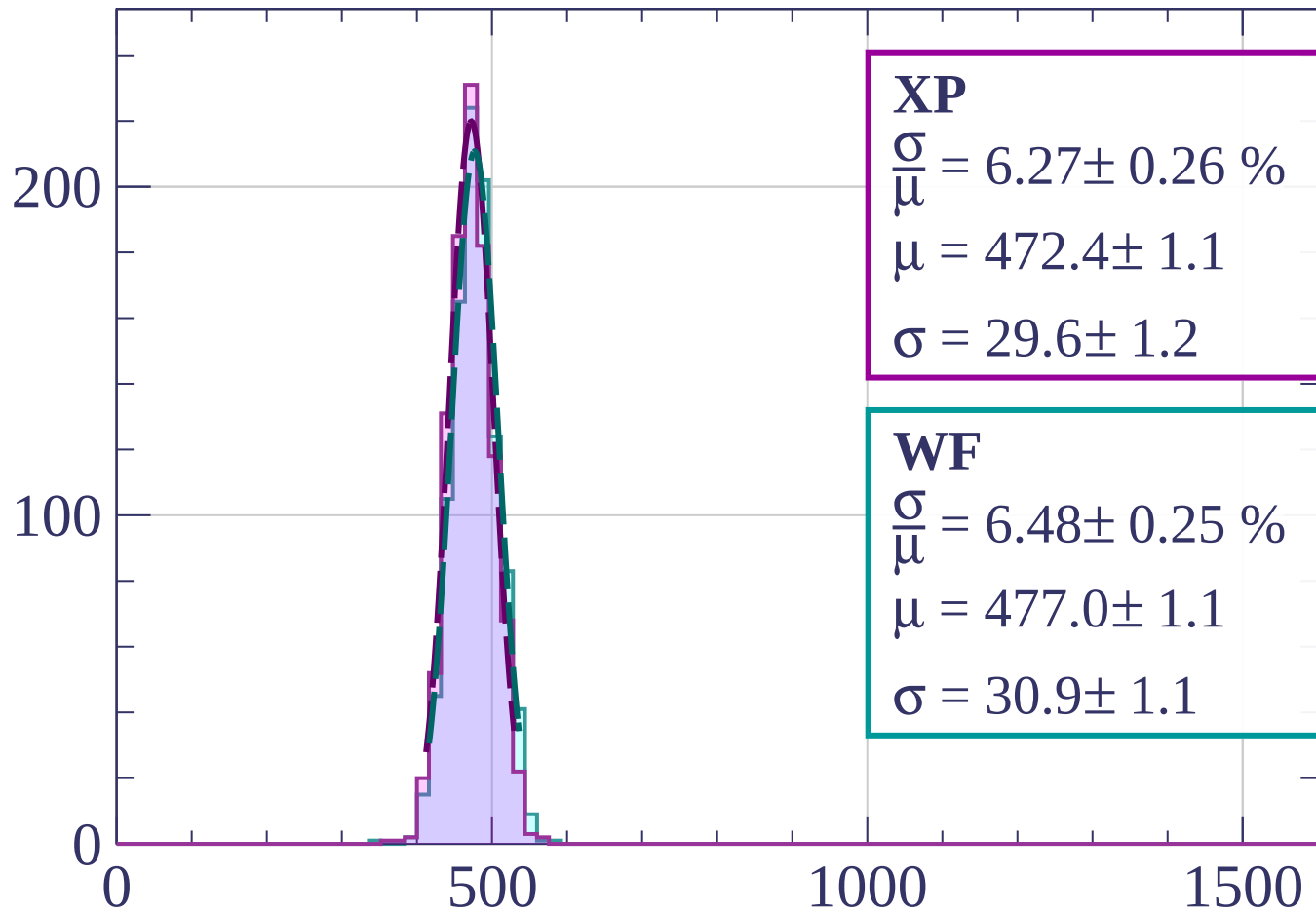
$$\mu = 473.2 \pm 1.1$$

$$\sigma = 29.9 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.27 \pm 0.26 \%$$

$$\mu = 472.4 \pm 1.1$$

$$\sigma = 29.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.25 \%$$

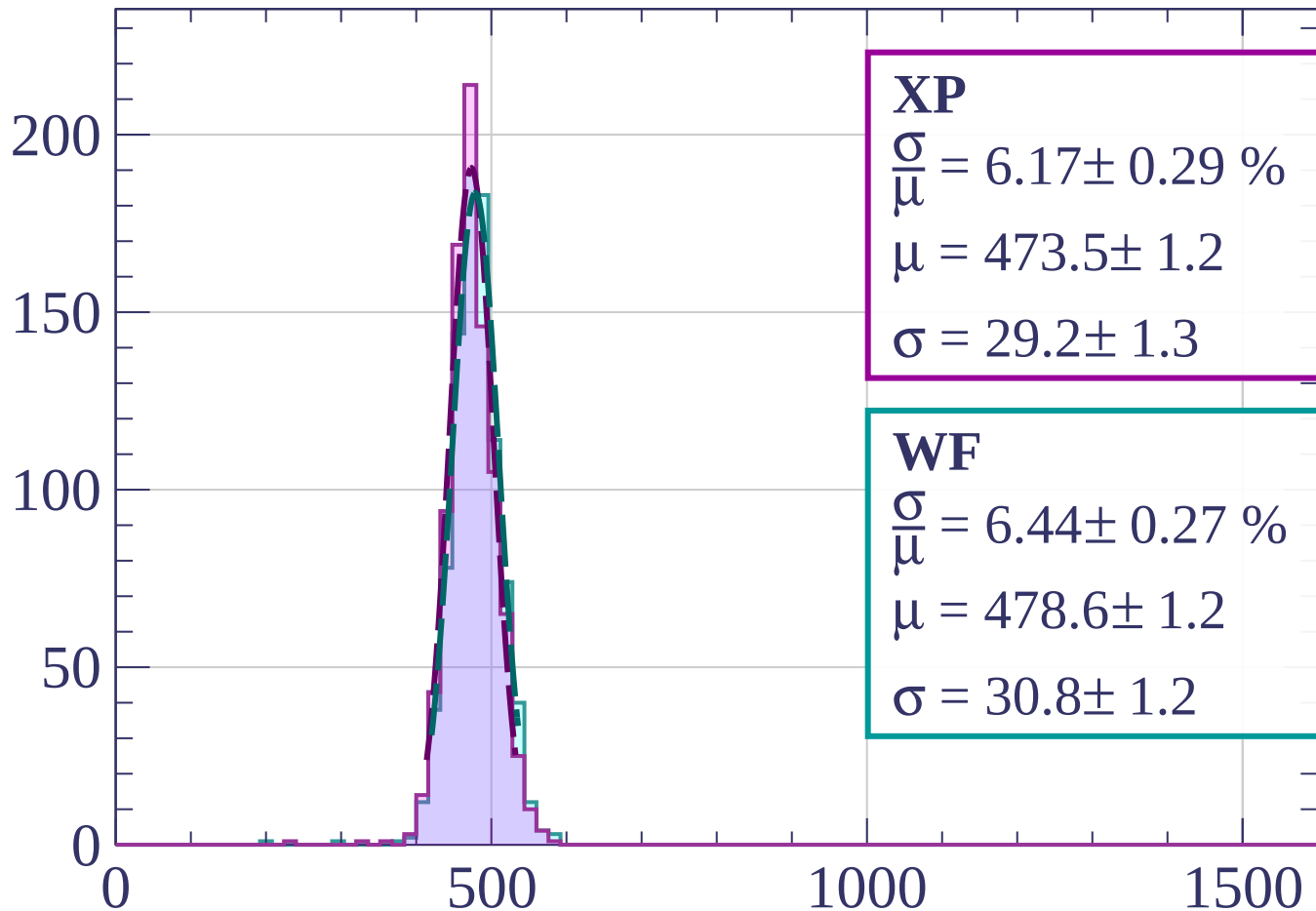
$$\mu = 477.0 \pm 1.1$$

$$\sigma = 30.9 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.17 \pm 0.29 \%$$

$$\mu = 473.5 \pm 1.2$$

$$\sigma = 29.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.27 \%$$

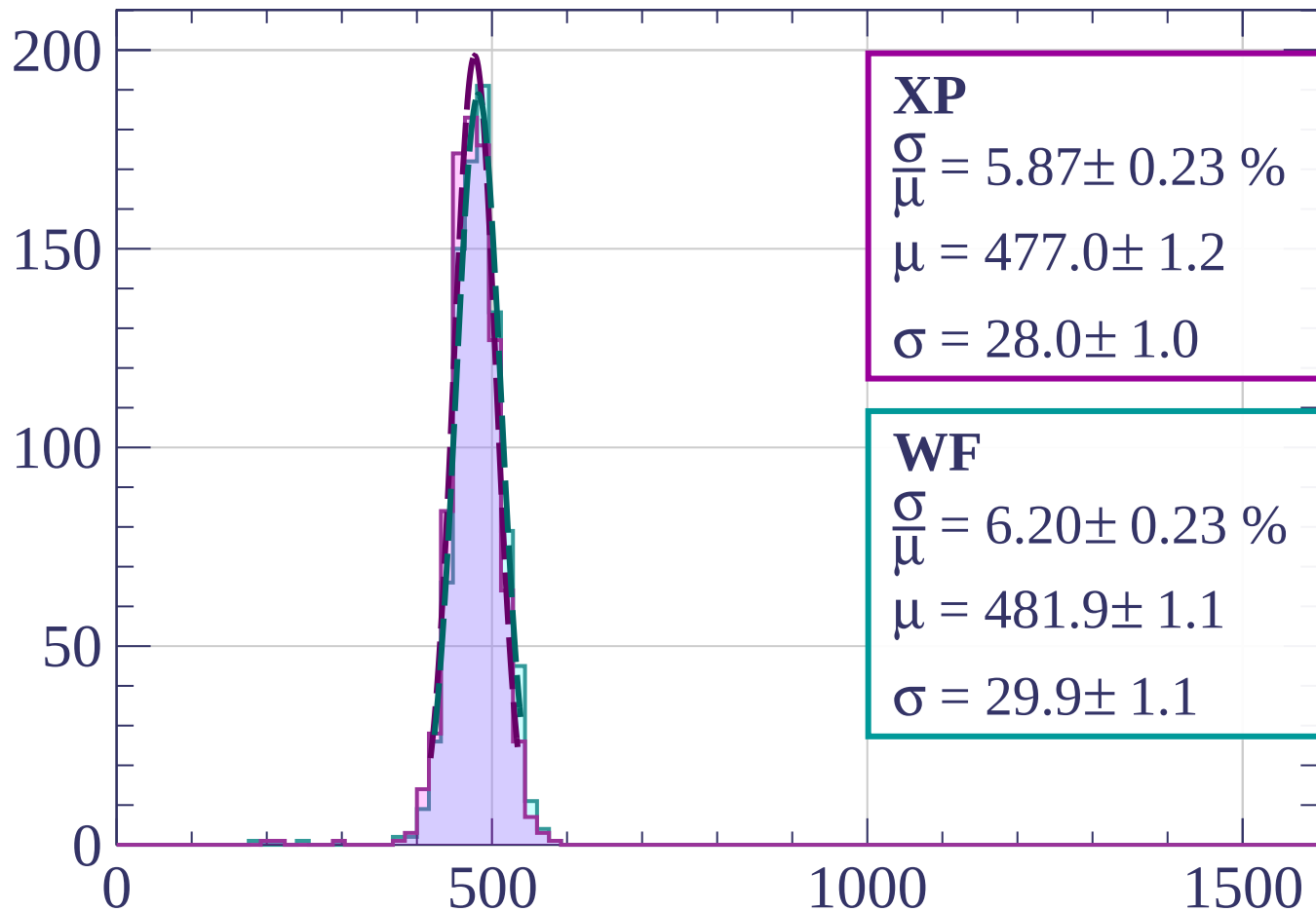
$$\mu = 478.6 \pm 1.2$$

$$\sigma = 30.8 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

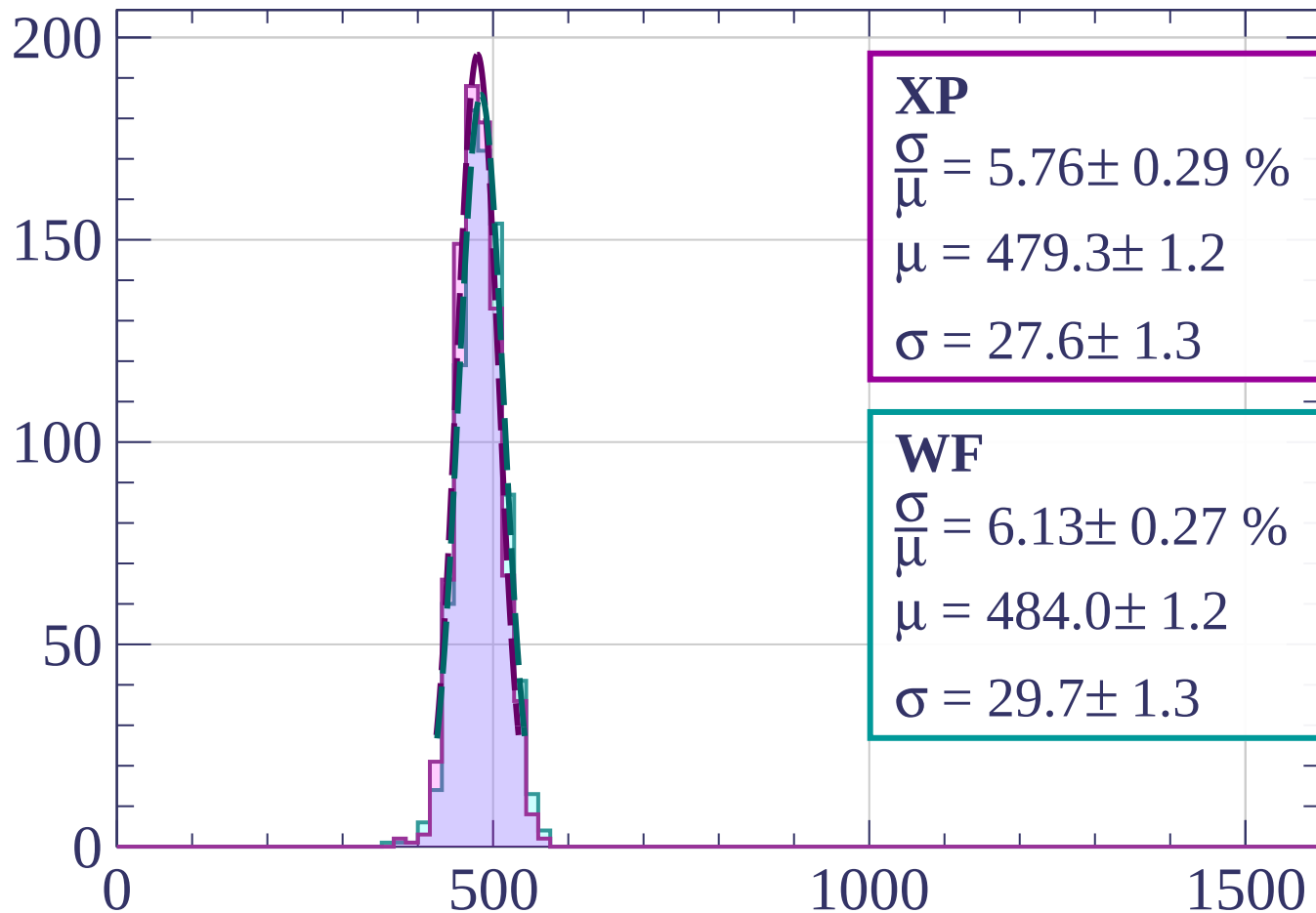
Count



dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 5.76 \pm 0.29 \%$$

$$\mu = 479.3 \pm 1.2$$

$$\sigma = 27.6 \pm 1.3$$

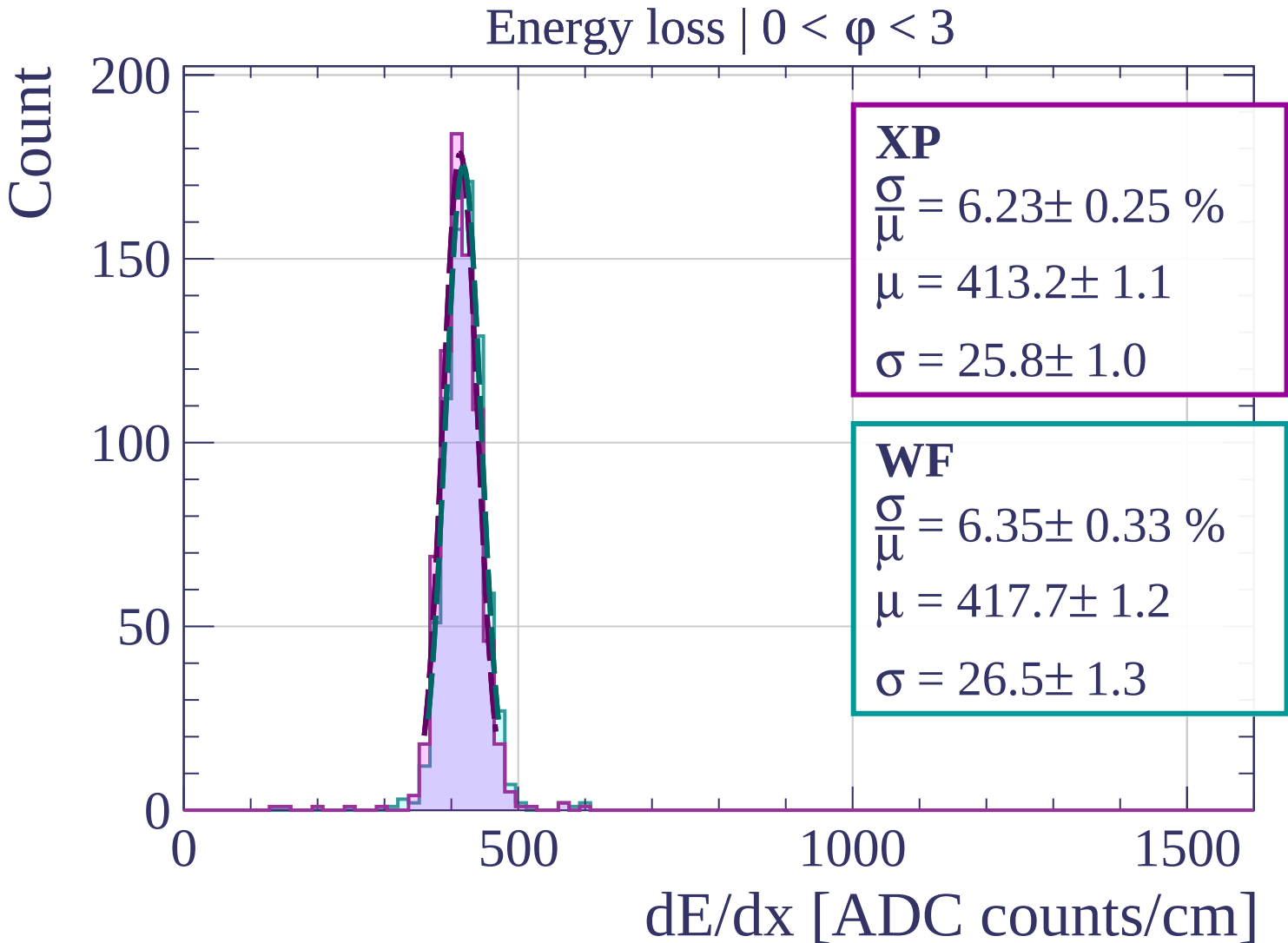
WF

$$\frac{\sigma}{\mu} = 6.13 \pm 0.27 \%$$

$$\mu = 484.0 \pm 1.2$$

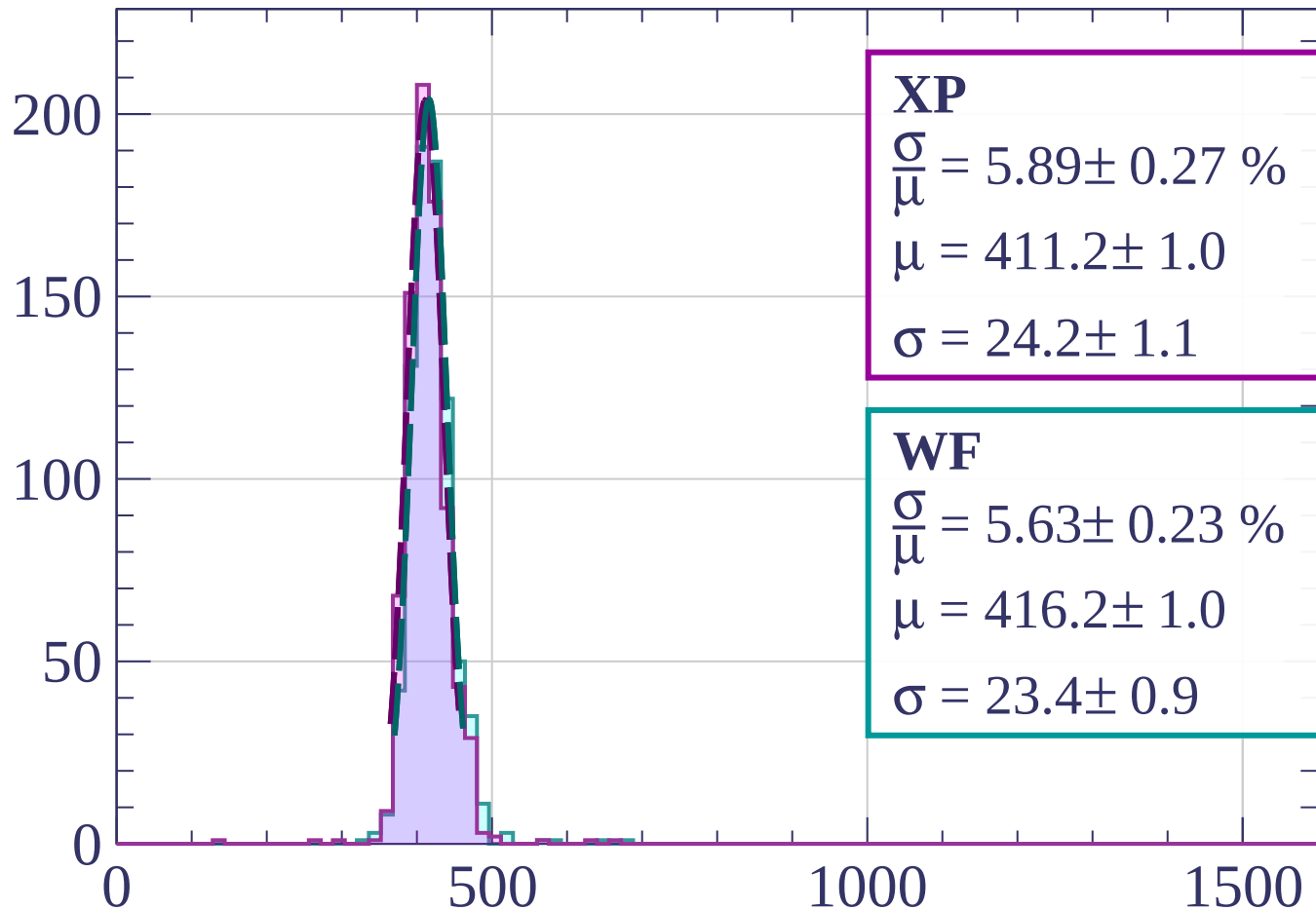
$$\sigma = 29.7 \pm 1.3$$

dE/dx [ADC counts/cm]



Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.89 \pm 0.27 \%$$

$$\mu = 411.2 \pm 1.0$$

$$\sigma = 24.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.63 \pm 0.23 \%$$

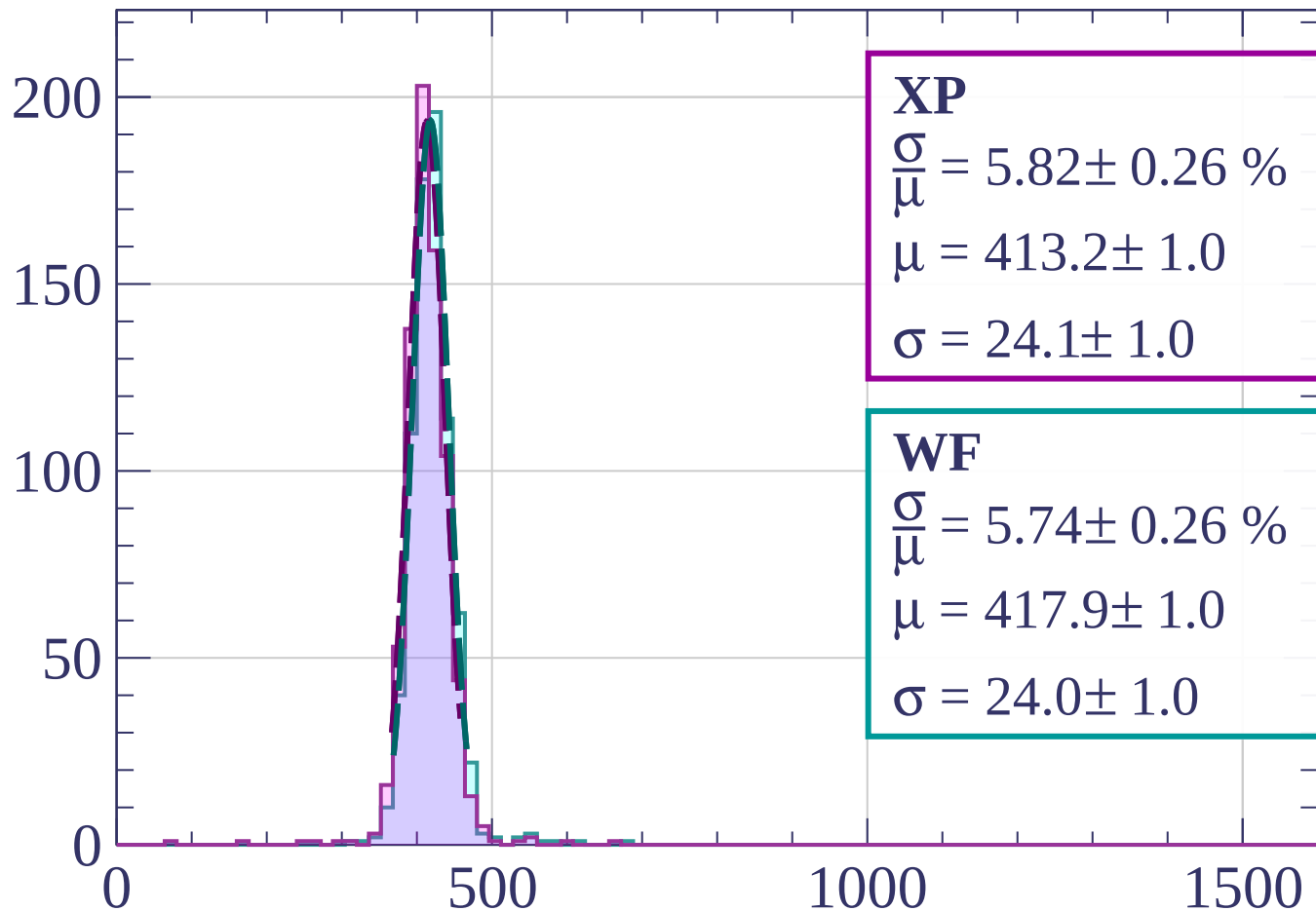
$$\mu = 416.2 \pm 1.0$$

$$\sigma = 23.4 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.82 \pm 0.26 \%$$

$$\mu = 413.2 \pm 1.0$$

$$\sigma = 24.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.74 \pm 0.26 \%$$

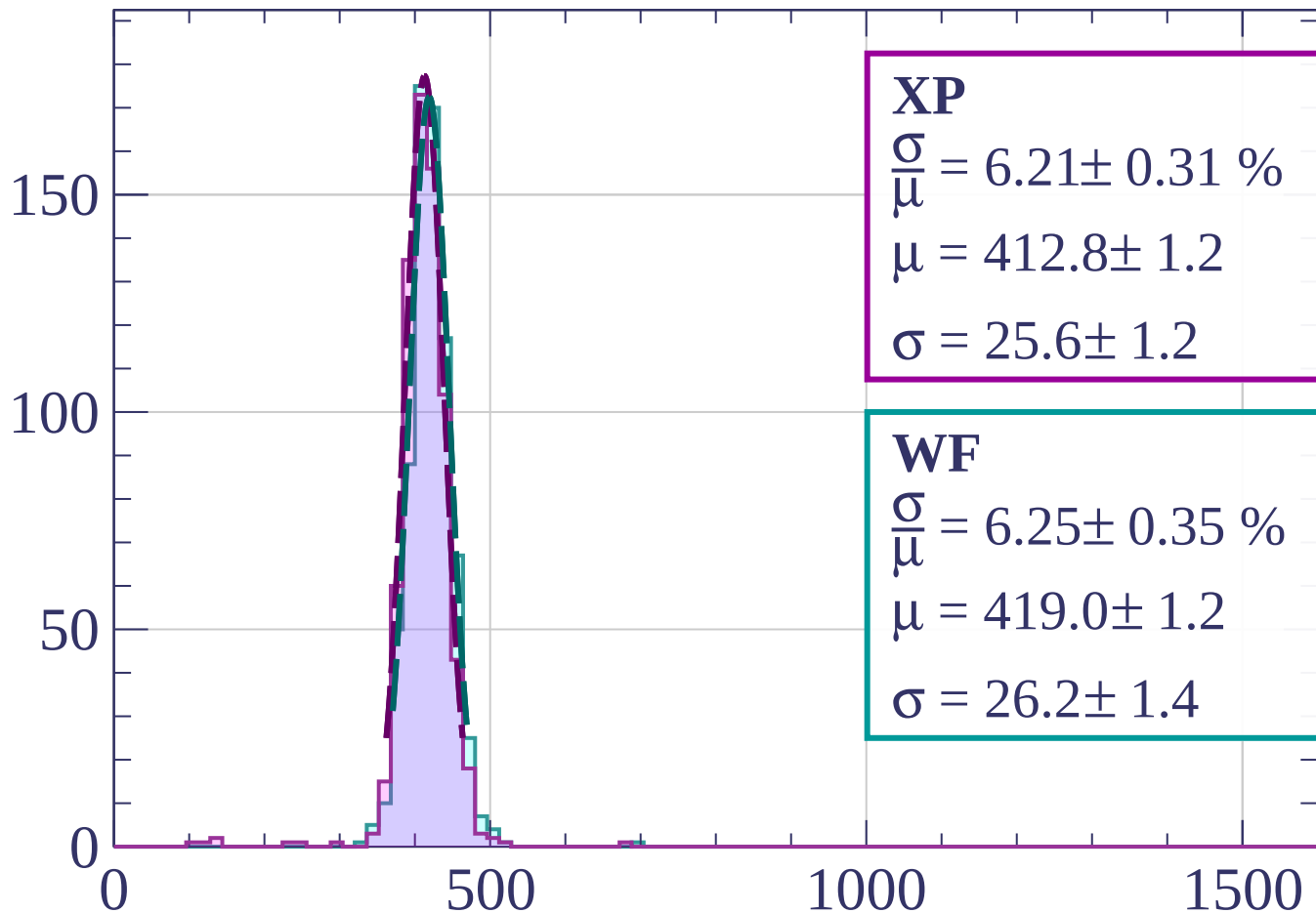
$$\mu = 417.9 \pm 1.0$$

$$\sigma = 24.0 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.21 \pm 0.31 \%$$

$$\mu = 412.8 \pm 1.2$$

$$\sigma = 25.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.25 \pm 0.35 \%$$

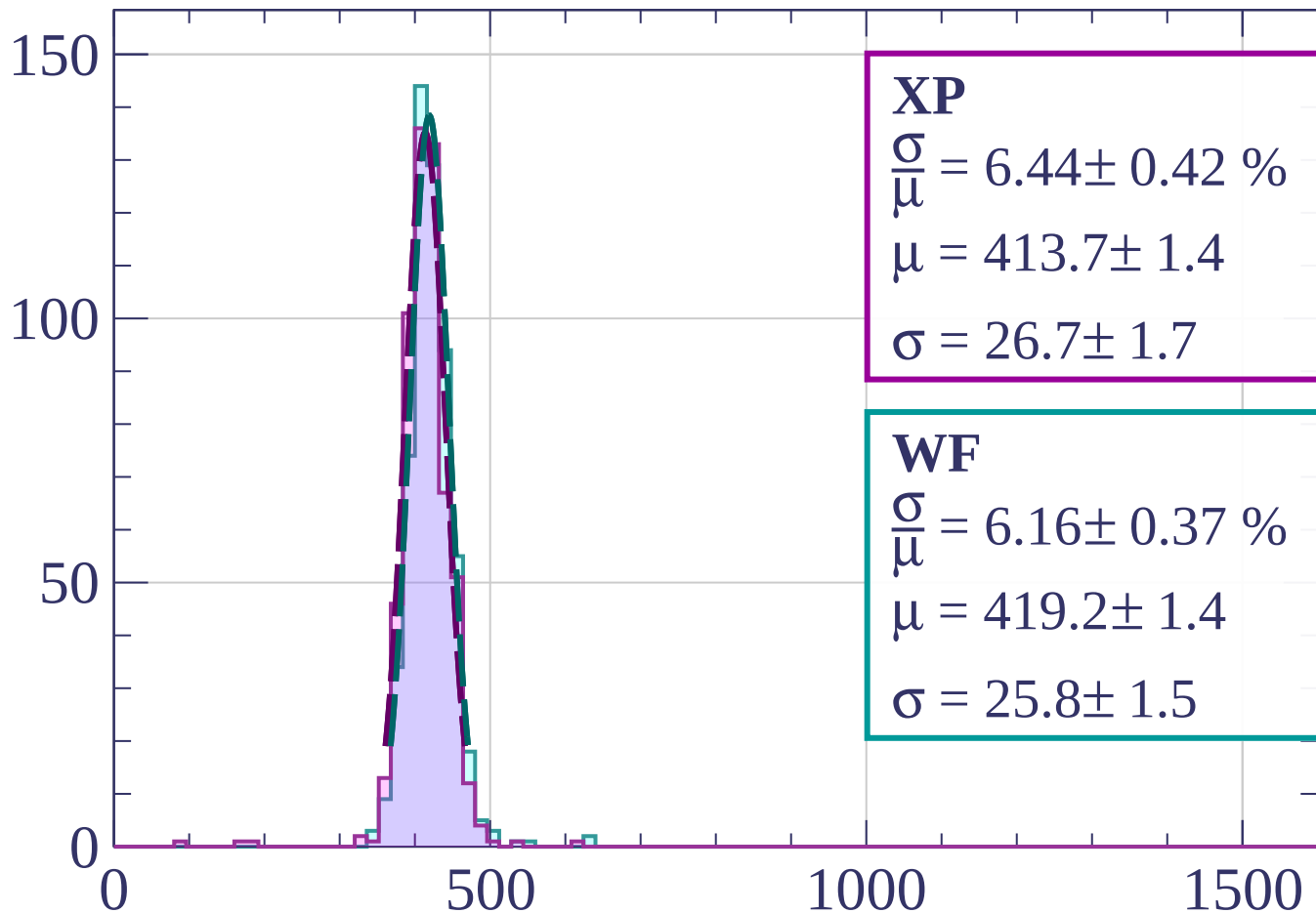
$$\mu = 419.0 \pm 1.2$$

$$\sigma = 26.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.44 \pm 0.42 \%$$

$$\mu = 413.7 \pm 1.4$$

$$\sigma = 26.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.16 \pm 0.37 \%$$

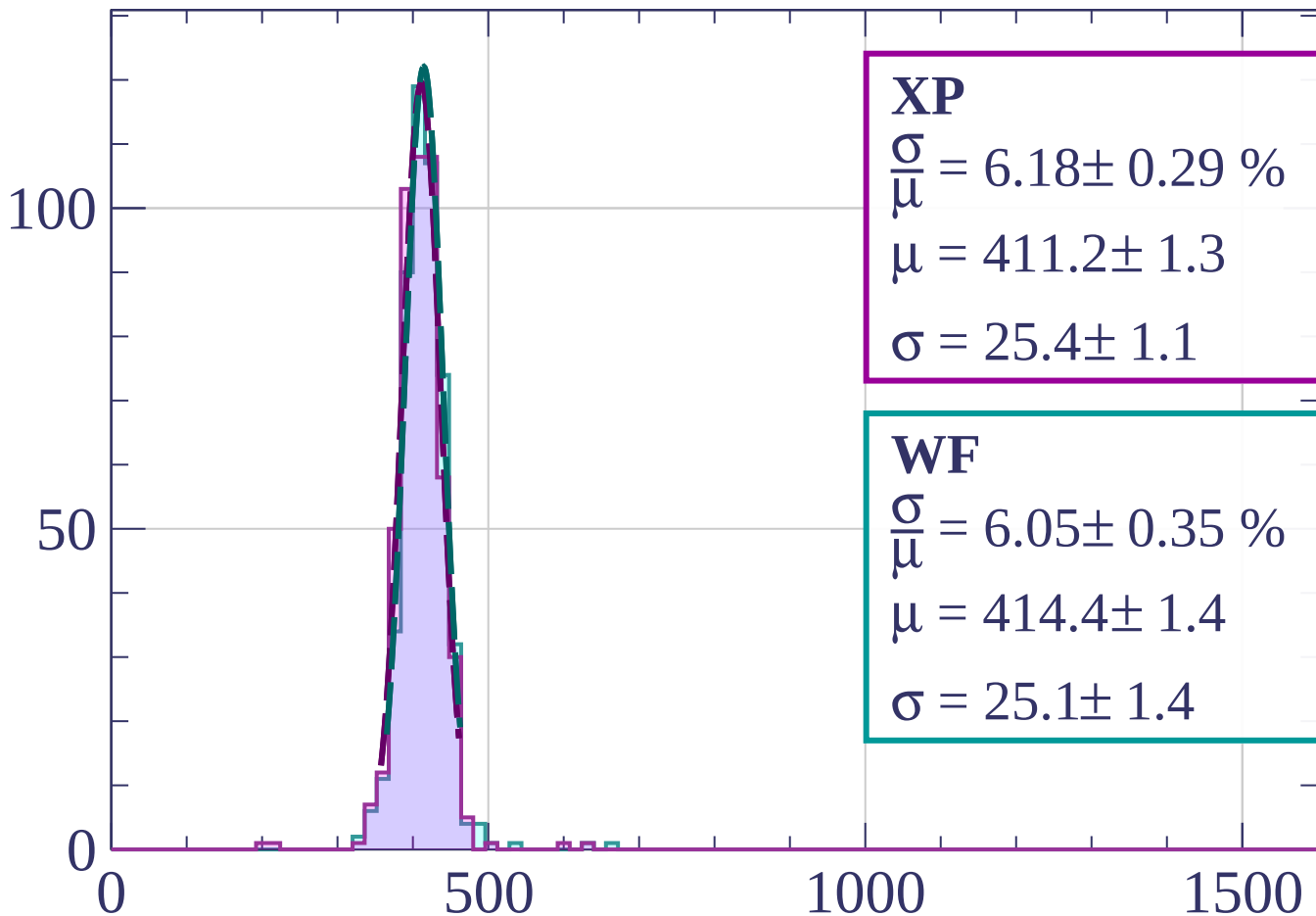
$$\mu = 419.2 \pm 1.4$$

$$\sigma = 25.8 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.18 \pm 0.29 \%$$

$$\mu = 411.2 \pm 1.3$$

$$\sigma = 25.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.05 \pm 0.35 \%$$

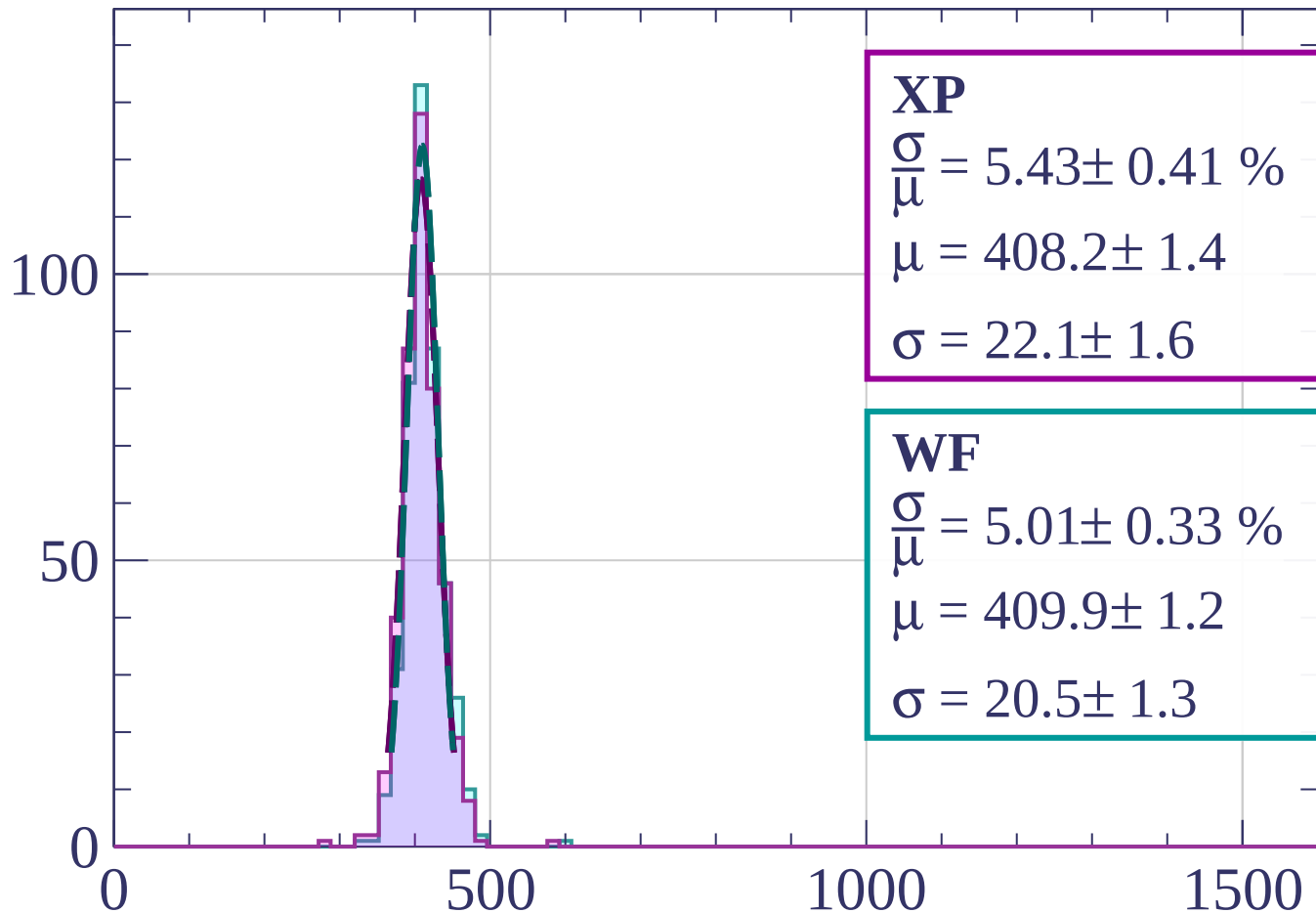
$$\mu = 414.4 \pm 1.4$$

$$\sigma = 25.1 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.43 \pm 0.41 \%$$

$$\mu = 408.2 \pm 1.4$$

$$\sigma = 22.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 5.01 \pm 0.33 \%$$

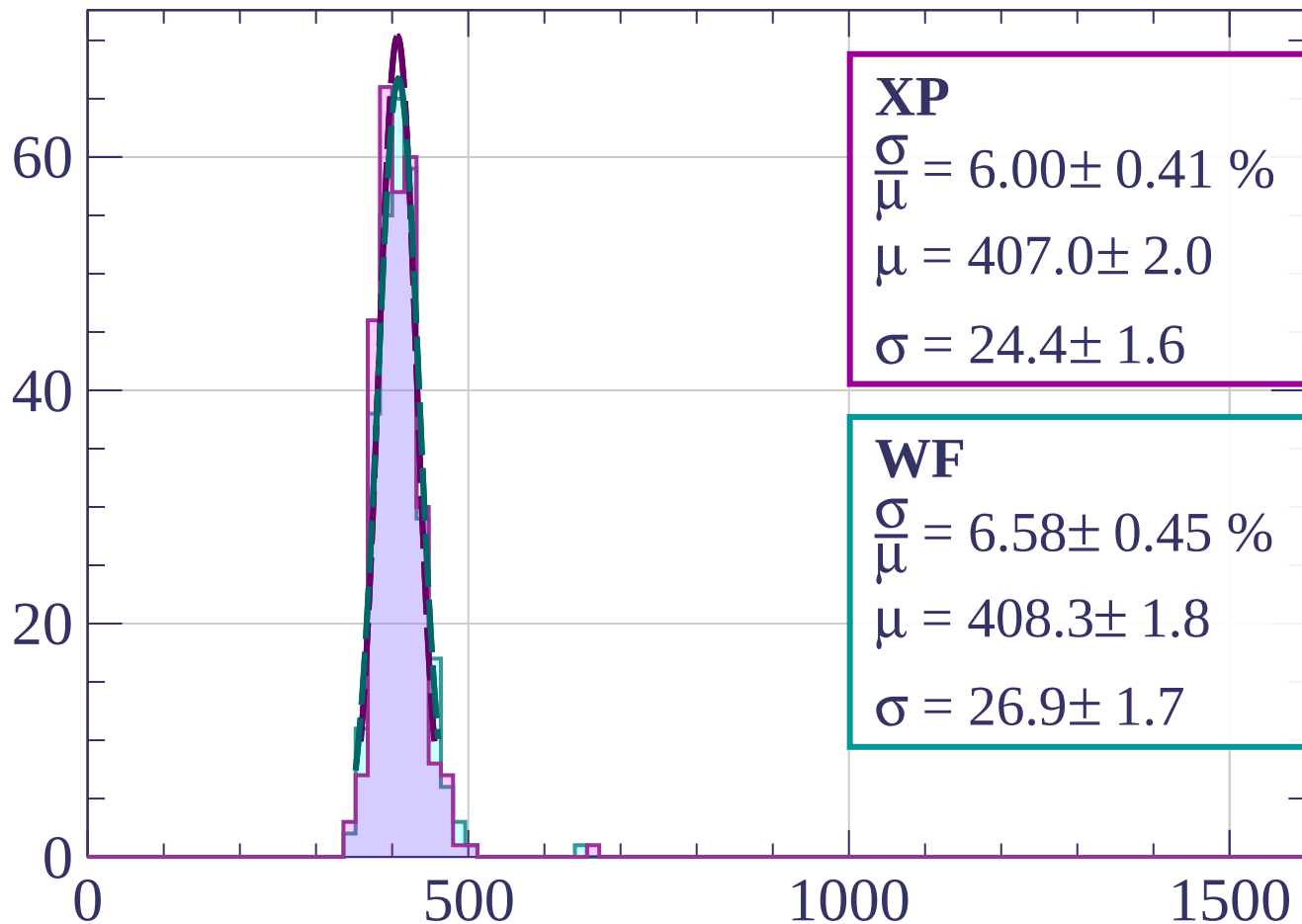
$$\mu = 409.9 \pm 1.2$$

$$\sigma = 20.5 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 6.00 \pm 0.41 \%$$

$$\mu = 407.0 \pm 2.0$$

$$\sigma = 24.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.45 \%$$

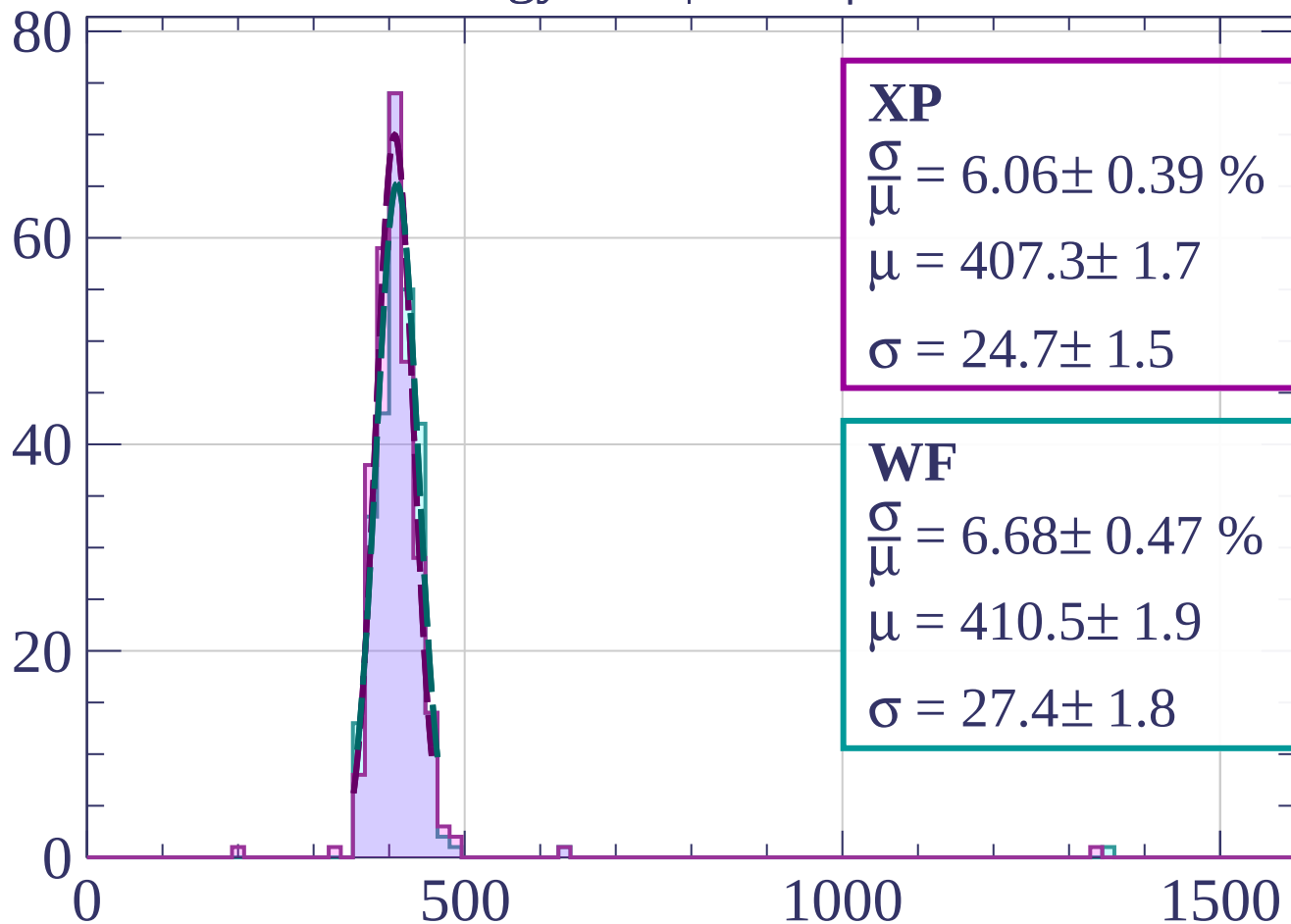
$$\mu = 408.3 \pm 1.8$$

$$\sigma = 26.9 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.39 \%$$

$$\mu = 407.3 \pm 1.7$$

$$\sigma = 24.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.47 \%$$

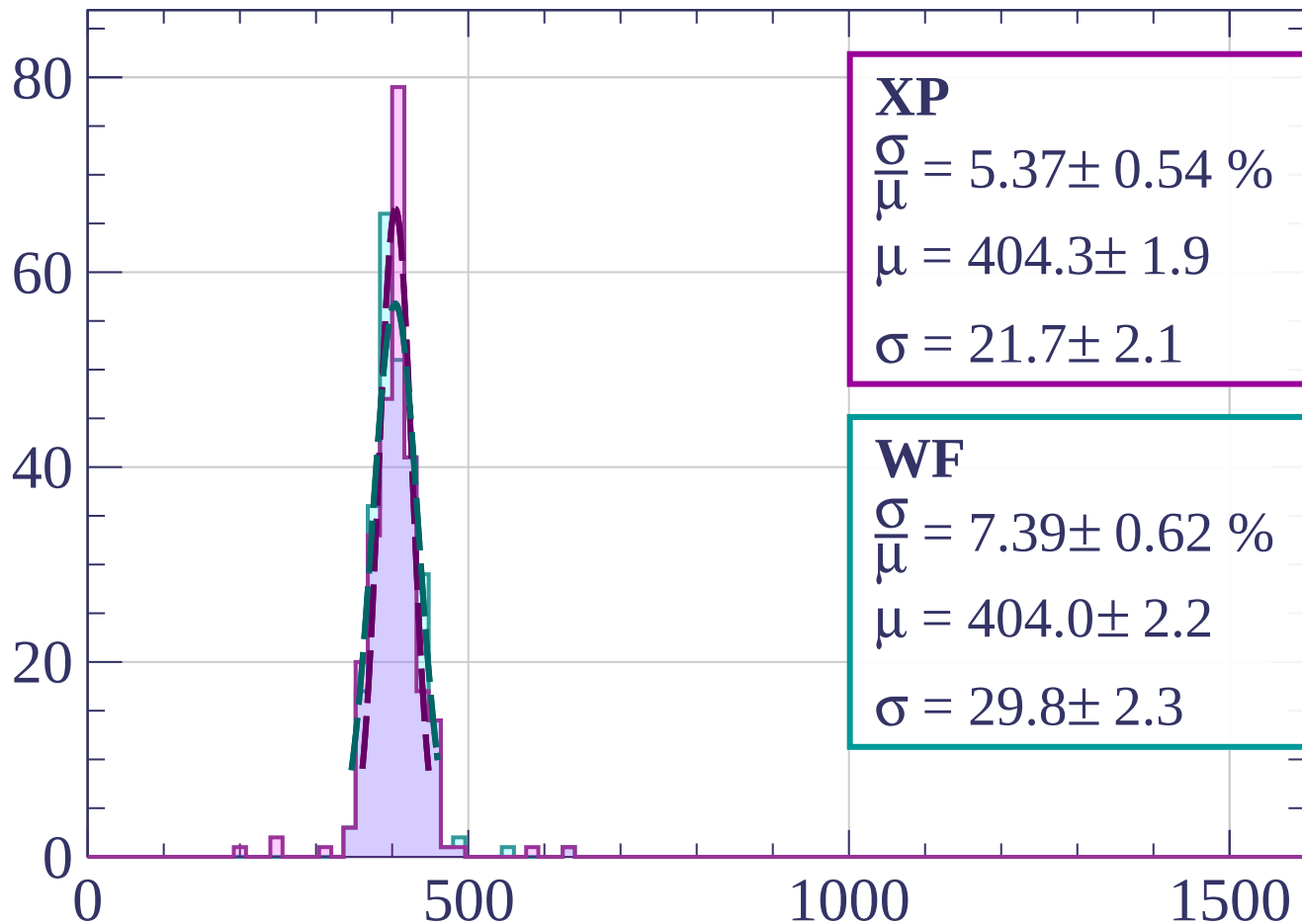
$$\mu = 410.5 \pm 1.9$$

$$\sigma = 27.4 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 5.37 \pm 0.54 \%$$

$$\mu = 404.3 \pm 1.9$$

$$\sigma = 21.7 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.62 \%$$

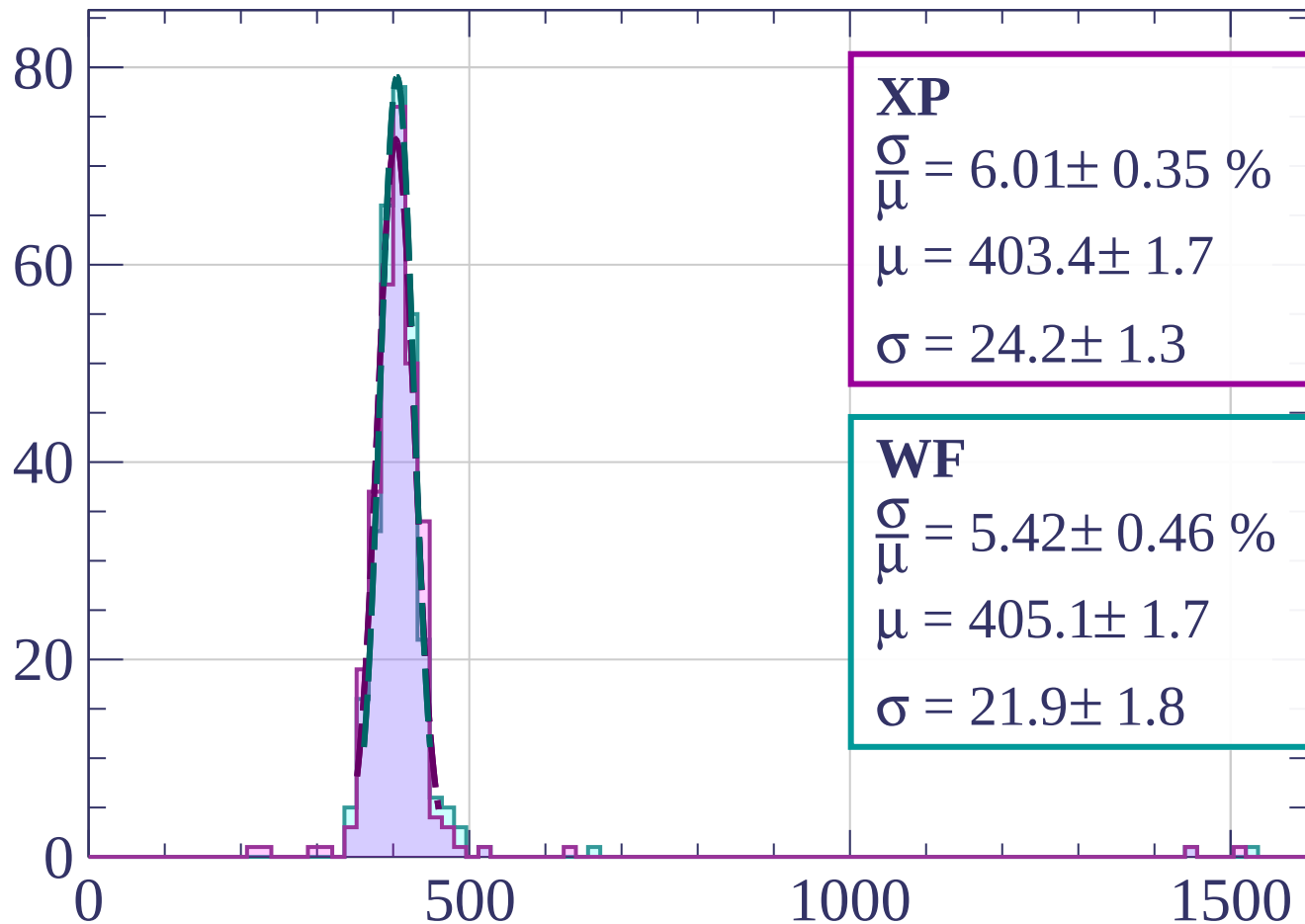
$$\mu = 404.0 \pm 2.2$$

$$\sigma = 29.8 \pm 2.3$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 6.01 \pm 0.35 \%$$

$$\mu = 403.4 \pm 1.7$$

$$\sigma = 24.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.46 \%$$

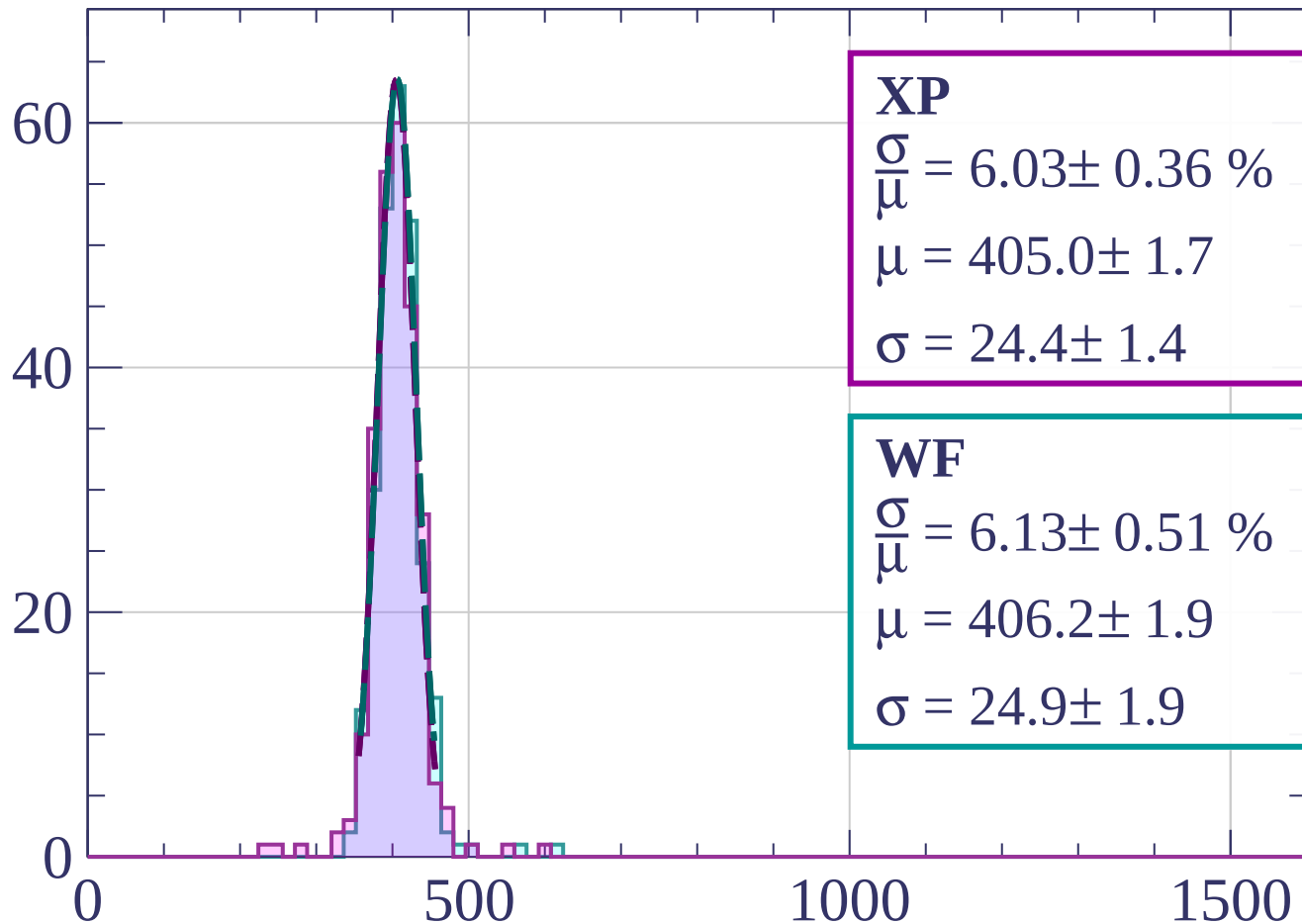
$$\mu = 405.1 \pm 1.7$$

$$\sigma = 21.9 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 6.03 \pm 0.36 \%$$

$$\mu = 405.0 \pm 1.7$$

$$\sigma = 24.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.13 \pm 0.51 \%$$

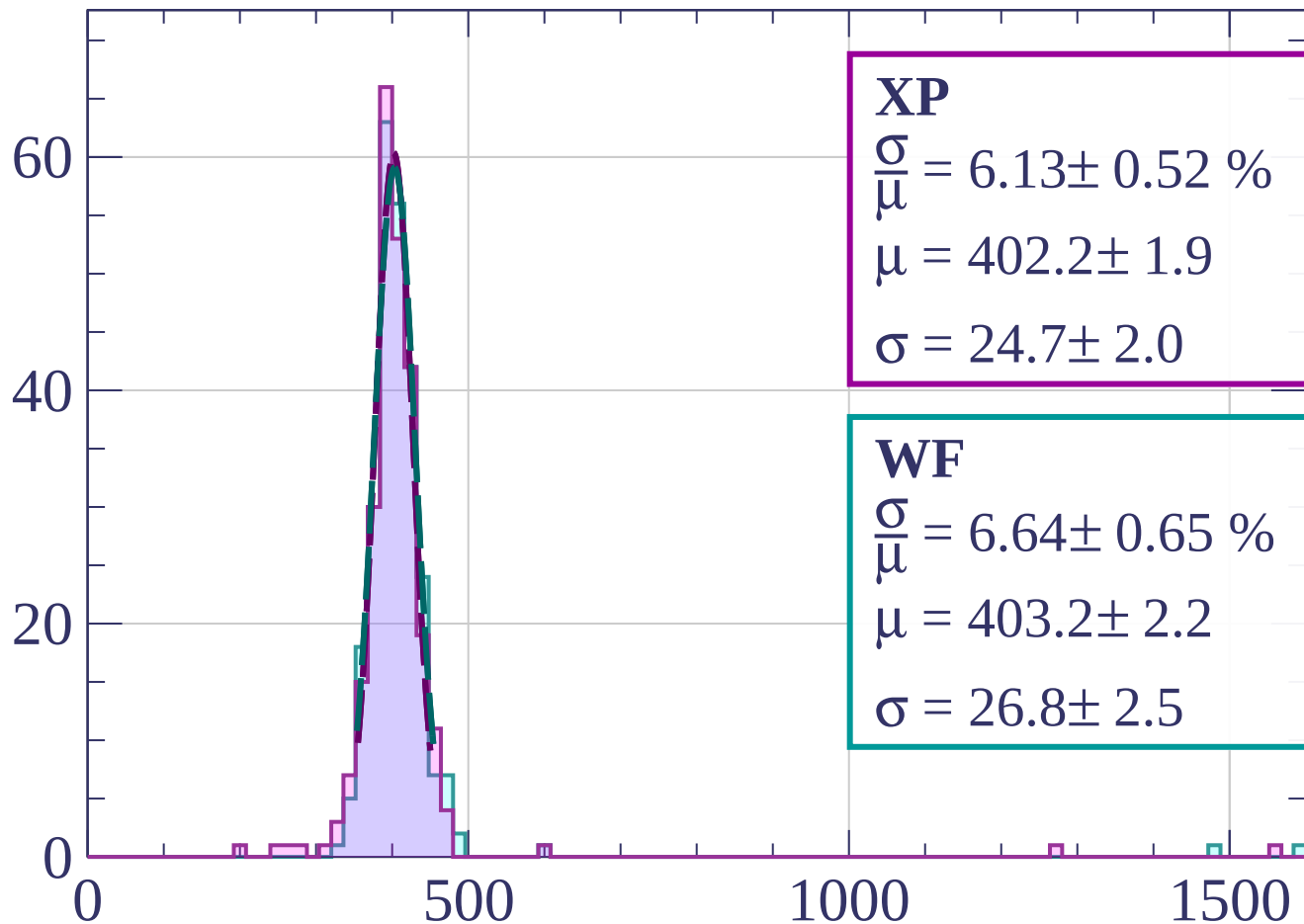
$$\mu = 406.2 \pm 1.9$$

$$\sigma = 24.9 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.52 \%$$

$$\mu = 402.2 \pm 1.9$$

$$\sigma = 24.7 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 6.64 \pm 0.65 \%$$

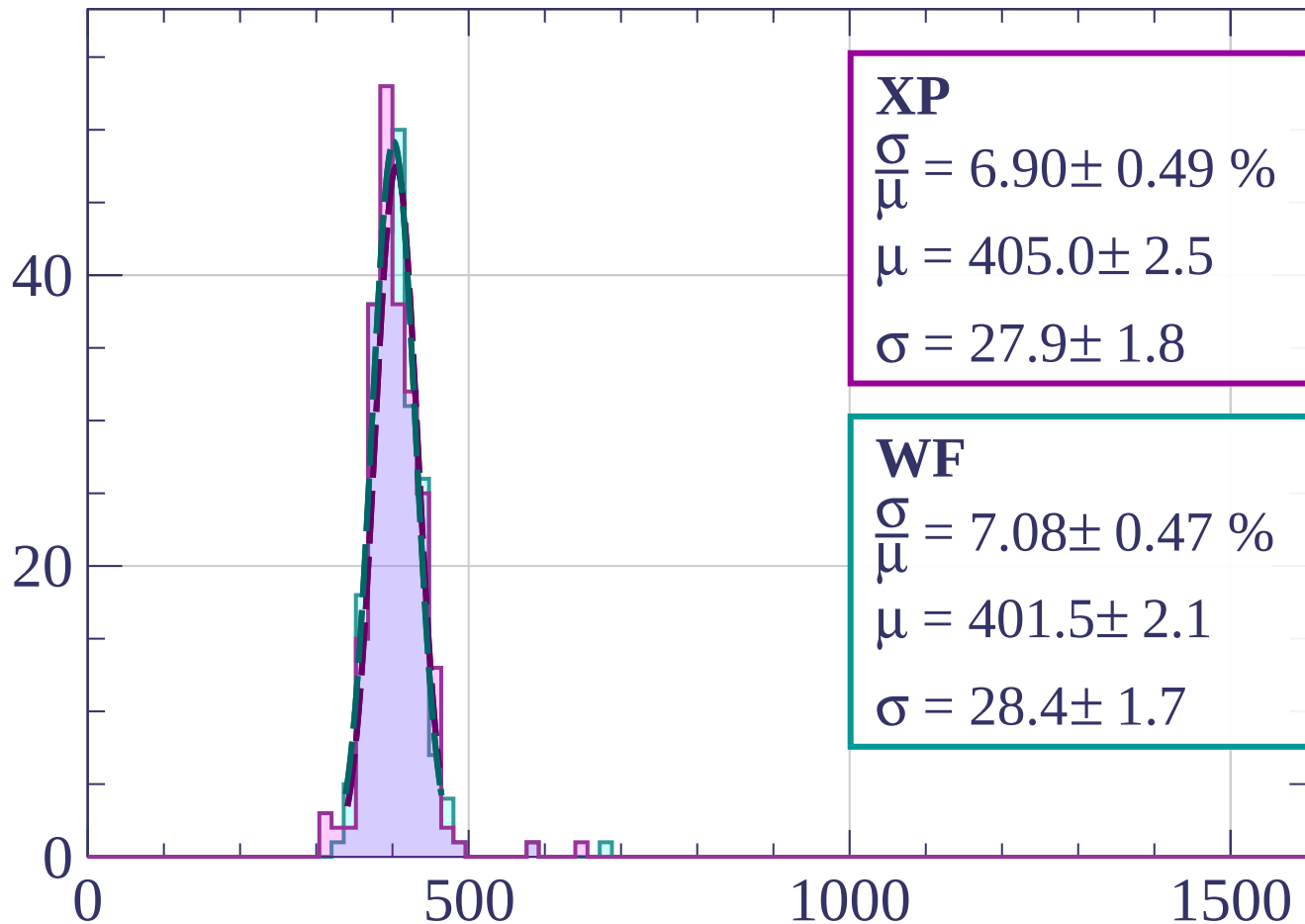
$$\mu = 403.2 \pm 2.2$$

$$\sigma = 26.8 \pm 2.5$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 6.90 \pm 0.49 \%$$

$$\mu = 405.0 \pm 2.5$$

$$\sigma = 27.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.08 \pm 0.47 \%$$

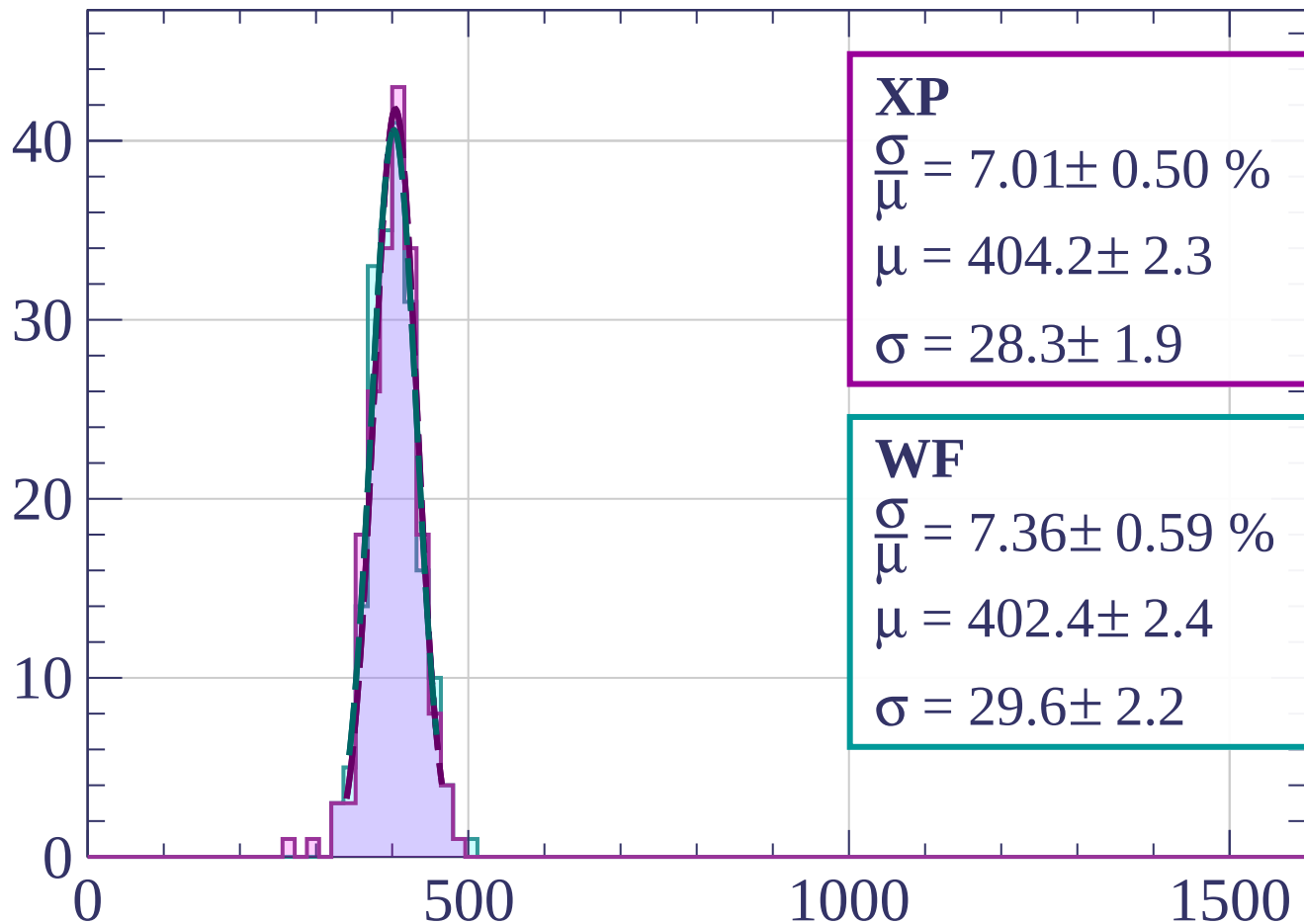
$$\mu = 401.5 \pm 2.1$$

$$\sigma = 28.4 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.01 \pm 0.50 \%$$

$$\mu = 404.2 \pm 2.3$$

$$\sigma = 28.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.36 \pm 0.59 \%$$

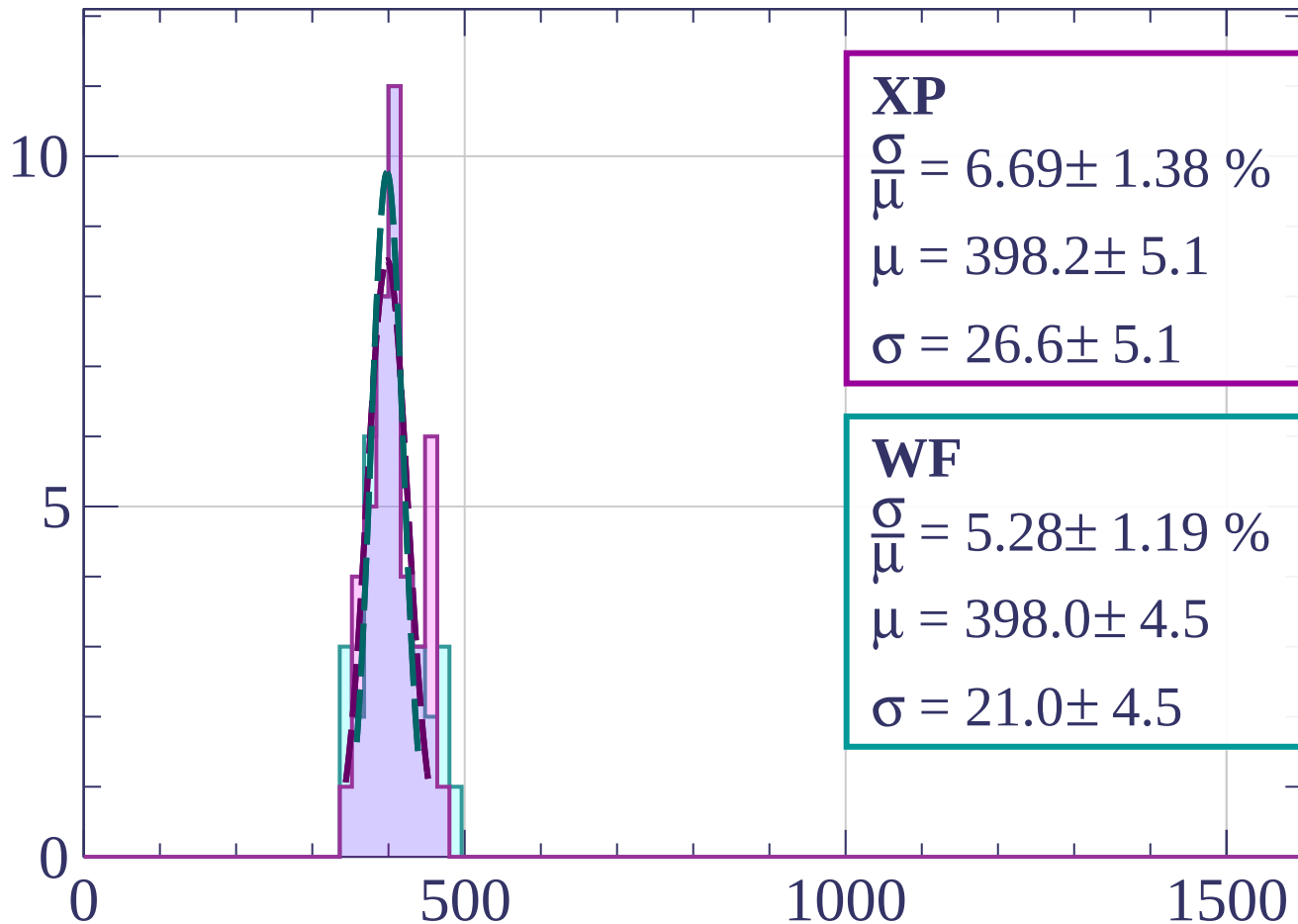
$$\mu = 402.4 \pm 2.4$$

$$\sigma = 29.6 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 6.69 \pm 1.38 \%$$

$$\mu = 398.2 \pm 5.1$$

$$\sigma = 26.6 \pm 5.1$$

WF

$$\frac{\sigma}{\mu} = 5.28 \pm 1.19 \%$$

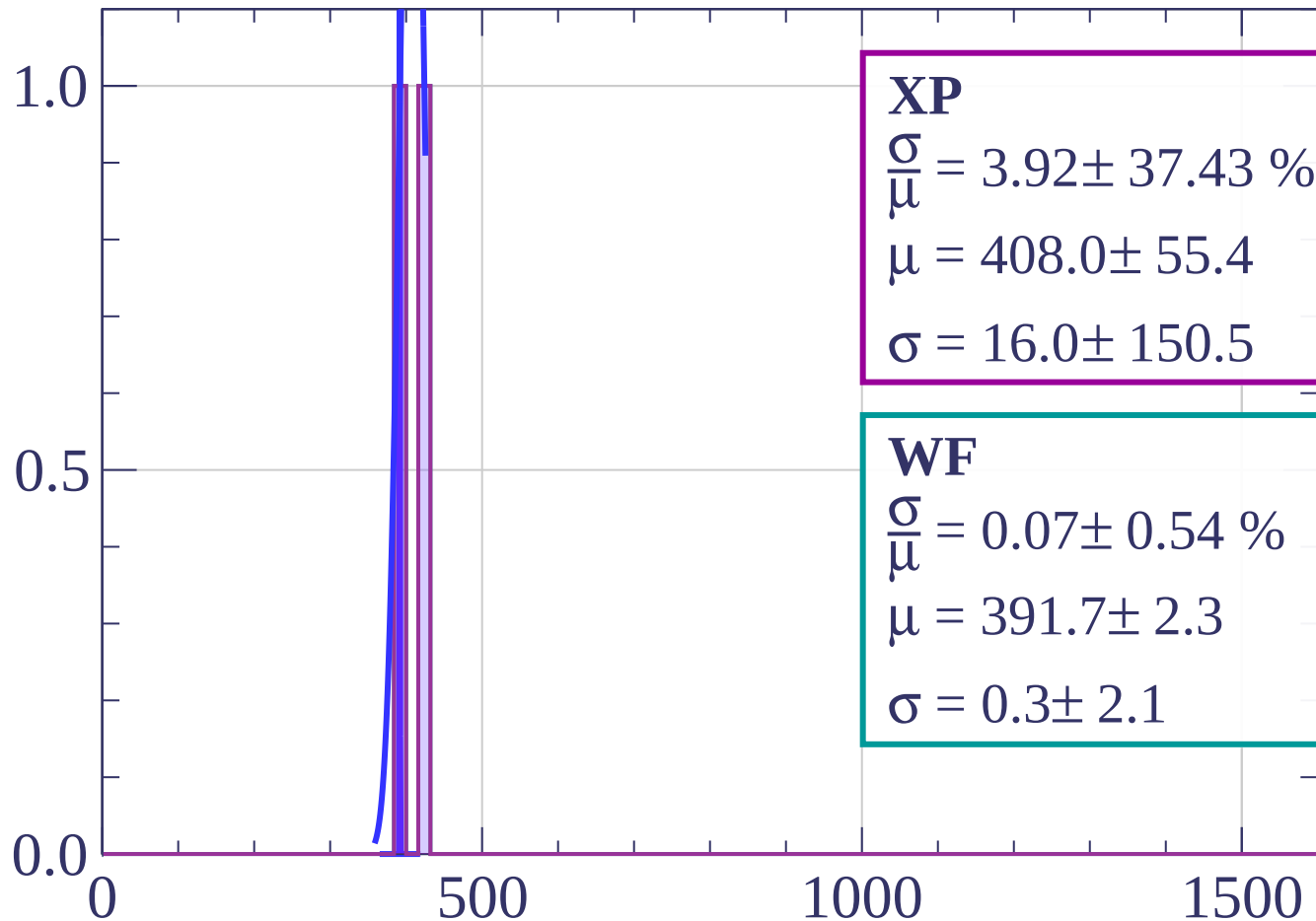
$$\mu = 398.0 \pm 4.5$$

$$\sigma = 21.0 \pm 4.5$$

dE/dx [ADC counts/cm]

Count

Energy loss | $48 < \phi < 51$



XP

$$\frac{\sigma}{\mu} = 3.92 \pm 37.43 \%$$

$$\mu = 408.0 \pm 55.4$$

$$\sigma = 16.0 \pm 150.5$$

WF

$$\frac{\sigma}{\mu} = 0.07 \pm 0.54 \%$$

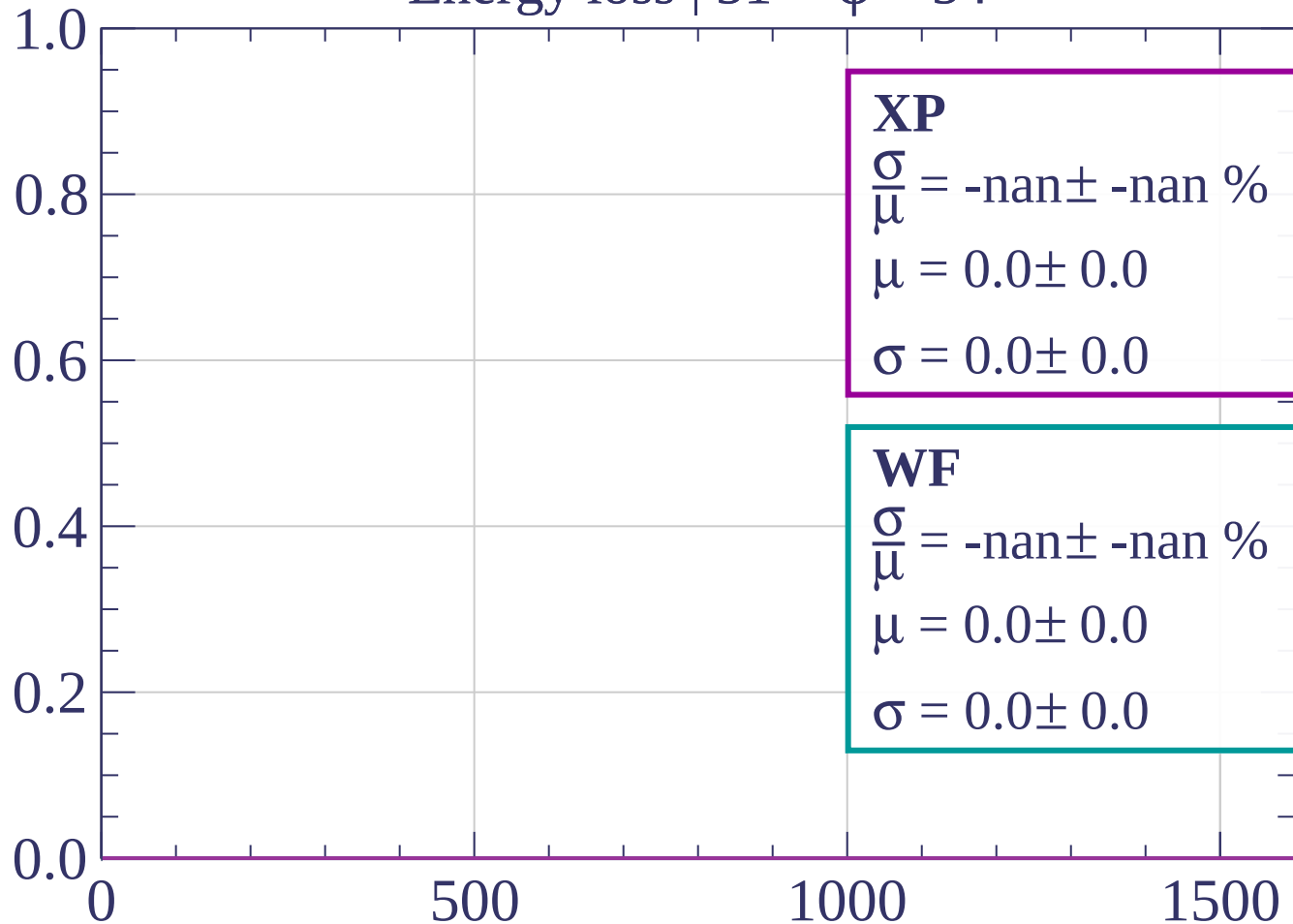
$$\mu = 391.7 \pm 2.3$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx [ADC counts/cm]

Count

Energy loss | $51 < \phi < 54$



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]

Count

Energy loss | $54 < \phi < 57$

XP

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $57 < \phi < 60$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $60 < \phi < 63$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

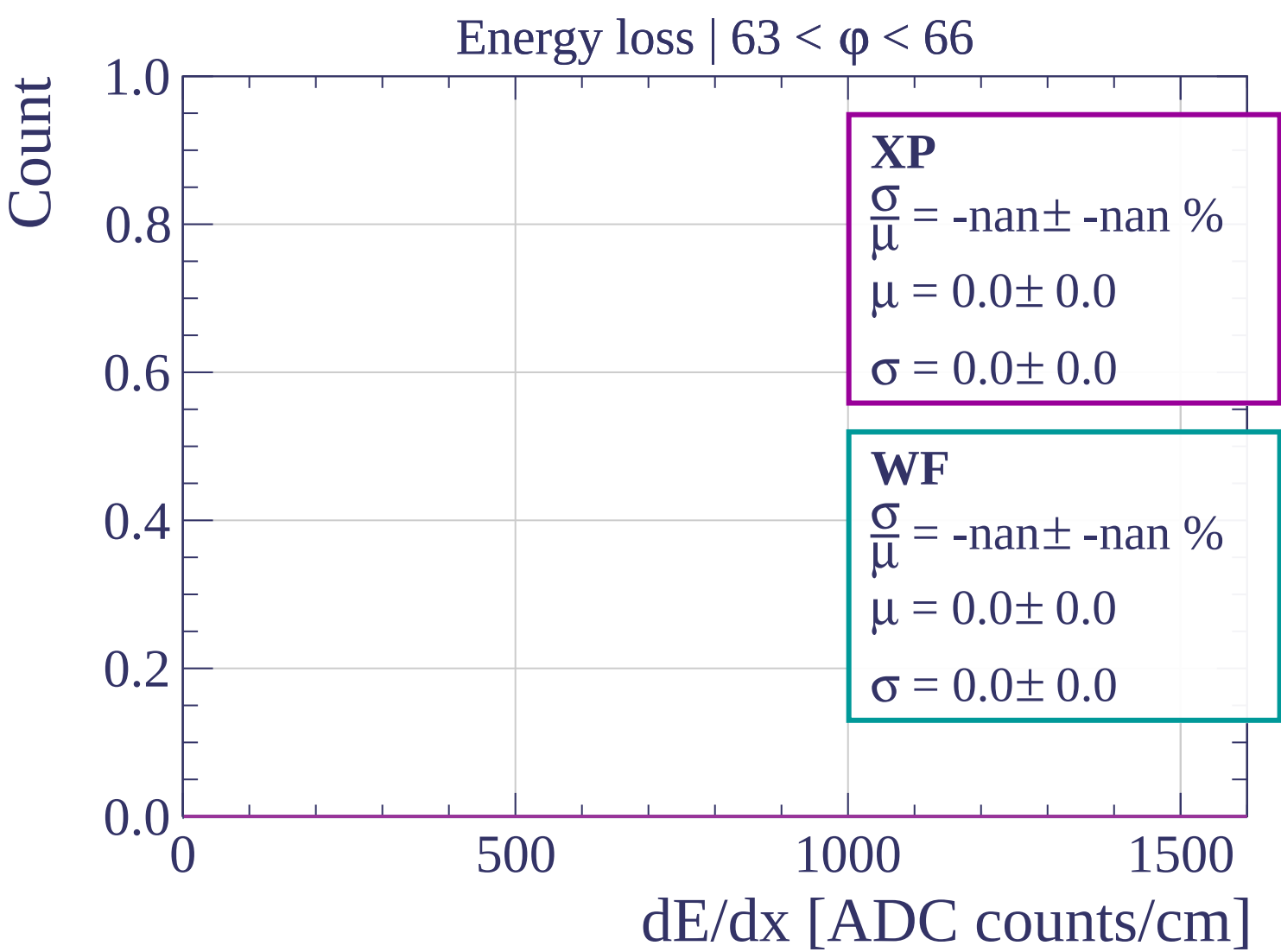
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $66 < \phi < 69$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $69 < \phi < 72$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $72 < \phi < 75$

XP

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $75 < \phi < 78$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

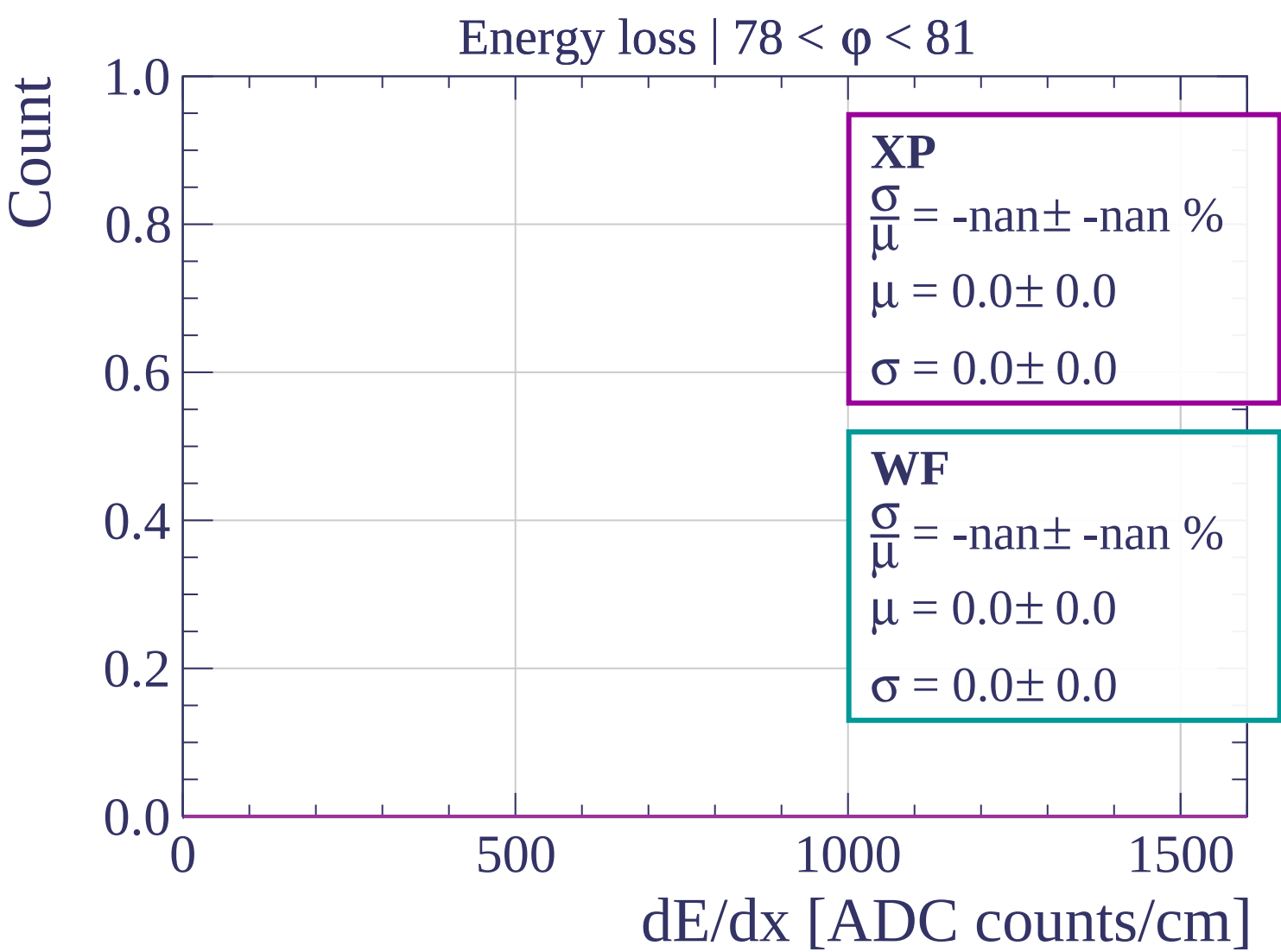
0

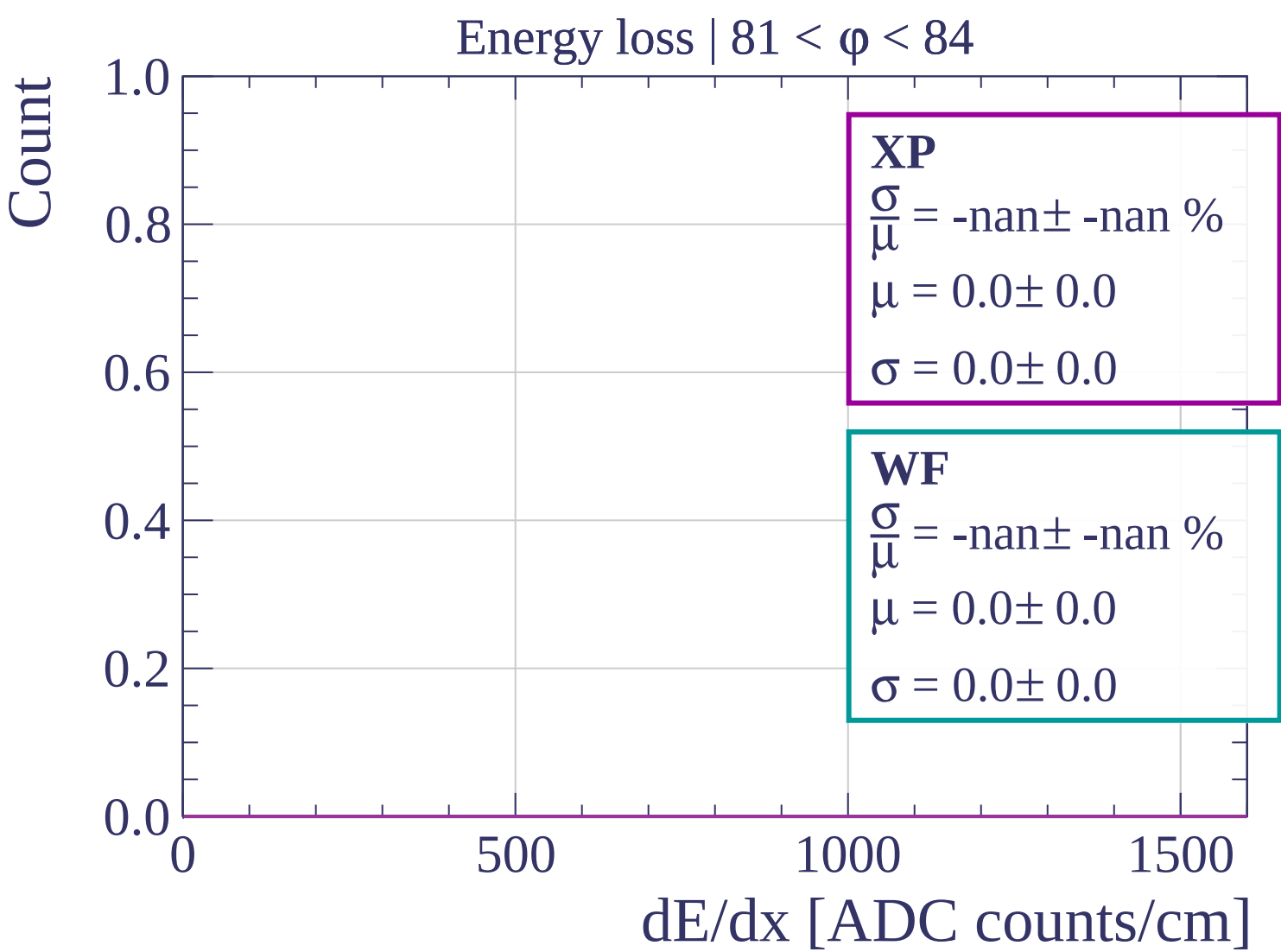
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $84 < \phi < 87$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $87 < \phi < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

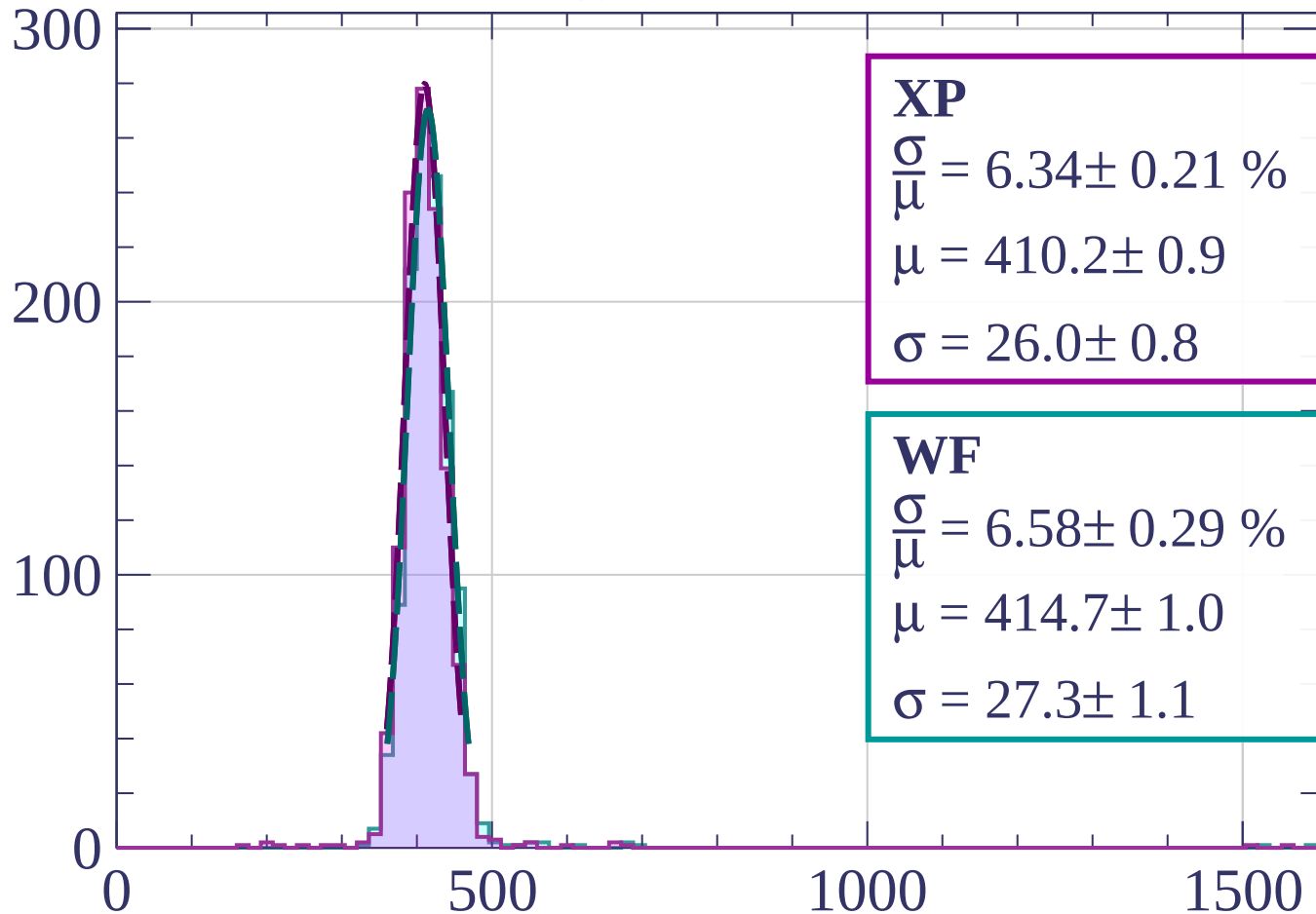
1000

1500

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 6.34 \pm 0.21 \%$$

$$\mu = 410.2 \pm 0.9$$

$$\sigma = 26.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.29 \%$$

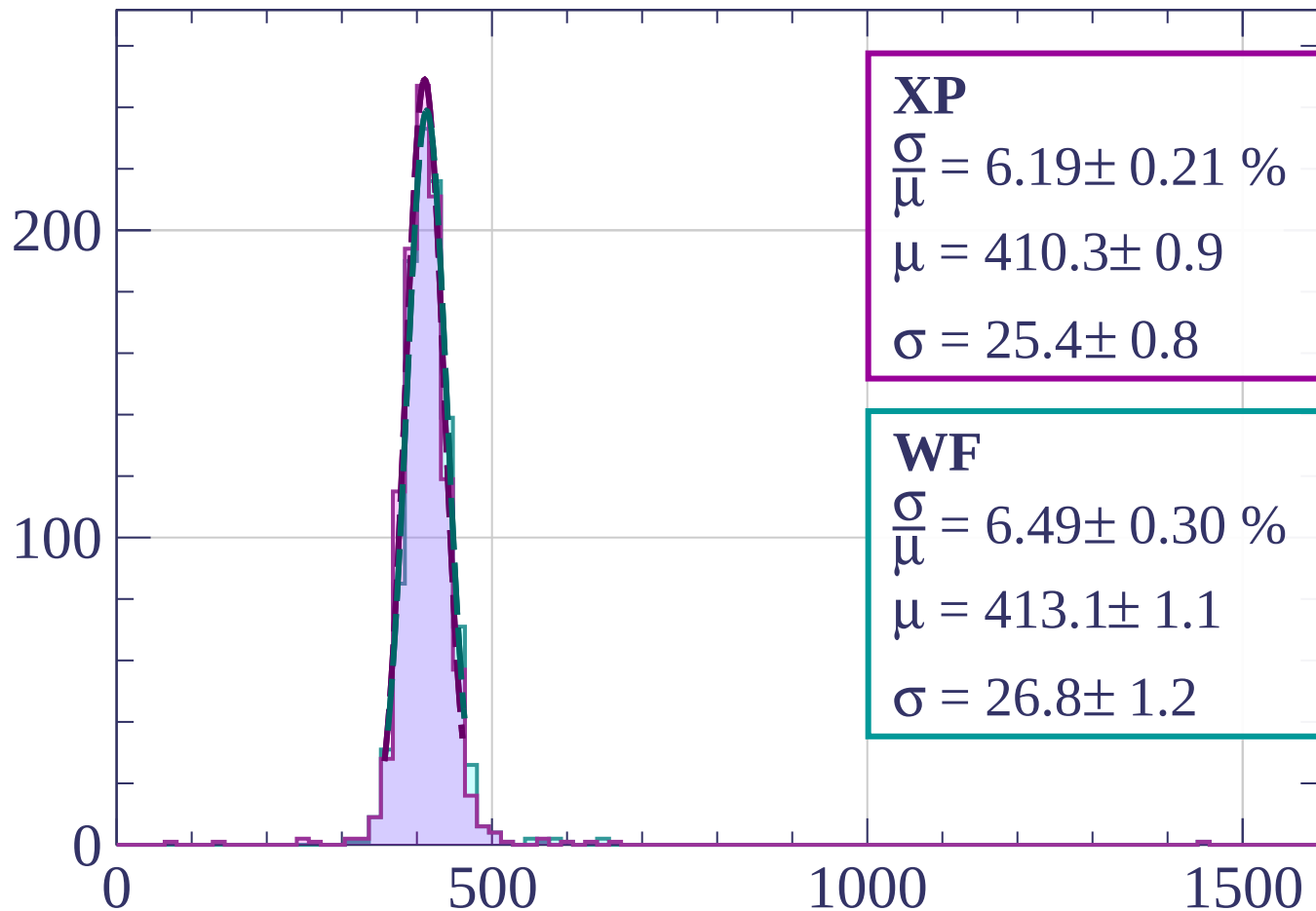
$$\mu = 414.7 \pm 1.0$$

$$\sigma = 27.3 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.21 \%$$

$$\mu = 410.3 \pm 0.9$$

$$\sigma = 25.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.49 \pm 0.30 \%$$

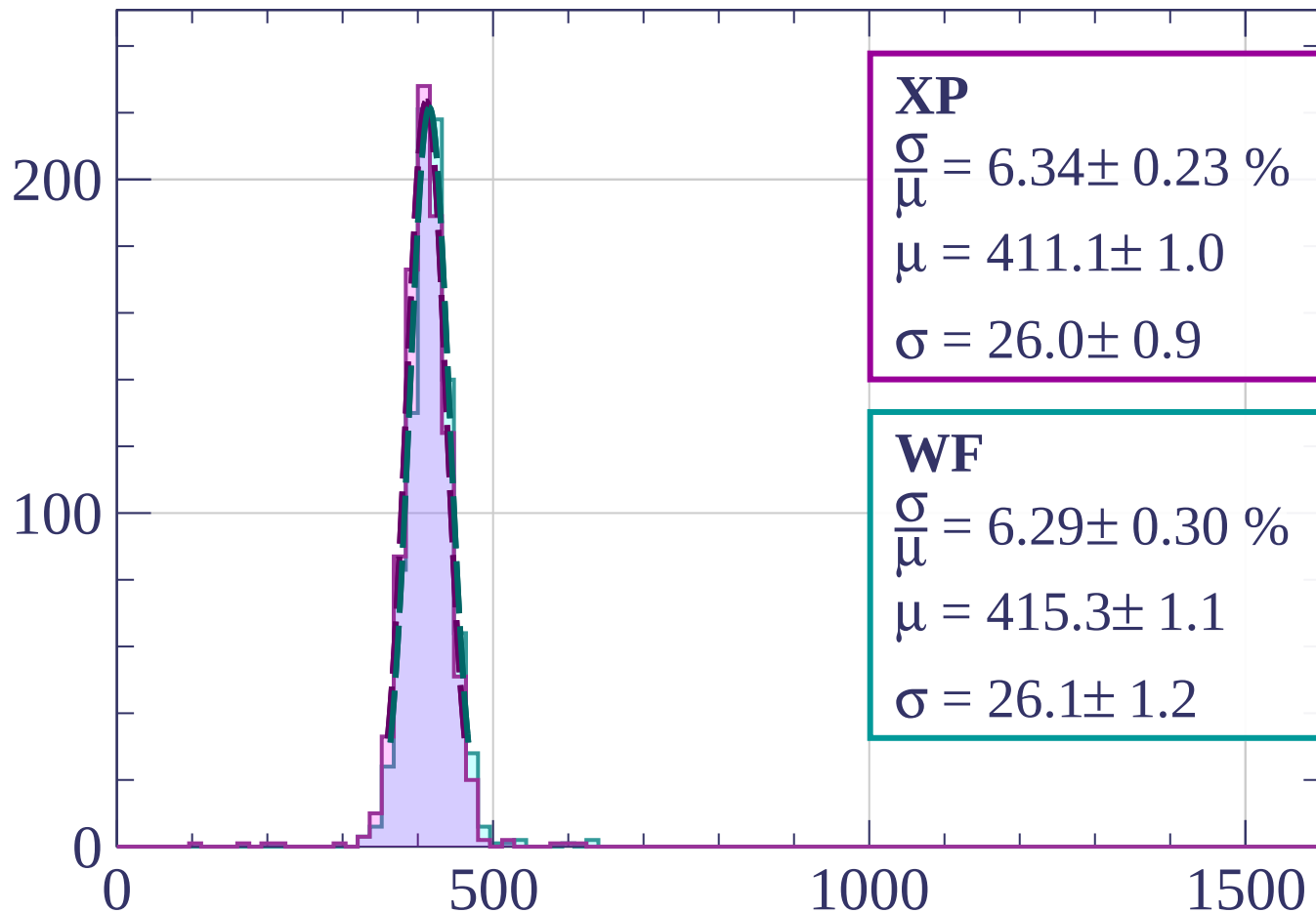
$$\mu = 413.1 \pm 1.1$$

$$\sigma = 26.8 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 6.34 \pm 0.23 \%$$

$$\mu = 411.1 \pm 1.0$$

$$\sigma = 26.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.29 \pm 0.30 \%$$

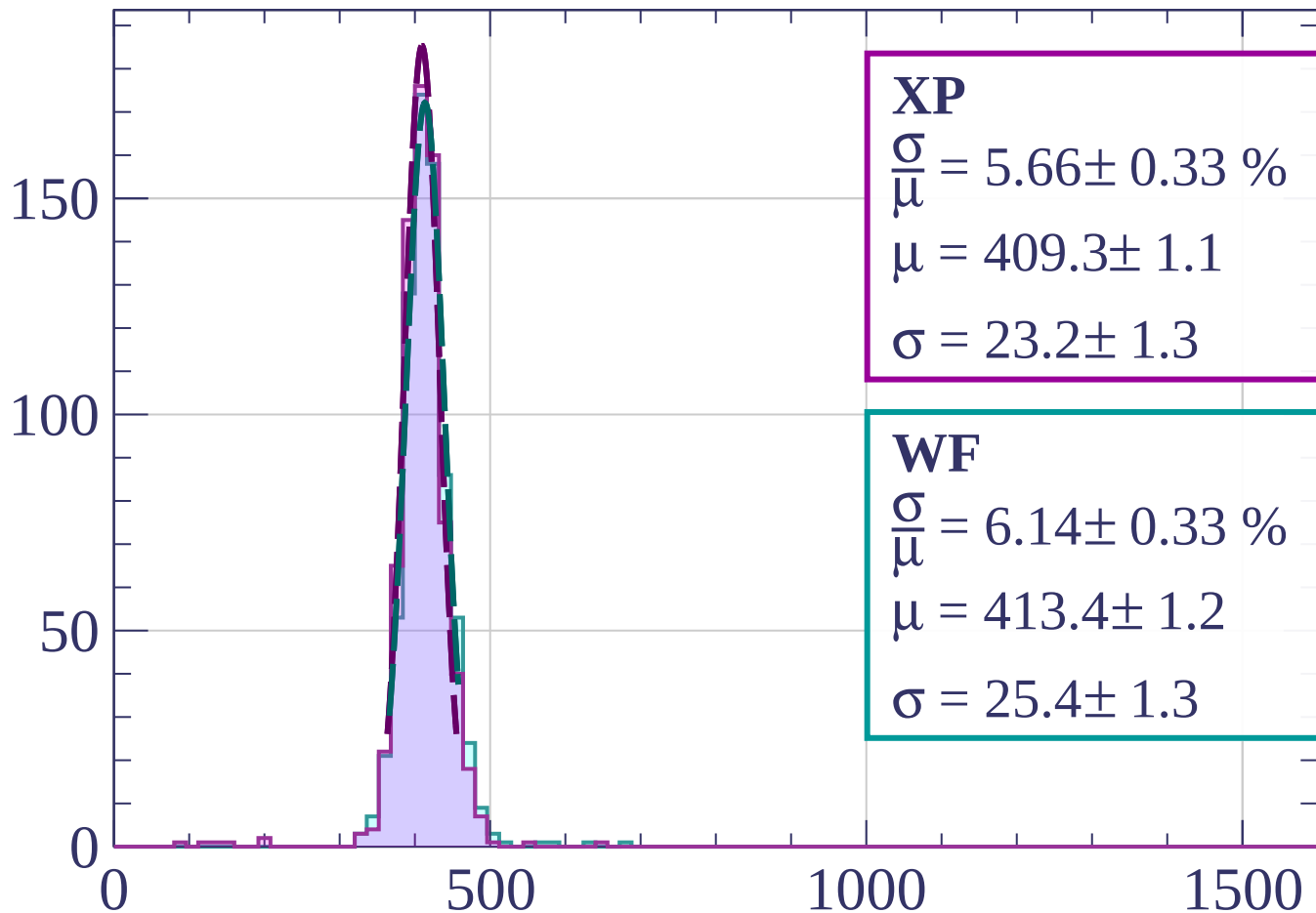
$$\mu = 415.3 \pm 1.1$$

$$\sigma = 26.1 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.66 \pm 0.33 \%$$

$$\mu = 409.3 \pm 1.1$$

$$\sigma = 23.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.14 \pm 0.33 \%$$

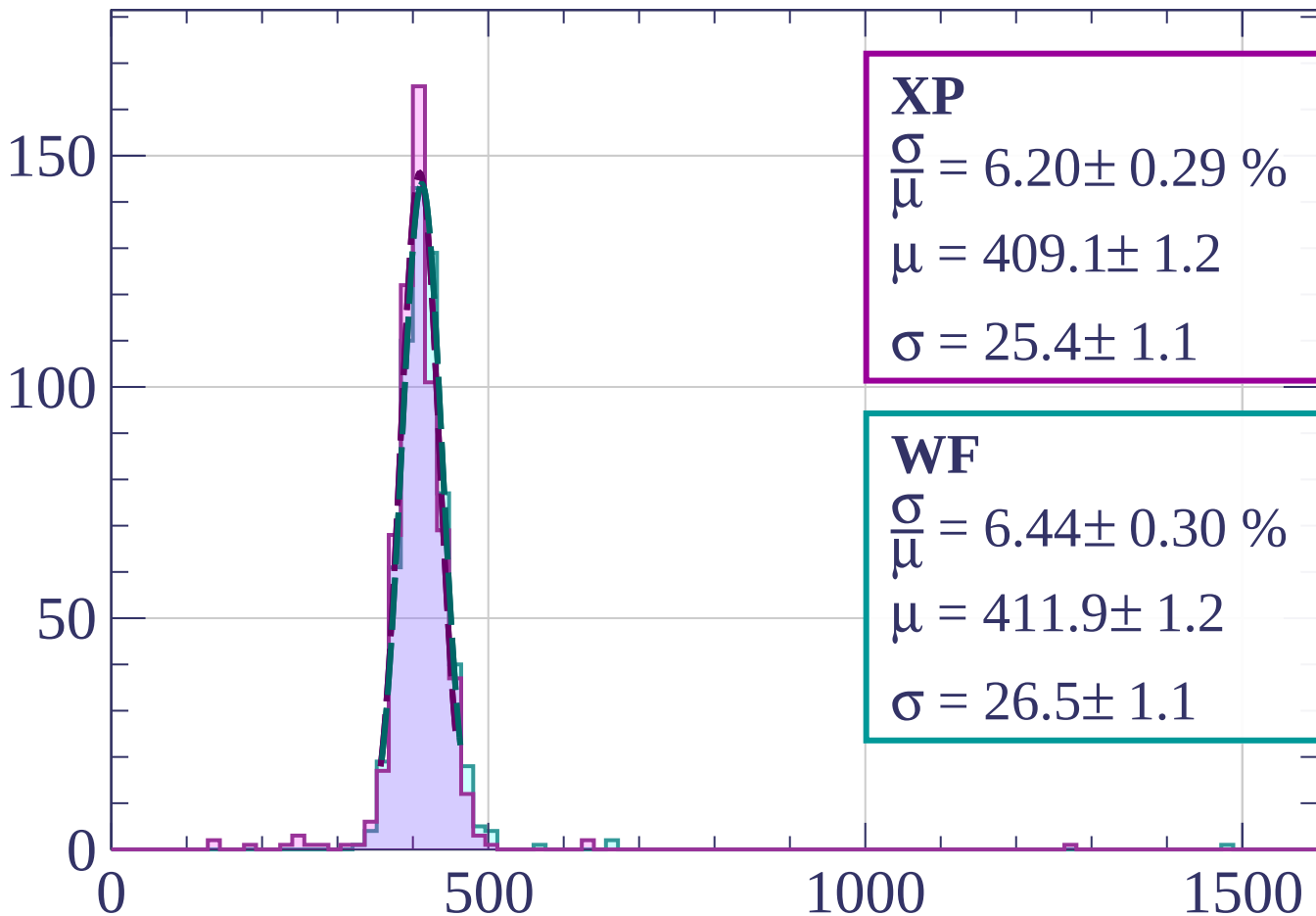
$$\mu = 413.4 \pm 1.2$$

$$\sigma = 25.4 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.20 \pm 0.29 \%$$

$$\mu = 409.1 \pm 1.2$$

$$\sigma = 25.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.30 \%$$

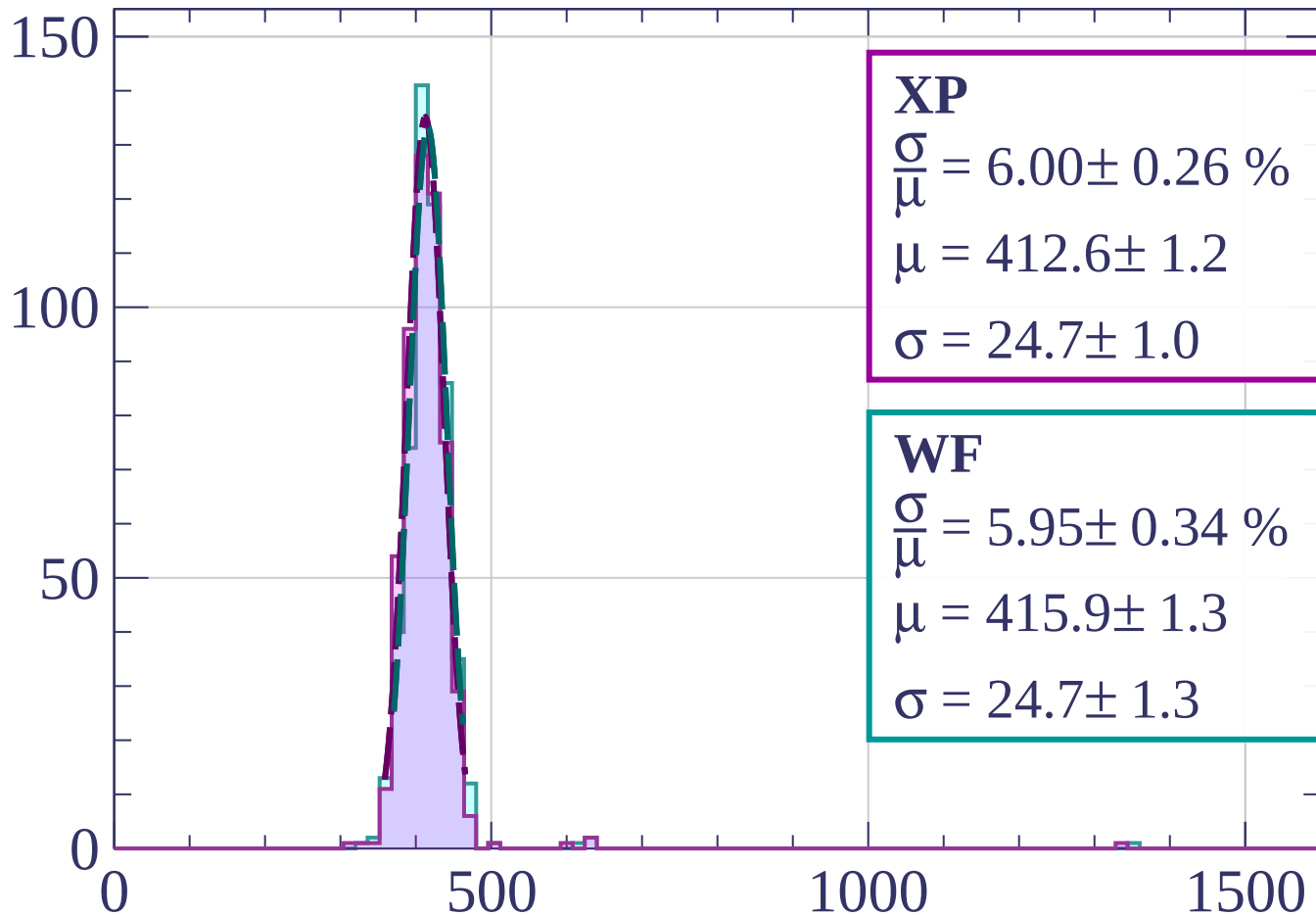
$$\mu = 411.9 \pm 1.2$$

$$\sigma = 26.5 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.00 \pm 0.26 \%$$

$$\mu = 412.6 \pm 1.2$$

$$\sigma = 24.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.95 \pm 0.34 \%$$

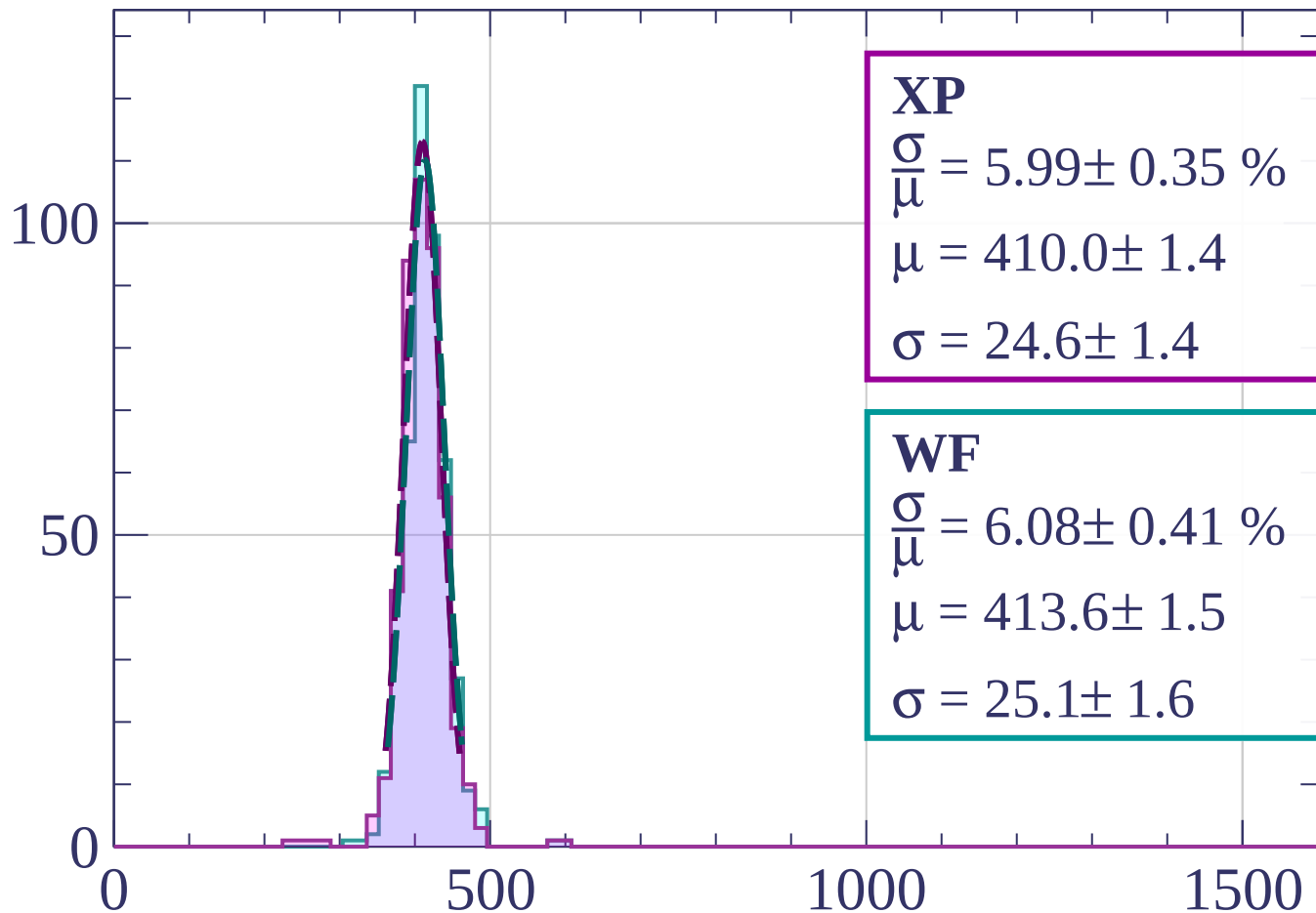
$$\mu = 415.9 \pm 1.3$$

$$\sigma = 24.7 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 5.99 \pm 0.35 \%$$

$$\mu = 410.0 \pm 1.4$$

$$\sigma = 24.6 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 6.08 \pm 0.41 \%$$

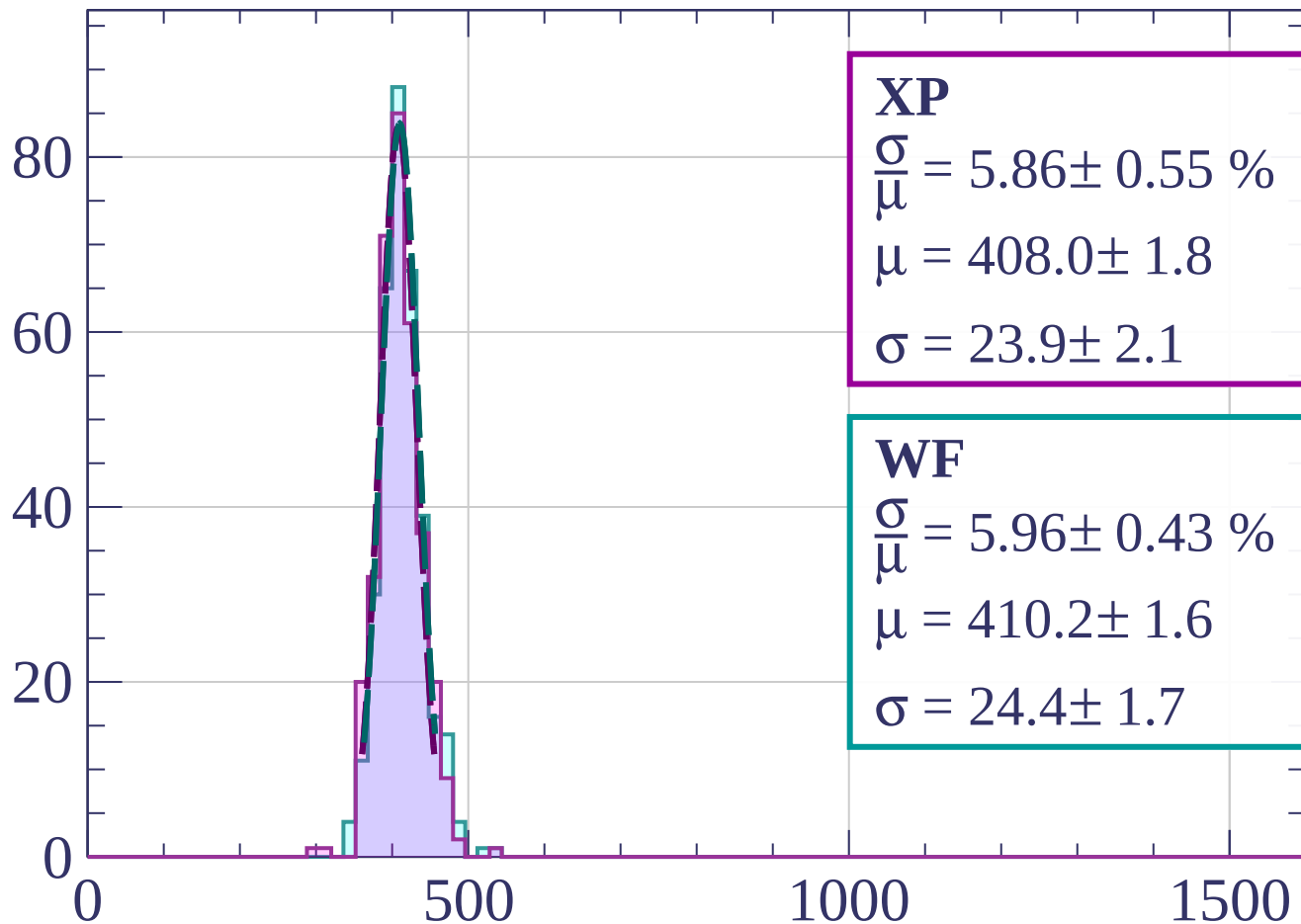
$$\mu = 413.6 \pm 1.5$$

$$\sigma = 25.1 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.86 \pm 0.55 \%$$

$$\mu = 408.0 \pm 1.8$$

$$\sigma = 23.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 5.96 \pm 0.43 \%$$

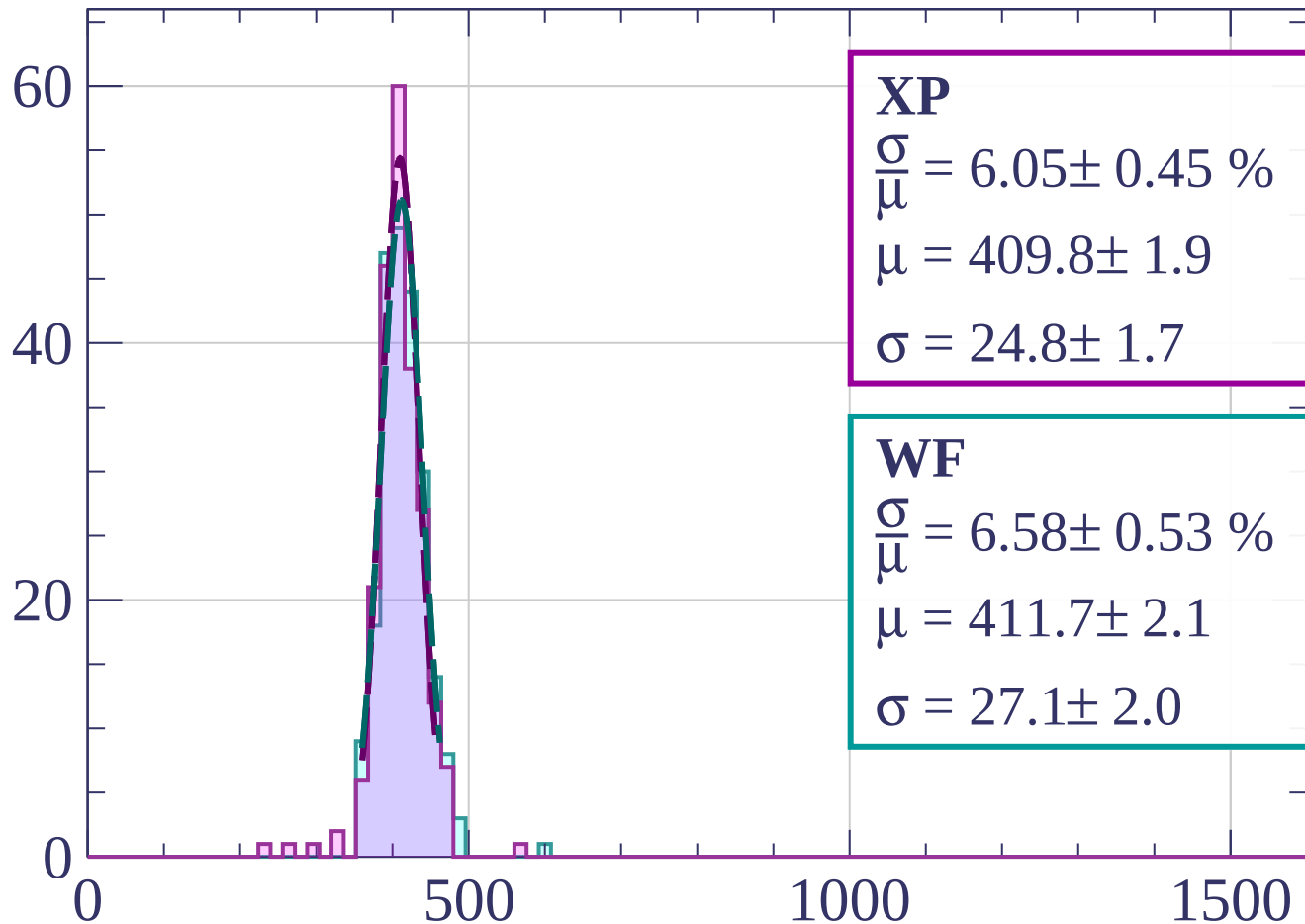
$$\mu = 410.2 \pm 1.6$$

$$\sigma = 24.4 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 6.05 \pm 0.45 \%$$

$$\mu = 409.8 \pm 1.9$$

$$\sigma = 24.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.53 \%$$

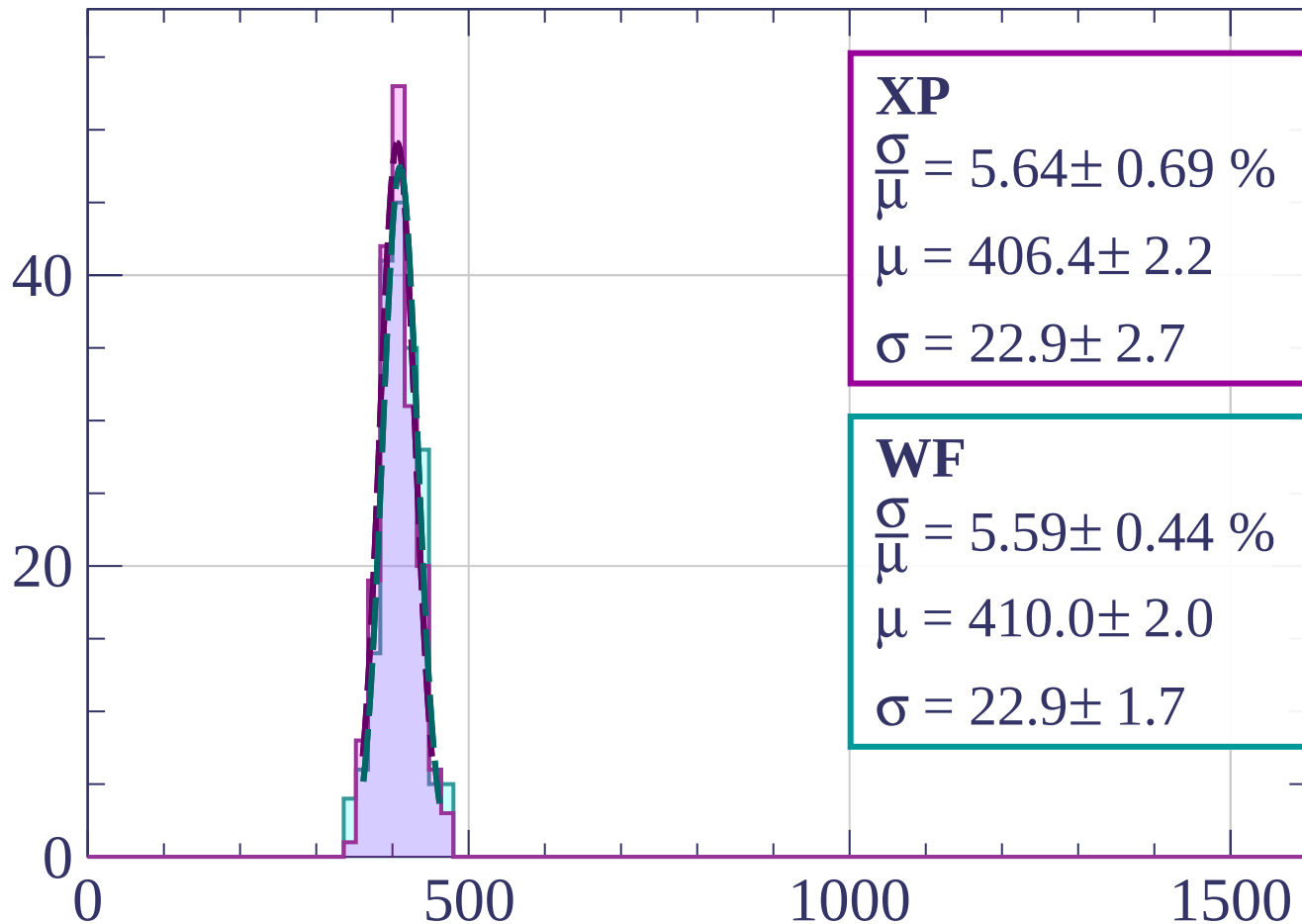
$$\mu = 411.7 \pm 2.1$$

$$\sigma = 27.1 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 5.64 \pm 0.69 \%$$

$$\mu = 406.4 \pm 2.2$$

$$\sigma = 22.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 5.59 \pm 0.44 \%$$

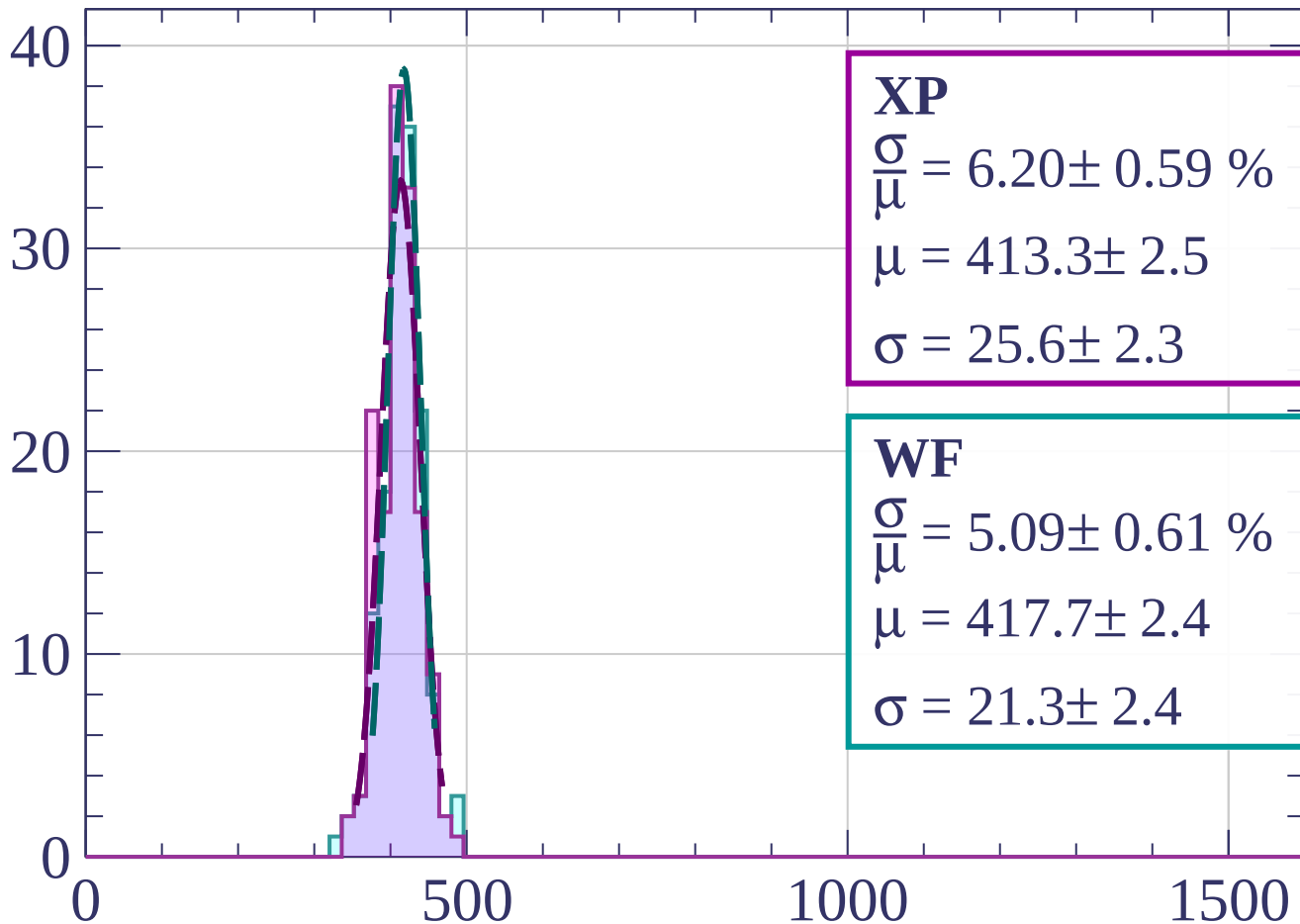
$$\mu = 410.0 \pm 2.0$$

$$\sigma = 22.9 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 6.20 \pm 0.59 \%$$

$$\mu = 413.3 \pm 2.5$$

$$\sigma = 25.6 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 5.09 \pm 0.61 \%$$

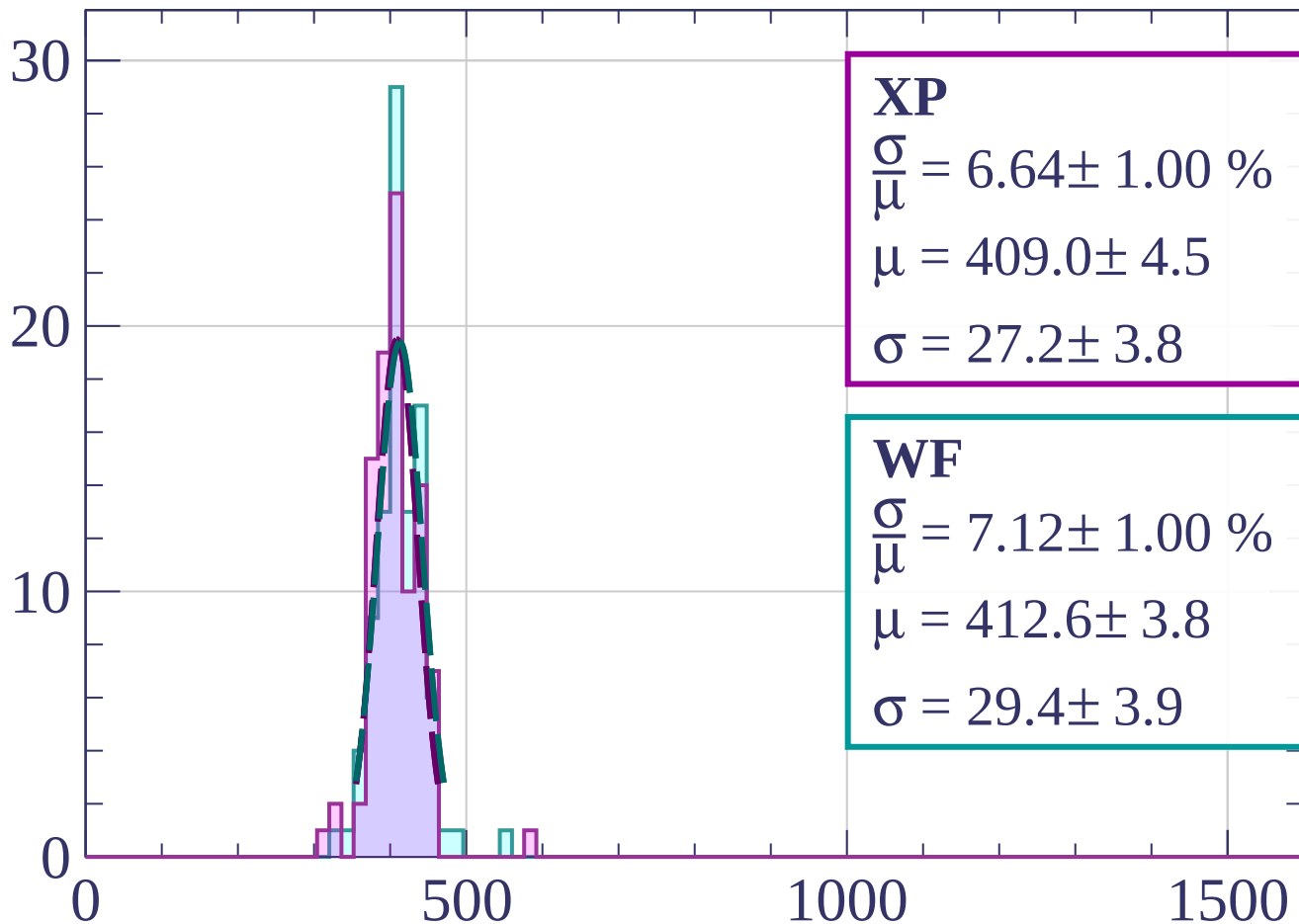
$$\mu = 417.7 \pm 2.4$$

$$\sigma = 21.3 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 6.64 \pm 1.00 \%$$

$$\mu = 409.0 \pm 4.5$$

$$\sigma = 27.2 \pm 3.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.12 \pm 1.00 \%$$

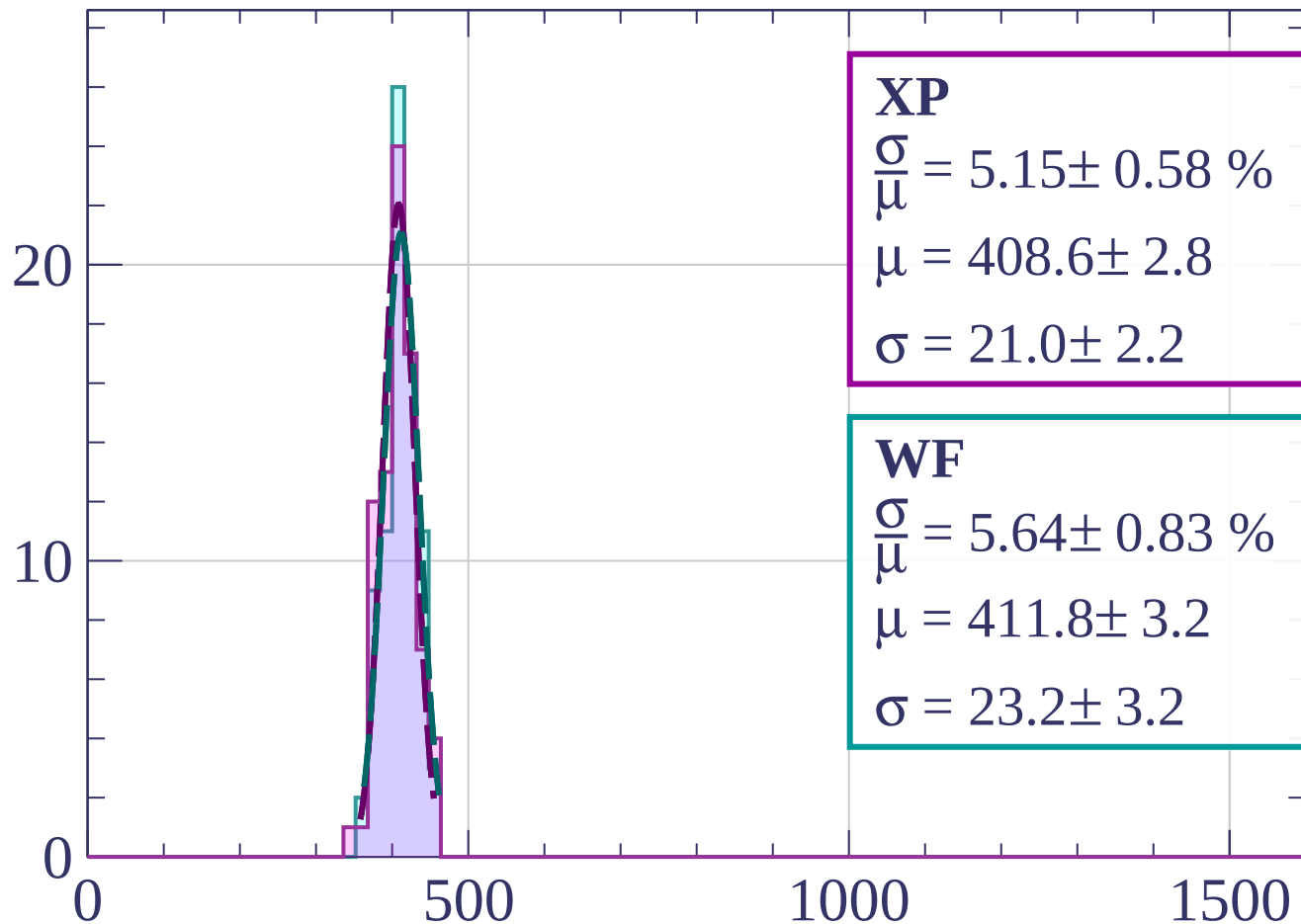
$$\mu = 412.6 \pm 3.8$$

$$\sigma = 29.4 \pm 3.9$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 5.15 \pm 0.58 \%$$

$$\mu = 408.6 \pm 2.8$$

$$\sigma = 21.0 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 5.64 \pm 0.83 \%$$

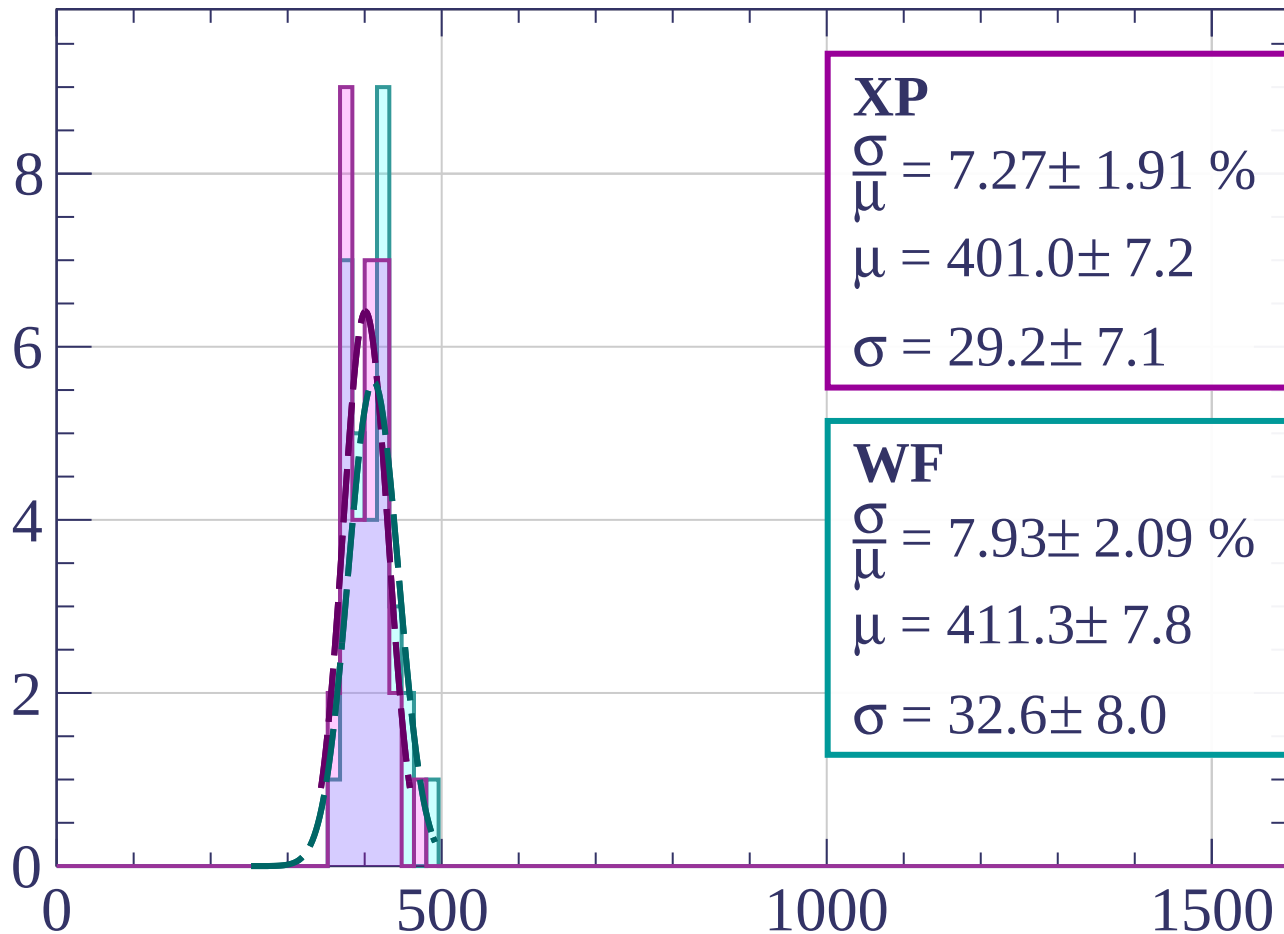
$$\mu = 411.8 \pm 3.2$$

$$\sigma = 23.2 \pm 3.2$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 1.91 \%$$

$$\mu = 401.0 \pm 7.2$$

$$\sigma = 29.2 \pm 7.1$$

WF

$$\frac{\sigma}{\mu} = 7.93 \pm 2.09 \%$$

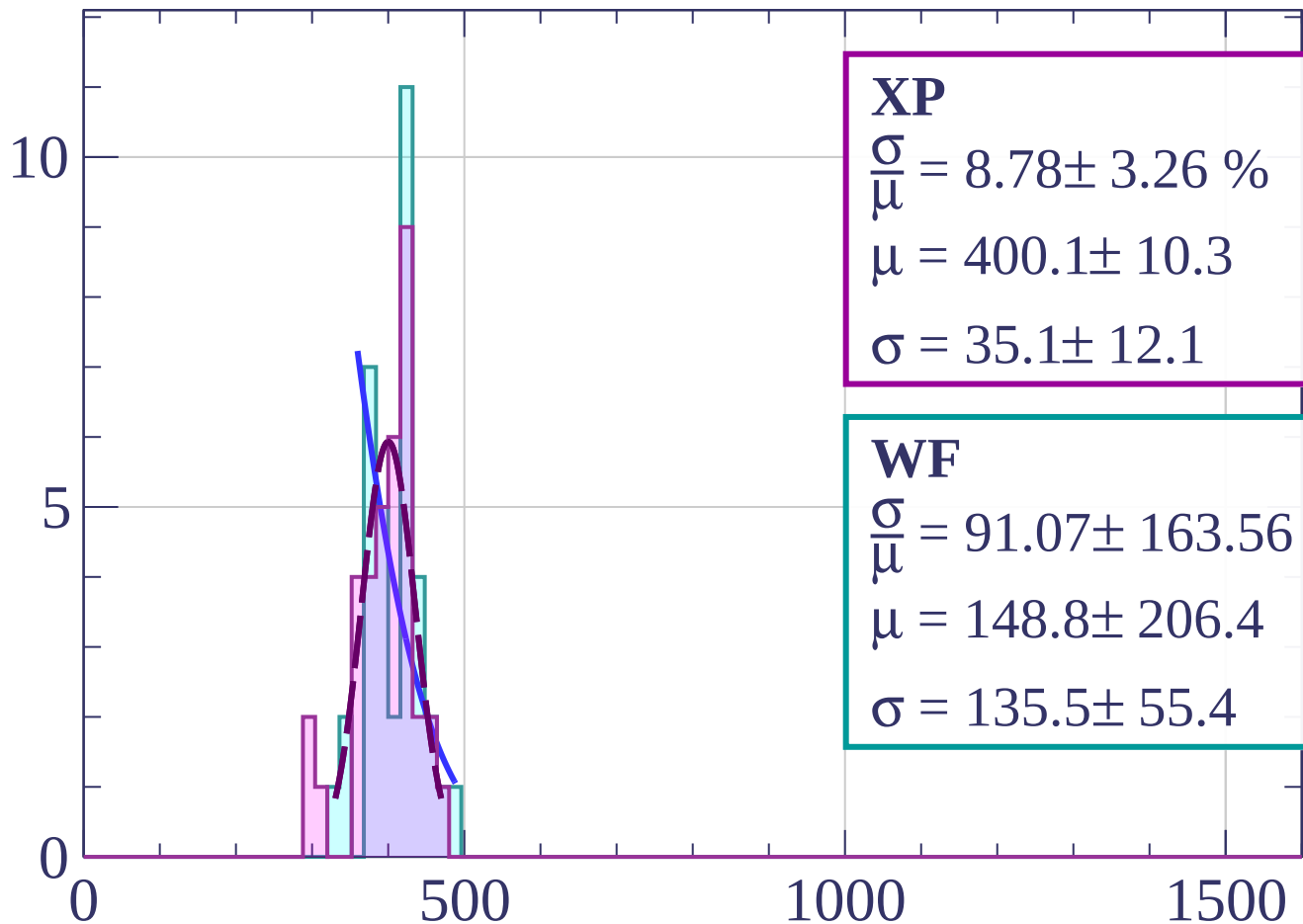
$$\mu = 411.3 \pm 7.8$$

$$\sigma = 32.6 \pm 8.0$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 3.26 \%$$

$$\mu = 400.1 \pm 10.3$$

$$\sigma = 35.1 \pm 12.1$$

WF

$$\frac{\sigma}{\mu} = 91.07 \pm 163.56 \%$$

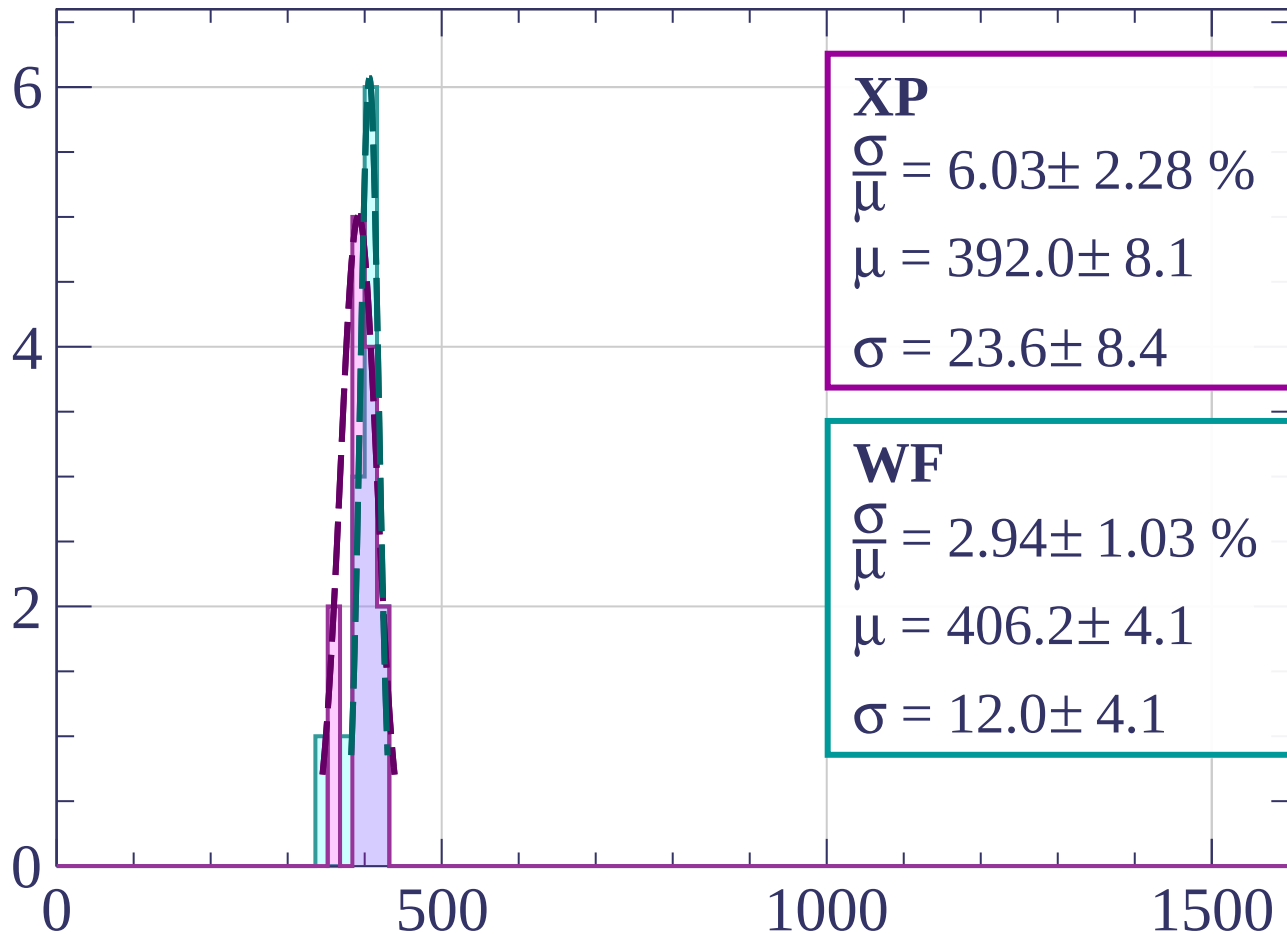
$$\mu = 148.8 \pm 206.4$$

$$\sigma = 135.5 \pm 55.4$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

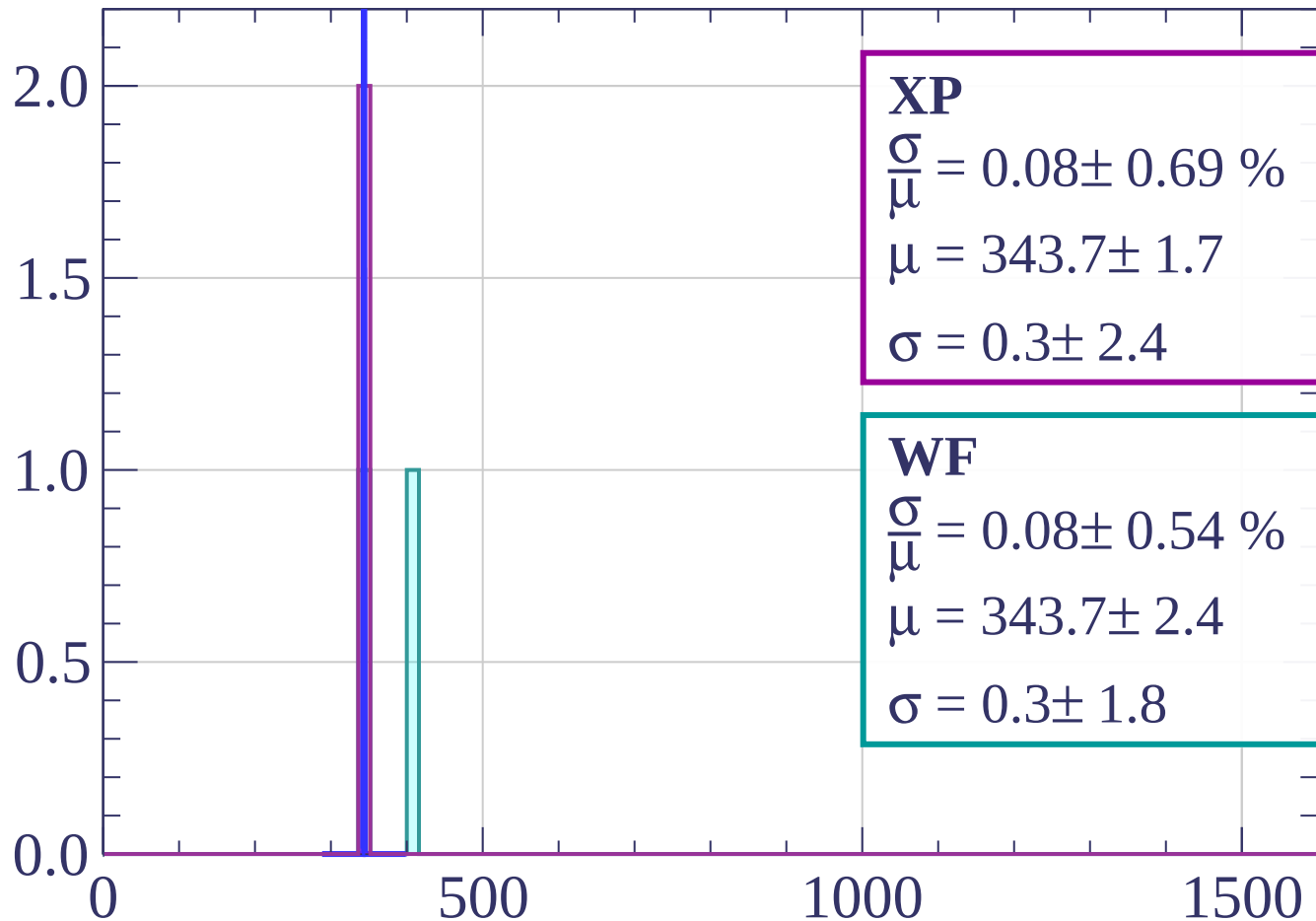
Count



dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

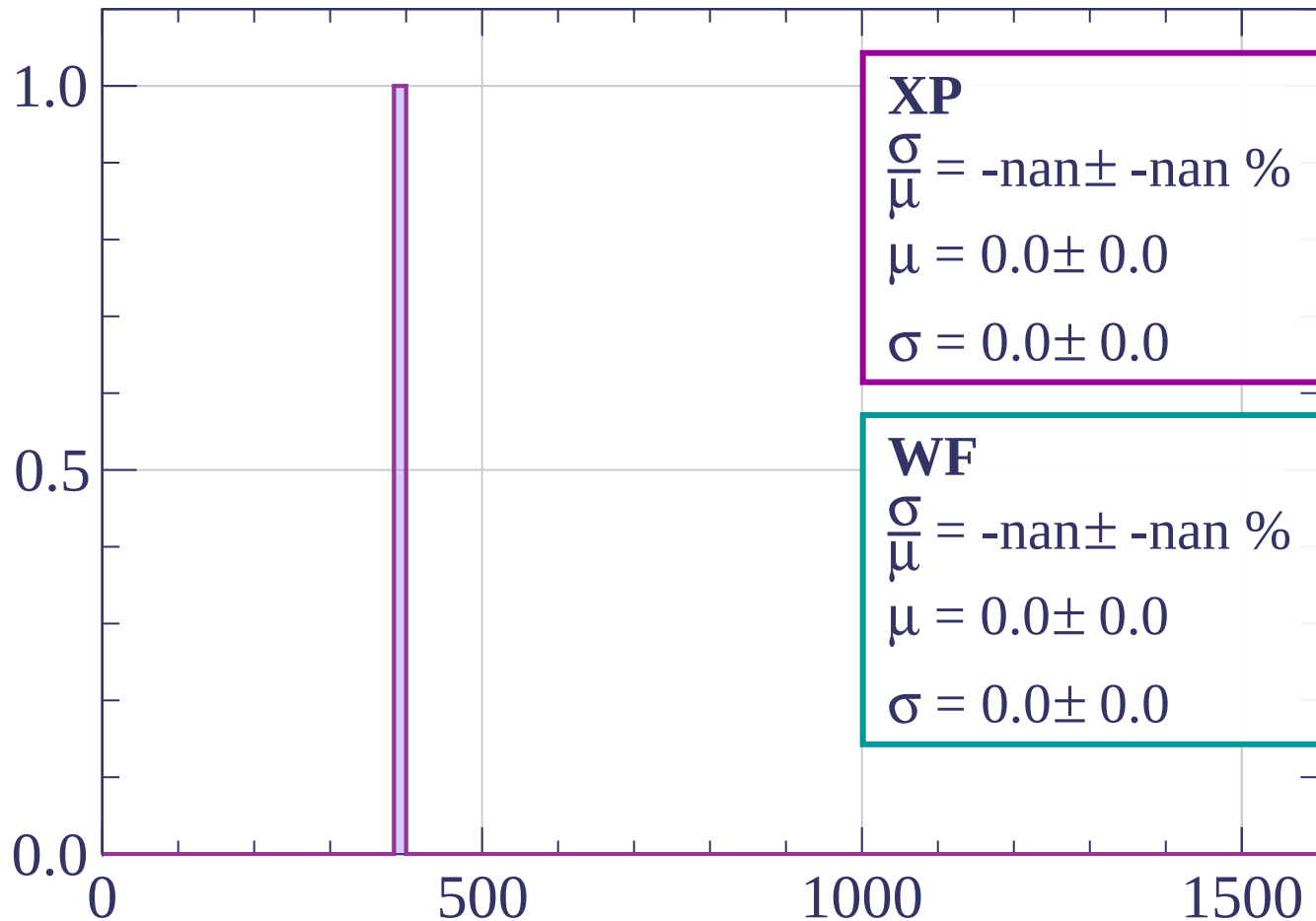
Count



dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

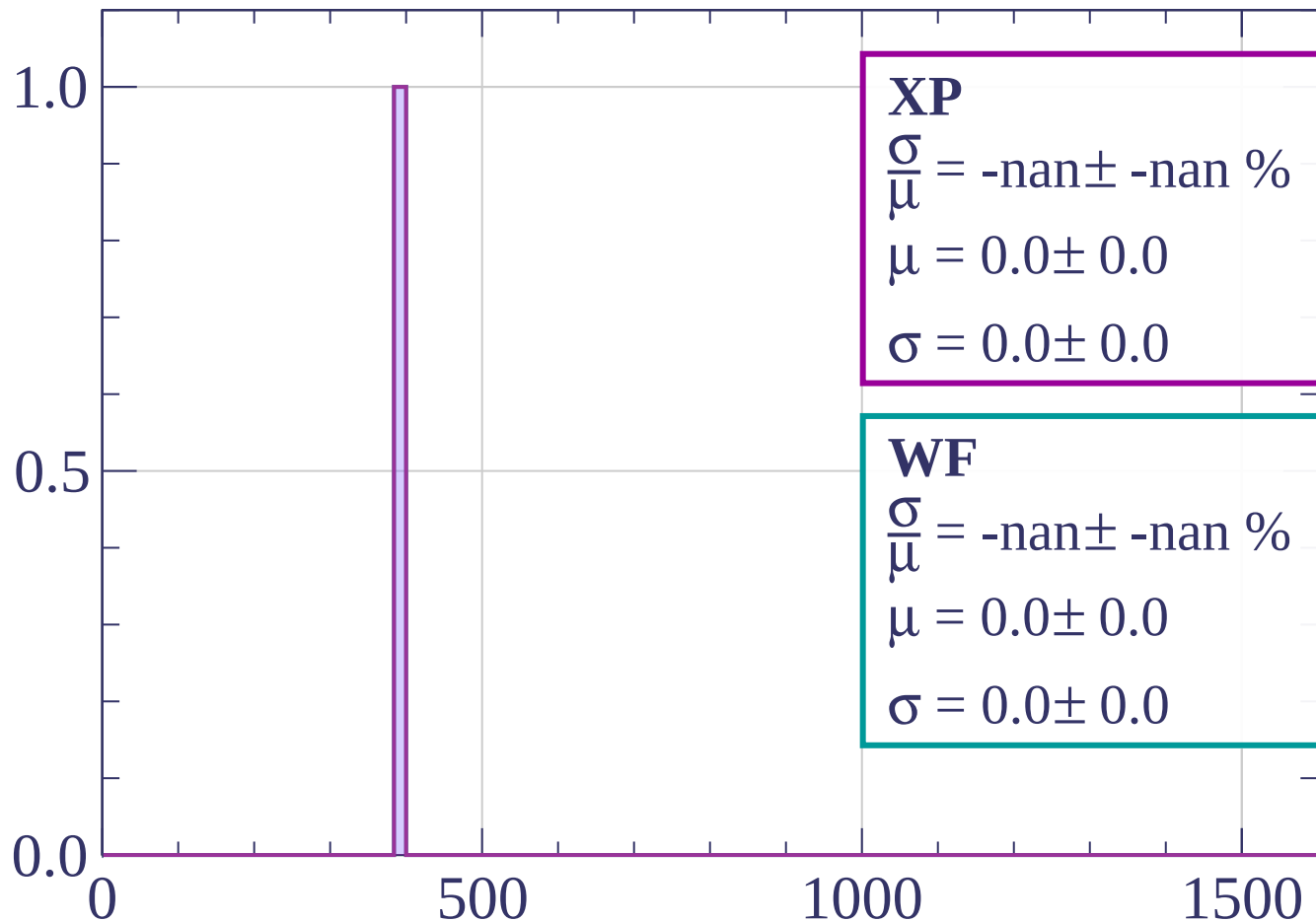
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

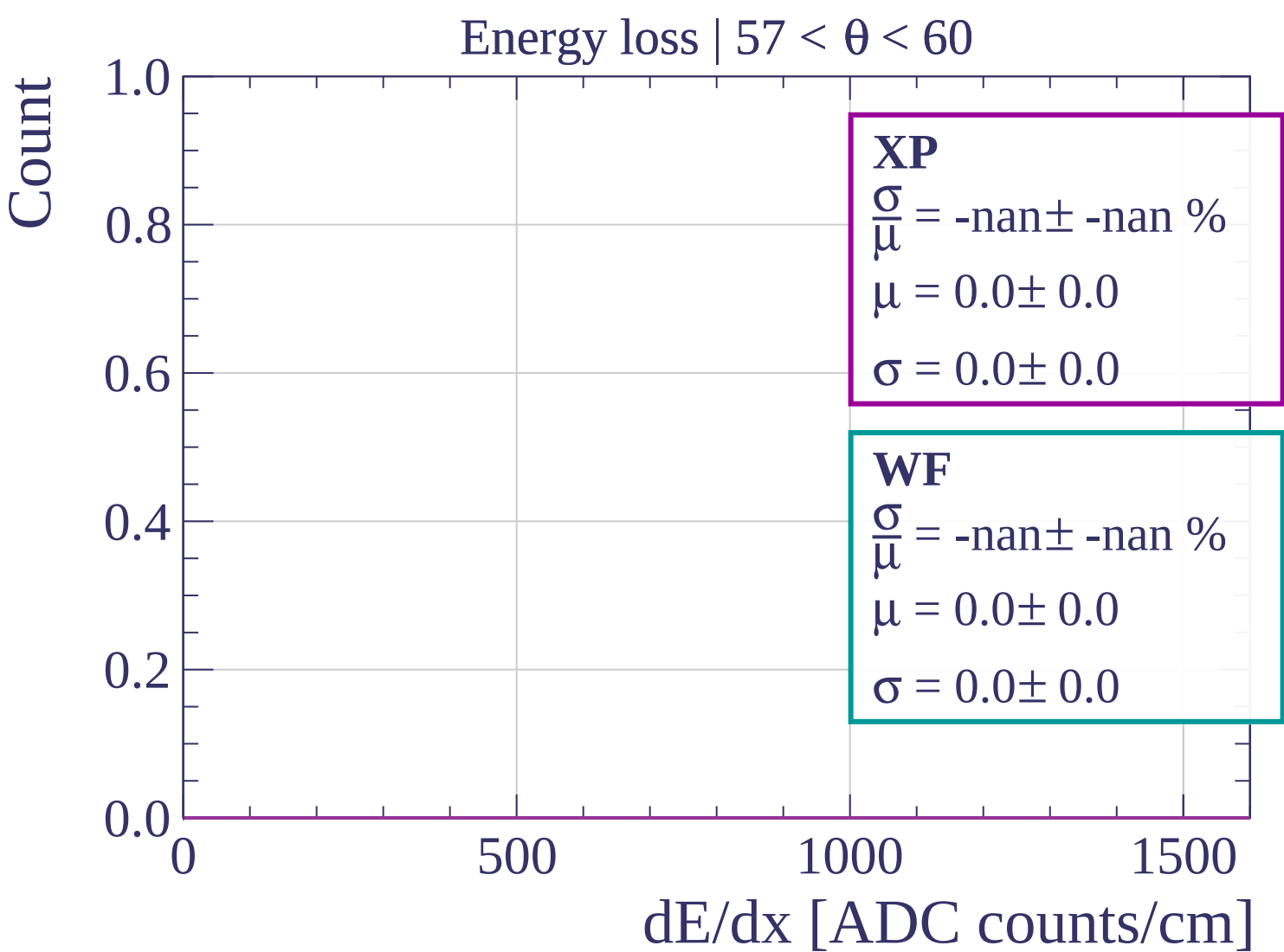
WF

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]



Count

Energy loss | $60 < \theta < 63$

XP

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

1.0
0.8
0.6
0.4
0.2
0.0

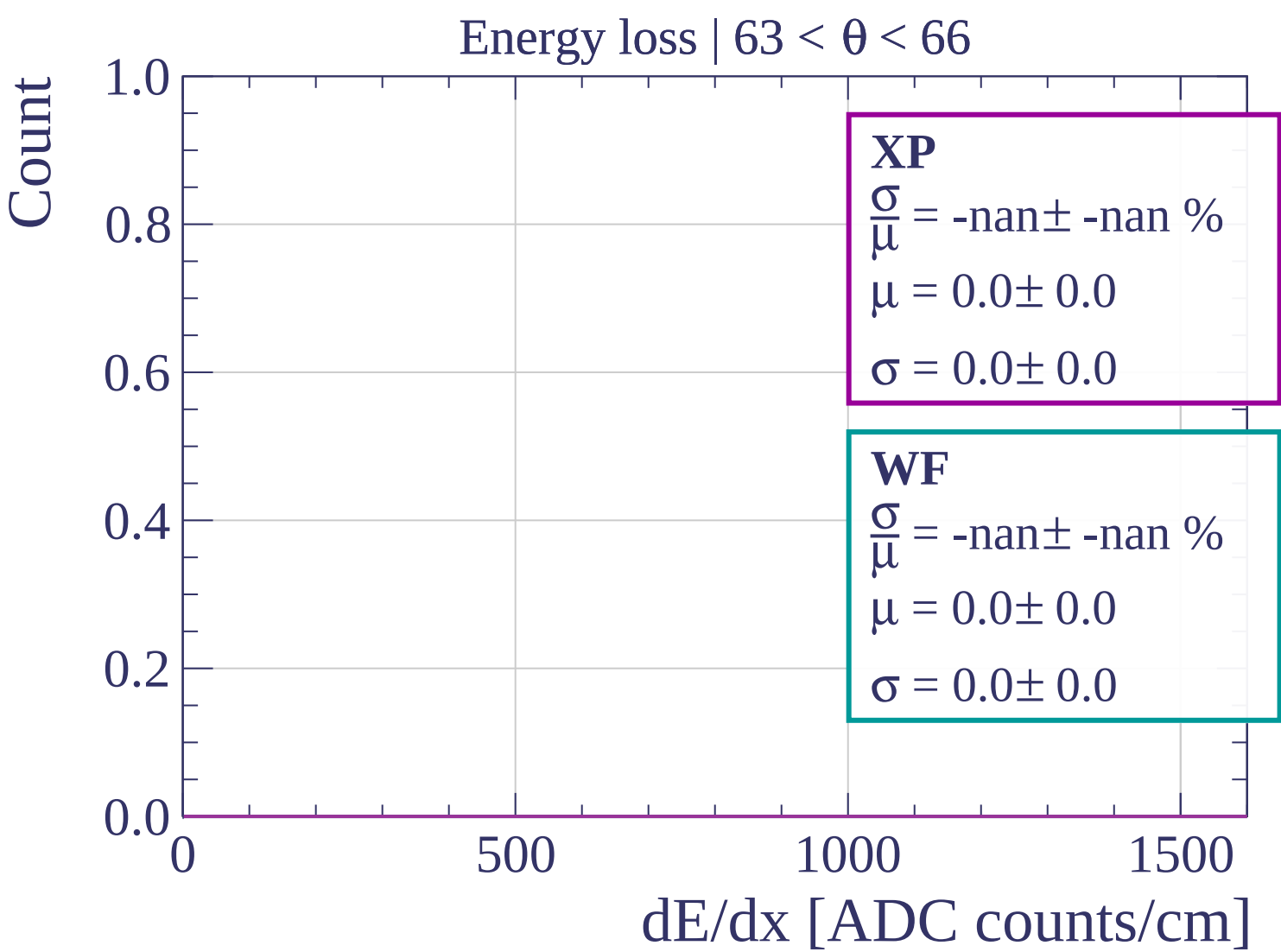
0

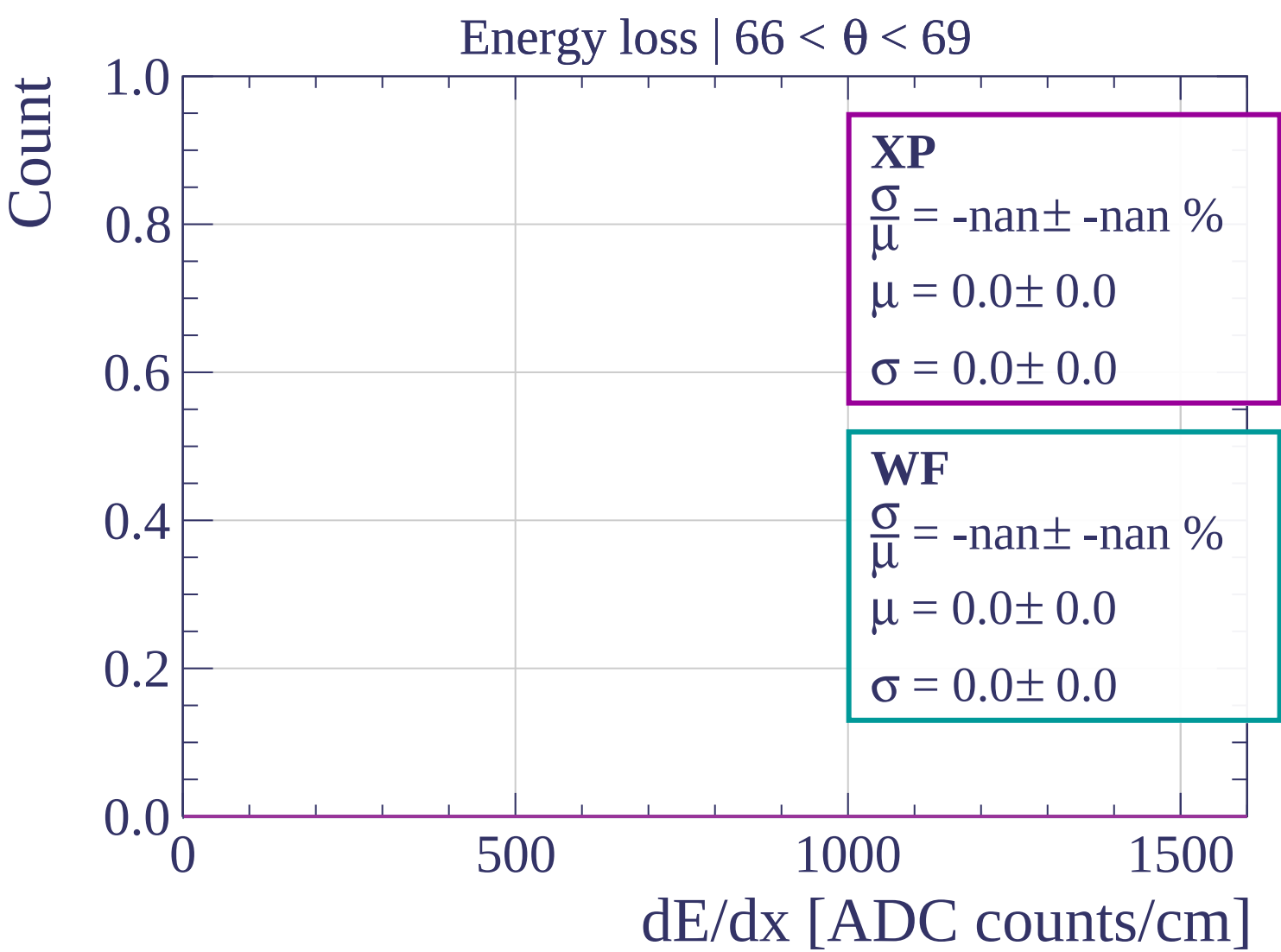
500

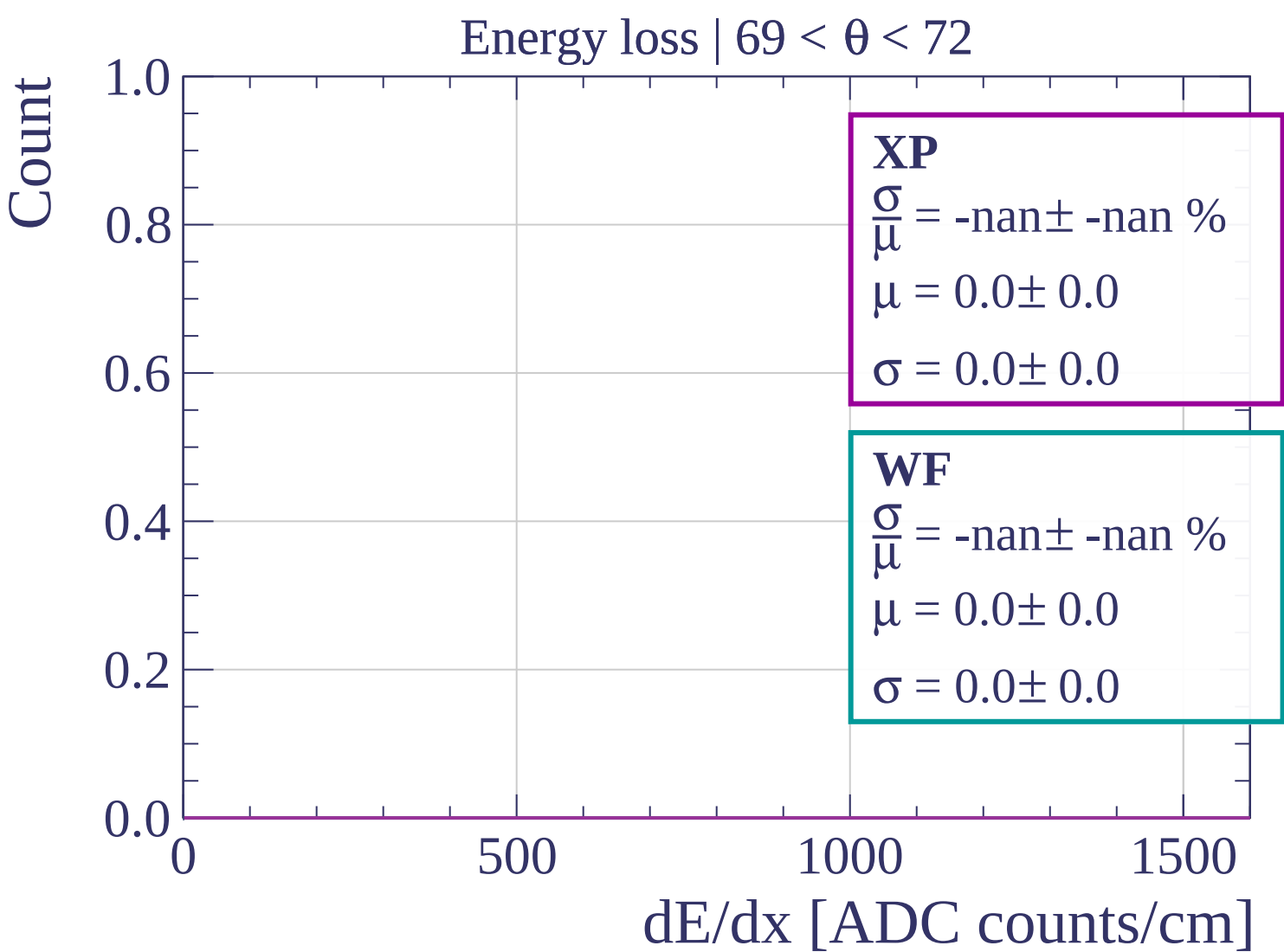
1000

1500

dE/dx [ADC counts/cm]







Count

Energy loss | $72 < \theta < 75$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

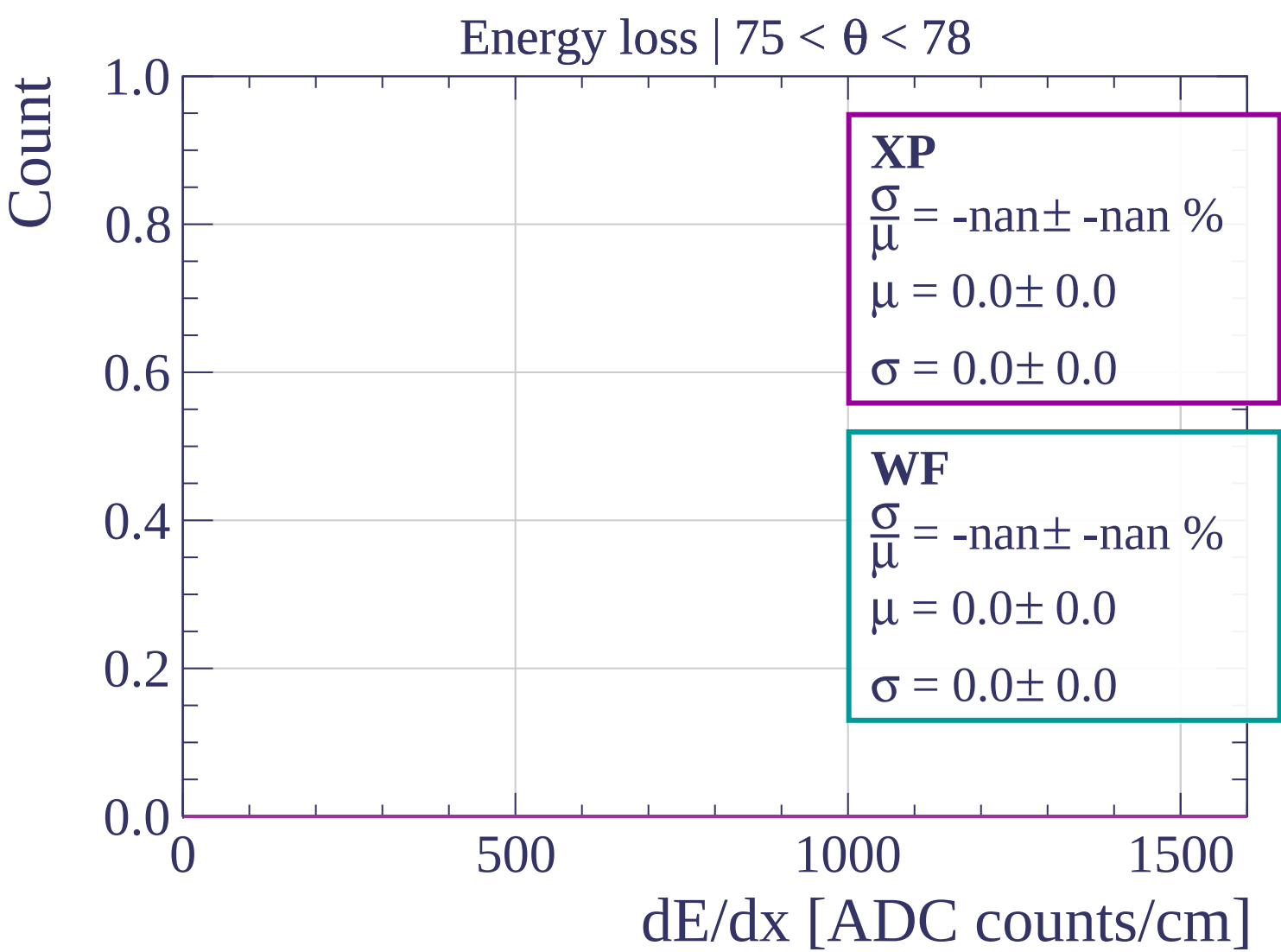
0

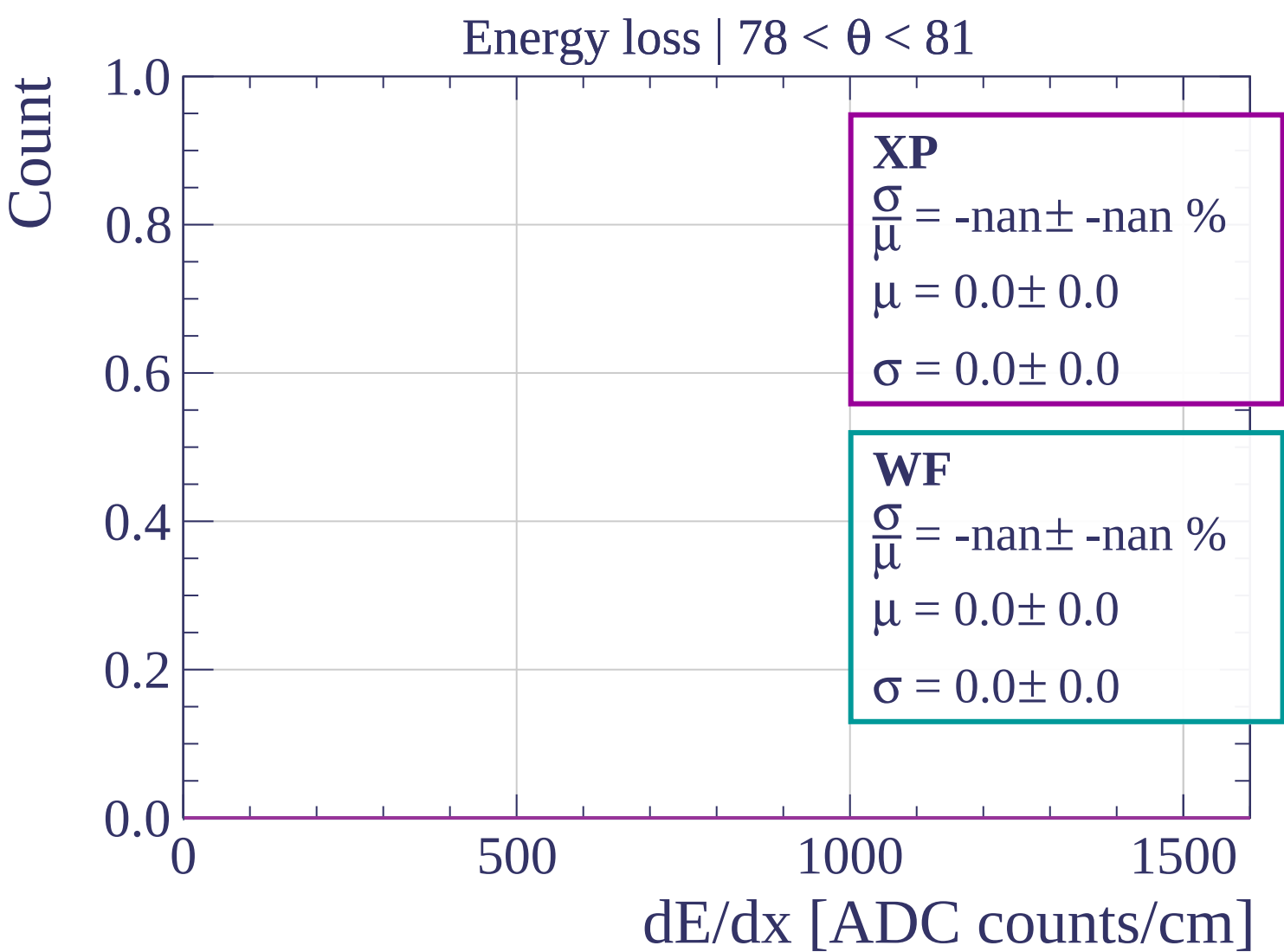
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $81 < \theta < 84$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

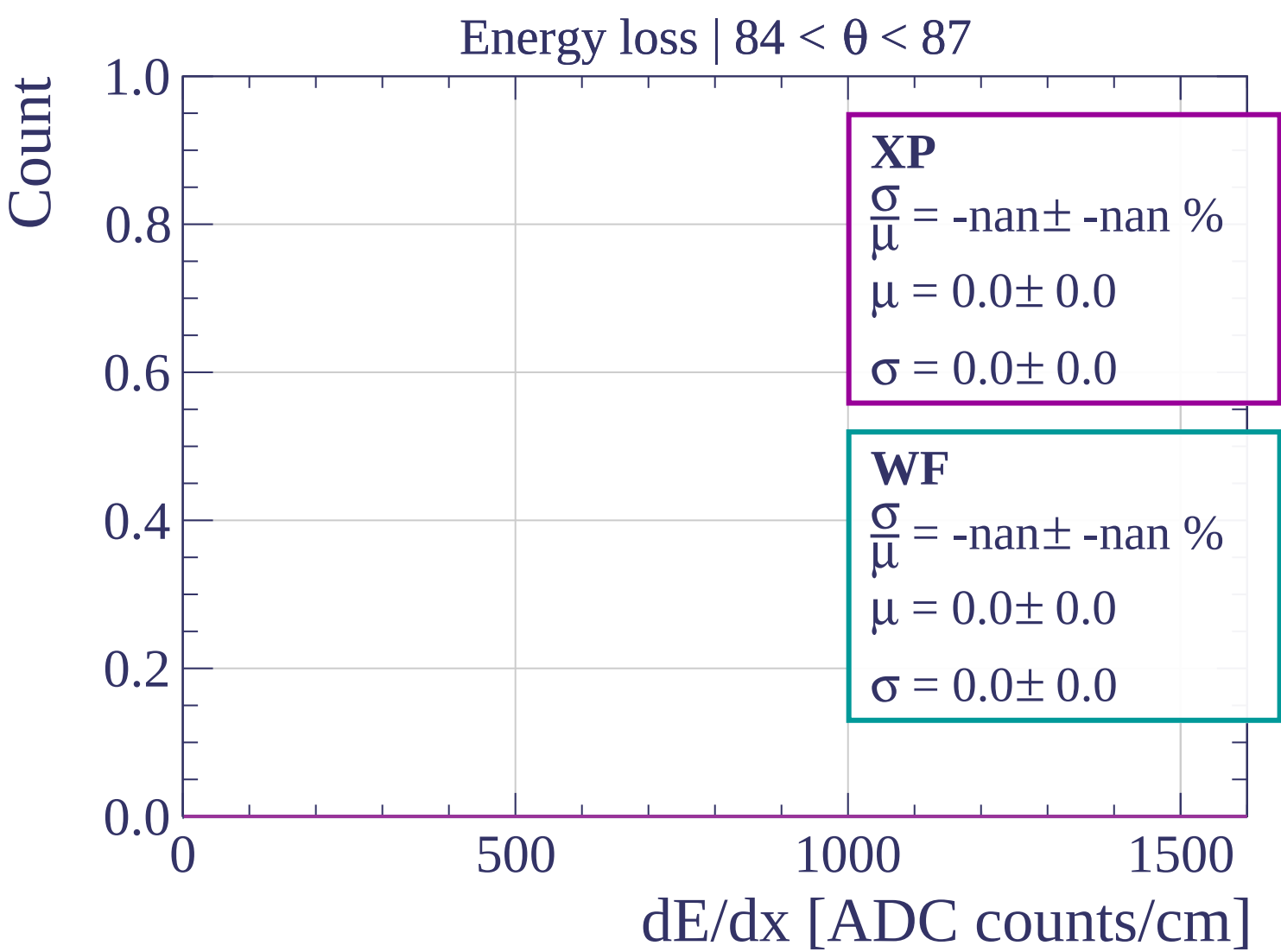
0

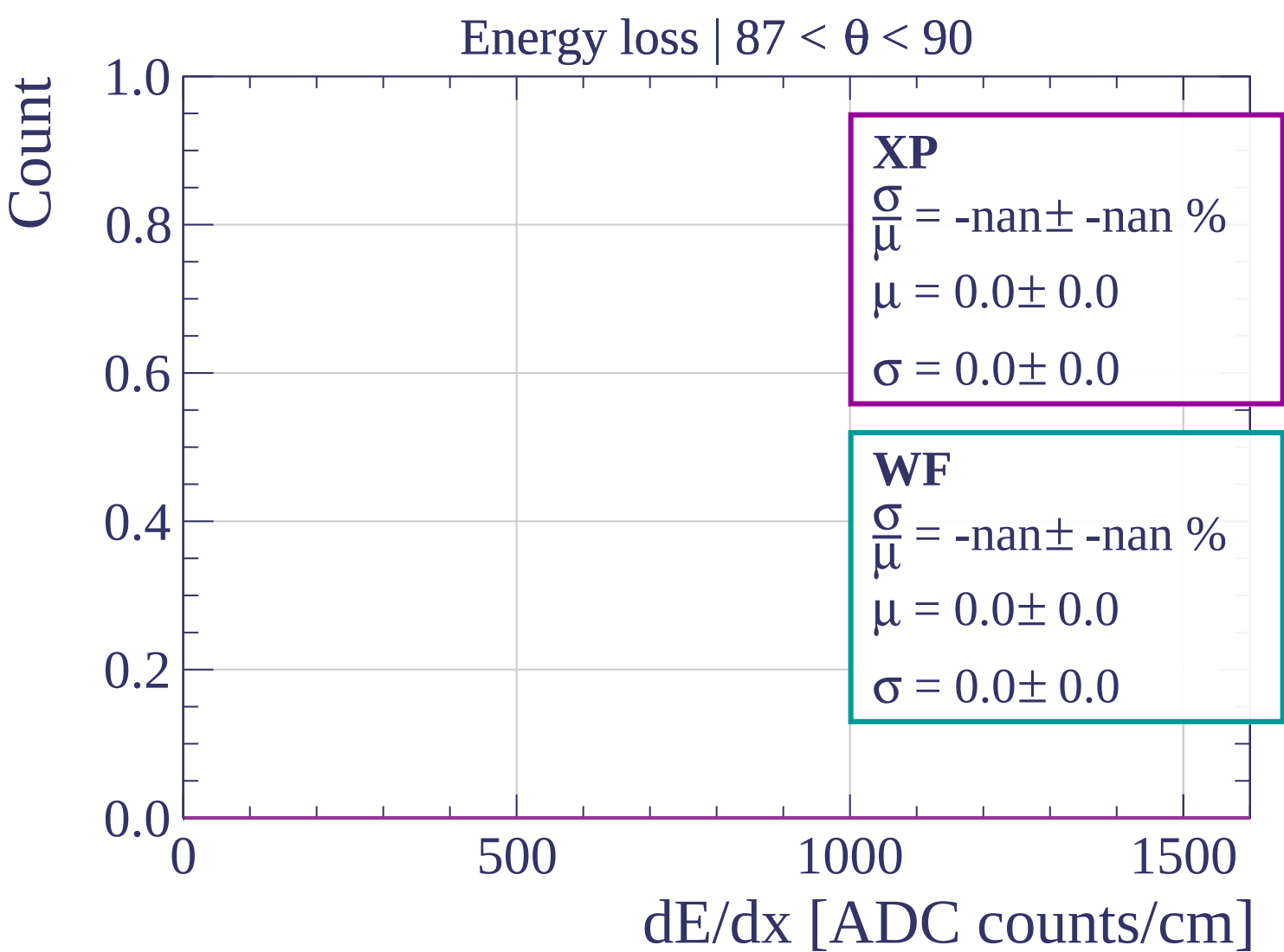
500

1000

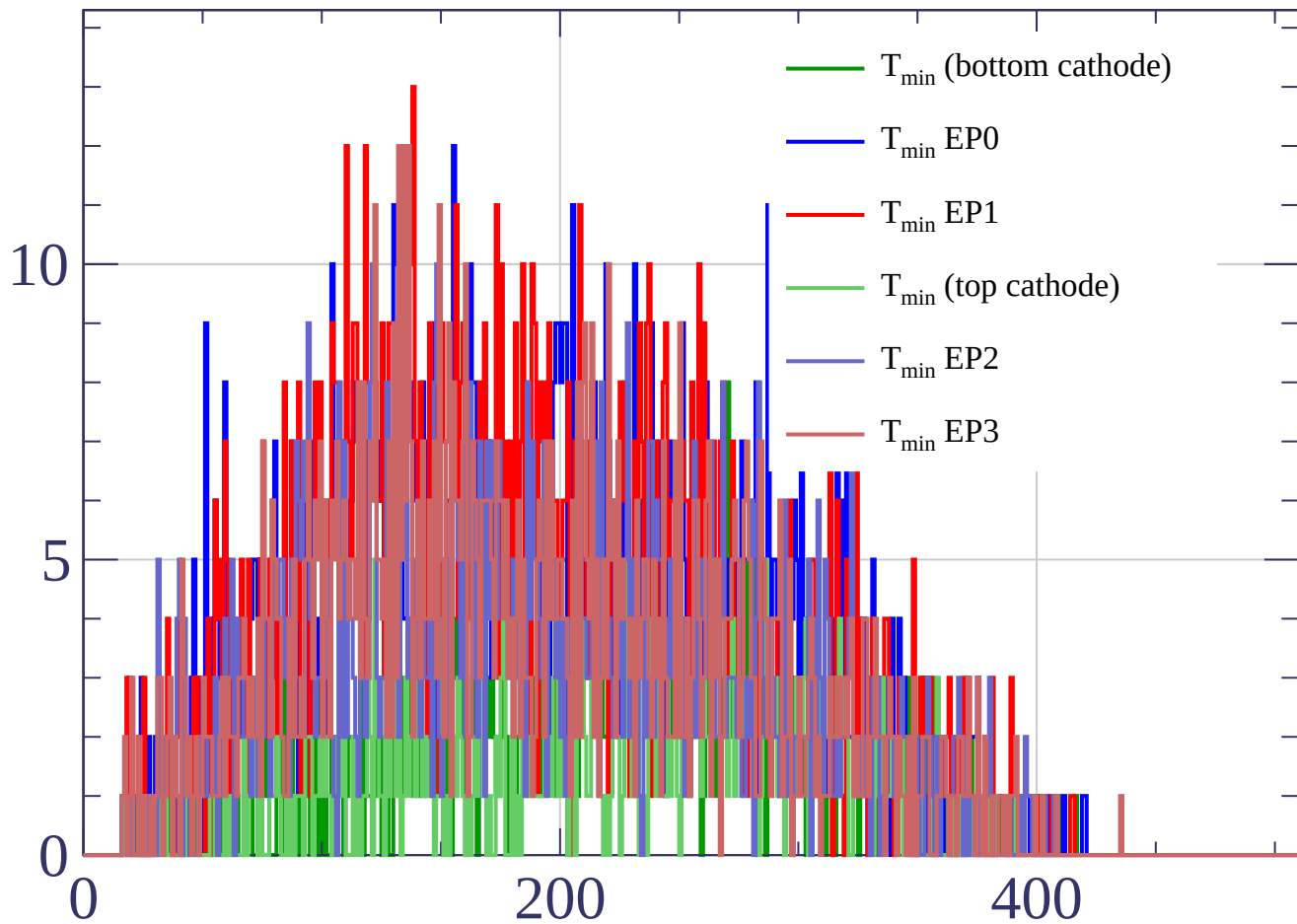
1500

dE/dx [ADC counts/cm]





Count



time bin

Count

